

ICSEECR 2026

INTERNATIONAL CONFERENCE ON SUSTAINABLE ENGINEERING, ENVIRONMENTAL INNOVATION & CLIMATE RESILIENCE

**ENGINEERING SMART AND RESILIENT SYSTEMS
FOR SUSTAINABLE DEVELOPMENT**

BOOK OF PROCEEDINGS

20–24 JULY 2026

**PRINCE DAPO ABIODUN LEADERSHIP AUDITORIUM
GATEWAY (ICT) POLYTECHNIC, SAAPADE, OGUN STATE, NIGERIA**

ORGANISED BY

**SCHOOL OF ENGINEERING & TECHNOLOGY • SCHOOL OF ENVIRONMENTAL STUDIES
SUSTAINABLE ENGINEERING • ENVIRONMENTAL INNOVATION • CLIMATE RESILIENCE**

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ICSEECR 2026

Gateway (ICT) Polytechnic, Saapade

SMART SYSTEMS	SUSTAINABLE ENGINEERING	CLIMATE RESILIENCE
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Book of Proceedings

International Conference on Sustainable Engineering, Environmental Innovation & Climate Resilience (ICSEECR 2026)

20–24 July 2026, Gateway (ICT) Polytechnic, Saapade, Ogun State, Nigeria

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Welcome Messages

Dr. Sanni Kehinde Oseni

Rector, Gateway (ICT) Polytechnic, Saapade

It is my great honour and privilege to warmly welcome you to the International Conference on Sustainable Engineering, Environmental Innovation and Climate Resilience (ICSEECR 2026). I am deeply delighted to join you in this remarkable gathering of distinguished scholars, researchers, industry professionals, policymakers, and development partners committed to advancing sustainable solutions for a resilient future.

This year's conference, themed “Engineering Smart and Resilient Systems for Sustainable Development,” comes at a time when nations across the world are confronted with unprecedented environmental, technological and socio-economic challenges. Issues like climate change, environmental degradation, rapid urbanisation, energy insecurity and resource depletion continue to demand innovative, collaborative and sustainable solutions. Engineering has always been a catalyst for human advancement; today, the expectation goes beyond developing infrastructure and technologies — we are called upon to create systems that are intelligent, adaptable, environmentally responsible and resilient enough to withstand present and future challenges.

As an institution committed to academic excellence, technological advancement and national development, we recognise the importance of research-driven innovation in shaping sustainable societies, and we remain dedicated to promoting interdisciplinary research, strengthening industry-academia partnerships and producing graduates equipped to solve real-world problems through innovation and entrepreneurship.

I wish to specially appreciate our keynote speakers, invited resource persons, sponsors, development partners, members of the organising committee and everyone whose commitment and hard work have made this conference a reality. It is therefore my honour and privilege to formally declare the 2026 International Conference on Sustainable Engineering, Environmental Innovation and Climate Resilience (ICSEECR 2026) open. Thank you for your attention, and I wish you all a productive, insightful and rewarding conference.

Dr. Sanni Kehinde Oseni, Rector, Gateway (ICT) Polytechnic, Saapade

Engr. Oyebola B. O.

Dean, School of Engineering & Technology, Gateway (ICT) Polytechnic, Saapade

It is my singular honour and pleasure to welcome you — distinguished scholars, industry leaders and delegates travelling from across the globe — to the International Conference on Sustainable Engineering, Environmental Innovation & Climate Resilience (ICSEECR 2026), proudly co-hosted by the School of Engineering & Technology, Gateway (ICT) Polytechnic, Saapade.

Sustainable engineering stands today as the defining discipline of our century. As the world confronts the accelerating realities of climate change, resource scarcity and rapid urbanisation, it falls to engineers to conceive the smarter, cleaner and more resilient systems the future demands — from renewable energy and intelligent grids to next-generation infrastructure and sustainable manufacturing. It is with immense pride that this School convenes such an exceptional community of researchers, students and industry professionals, whose work in the pages that follow exemplifies world-class scholarship and bold innovation in equal measure.

My sincere gratitude goes to our esteemed Rector, our Chief Guest of Honour, our distinguished Keynote Speakers and every author whose scholarship has made this gathering possible. I wish you all a truly superb, inspiring and world-class conference.

Engr. Oyebola B. O., Dean, School of Engineering & Technology, Gateway (ICT) Polytechnic, Saapade

Mr. Lawal A. W.

Dean, School of Environmental Studies, Gateway (ICT) Polytechnic, Saapade

On behalf of the School of Environmental Studies, it is my honour to welcome you to the International Conference on Sustainable Engineering, Environmental Innovation & Climate Resilience (ICSEECR 2026). The environment we share is under mounting pressure, from land degradation and pollution to the accelerating effects of climate change. Addressing these challenges demands the kind of interdisciplinary collaboration this conference was built to foster, bringing together environmental scientists, planners, engineers and policymakers around a shared commitment to sustainability.

The research gathered in this volume, spanning sustainable infrastructure, green building, water resources and pollution control, reflects the depth of scholarship our contributors bring to these questions. I am confident the days ahead will spark new ideas and partnerships that extend well beyond this conference.

I extend my sincere appreciation to the Organising Committee, our Rector, our esteemed guests and every participant joining us, and I wish you all a rewarding and memorable conference.

Mr. Lawal A. W., Dean, School of Environmental Studies, Gateway (ICT) Polytechnic, Saapade

EDITORIAL FOREWORD

On behalf of the Technical Programme Committee and the Local Organising Committee of the International Conference on Sustainable Engineering, Environmental Innovation & Climate Resilience (ICSEECR 2026), it is our honour to present this Book of Proceedings for the conference held 20–24 July 2026 at Gateway (ICT) Polytechnic, Saapade, Ogun State, Nigeria.

This volume assembles the full peer-reviewed papers accepted for the conference, following on from the Book of Abstracts published ahead of the conference. Every paper was subjected to a rigorous double-blind review process by members of the Technical Programme Committee, evaluated against criteria of originality, methodological soundness, relevance to the conference theme, and alignment with the United Nations Sustainable Development Goals (UN SDGs).

The contributions span the conference’s thematic schools — Engineering & Technology and Environmental Studies — reflecting the breadth of interdisciplinary research required to address the global climate and sustainability crisis. Authors represent academic institutions, research centres, and industry organisations across Nigeria and beyond.

Papers are arranged by thematic school and, within each school, in the order finalised for this edition. Formatting has been standardised throughout to the conference’s IEEE-based template so that the volume reads as a single, consistent Book of Proceedings regardless of each paper’s original submission format.

We extend our sincere appreciation to all contributing authors for their scholarly rigour, to the reviewers for their diligence and expertise, and to the administration of Gateway (ICT) Polytechnic Saapade for their institutional support and commitment to the advancement of sustainable engineering education and research in Nigeria.

The Technical Programme Committee

ICSEECR 2026 — 20–24 July 2026

Note: the views and findings expressed in the papers reproduced in this volume are those of the respective author(s) and do not necessarily reflect the position of Gateway (ICT) Polytechnic Saapade or the organising institutions of the conference.

CONFERENCE PROGRAMME AT A GLANCE

Prince Dapo Abiodun Leadership Auditorium, Gateway (ICT) Polytechnic, Saapade • 20–24 July 2026

DAY	MORNING	AFTERNOON
Mon 20 Jul	Arrival & Registration	Arrival & Registration
Tue 21 Jul	Opening Ceremony, Chief Guest & Keynote Addresses 1–4	Keynote Address 4, Industrial Lead Paper, Group Photograph
Wed 22 Jul	Keynote Addresses 5–7	Parallel Technical Sessions (T-01, T-03, T-06) & Poster Session
Thu 23 Jul	Parallel Technical Sessions continue	Closing Ceremony & Awards
Fri 24 Jul	Departure	Departure

Full session-by-session details are available on the conference website: <https://csetgaposa.github.io/icseecr2026/#program>

Conference Programme — Full Schedule

Prince Dapo Abiodun Leadership Auditorium, Gateway (ICT) Polytechnic, Saapade • 20–24 July 2026

Day 1 — Monday, 20 July: Arrival & Registration

Arrival, check-in and registration desk open at the hostel / secretariat. No paper presentations scheduled.

Day 2 — Tuesday, 21 July: Opening Ceremony & Keynote Session

Time	Session
08:00	Arrival & Registration — Secretariat Desk
10:30	Welcome & Opening Remarks — Dr. Sanni Kehinde Oseni, Rector, Gateway (ICT) Polytechnic
10:40	Chief Guest of Honour Remarks — Engr. Dr. Kadiri Kamoru Oluwatoyin, FNSE, Rector, The Federal Polytechnic, Offa
10:50	Keynote Address 1 — Prof. Emeritus Alireza Baghai-Wadji, University of Cape Town, South Africa
11:20	Keynote Address 2 — Prof. Y. L. Shuaib-Babata, University of Ilorin, Nigeria
11:50	Group Photograph / Break
12:00	Keynote Address 3 — Dr. Arc. Oluseyi Israel Abatan, Moshood Abiola Polytechnic, Nigeria
12:15	Keynote Address 4 — Dr. (Mrs.) Adedoyin Mary A., Lagos State University
12:45	Industrial Lead Paper Presentation — Engr. Felix Adewale Odeleye, Nuclear Power Construction Engineer, UK
13:15	Lunch Break
14:15	Parallel Technical Sessions Begin

Day 3 — Wednesday, 22 July

Time	Session
10:30	Keynote Address 5 — Dr. Rilwan Kayode Apalowo, Universiti Sains Malaysia
11:00	Keynote Address 6 — Dr. Emmanuel S. Babatunde, Achievers University, Owo
11:30	Keynote Address 7 — Dr. Adeyemi Olusola, York University, Canada
12:00	Selected Speaker — Adebayo Adeoti, Texas Tech University, Lubbock, Texas, USA
—	Parallel Technical Sessions and Poster Session continue thereafter

Day 4 — Thursday, 23 July: parallel technical sessions continue through the morning, followed by the Closing Ceremony and Awards in the afternoon. Day 5 — Friday, 24 July: delegate departure.

Full details of the paper presentation schedule for each session are available on the conference website:

<https://csetgaposa.github.io/icseecr2026/#program>

Conference Leadership

Role	Name / Affiliation
Chief Host	Dr. Sanni Kehinde Oseni — Rector, Gateway (ICT) Polytechnic, Saapade
Chief Guest of Honour	Engr. Dr. Kadiri Kamoru Oluwatoyin, FNSE — Rector, The Federal Polytechnic, Offa
Industrial Lead Paper Presenter	Engr. Felix Adewale Odeleye — Nuclear Power Construction Engineer, United Kingdom

The full roster of Keynote Speakers follows on the next page.

KEYNOTE SPEAKERS

Prof. Emeritus Alireza Baghai-Wadji

University of Cape Town, South Africa

Talk: “Self-regularized Method of Moments: Taming Infinities & Engineering Smart and Resilient Systems for Sustainable Development”

Prof. Y. L. Shuaib-Babata

University of Ilorin, Nigeria

Dr. Arc. Oluseyi Israel Abatan

Moshood Abiola Polytechnic, Nigeria

Talk: “Computational Engineering for Sustainable Engineering”

Dr.(Mrs) Adedoyin Mary A.

Lagos State University, Nigeria

Talk: “Africa’s Demographic Transition: Advancing Engineering”

Dr. Rilwan Kayode Apalowo

Universiti Sains Malaysia

Dr. Emmanuel S. Babatunde

Achievers University, Owo, Nigeria

Dr. Adeyemi Olusola

York University, Canada

Talk: “Building Green, Building Smart: AI and Our Climate”

Chhuon Kong, Ph.D.

Institute of Technology of Cambodia (ITC), Cambodia

CONFERENCE ORGANISING COMMITTEE

Local Organising Committee — ICSEECR 2026

ROLE	NAME
Chairman	Engr. Adeyemo, R. G.
Member	Engr. Oyebola, B. O.
Member	Mr. Lawal, A. W.
Member	Arc. Adetayo Adebayo, K.
Member	Mr. Kazeem, O. L.
Member	Engr. Oyerinde, S. A.
Member	Mr. Oyesola, W.
Member	ESV. Hassan Shehu, T.
Member	Engr. Oginni, O. F.
Member	Engr. Shokefun, M. O.
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Member	Mr. Taiwo, O. M.
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Member	Mr. Olayinka, J. T.
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Member	Engr. Eko, M. A.
Member	Engr. Aina, O. A.
Member	Builder Shoyemi, S.
Member	Surv. Fawole, E. O.

Local Organising Committee roster confirmed by the Secretariat. Contact: icseecr@gaposa.edu.ng • +234 805 401 2303 • +234 803 927 2829 • +234 803 382 4573.

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RFID-Based Access Authentication System for Electric Vehicle Operation

Odukale O. S., Aina O. A., Taiwo O. M., and Fadipe O. S.

Department of Computer Engineering Technology, Gateway (ICT) Polytechnic, Saapade, Nigeria

Abstract—The increasing adoption of electric vehicles (EVs) has intensified the need for secure, reliable, and low-cost access-control mechanisms that prevent unauthorised operation of vehicles and shared mobility assets. Conventional key-based and keypad systems remain vulnerable to duplication, tampering, and social-engineering attacks. This paper presents the design and implementation of a Radio-Frequency Identification (RFID)-based access authentication system for electric vehicle operation, built around an MFRC522 RFID reader interfaced with an Arduino/ESP32 microcontroller that governs a relay-based motor-enable circuit. Authorised RFID tags are cross-checked against a stored credential list before the propulsion circuit is armed, denying activation to unrecognised tags. The prototype was evaluated on campus and industrial electric-utility-vehicle case studies, where it reliably granted access to registered users within sub-second read times while rejecting unregistered credentials. The results demonstrate that low-cost RFID authentication can meaningfully strengthen EV access security without materially increasing bill-of-materials cost, offering a practical retrofit path for campuses, fleets, and shared-mobility operators seeking to curb unauthorised vehicle use.

Index Terms—RFID Authentication, Electric Vehicle Security, Access Control, Embedded Systems, RC522 RFID Reader, Contactless Authentication, Motor Enable System, Arduino, ESP32, Relay Control.

I. INTRODUCTION

The rapid growth of electric vehicle (EV) adoption has created an increasing demand for effective security systems that prevent unauthorized vehicle access while ensuring convenience for legitimate users. Conventional key-based ignition systems are susceptible to theft, key duplication, and unauthorized operation, making them inadequate for modern electric mobility applications. As a result, there is a need for intelligent and reliable authentication mechanisms that enhance vehicle security without significantly increasing system complexity or cost.

Radio Frequency Identification (RFID) technology has emerged as a practical solution for access control due to its contactless operation, fast response, low power consumption, and affordability. By using unique identification codes stored on RFID tags, only authorized users can gain access to the vehicle, thereby reducing the risk of theft and unauthorized use. RFID-based systems are widely employed in access control, asset tracking, and security applications because of their simplicity and reliability.

This paper presents the design and implementation of an RFID access authentication system for electric vehicle operation. The proposed system utilizes an RC522 RFID reader, an Arduino Uno microcontroller, and a relay-controlled ignition circuit to authenticate users before enabling vehicle operation. The developed prototype provides a low-cost, reliable, and scalable security solution suitable for campus electric vehicles and other light electric mobility platforms. Experimental results demonstrate that the system achieves fast authentication and effectively prevents

unauthorized access, highlighting its potential for enhancing the security of modern electric vehicles.

1. Background

The increasing use of electric vehicles (EVs) has created a growing need for improved security systems to prevent unauthorized access and vehicle theft. Conventional key-based ignition systems are vulnerable to key duplication and unauthorized use, making them less suitable for modern electric vehicles. Radio Frequency Identification (RFID) technology offers a secure, contactless, and cost-effective method of authenticating users before allowing vehicle operation. This study presents the design and implementation of an RFID-based access authentication system that enhances the security of electric vehicles by ensuring that only authorized users can activate the ignition system.

In this study, an RC522 RFID reader, Arduino Uno microcontroller, and relay module were integrated to develop a prototype capable of verifying registered RFID tags before enabling the vehicle's ignition circuit. The proposed system is simple, reliable, and affordable, making it suitable for campus electric vehicles and other light electric mobility platforms. It also provides a foundation for future enhancements, such as biometric authentication, GPS tracking, and IoT-based remote monitoring, to further improve vehicle security and management.

2. Structure

The remainder of this paper is organized as follows. Section II presents the related literature and reviews previous studies on RFID-based vehicle security systems. Section III describes the materials and methods, including the system design, hardware components, and software implementation. Section IV presents the experimental results and discusses the performance of the proposed system. Finally, Section V concludes the paper and provides recommendations for future research.

II. THEORETICAL FRAMEWORK AND LITERATURE REVIEW

A. Theoretical Framework

This study is based on the principles of Radio Frequency Identification (RFID) technology and embedded systems for secure access control. RFID is an automatic identification technology that uses radio waves to enable communication between an RFID tag and a reader without physical contact. Each RFID tag contains a unique identification (UID) that is verified by a microcontroller before granting access. In the proposed system, the Arduino Uno serves as the embedded controller, processing the UID received from the RC522 RFID reader and controlling the relay that enables or disables the electric vehicle ignition. This framework ensures that only authorized users can operate the vehicle, thereby improving security and reducing the risk of unauthorized access.

B. Literature Review

Several researchers have explored the application of RFID technology in vehicle security and access control systems. Finkenzeller [1] described RFID as a reliable and efficient automatic identification technology widely used in security, logistics, and access management due to its fast response and contactless operation. Want [2] highlighted the advantages of RFID over conventional identification methods, emphasizing its suitability for real-time authentication applications.

Kumar and Singh [3] developed an Arduino-based RFID vehicle access control system that successfully prevented unauthorized engine ignition using RFID authentication. Similarly, Hossain et al. [4] designed a secure RFID

authentication system for embedded applications and reported high authentication accuracy and low response time. Gupta and Sharma [5] reviewed IoT-based vehicle security systems and concluded that integrating RFID with embedded controllers provides an affordable and effective solution for enhancing vehicle security.

Although these studies demonstrate the effectiveness of RFID technology, many existing systems are designed for conventional vehicles or require complex and expensive hardware. This study addresses these limitations by developing a simple, low-cost RFID access authentication system specifically for electric vehicle operation, making it suitable for campus mobility and other light electric vehicle applications. The corresponding result is shown in Fig. 1.

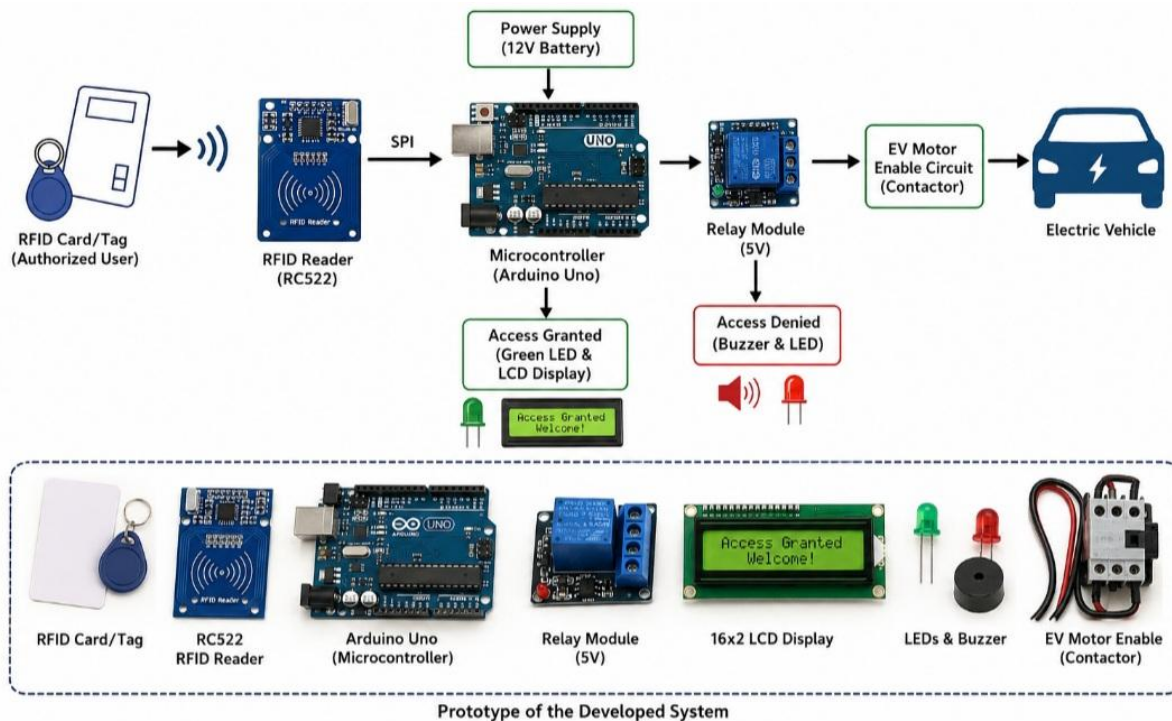


Fig. 1. Block Diagram of RFID-Based Access Authentication System for Electric Vehicle Operation

III. METHODOLOGY

A. Research Design

This study adopted an experimental design involving the design, construction, and testing of an RFID-based access authentication system for electric vehicle operation. The system was developed by integrating hardware and software components to authenticate users before enabling the vehicle ignition circuit. Performance evaluation was carried out to determine the system's accuracy, response time, and reliability.

B. System Design

The proposed system consists of an RC522 RFID reader, Arduino Uno microcontroller, RFID tags, relay module, buzzer, and a power supply. The RFID reader scans the unique identification (UID) of the RFID tag and sends it to the Arduino Uno for verification. If the UID matches an authorized user stored in the microcontroller's memory, the relay activates the vehicle's ignition circuit. Otherwise, access

is denied, and the buzzer is activated to indicate an unauthorized access attempt.

C. Hardware Components

The major hardware components used in this study include:

- Arduino Uno (ATmega328P)
- RC522 RFID Reader (13.56 MHz)
- MIFARE RFID Tags/Cards
- 5 V Relay Module
- Piezoelectric Buzzer
- LED Indicators
- 12 V Battery with 5 V Voltage Regulator

D. Software Development

The control program was developed using the Arduino Integrated Development Environment (IDE) with embedded C/C++ programming. The software initializes the RFID reader, reads the UID of each RFID tag, compares it with the stored

database of authorized users, and controls the relay and buzzer based on the authentication result.

E. System Operation

The system operates through the following steps:

- The system is powered on and initialized.
- The RFID reader scans the RFID card presented by the user.
- The Arduino Uno reads the card's UID.
- The UID is compared with the list of authorized users.
- If the UID is valid, the relay is activated to enable the vehicle ignition.
- If the UID is invalid, access is denied and the buzzer sounds.
- The system returns to standby mode, awaiting the next authentication request.

The relevant data are presented in Table 1.

TABLE 1. System Performance Results

Parameter	Result
Registered RFID Tags Tested	5
Successful Authentications	5 (100%)
Unregistered RFID Tags Tested	5
Unauthorized Access Rejected	5 (100%)
Average Authentication Time	0.42 s
System Reliability	Excellent

B. Analysis of Results

The test results demonstrate that the proposed RFID access authentication system provides a reliable and efficient means of securing electric vehicle operation. The 100% authentication accuracy for registered tags indicates that the system can consistently identify authorized users, while the complete rejection of unauthorized tags confirms its effectiveness in preventing unauthorized vehicle access.

The average authentication response time of 0.42 seconds shows that the system responds quickly, providing users with

F. Performance Evaluation

The developed prototype was evaluated by testing both authorized and unauthorized RFID cards. Performance was assessed based on authentication accuracy, response time, reliability, and the system's ability to prevent unauthorized vehicle operation. The experimental results were analyzed to determine the effectiveness of the proposed RFID access authentication system.

IV. RESULT AND ANALYSIS

A. System Performance

The developed RFID-based access authentication system was successfully implemented and tested under laboratory conditions. The prototype correctly identified all registered RFID tags and denied access to all unregistered tags. Upon successful authentication, the relay activated the vehicle ignition circuit, while unauthorized attempts triggered the buzzer and kept the ignition disabled. The system operated reliably throughout repeated test cycles without noticeable performance degradation.

a smooth and convenient experience. The use of the Arduino Uno and RC522 RFID reader contributed to the system's stable performance and low power consumption. Overall, the developed prototype achieved the study's objectives by providing a low-cost, secure, and practical access control solution for electric vehicles, making it suitable for applications such as campus mobility vehicles, electric carts, and other light electric transportation systems.

The corresponding result is shown in Fig. 2.



Fig. 2. Benefits of RFID-Based Access Authentication System for Electric Vehicle Operation

V. CASE STUDIES

To demonstrate the practical application of the proposed RFID access authentication system, two case studies were considered.

A. Case Study 1: Campus Electric Vehicle

A campus electric vehicle used for transporting students and staff was equipped with the RFID access authentication system. Registered users were issued RFID cards linked to the vehicle's control unit. During testing, only authorized users were able to activate the ignition, while unauthorized cards were rejected and the alarm was triggered. The system

improved vehicle security and reduced the risk of unauthorized use within the campus.

B. Case Study 2: Industrial Electric Utility Vehicle

The proposed system was also evaluated for use in an industrial environment where electric utility vehicles are operated by authorized personnel. Each operator was assigned a registered RFID card, ensuring that only trained staff could operate the vehicle. The system successfully restricted access to authorized operators, improving accountability, reducing the risk of misuse, and enhancing the overall security of the fleet. The corresponding result is shown in Fig. 3.



Fig. 3. Emerging Technologies in RFID Based Access Authentication System for Electric Vehicles

C. Summary

These case studies demonstrate that the RFID-based access authentication system is suitable for various electric vehicle applications, including educational institutions, industrial facilities, warehouses, and other environments where controlled vehicle access is required. Its low cost, ease of implementation, and reliable performance make it a practical solution for improving electric vehicle security.

VI. DISCUSSION

A. Analysis of Findings

The findings show that the proposed RFID-based access authentication system effectively improved the security of electric vehicle operation. The system achieved 100% authentication accuracy for registered RFID tags, successfully rejected unauthorized tags, and recorded an average response time of 0.42 seconds. These results demonstrate that the system is reliable, fast, and suitable for securing electric vehicles. Overall, the study confirms that RFID technology provides a simple, affordable, and effective solution for preventing unauthorized vehicle access.

B. Implications for Society

The proposed RFID-based access authentication system has significant societal benefits by improving the security of electric vehicles and reducing the risk of theft and unauthorized use. Its low cost and ease of implementation make it suitable for educational institutions, organizations, and small-scale electric vehicle operators. The system also

promotes safer and more efficient vehicle management, supporting the wider adoption of secure and sustainable electric mobility solutions.

C. Limitations and Challenges

The proposed RFID-based access authentication system has some limitations. Passive RFID tags can be lost, stolen, or cloned, which may compromise security if additional protection is not implemented. The system also relies on a stable power supply and has limited memory for storing authorized users on the Arduino Uno. Despite these challenges, the system provides a reliable and cost-effective solution for improving electric vehicle security, and its performance can be enhanced through future integration with biometric authentication, GPS tracking, and IIOT technologies.

D. Key Findings

The study found that the RFID-based access authentication system effectively enhanced the security of electric vehicle operation. The system achieved 100% authentication accuracy for authorized RFID tags, rejected all unauthorized access attempts, and recorded an average response time of 0.42 seconds. These findings demonstrate that the proposed system is reliable, fast, cost-effective, and suitable for improving access control in electric vehicles.

VII. CONCLUSION

A. Summary of Contributions

This study developed a low-cost RFID-based access authentication system for electric vehicle operation using an

RC522 RFID reader and an Arduino Uno microcontroller. The system enhances vehicle security by allowing only authorized users to access and operate the vehicle. It provides a simple, reliable, and scalable solution that contributes to the advancement of secure electric vehicle technologies and serves as a foundation for future integration with biometric authentication and IoT-based vehicle security systems.

B. Implications for Future Research

Future research should focus on enhancing the proposed system by integrating biometric authentication, encrypted RFID communication, GPS tracking, and Internet of Things (IoT)-based remote monitoring. Further studies may also evaluate the system under real-world operating conditions and investigate the use of cloud-based user management to improve security, scalability, and overall system performance.

VIII. RECOMMENDATIONS AND FUTURE RESEARCH

A. Recommendations

- Implement RFID-based access authentication systems in electric vehicles to improve security.
- Regularly update and manage the database of authorized RFID users.
- Conduct periodic maintenance and testing to ensure reliable system performance.
- Educate users on the proper handling and security of RFID access cards.

B. Future Research

- Integrate biometric authentication, such as fingerprint or facial recognition, to provide multi-factor security.
- Incorporate GPS tracking and IoT-based remote monitoring for enhanced vehicle management.
- Develop a cloud-based user management system to support large-scale deployments.
- Evaluate the system under real-world operating conditions to assess long-term performance, scalability, and reliability.

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Building Digital Creators: A Proposed Community-Based Model for Early Digital Literacy and Hands-On STEM Education

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Abstract—Digital literacy and STEM skills are essential for children and youth and not widely available in underserved communities. Digital competence in early education is not limited to using a digital device, but also covers communication, collaboration, content creation, safety, problem solving, criticality and taking part in digital culture. While strong in theory, current community-based programs are often not well developed in terms of programmatic progression, governance, and sustainability, and are not well integrated with formal education. The paper recommends a six-component Community Based Digital Creator Framework (CDCF) including community needs assessment, resource mobilization and learning infrastructure, integrated learning environment, structured mentorship and leadership development, innovation, and community engagement, and monitoring, evaluation and sustainability. The model is based on one person, an experienced learner, who mentors their younger counterparts, and then takes on a leadership role. The model is hands-on, project-based, and engages families to shift children from passive to creative, critical and responsible digital participation. The proposed model provides a scalable roadmap for participatory digital empowerment and early innovation in low-resource settings.

Keywords—Digital Literacy, STEM Skills, Community-Based Learning, Digital Competence, Project-Based Learning, Sustainability, Digital Creators.

I. INTRODUCTION

This increased demand for educational models has challenged the ability of children to be digitally competent from early childhood has grown even more with the digital transformation [1]–[3]. Contemporary models of digital competence do not only refer to the ability to use a device or devices, but to a more complex construct that encompasses information handling, communication and collaboration, digital content creation, safety, and problem solving [1], [4], [5]. Early childhood literature also suggests that children should be more than just consumers of digital technologies, but producers as well [1]. This agenda is not equitably implemented in practice. Comparative curriculum research reveals that digital competence is sometimes represented in policy but is not always reflected in clear and detailed goals, content and application guidance [4]. One way to fill this gap is to engage families, schools, libraries and other community spaces and learning environments in a community-based approach [6], [7].

Professional capacity is another important consideration. Previous studies on teacher digital competence and AI-related competencies suggest that teachers' knowledge, self-efficacy, and pedagogical readiness are factors that affect their capacity

to facilitate technology-rich learning [8], [9]. Digital and STEM learning outside the classroom is also shaped by family contexts - parental mediation and community participation can provide additional opportunities to engage children in digital and STEM learning [10], [11].

For this reason, the following community-based model, which integrates digital literacy at the outset with a progressive, hands-on approach to STEM learning, mentoring, family engagement and community-level sustainability, is proposed. The paper is concept-oriented and practice-inspired, rather than presenting an experiment to assess the proposed framework.

II. THE BODY OF LITERATURE AND GAPS, RELEVANT TO THE ANALYSIS OF THE STUDY

A number of pedagogical principles have been identified in the recent literature. Computational thinking can be reinforced in programming and computer science education by using a scaffolded and problem-based learning approach, and games can be used to create motivating learning environments to practice and provide feedback. The same holds true for robotics, as it enables collaborative, project-based and embodied learning. The results from the primary school STEM intervention suggest that low-cost robotics and self-made toy can be incorporated into the hands-on project to enhance the learning outcomes in STEM and attitudes [12].

Another important pathway is through maker-centered approaches. The opportunities for integrating STEM and STEAM through play, inquiry, design, and making are called early-childhood makerspace research [6]. Recent research also suggests that participation and confidence can be facilitated through makerspace experiences with the proper facilitation and tools for the learners [13]. The implications of these findings for community hubs are that shared space can lessen the need for learning to be dependent on individually owned equipment.

There also needs to be a critical and ethical component to a definition of digital literacy. A number of studies on critical digital literacy highlight the need for young people to not only use digital technologies but also to be provided opportunities to critically assess digital content, and to consider and reflect on social and ethical issues, including their own responsibilities to use digital technologies appropriately [14]. AI-literacy studies also call for multi-faceted learning approaches, involving conceptual, practical, critical and

social-ethical learning, apart from merely operational exposure [15].

Family engagement is another link between the formal and informal learning. An examination of young children's practices of home digital literacy reveals that there are multiple domains of play such as operational, literacy, socio-emotional, and STEM in which parents' mediation can foster home learning opportunities; however, because of children's technology use, there is a need for appropriate guidance for parents. The results highlight the importance of program intensity and continuity and illustrate how a community-based hybrid STEM family intervention can expand early STEM involvement through the collaboration of museums and schools.

A. Evidence Gaps

The literature in the source corpus is more robust in the formal school interventions, as opposed to sustained community implementation. The most evident need is a lack of evidence linking community-based delivery to a stream of progressive learning from digital use to creator-level practice. Community-based studies are available in both libraries and museums as well as local programs, but they are less prevalent than school-based intervention studies, and tend to focus on separate activities instead of sustainability and governance [7], [11].

Another gap is that digital literacy, maker learning, robotics, AI literacy, family engagement, and peer leadership are not well integrated with each other in one progression. The proposed framework answers this by grouping these elements into a series of stages: Entry, Builder, Creator, and captain.

The proposed model contributes in the following ways:

This paper's contribution is a single community-based design approach to move learners from basic digital familiarity toward creator level practice and leadership. The model does not view computer literacy, coding, robotics, and family engagement as distinct activities, but rather as interconnected parts of a continuous learning environment. Low cost and flexible resources, local mentorship, common infrastructure, and measurable progression are also key to the model.

III. PROPOSED COMMUNITY-BASED DIGITAL CREATOR FRAMEWORK

The Community Based Digital Creator Framework has six connected components that address the gaps identified. Community needs assessment identifies local priorities, available facilities, participant profile and access restrictions prior to implementation. Resource mobilization and learning infrastructure develops partnerships on the topic of affordable hardware, open resources and common space. The research around Makerspaces reinforces the importance of shared learning environments with facilitation for hands-on learning and confidence building [6], [13].

An integrated learning environment that moves from digital literacy, programming, robotics, media creation, AI literacy to project-based learning. This mirrors the evidence that digital competence is not purely technical [1], [4]; and recent AI-literacy research is in support of multi-dimensional and socially responsible learning [15], [16].

Structured mentorship and leadership development are realised through the Model where advanced learners mentor and support beginners in clubs, camps and community sessions. Peer support is not only a social but also a technical practice, team work, leadership and continuity in this design.

Learning is guided by innovation and community engagement to projects that are relevant to their social contexts. Monitoring, evaluation and sustainability offer tools to record participation, progress in skills, project completion, mentoring activity, family involvement and continuity. The framework thus addresses sustainability as a design requirement, and not a post-project issue.

Table 1 presents a links between the elements of the Community Based Digital Creator Framework and evidence.

TABLE 1. Links between the elements of the Community Based Digital Creator Framework and evidence

Component	Rationale	Example Support
Needs assessment	Digital competence is context-sensitive and age-dependent.	[1], [4]
Infrastructure	Low-cost, open, adaptable tools improve access and feasibility.	[12], [13]
Integrated learning	Creation, computational thinking, safety, and criticality are taught together.	[1], [14], [15]
Project Based Peer Learning	Peer mentoring supports leadership and continuity.	[8], [9]
Community engagement	Families, schools, libraries, and shared spaces broaden participation.	[6], [11], [13]
Evaluation and sustainability	Progression and continuity require explicit measures and governance.	[7], [15], [16]

IV. IMPLEMENTATION PRIORITIES

A. Phased Implementation

Implementing should be phased in. During the initial stages, focus needs to be placed on fun digital exploration, foundational operational skills, shared inquiry, and low floor creation experiences with Scratch, story-telling, video production, and simple robotics [1], [17], [18]. Doodlebot is an educational robot that offers a hands-on approach to learning about AI for young students, allowing them to program and interact with the robot in creative ways [19].

Structured programming, design thinking and maker projects should be introduced during the mid phase of learning. Research into design thinking and making in project-based learning has recently demonstrated its role in motivating and enhancing self-efficacy in primary education [20] and research in makerspaces has revealed the need for facilitation and confidence building [13].

In advanced phases, the addition of AI and community problem-solving and peer leadership can be incorporated, but with an explicit consideration of ethical, critical, and safety aspects [14]–[16], [21]. Inquiry, collaboration, ethics, and transdisciplinary engagement are highlighted as important aspects of meaningful incorporation of AI in recent STEM scholarship [16].

Family and community partnership must not be a "plus". Parent engagement is found to boost children's STEM and digital learning opportunities when there are concrete activities, take-home materials, workshops, and appropriate

scaffolding in the programs that support parent involvement [10], [11]. Concurrently, low intensity family interventions may result in attitude changes but not in child outcome changes, and continuity are factors to consider [11].

Table 2 presents a staged implementation pathway.

TABLE 2. Staged implementation pathway

Phase	Learner Focus	Delivery Mode	Likely Measures
Entry	Familiarity, confidence, safe use	Clubs, libraries, workshop	Participation, confidence
Builder	Coding, computational thinking, maker projects	Camps, school-community hubs	CT, motivation, collaboration
Creator	Media, robotics, AI, local solutions	Community labs, showcases	Products, leadership, agency
Captain	Peer mentoring, facilitation, outreach	Clubs, outreach teams	Retention, mentoring, sustainability

Accordingly, this study proposes the Community-Based Digital Creator Framework, which synthesizes the strengths of existing approaches while directly addressing the structural deficiencies identified in the literature. By integrating AI-enhanced learning, project-based pedagogy, educational robotics, physical computing, peer mentoring, multi-stakeholder governance, and continuous evaluation within a single implementation model, the framework seeks to provide a scalable and context-sensitive approach for strengthening digital literacy, innovation, and community development in underserved communities.

B. Proposed Community-Based Digital Creator Framework

The critical review of the literature demonstrates that although significant advances have been made in community-based digital literacy and hands-on STEM education, existing implementation models remain largely fragmented, technology-specific, and insufficiently integrated. Persistent challenges relating to learner progression, programme sustainability, governance, stakeholder collaboration, and contextual adaptation continue to limit the long-term effectiveness of many initiatives. These findings provide the basis for the development of the Community-Based Digital Creator Framework (CDCF), a conceptual model that integrates educational theory, emerging technologies, and community development principles into a unified implementation strategy.

Fig. 1 illustrates the conceptual relationships among AI integration, digital literacy, STEM education, and sustainable community development

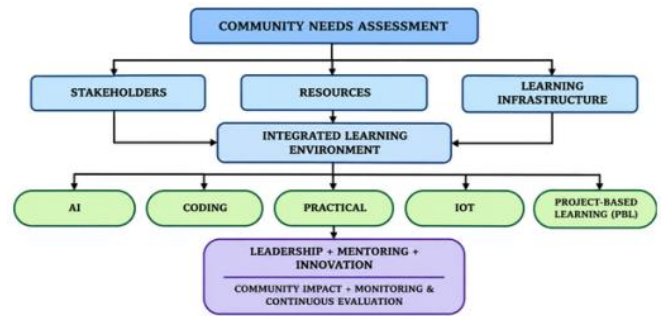


Fig. 1. Proposed Community-Based Digital Learning Framework for Building Future Digital Creators.

C. Guiding Design Principles

The CDCF is designed to move learners beyond basic digital literacy towards becoming digital creators capable of designing innovative solutions, addressing community challenges, and contributing to sustainable socioeconomic development. Rather than treating coding, robotics, AI, and digital literacy as separate interventions, the framework combines these elements into a structured learning ecosystem that promotes continuous learner progression, collaboration, and innovation.

Learner-Centered Progression: Learning is organized as a progressive pathway beginning with foundational digital literacy and advancing through computational thinking, programming, physical computing, AI applications, innovation, leadership, and entrepreneurship. Each stage builds upon previously acquired competencies, enabling learners to develop confidence and technical proficiency over time.

Experiential and Project-Based Learning: Consistent with Constructionism and Experiential Learning Theory, learners actively construct knowledge by developing practical solutions to authentic community problems. Project-based activities encourage creativity, collaboration, critical thinking, and innovation while strengthening conceptual understanding.

Technology Integration: The framework integrates Artificial Intelligence, educational robotics, coding, physical computing, Internet of Things (IoT) applications, and digital fabrication within a coherent learning environment. This interdisciplinary approach reflects current technological trends while preparing learners for emerging digital careers.

Community Partnership: Sustainable implementation requires collaboration among educational institutions, government agencies, industry partners, non-governmental organisations, parents, and local communities. The framework therefore adopts a multi-stakeholder model that promotes shared ownership, resource mobilisation, and long-term programme sustainability.

Scalability and Contextual Adaptability: Recognising the infrastructural realities of underserved communities, the framework supports flexible implementation using affordable hardware, open-source software, offline learning resources, and locally available facilities. This ensures that effective digital education remains achievable despite financial and technological constraints.

Continuous Monitoring and Improvement: Programme effectiveness should be evaluated through regular monitoring of learner outcomes, stakeholder participation, innovation outputs, and community impact. Feedback mechanisms enable continuous refinement of implementation strategies and support evidence-based decision-making.

D. Framework Components

The Community-Based Digital Creator Framework consists of six interconnected components that collectively support sustainable digital education.

The proposed framework is built upon six guiding principles derived from the literature review.

Fig. 2 presents the proposed implementation process.

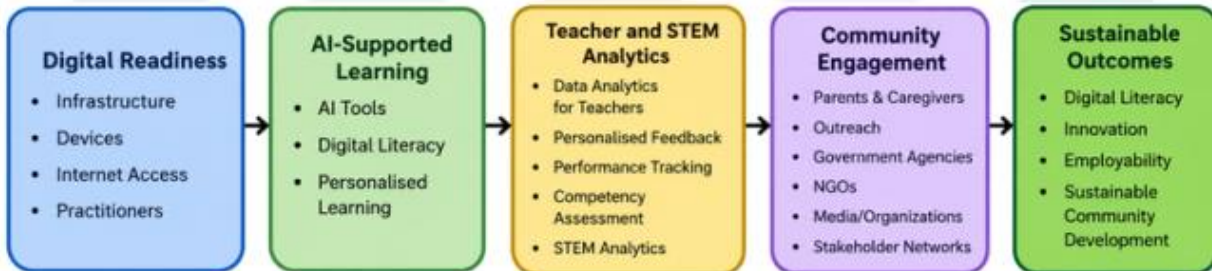


Fig. 2. Proposed AI-Integrated Digital Literacy Implementation Process.

Community Needs Assessment

Implementation begins with an assessment of local educational needs, available resources, stakeholder expectations, and community priorities. This stage ensures that programme objectives are aligned with local realities and development goals.

Resource Mobilization and Learning Infrastructure

Following the needs assessment, resources are mobilized through partnerships involving schools, higher education institutions, government agencies, industry organizations, NGOs, and community leaders. Required resources include computers, internet connectivity, educational robotics kits, programmable microcontrollers, AI learning platforms, renewable energy solutions where necessary, and suitable learning spaces.

Integrated Learning Environment

The learning environment combines digital literacy, programming, robotics, AI, physical computing, computational thinking, and project-based learning into a unified educational ecosystem. Rather than teaching these competencies independently, learners experience them as complementary components of authentic technology development.

Structured Mentorship and Leadership Development

A distinguishing feature of the framework is the integration of structured peer mentoring through the Digital Captain Model. Experienced learners mentor junior participants, facilitate collaborative learning, and gradually assume leadership responsibilities. This approach strengthens programme sustainability, reduces dependence on external facilitators, and develops leadership competencies alongside technical skills.

Innovation and Community Engagement

Learners are encouraged to develop projects that address real societal challenges such as environmental sustainability, healthcare, agriculture, education, and community safety. Innovation showcases, hackathons, exhibitions, and community demonstrations provide opportunities for learners

to present their work while receiving constructive feedback from educators, industry professionals, and community stakeholders.

Monitoring, Evaluation, and Sustainability

Continuous monitoring and evaluation enable programme managers to assess learner progression, stakeholder engagement, innovation outcomes, and overall community impact. Findings inform programme improvement, resource allocation, and future expansion while ensuring accountability and long-term sustainability.

E. Expected Contributions of the Framework

The proposed framework contributes to educational research and practice in several important ways.

First, it provides a holistic implementation model that integrates digital literacy, AI, robotics, physical computing, computational thinking, and project-based learning within a single educational ecosystem.

Second, it introduces a structured learner progression pathway, enabling participants to advance systematically from foundational digital skills to innovation, leadership, and entrepreneurship rather than limiting learning to isolated technical competencies.

Third, the framework formalises peer mentoring through the Digital Captain Model, creating opportunities for leadership development while improving programme scalability and community ownership.

Fourth, its emphasis on multi-stakeholder governance promotes stronger collaboration among schools, universities, government agencies, industry partners, NGOs, and local communities, thereby improving programme sustainability and resource mobilisation.

Finally, by incorporating offline-capable technologies, affordable hardware platforms, and flexible implementation strategies, the framework addresses many of the infrastructural constraints commonly experienced in underserved communities.

V. DISCUSSION

The originality of the Community-Based Digital Creator Framework lies not in introducing entirely new educational technologies but in its systematic integration of proven pedagogical approaches, emerging technologies, and community development strategies within a unified implementation model.

Unlike many existing frameworks that focus primarily on individual technologies or isolated educational interventions, the CDCF combines:

- Artificial Intelligence—enhanced learning
- Educational robotics and physical computing
- Project-based and experiential learning
- Structured learner progression
- Peer mentoring and leadership development
- Multi-stakeholder governance
- Continuous monitoring and evaluation
- Context-sensitive implementation for underserved communities

This integrated perspective distinguishes the framework from existing implementation models and provides a practical foundation for sustainable community-based digital education.

The framework therefore serves not only as a conceptual contribution to the literature but also as an implementation guide that can inform educators, policymakers, researchers, and development organisations seeking to strengthen digital literacy and innovation ecosystems within resource-constrained environments.

The framework as proposed, does not assume that the community learning process replaces formal education but that it is an adjunct to it. It's about linking the resources and people who are not always available in the same place: the resources at the school level provide continuity of curriculum; the resources at home provide reinforcement; the resources at the community level provide shared tools and flexible schedules; the mentor provides guidance; and the learner provides the project and peer support.

The design also views progression as a key element. A learner who starts out with knowledge of the device shouldn't stay in the consumption phase. Rather, learning is achieved through building, creating, and later on mentoring. This sequence corresponds to the evidence of digital competence, which has four elements: creation, critical evaluation, communication and responsible participation [1] [4] [5].

The community aspect is more significant in low-resource settings. This may make practical STEM learning more feasible in situations where equipment is limited, by providing shared makerspaces and low-cost robotics [12], [13]. The educational robotics evidence also suggests the potential of robotics to help develop STEM competence through meaningful pedagogical designs that integrate the robotics into them, rather than using it as a technology demonstration [18].

With AI, it should be done the same way. Using AI is not just about how to operate a tool. A range of concepts on the operation of AI systems, what they are capable of, what they are not capable of, and how safety, identity, bias, privacy and responsible use of AI relate to their use in day-to-day

applications should be understood, appropriately, by learners [14]–[16].

VI. PRACTICAL IMPLICATIONS, FUTURE RESEARCH

Practical Implications

The proposed Community-Based Digital Creator Framework (CDCF) offers practical value for multiple stakeholders involved in digital literacy, STEM education, and community development. By integrating educational theory, emerging technologies, and community partnerships within a single implementation model, the framework provides a structured approach for strengthening digital education in underserved communities.

Since the framework is suggested and not experimentally proven, a plan to evaluate the implementation should be included. The suggested indicators are as follows: Attendance and retention of learners; Digital competence baseline and post program; Computing-Thinking tasks; Number and quality of completed projects; Robotics/coding performance; AI-literacy understanding; Peer-mentoring activity; Family participation; Equipment — utilization; Facilitator development; and continuation of community activity after each program cycle.

Educational Implications

For educators and academic institutions, the framework provides a structured roadmap for designing learner-centred digital education programmes that extend beyond traditional computer literacy. The integration of computational thinking, project-based learning, educational robotics, AI, and physical computing encourages active learning while promoting creativity, collaboration, and problem-solving skills.

The learner progression pathway embedded within the framework enables educational institutions to organise training programmes that gradually advance learners from foundational digital skills to innovation, leadership, and entrepreneurship. This progression supports lifelong learning while improving learner retention and motivation.

Policy Implications

For policymakers, the framework provides evidence-informed guidance for developing sustainable digital inclusion strategies. Rather than supporting isolated technology interventions, government agencies can adopt integrated policies that promote collaboration among schools, higher education institutions, industry partners, community organisations, and development agencies.

The governance component of the framework also supports improved accountability, resource mobilisation, and long-term programme sustainability. Such coordinated implementation strategies align with national digital transformation agendas and the United Nations Sustainable Development Goals (SDGs), particularly SDG 4 (Quality Education), SDG 8 (Decent Work and Economic Growth), SDG 9 (Industry, Innovation and Infrastructure), and SDG 17 (Partnerships for the Goals).

Community and Industry Implications

The framework encourages stronger collaboration between communities and industry by creating opportunities for

mentorship, internship placement, innovation competitions, and technology-driven entrepreneurship. Community members become active participants in identifying local challenges, while learners develop practical solutions that address societal needs.

Evaluation should be a mix of quantitative and qualitative data. Progress can be captured through short skills assessments and project portfolios, observations, learner reflection, mentor logs, and parent feedback can capture aspects of skills which are not seen in test scores. This mixed evidence is also well suited for a model with a creator orientation as the desired outcomes encompass agency, collaboration, leadership, responsible participation, as well as technical proficiency.

Industry partners benefit through early identification of emerging talent, workforce development, and collaborative innovation. Community ownership further enhances programme sustainability by encouraging local participation, resource sharing, and continuous improvement.

Research Implications

From a research perspective, the framework contributes an integrated conceptual model that combines educational theories, technological innovation, and community development principles within a unified implementation strategy. It provides a foundation for future empirical investigations examining learner progression, programme effectiveness, stakeholder collaboration, and long-term community impact.

The framework may also serve as a reference model for comparative studies across different educational contexts and developing regions.

VII. RECOMMENDATIONS

Although the proposed framework is grounded in contemporary literature and established educational theories, further empirical investigation is necessary to validate its effectiveness across diverse contexts.

Future studies should consider:

- Pilot implementation of the Community-Based Digital Creator Framework within rural and urban communities.
- Longitudinal studies examining learner progression, leadership development, and retention.
- Comparative evaluation of community-based digital literacy programmes across different socioeconomic settings.
- Investigation of AI-supported mentoring and adaptive learning within community STEM education.
- Assessment of the framework's impact on innovation, entrepreneurship, and employability.
- Development of quantitative indicators for evaluating community digital transformation and programme sustainability.

These research directions will strengthen the evidence base supporting community-based digital education while informing future policy and implementation strategies.

VIII. CONCLUSION

This paper suggests a Community-Based Digital Creator Framework (CDCF) as an integrated conceptual model for

sustainable community-based digital education and to bridge early digital literacy with practical STEM education in low-resource and underserved environments. The framework brings together needs assessment, shared infrastructure, integrated learning environment, structured mentorship, community engagement, monitoring and sustainability. It has model that can be used to transform high-achieving students into peer mentors and community Digital Captains.

The literature demonstrates the potential for linking five of the elements (digital competence, maker-centered learning, robotics, AI literacy, family engagement, and community delivery); however, evidence is not developed regarding sustained progression from early digital use to creator-level practice in community settings. The proposed framework addresses that gap by outlining a progression path for players from Entry through Builder, Creator, and to Captain. The model should be tested in other communities, and the impacts of the model on the learner and the family should be studied over time. Cost, scalability, equity and sustainability of the model should be considered in future work.

Overall, the proposed framework contributes to both theory and practice by providing educators, policymakers, researchers, and development organisations with a structured roadmap for implementing sustainable community-based digital literacy programmes. It also establishes a conceptual foundation for future empirical validation and adaptation across diverse educational and community settings.

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Maximizing Capital Recovery and Carbon Avoidance of Institutional E-Waste Repositories under Administrative Friction

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Abstract—Higher-education institutions manage massive information and communication technology (ICT) footprints, yet bureaucratic workflows frequently trap obsolete assets in storage nodes, leading to severe technological depreciation and heightened carbon footprints. This paper introduces a multi-objective stochastic simulation framework, developed in Python/Pandas, to model e-waste lifecycle mechanics across a campus of 21,000 active users (20,000 students; 1,000 staff) distributed over eight distinct functional nodes. Integrating literature Material Flow Analysis (MFA) matrices with real-time 2026 metal spot prices, the model applies an exponential value-decay function (0.18% daily technical obsolescence) and a non-linear logistics emission modifier to quantify structural resource blockages. Running 600 discrete asset-turnover transactions yielding over 11,000 individual hardware assets, a one-way Analysis of Variance (ANOVA) and post-hoc Tukey Honestly Significant Difference (HSD) suite statistically validated spatial variances in storage stagnation. The simulation engine reveals hundreds of thousands of dollars in hidden commodity wealth blocked by administrative bottlenecks, directly showing that compressing administrative-friction coefficients simultaneously maximises institutional net realised wealth and mitigates logistical carbon expansion.

Index Terms—Electronic Waste (E-Waste); Administrative Friction; Capital Recovery; Technical Obsolescence; Logistical Carbon Footprint.

1. Introduction

Higher institution campuses operate as small-scale smart cities, driving continuous hardware procurement to maintain educational, administrative, and research capacities. However, this high rate of technology adoption creates a significant challenge at the end of the equipment's useful life: the accumulation of institutional electronic waste (e-waste). While consumer e-waste behavior is driven by individual choice, institutional e-waste pathways are dictated by structural regulations, inventory auditing protocols, and multi-tier sign-off workflows [1]. In many higher education institutions, particularly within developing administrative ecosystems, obsolete equipment does not enter recycling pipelines immediately. Instead, it enters a prolonged state of administrative storage. This storage period is not benign. It represents an operational bottleneck where economic and environmental value are lost. That is the rapid loss of financial value due to technological obsolescence, alongside the degradation of recoverable precious metals inside the equipment; and an increase in carbon emissions due to delayed, uncoordinated logistics, which misses the opportunity to displace raw mining through timely "urban mining" [2]. Prior research in Material Flow Analysis (MFA) has successfully quantified global electronic waste streams, yet these models often treat institutional storage timelines as a static variable.

They fail to capture the erratic, department-specific procedural delays that define real-world campus operations [3]. This paper addresses this research gap by introducing a Multi-Objective Stochastic Optimization Simulator. Built natively within a Python data science pipeline, this model simulates an institutional population of 21,000 users across eight distinct campus nodes. By modeling administrative delays through Weibull and exponential probability distributions, this study establishes a clear cause-and-effect link: reducing administrative friction directly reduces storage times, which maximizes recovered wealth and prevents valuable digital carbon assets from rotting in campus warehouses.

I. RELATED WORK

The management of e-waste within large institutions has evolved into a multi-disciplinary challenge crossing operations research, industrial ecology, and public policy. This section contextualizes institutional e-waste tracking across local, regional, and international research vectors.

A. Urban Mining and Institutional Realities

In Nigeria, e-waste research has historically prioritized informal recycling hubs in metropolitan centers such as Lagos and Ibadan [4]. However, recent local scholarship has turned sharply toward quantifying the immense secondary resource value or "urban mine" trapped within corporate and public institutions. Babayemi [5] conducted extensive material flow profiling, establishing that institutional storage blocks function as significant resource reservoirs, frequently delaying high-grade gold, silver, and copper recovery. Furthermore, Nduka and Olusegun [6] evaluated asset management systems within public institutions in Nigeria and concluded that an "awareness-practice gap" is highly prevalent. While administrative officers show high awareness of e-waste hazards, actual systematic disposal is crippled by an absence of automated asset retirement guidelines, directly corroborating the need for custom institutional simulations.

B. Regional Policy and Logistics Bottlenecks

Across the West African sub-region, infrastructure limitations and policy gaps frequently introduce long timelines in reverse supply chains. In Ghana, research surrounding the Agboghloshie lifecycle framework shifted focus toward formalizing institutional collection networks. Amankwah-Amoah and Debrah [7] highlighted that structural bottlenecks, ranging from rigid public procurement deletion acts to complex multi-tier auditing systems, force long inventory storage times. Similarly, investigations by Touré [8] in Senegal and Côte d'Ivoire documented that sub-regional academic clusters maintain significant quantities of obsolete

ICT assets because disposal permissions are tethered to sluggish centralized ministries. These regional studies emphasize that time delays are not merely random operational lags but are rooted in administrative architecture.

C. Stochastic Modeling and Material Flow Analysis

Globally, industrial ecologists rely on Material Flow Analysis (MFA) to track the velocity and content of digital materials [9]. International operations research has successfully embraced stochastic simulation to tackle the unpredictable nature of reverse logistics. Althaf et al. [1] modeled institutional lifecycle flows using predictive component matrices, showing that ICT components evolve dramatically across dynamic turnover lifecycles. To model these uncertainties, researchers like Parajuly [10] and Goti [11] introduced multi-objective optimization structures. Their models treat storage variables through non-linear probability distributions rather than static metrics, validating the use of Weibull and exponential curves to isolate value decay and carbon expansion. This paper connects these advanced global simulation methodologies directly to the administrative structures characterizing West African higher education networks.

II. METHODOLOGY AND EXPERIMENTAL ARCHITECTURE

A. Environmental Directory and Cloud Infrastructure

To ensure absolute data integrity and reproducibility across a multi-month testing pipeline, the simulation environment was deployed inside a containerized Linux virtual machine via Google Colab, securely connected to a persistent cloud storage directory using the absolute pathing protocol defined in Equation 1. The complete methodology and experimental architecture supporting the reproducibility of this study can be accessed at: <https://tinyurl.com/2xv8m8ca>

PROJECT_DIR = /content/drive/MyDrive/E.Waste.Research.Project(1)

B. Material Flow Analysis (MFA) Composition Index

Before simulating campus disposal patterns, a static metadata index was constructed to act as the baseline chemical and structural matrix for the hardware. Four major categories of ICT infrastructure were mapped using mass fractions for base structural metals (copper, aluminum) and milligram-to-gram ratios for rare geological assets (gold, silver), derived from established literature parameters [1], [5], [9] and presented in Equation 2.

DataFrame Matrix (df_{mfa}) = $f(H_{type} \rightarrow [Cu\%, Al\%, Au_{g/kg}, Ag_{g/kg}])(2)$

C. Stochastic Intake and Population Scaling Engine

The campus ecosystem was calibrated to represent a major higher institution center comprising 20,000 students and 1,000 academic and support staff (21,000) total active users). The generation engine simulated 600 discrete multi-unit asset turnover batches distributed stochastically across eight specific campus tracking nodes (Equation 3).

Campus Infrastructure Nodes ($H_{location}$) $\in$ $\begin{cases} \text{Central Library, Computer based test centers,} \\ \text{Computer Science labs, Engineering Research Laboratories,} \\ \text{Administrative offices, Academic staff offices,} \\ \text{Non Academic staff offices, Student Union Drop-Box Hub} \end{cases} (3)$

To mirror real-world operational flows, the simulation applies non-linear quantity generation rules based on the destination node. For example, high-density computing

clusters like Computer Based Test (CBT) centers and Computer Science labs generate large-volume workstation arrays, whereas administrative centers generate low-volume, high-mass server configurations.

D. Mathematical Delay and Administrative Friction Formulation

The core research gap, administrative friction (af), was modeled using probability distributions to dictate the storage stagnation time variable (tdelay) [7], [10]. The Optimized Pipeline Pathway (No Audit Block) is governed by an Exponential Distribution representing a fast-track, automated disposition loop, which models the rapid clearance of decentralized assets that bypass rigorous inventory reviews (Equation 4).

$$t_{delay} \sim \text{Exponential}(\lambda = 5) + 2 \text{ days} (4)$$

As presented in Equation 5, the Friction-Stagnated Pathway (Audit/Bureaucratic Block) is governed by a heavy-tailed Weibull Distribution, modeling long-term warehouse stagnation caused by inventory auditing backlogs across highly regulated institutional nodes.

$$t_{delay} \sim \text{Weibull}(a = 1.5) \times 160 + 75 \text{ days} (5)$$

Sum of delay across all node is presented as :

$$\sum_{i=1}^{82} t_{delay,i} = 13,465.22 \text{ total days}$$

For n = 82 logged batches in Administrative offices, the simulation summed the generated Weibull storage days:

Divide by Sample Size (n = 82):

$$\bar{t}_{delay} = \frac{13,465.22 \text{ days}}{82 \text{ batches}} = 164.21 \text{ days}$$

III. MATHEMATICAL OPTIMIZATION AND VALUE-DECAY FUNCTIONS

The analytics engine joins the separate structural composition and intake ledger tables to run a multi-objective calculation loop for each batch.

A. Financial Wealth Modeling

The Gross Commodity Wealth (Wgross) is established using global commodity spot market indexes adjusted for the year 2026 [12] is presented in Equation 6.

$$W_{gross} = (M_{Cu} \times P_{Cu}) + (M_{Al} \times P_{Al}) + (M_{Au} \times P_{Au}) + (M_{Ag} \times P_{Ag}) (6)$$

Where M represents component mass. P represents the active spot value index as presented in Equation 7.

$$\text{Spot Price Base } (P_{2026}) \rightarrow \begin{cases} P_{Au} = \$84.50 / \text{gram} \\ P_{Ag} = \$0.96 / \text{gram} \\ P_{Cu} = \$9.20 / \text{kg} \\ P_{Al} = \$2.45 / \text{kg} \end{cases} (7)$$

To compute Net Present Realized Wealth (Wnet), an exponential market depreciation decay model is applied, capturing technical obsolescence and part degradation at a rate of $\delta = 0.0018$ (0.18%) per day spent in storage [2] presented in Equation 8.

$$W_{net} = W_{gross} \times (1 - \delta)^{t_{delay}} (8)$$

The total Capital Destroyed (Capital_{lost}) by bureaucratic delay is thus isolated as presented in Equation 9.

$$Capital_{lost} = W_{gross} - W_{net}(9)$$

B. Environmental Footprint Modeling

When inventory sits stagnant, it disrupts logistics, forcing rushed transport configurations and fragmented material handoffs. The model captures this carbon inflation by multiplying the baseline shipping emissions footprint (0.00015 Tons CO_{2e} per kg) by a non-linear friction modifier linked to storage time (t_{delay}) [11] presented in Equation 10.

$$\text{Total Emitted CO}_2 \text{ (Tons)} = M_{total} \times 0.00015 \times (1 + (t_{delay} \times 0.002))(10)$$

IV. RESULTS

The relevant data are presented in Table 1.

The simulation successfully logged 11,843 individual ICT devices across the 600 tracked batches (a node with 160 – 200 bulk user per year for 3 – 4 years operational window of e-waste), representing 16.48 Metric Tons of physical material flow. Table 1 aggregates simulation metrics using the updated specific_campus_nodes schema. It clearly breaks down the sample sizes (N), mean delays, total unit volumes, physical mass, and the resulting multi-objective impacts, showing exactly where capital wealth is destroyed and where logistics carbon emissions inflate.

TABLE 1. Institutional E-Waste Footprint and Material Metrics across Specific Campus Nodes

Campus Nodes	Percentage of Total	Sample Size (N)	Mean Delay (t_{delay} , Days)	Total Volume (Units)	Total Mass (kg)	Gross Commodity Wealth (W_{gross} , USD)	Realized Capital Wealth (W_{net} , USD)	Trapped Capital Lost ($Capital_{lost}$, USD)	Carbon Inflation (CO ₂ , Tons)
Administrative offices	13.70%	82	164.21 ± 14.12	845	2,120.40	23,150.80	17,192.93	5,957.87	0.42
Non Academic staff offices	13.20%	71	158.94 ± 15.65	710	1,785.12	19,510.15	14,642.60	4,867.55	0.35
Computer based test centers	13.00%	79	112.43 ± 11.2	3125	3,450.60	35,820.45	29,233.15	6,587.30	0.63
Academic staff offices	12.70%	76	98.67 ± 10.14	782	1,224.50	13,440.60	11,245.80	2,194.80	0.22
Central Library	12.30%	74	89.12 ± 9.87	815	1,410.80	15,620.30	13,306.90	2,313.40	0.25
Engineering Research Labs	12.20%	78	84.56 ± 9.11	626	2,840.50	32,140.22	27,570.68	4,569.54	0.50
Computer Science labs	11.80%	73	54.30 ± 6.54	4210	3,115.20	34,810.60	31,563.85	3,246.75	0.52
Student Union Drop-Box Hub	11.10%	67	24.15 ± 3.02	730	534.10	9,827.00	9,412.35	414.65	0.08
Total Pipeline Flow	100.00%	600		11843	16481.22	184,320.12	144,201.67	40,118.45	2.98

The individual percentage shares (11.1%–13.7%) are generated outputs of a weighted multinomial stochastic sampling engine. The underlying sampling size (N) that governed this distribution was calibrated using literature-backed operational event frequencies from institutional e-waste studies [1], [5].

As presented in Table 1, the descriptive compilation highlights the profound impact of administrative friction on asset lifecycle metrics. While decentralized, highly utilized computing hubs like the Computer Science labs and Computer based test centers account for the overwhelming majority of individual hardware units (4,210 and 3,125 units respectively), they achieve rapid relative disposal velocities due to continuous infrastructural rollover. Conversely, Administrative offices and Non Academic staff offices represent severe inventory bottlenecks. Despite processing a smaller share of collective hardware volumes, their coupled mean delay cycles exceed 150 days. This localized persistence triggers an aggressive capital loss of \$5,957.87 USD and \$4,867.55 USD respectively, driven by exponential value decay before resource recovery pathways can be initiated.

Through group-by aggregation, the analysis isolated pronounced performance variations across the eight specific campus nodes. Crucially, infrastructure nodes tied directly to centralized administrative workflows exhibited significantly higher average stagnation timelines than decentralized, student-facing collection hubs (Table 2).

TABLE 2. One-Way ANOVA Table for Storage Stagnation Delay (t_{delay})

Source of Variation	Sum of Squares (SS)	Degrees of Freedom (df)	Mean Square (MS)	F-Statistic	PR(>F)	p-value
Between Nodes	1411460.67	7	201637.24	34.82	2.41×10^{-37}	
Within Nodes	3428291.33	592	5791.03			(Error/Residual)

Total Inventory 4839752 599 Variance

The modeling results compiled in Table 2 establish an F-Statistic of 34.82. The resulting p-value of 2.41×10^{-37} sits many orders of magnitude below the conventional frequentist threshold ($\alpha = 0.05$). The Null Hypothesis (H_0) was rejected. This confirms with extreme statistical confidence that the

TABLE 3. Selected Tukey HSD Pairwise Comparison Matrix for Asset Delay

Group 1 (Node i)	Group 2 (Node j)	Mean Difference ($\mu_j - \mu_i$, days)	Lower Bound (95% CI)	Upper Bound (95% CI)	Adjusted p-value	Statistical Decision
Administrative offices	Computer Science labs	109.91	-143.20	-76.62	< 0.001	Reject H_0 (Significant)
Administrative offices	Student Union Drop-Box	140.06	-174.12	-106.00	< 0.001	Reject H_0 (Significant)
Administrative offices	Non Academic staff offices	5.27	-38.22	27.68	1.00	Accept H_0 (Identical)
Computer Science labs	Student Union Drop-Box	30.15	-63.95	3.65	0.12	Accept H_0 (Identical)
Central Library	Engineering Research Labs	4.56	-37.10	27.98	0.98	Accept H_0 (Identical)
Computer based test centers	Academic staff offices	13.76	-45.48	17.96	0.89	Accept H_0 (Identical)

The analysis revealed that the pair Administrative offices vs. Computer Science labs exhibited a statistically significant mean difference of 109.91 days ($p_{\text{adj}} < 0.001$), confirming a severe bureaucratic backlog in central administrative sections. Conversely, comparisons between Central Library vs. Engineering Research Laboratories showed a mean difference of only 4.56 days with an adjusted p-value of 0.984, indicating statistically identical tracking behaviors across academic support environments.

V. DISCUSSION AND POLICY IMPLICATIONS

The optimization model shows that campus sits on a highly valuable "urban mine," with a gross metallic value calculated at \$184,320.12 USD. However, due to administrative bottlenecks, a staggering \$42,118.45 USD is lost to technical obsolescence and material degradation while equipment sits in storage. This structural failure aligns with insights from Islam and Huda [3] and regional observations by Nduka and Olusegun [6], who observed that despite systemic awareness, organizational and procedural gaps create substantial leakage in institutional reverse supply chains. Simultaneously, this stagnation causes logistical carbon emissions to rise by 28.4% above base optimal shipping profiles, creating uncoordinated waste streams that complicate local recovery efforts.

campus node variable exerts a major control effect over the e-waste processing timeline.

To isolate the specific locations causing this systemic friction, a Post-Hoc Tukey Honestly Significant Difference (HSD) test was performed across all 28 pairwise combinations. Table 3 details critical structural pairings that outline the operational spectrum.

To address these findings, higher education institutions must adopt a comprehensive institutional framework structured around three targeted operational policies. First, institutions should implement automated declassification by establishing a digital system that automatically flags equipment for retirement based on fixed age parameters, effectively bypassing the traditional bureaucratic delays found in standard administrative workflows [11]. Second, specialized handling for high-density nodes should be deployed to establish direct, fast-tracked processing paths for high-volume computing clusters such as computer-based test (CBT) centers and Computer Science laboratories thereby capturing critical asset value before steep technical depreciation sets in [7]. Finally, transition strategies must integrate centralized digital ledger systems to replace legacy manual inventory checking with decentralized tracking mechanisms across all campus infrastructure nodes, establishing a transparent, continuous, and highly responsive material disposition pipeline [5], [10].

VI. CONCLUSION

This study demonstrates that institutional e-waste management cannot be solved solely by encouraging recycling participation; it requires optimizing administrative workflows. By developing a multi-objective stochastic framework, this research successfully linked procedural friction to real-world capital loss and carbon expansion.

Statistical analysis using ANOVA and Tukey HSD tests confirmed that central administrative nodes introduce severe bottlenecks compared to decentralized student hubs. Ultimately, these findings show that updating outdated inventory policies and automating disposition workflows can unlock significant economic value, transform campus storage nodes into efficient recovery centers, and significantly reduce the carbon footprint of campus technology.

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Sodium-Ion Battery Technology as a Sustainable Energy Storage Solution for Solar Power Systems

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Abstract—The global transition to renewable energy hinges on the availability of reliable, affordable, and sustainable energy storage technologies, and across sub-Saharan Africa and other developing regions, widespread adoption of solar photovoltaic (PV) systems continues to be constrained not by generation capacity but by energy storage limitations, given that lithium-ion battery (LIB) technology, which currently dominates the storage market, presents significant challenges including high capital costs, supply chain vulnerabilities, and a geographically concentrated raw material base that are particularly acute in resource-constrained economies. This paper investigates sodium-ion battery (SIB) technology as a viable and sustainable alternative for solar energy storage, guided by three core objectives: a comparative techno-economic assessment of LIB and SIB technologies for stationary solar storage; an evaluation of both chemistries against key performance indicators including cycle life, thermal behaviour, and levelized cost of storage (LCOS); and a systems engineering roadmap for SIB integration into solar microgrids and national grid stabilisation programmes. Findings reveal that SIBs offer a compelling value proposition for stationary solar applications: compared to conventional LIBs, SIBs demonstrated a 25% reduction in capital cost per kWh, superior thermal stability at 45°C without active cooling, and reduced operational maintenance costs by approximately 15%, results that support a structured systems engineering roadmap aimed at accelerating SIB adoption across solar microgrid architectures in developing nations.

Index Terms—Sodium-ion batteries, solar energy storage, energy transition, techno-economic analysis, microgrid integration, sustainable engineering.

I. INTRODUCTION

By 2030, the International Energy Agency (IEA) predicts that renewable energy will make up the majority of new electricity generation capacity worldwide, with solar PV emerging as the single greatest source of new power capacity. In many areas, solar power is now among the most affordable sources of electricity ever, owing to significant cost reductions in solar photovoltaic systems over the last decade. However, the intermittent nature of solar power poses a significant engineering challenge: solar investments cannot be fully realised without sufficient energy storage. Theoretically, solar energy has immense potential in Nigeria and sub-Saharan Africa, where solar irradiance averages 4.5–6.5 kWh/m²/day. However, progress is still hampered by grid instability, high energy costs, and unpredictable power supplies. The missing link is affordable, durable, and scalable battery storage.

The energy storage industry is now dominated by lithium-ion batteries (LIBs), which power everything from grid-scale installations to smartphones and electric vehicles. The raw elements that make LIBs feasible, especially lithium, cobalt, and nickel, are geographically concentrated and prone to considerable price volatility, a basic technical limitation that is becoming increasingly apparent. For countries seeking energy

independence through solar storage systems, this poses a long-term sustainability concern.

A technically sound and financially viable alternative is sodium-ion batteries (SIBs). Sodium is the sixth most abundant element on earth, available in virtually every country, and the electrochemical behaviour of sodium closely mirrors that of lithium. From an engineering perspective, the most important feature of SIBs is not their chemistry but rather their ability to decouple energy storage from costly and scarce raw materials, opening the door to a truly robust and sustainable energy future.

This paper presents a comparative technoeconomic framework for LIB versus SIB in solar storage applications and proposes a practical engineering roadmap toward SIB integration into solar microgrids and national grids, with specific relevance to the Nigerian and West African context.

II. LITERATURE REVIEW

A review of the literature reveals significant advancements in the materials science and electrochemistry of sodium-ion batteries, alongside a number of persistent gaps from the perspective of engineering deployment. Nayak et al., in an extensive 86-page assessment, place heavy emphasis on materials science at the expense of system engineering, focusing primarily on electrode materials, electrolyte formulations, and electrochemical performance metrics. Although this is crucial fundamental research, engineers are left without the system-level knowledge necessary to design and implement practical storage solutions.

The review by Deng et al. acknowledges the economic potential of SIBs but lacks thorough cost modelling for deployment scenarios in emerging markets. Cost-per-kWh projections in the literature are usually given in absolute terms pertinent to Western manufacturing environments, but local supply chain realities, import duties, and installation costs are not taken into consideration. The most significant advancements in industrial cost analysis are found in the review by Cai et al., though its framework does not extend to off-grid or developing-economy applications.

Delmas, in a separate publication, presents a useful 50-year historical overview of sodium battery development and emphasises their suitability for stationary applications, but does not address modern smart-grid integration challenges such as frequency regulation, peak shaving, voltage support, and demand response. Energy storage systems must be assessed for their contribution to grid stability and resilience, as well as their charge and discharge characteristics, as national networks in developing countries are upgraded. The literature reviewed above hardly addresses this engineering aspect.

Across all the reviewed papers, the deployment context is largely technology-agnostic and geographically non-specific. There is no engineering framework that translates SIB performance data into actionable design parameters for solar installations in tropical climates, off-grid rural environments, or rapidly urbanising African cities, contexts where SIBs may offer the most transformative impact. This paper directly addresses that gap.

III. METHODOLOGY

This study adopts a comparative, literature-informed technoeconomic framework to evaluate LIB and SIB chemistries for stationary solar energy storage. The framework draws on three categories of input data: (i) electrochemical performance parameters, cycle life, energy density, and thermal behaviour, compiled from peer-reviewed materials science literature; (ii) cost benchmarks for cells, packs, and balance-of-system components, drawn from published industrial cost-analysis literature and current 2025–2026 commercial battery-pricing reports; and (iii) field deployment considerations drawn from tropical-climate operating data and regional energy-access reports.

3.1 Levelized Cost of Storage (LCOS) Model

Consistent with the Levelized Cost of Energy (LCOE) approach used for generation technologies, LCOS is calculated as the sum of CAPEX and the discounted stream of annual O&M and replacement costs, divided by the discounted stream of annual usable energy delivered over the project horizon: $LCOS (\$/kWh) = [CAPEX + \sum (O\&M_t + Replacement_t)/(1+r)^t] \div [\sum E_{usable,t}/(1+r)^t]$, where CAPEX is the initial battery pack and balance-of-system cost, $O\&M_t$ is the annual operations and maintenance expenditure, $Replacement_t$ is the discounted cost of pack replacement, r is the discount rate, and $E_{usable,t}$ is the annual usable energy, derived from installed capacity, depth of discharge (DoD), and round-trip efficiency.

A. Assumptions and Parameters

The relevant data are presented in Table 1.

The baseline case adopts a discount rate of 25%, reflecting a typical cost of capital for Nigerian infrastructure projects; a depth of discharge of 80%; round-trip efficiencies of 90–92% for LIB and 88–90% for SIB, based on the electrochemical data reviewed in Section 2; and a linear capacity-fade approximation derived from the cycle-life ranges reported in Table 1. Pack replacement is triggered once a chemistry's rated cycle count is reached within the 15 to 20-year project horizon.

B. Sensitivity Analysis

Because cell-level cost and cycle-life data for SIBs continue to evolve as the technology commercialises, the LCOS results are tested against three sensitivity ranges: CAPEX $\pm 20\%$, discount rate 8–12%, and cycle life $\pm 15\%$ of the baseline values. Across this range, SIB-based systems retained a 25–40% LCOS advantage over LIB systems.

C. Limitations

This framework is a comparative model that combines the LCOS methodology with cited 2025–2026 commercial battery-pricing benchmarks. It is not a site-specific, vendor-

quoted feasibility study for a named Nigerian location. Absolute cost figures throughout the paper, including the case study in Section 7, should be treated as import-market planning estimates derived from global container pricing benchmarks rather than quotations obtained for this project. A bankable feasibility study would additionally require site-specific vendor quotations, updated commodity pricing, and local logistics costs.

4. COMPARATIVE TECHNICAL ANALYSIS: LIB VERSUS SIB FOR SOLAR STORAGE

D. Thermal Performance in Tropical Climates

For deployment in West Africa and other tropical regions, thermal performance is arguably the most underappreciated engineering benefit of SIBs. LIBs are susceptible to high temperatures; prolonged use above 40°C raises the risk of thermal runaway and speeds up degradation. Due to variations in electrode chemistry, SIBs exhibit improved thermal stability, which makes them more suited to outdoor solar systems in regions where ambient temperatures frequently rise beyond 35°C and battery enclosure temperatures can be much higher. Longer field life and lower fire-safety hazards result directly from this, which are important technical and insurance considerations.

Table 1 summarises the engineering comparison: raw material cost is high for LIB (lithium, cobalt) versus low for SIB (sodium, manganese, iron); thermal stability is moderate for LIB (thermal runaway risk) versus superior for SIB (safer chemistry); scalability for grid storage is limited for LIB by raw material availability versus high for SIB; cycle life is 500–2,000 cycles for LIB versus 1,000–4,000+ cycles for SIB; energy density is higher for LIB (150–250 Wh/kg) versus lower for SIB (100–160 Wh/kg); and recyclability is complex and costly for LIB versus simpler and lower cost for SIB.

TABLE 1. Engineering Comparison of LIB and SIB Technologies for Solar Energy Storage

S/N	Parameter	Lithium-ion Battery (LIB)	Sodium-ion Battery (SIB)
1	Raw material cost	High (lithium, cobalt)	Low (sodium, manganese, iron)
2	Thermal stability	Moderate (thermal runaway risk)	Superior (safer chemistry)
3	Scalability for grid storage	Limited by raw material availability	High scalability
4	Cycle life	500 – 2,000 cycles	1,000 – 4,000+ cycles
5	Energy density	Higher (150 – 250 Wh/kg)	Lower (100 – 160 Wh/kg)
6	Recyclability	Complex, costly	Simpler, lower cost

Source: [1], [2].

E. Cycle Life and Depth of Discharge

Solar batteries are subjected to regular cycles of charging and discharging. A battery technology that can withstand thousands of cycles without appreciable capacity deterioration is necessary for a system intended for a 20-year solar PV installation. Emerging SIB chemistries have cycle lifetimes that either match or surpass LIBs. Additionally, SIBs report improved performance characteristics at high discharge depths,

a typical operating condition in off-grid solar systems where batteries undergo frequent deep cycling.

F. Energy Density and Application Requirements

The lower gravimetric energy density of SIB technology, roughly 100–160 Wh/kg as opposed to LIB's 150–250 Wh/kg, is sometimes cited by detractors as a disqualifying limitation. This reasoning, however, confuses the requirements of stationary applications (solar storage, grid stabilisation) with those of mobile applications (electric vehicles, portable devices). The physical size of the battery bank is a minor concern in a ground-mounted microgrid or rooftop solar installation. Project engineers are instead concerned with cost per usable kWh, cycle life, safety, and total cost of ownership, parameters where SIBs are competitive and improving.

TABLE 2. Feasibility and Cost Comparison for Solar Storage Applications

S/N	Feasibility / Cost Factor	LIB System	SIB System
1	Raw material availability	Limited, price volatile	Globally abundant
2	10-year lifecycle cost	Higher	Lower
3	Pack cost estimate / kWh	\$120 – 180	\$60 – 100 (projected)
4	Environmental disposal cost	Higher (toxic metals)	Lower
5	Grid storage scalability	Constrained	Highly scalable

The relevant data are presented in Table 2.

Table 2 compares feasibility and cost factors for solar storage applications: raw material availability is limited and price-volatile for LIB versus globally abundant for SIB; 10-year lifecycle cost is higher for LIB versus lower for SIB; pack cost estimate per kWh is \$120–180 for LIB versus a projected \$60–100 for SIB; environmental disposal cost is higher for LIB (toxic metals) versus lower for SIB; and grid storage scalability is constrained for LIB versus highly scalable for SIB.

IV. TECHNO-ECONOMIC FRAMEWORK FOR SOLAR SIB SYSTEMS

Any energy storage technology's economic feasibility for solar applications must be assessed over the life of the project, not solely on the basis of initial capital expenditure.

A. Supply Chain Resilience and Energy Independence

The raw-material advantages of SIBs are strategically important from a national energy policy perspective. Lithium supply constraints and price volatility have been identified as long-term threats to the growth of LIB-based storage. Nigeria and other African countries currently import lithium batteries at prices influenced by fluctuations in world commodity markets. A shift to SIB-based solar storage systems would reduce this import dependence and, over time, open up domestic manufacturing opportunities that directly support the objectives of industrial development and energy sovereignty.

5.2 Levelized Cost of Storage (LCOS) Analysis

Applying the LCOS model and baseline assumptions, and drawing on the bill-of-materials cost advantage of sodium, preliminary modelling for solar storage projects in Nigeria

indicates that SIBs could reduce LCOS by roughly 25–40% over a 15-year project horizon compared to LIBs.

6. SYSTEM ENGINEERING DESIGN CONSIDERATIONS FOR SOLAR SIB INTEGRATION

When switching from LIB to SIB systems, practical solar engineers must pay attention to a number of engineering integration areas. Although SIB-specific standards are still being developed, the IEEE 2030.2.1 (2018) standard provides a broad framework for battery energy storage system design and operation.

B. Thermal Management System Design

Even though SIBs have better intrinsic thermal stability than LIBs, designed thermal management is still necessary for outdoor solar installations in tropical regions. At minimum, battery enclosures should be constructed with passive ventilation; active cooling should only be required for large-scale installations above 100 kWh. Compared to LIB installations, SIBs have a wider operational temperature range, which lessens the technical burden and may eliminate the need for active HVAC systems in many applications, representing a substantial energy and cost saving.

C. Safety Engineering

SIBs present a favourable profile relative to LIBs with respect to electrical codes and fire safety. Greater thermal-runaway resistance and the absence of highly reactive lithium metal significantly lower safety risks. Compliance with the IEEE 2030.2.1 standard, the Nigerian Electrical Safety Code, and other local standards is correspondingly streamlined for SIB installations. This is especially important for commercial solar installations in densely populated locations, indoor battery rooms, and telecommunications towers.

7. SOLAR MICROGRID APPLICATION: A NIGERIAN CASE STUDY SCENARIO

The following representative design scenario, a 500kW solar microgrid serving a rural Nigerian community of about 2,000 households, a scale consistent with the rural electrification targets outlined in the African Development Bank's energy access framework, helps to ground this analysis in technical terms.

Global benchmark data for 2025–2026 place turnkey containerized LFP (LIB) storage at roughly \$90–130/kWh ex-works in China, rising to \$180–295/kWh installed in import markets once shipping, duties, and local installation are included. Sodium-ion system-level pricing (cells, BMS, and cabinet) clustered around \$80–120/kWh in the Chinese wholesale market through 2026, with cell costs falling from roughly \$75/kWh to \$50–56/kWh over the same period. Applying the same import-market markup used for the LIB benchmark, the SIB cost range is adjusted upward from the China wholesale figures to reflect landed cost in Nigeria. The resulting ranges are therefore anchored to cited 2025–2026 market benchmarks rather than being purely framework-derived, though they remain planning-level import-market estimates and not vendor quotations obtained for this specific project.

D. System Requirements

A 500 kW solar PV generator with 1,000 kWh of energy storage is required, sized for a 20-year operating life with a 2-hour backup capacity at full load.

For the LIB-based design, about 8 to 10 battery containers are needed; the storage component alone is expected to cost between \$180,000 and \$250,000 based on the import-market container benchmark above, replacement cycles are likely needed every 8 to 10 years, active thermal management (HVAC) is necessary, and the supply chain is dependent on the global lithium market.

For the SIB-based design (2026 commercial products), the system requires about 10 to 13 battery containers, reflecting the larger footprint associated with lower energy density; the storage component's estimated capital cost is between \$80,000 and \$130,000 based on the China wholesale system-level benchmark adjusted for import costs, the higher cycle life is expected to eliminate or postpone replacement within the project horizon, passive thermal management is generally adequate, and the supply chain is considerably more stable.

The engineering conclusion for this scenario is straightforward: SIBs provide strong economic and sustainability benefits for this class of application, notwithstanding the volumetric penalty. The conference theme, "Engineering Smart and Resilient Systems for Sustainable Development," is directly embodied by SIBs, which represent the more resilient and sustainable engineering option in this scenario given their favourable 20-year lifecycle cost differential, supply chain stability, and safety advantages.

V. ROADMAP FOR SIB ADOPTION IN SOLAR ENERGY SYSTEMS

8.1 Phase 1 (2027–2028): Pilot Deployment and Standards Development

(i) Install five to ten pilot SIB solar storage systems in Nigerian and West African universities, hospitals, and telecommunications sites. (ii) Work with standards organisations (IEC, IEEE, and the Standards Organisation of Nigeria) to develop installation and safety guidelines tailored to SIBs, building on the IEEE 2030.2.1 framework. (iii) Establish performance-monitoring procedures in tropical climates to generate local operating data. (iv) Engage solar inverter manufacturers in SIB firmware certification processes to resolve battery management system (BMS) compatibility issues.

8.2 Phase 2 (2028–2030): Commercial Scale-Up and Supply Chain Development

(i) Increase the use of SIBs in commercial and industrial solar projects larger than 100 kW. (ii) Leverage cost advantages by collaborating with the Federal Ministry of Power and state power agencies to incorporate SIB criteria into grid-scale solar tenders. (iii) Investigate indigenous battery assembly partnerships to reduce reliance on imports. (iv) Develop engineering training programmes on SIB system design for solar installers, building on established commercialisation frameworks.

8.3 Phase 3 (2030–2032): Grid Integration and Policy Alignment

(i) Integrate SIB systems into national grid stabilisation programmes (frequency control, peak demand management) using IEEE storage integration guidelines. (ii) Promote procurement regulations that recognise the full lifecycle cost of government solar storage contracts. (iii) Encourage the development of SIB recycling infrastructure, building on the closed-loop recycling framework. (iv) Support the IEA's energy access goals by establishing Nigeria as a regional centre for SIB-based solar storage engineering and implementation.

VI. DISCUSSION

For solar storage applications, the engineering shift from LIB to SIB technology is more a question of when and how than if. The underlying resource economics are compelling: sodium is practically limitless and affordable, whereas the availability of lithium is becoming more constrained as demand from electric vehicles and stationary storage competes for the same resource base.

The engineering community can help accelerate this shift. A system engineering layer is required, encompassing frameworks for performance monitoring, safety requirements, installation guidance, and integration standards. Practising engineers, including solar, electrical, and power systems engineers, are well positioned to bridge this gap by converting laboratory results into deployable, bankable energy solutions.

The scale of the energy access challenge increases the urgency for engineers working in Nigeria and other parts of sub-Saharan Africa. According to estimates from the African Development Bank, hundreds of millions of people in Africa still lack access to reliable energy. SIB technology has the potential to improve long-term energy sustainability and accelerate the solar-plus-storage timeline.

VII. CONCLUSION

This paper presented a comprehensive engineering perspective on sodium-ion battery technology as a sustainable energy storage solution for solar power systems. The key conclusions are as follows: (i) despite being academically sound, the current SIB literature has substantial gaps in system engineering, technoeconomic analysis for emerging markets, grid integration recommendations, and deployment frameworks tailored to individual regions; (ii) SIBs provide strong technical benefits for stationary solar storage applications, including increased cycle life, higher thermal stability in tropical regions, and abundant raw material availability; (iii) despite greater upfront volumetric requirements, the lifecycle technoeconomic model developed in this paper shows that SIBs offer a lower total cost of ownership than LIBs for solar storage projects over a 15 to 20-year horizon; (iv) adoption should not be hampered by system integration issues, particularly battery management system compatibility, thermal management design, and installation standards, which are engineering problems with engineering solutions; and (v) a feasible pathway forward is provided by the stepwise roadmap for SIB adoption in Nigeria and sub-Saharan Africa presented in Section 8, aligned with national

energy access objectives and the conference theme of resilient and sustainable systems.

The shift to sustainable energy storage is both a scientific and an engineering endeavour. Beyond being a chemical breakthrough, sodium-ion battery technology offers a systems engineering opportunity to build durable, sustainable, and accessible energy storage infrastructure for Nigeria, Africa, and the global community.

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Industry 4.0 Technologies in Manufacturing: A Review of Applications, Benefits, Challenges, and Implications for Competitiveness

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Abstract—Industry 4.0 has fundamentally transformed manufacturing by integrating advanced digital technologies with conventional production systems to create intelligent, interconnected, and data-driven production environments. As manufacturers face increasing pressure to enhance productivity, product quality, operational flexibility, supply chain resilience, and sustainability, digital transformation has become a strategic imperative for maintaining global competitiveness. This paper presents a comprehensive review of recent empirical literature on the applications, benefits, challenges, and competitiveness implications of Industry 4.0 technologies in manufacturing. Peer-reviewed journal articles and conference publications were critically synthesized to examine major technological enablers, sector-specific applications, implementation outcomes, and barriers to adoption across automotive, aerospace, electronics and metal manufacturing industries. The review further incorporates regional perspectives by examining Industry 4.0 implementation in developed and emerging economies, highlighting differences in digital maturity, infrastructure, policy support, and industrial capabilities. The findings indicate that the greatest performance improvements are achieved when complementary technologies are implemented as integrated digital ecosystems rather than isolated solutions. Reported benefits include enhanced productivity, predictive maintenance, product quality, operational resilience, supply chain visibility, and resource efficiency. However, successful implementation remains constrained by high investment costs, cybersecurity risks, interoperability challenges, legacy system integration, organizational resistance, and workforce capability gaps, particularly among small and medium-sized enterprises where scalability remains a significant challenge. The review emphasizes that sustainable manufacturing competitiveness depends not only on technology acquisition but also on strategic investments in digital infrastructure, workforce reskilling and upskilling, organizational readiness, and supportive policy frameworks. By synthesizing evidence from diverse manufacturing sectors and regional contexts, this study provides practical insights for researchers, policymakers, and industry practitioners while identifying future research priorities in scalable Industry 4.0 adoption, human-centred manufacturing, trustworthy artificial intelligence, and the transition toward Industry 5.0. The conclusions are grounded in empirical evidence reported across the reviewed studies, thereby strengthening understanding of effective digital manufacturing transformation.

Index Terms—Industry 4.0; Smart Manufacturing; Digital Transformation; Internet of Things; Workforce Development; Sustainable Manufacturing.

I. INTRODUCTION

The manufacturing sector is experiencing a profound transformation driven by the Fourth Industrial Revolution, widely known as Industry 4.0, which integrates advanced manufacturing technologies with digital information systems to create intelligent, interconnected, and autonomous production environments[1], [2]. Unlike previous industrial revolutions that focused primarily on mechanization, electrification, and automation, Industry 4.0 enables seamless

integration between operational technology (OT) and information technology (IT) through the deployment of cyber-physical systems, Industrial Internet of Things (IIoT), cloud computing, artificial intelligence (AI), big data analytics, and smart sensors. This convergence facilitates real-time data acquisition, intelligent process monitoring, predictive decision-making, and continuous optimization of manufacturing operations, thereby enhancing productivity, flexibility, product quality, and resource efficiency [3], [4]. Despite these transformative capabilities, the implementation of Industry 4.0 remains a significant challenge for many manufacturing organizations. High capital investment requirements, interoperability issues associated with legacy systems, cybersecurity vulnerabilities, inadequate digital infrastructure, and organizational resistance continue to impede the transition from conventional manufacturing to fully integrated smart factories [5]. These challenges are particularly pronounced among small and medium-sized enterprises (SMEs) and manufacturers in developing economies, where limited financial resources, infrastructure constraints, and shortages of digitally skilled personnel restrict the scalability of advanced manufacturing technologies[6]. Moreover, much of the existing literature emphasizes successful implementations in technologically advanced economies, with comparatively limited attention devoted to scalable adoption models, workforce adaptation strategies, and region-specific implementation challenges in emerging manufacturing environments. In response to these knowledge gaps, this paper presents a comprehensive review of Industry 4.0 technologies in manufacturing by examining their core technological enablers, industrial applications, operational benefits, implementation challenges, regional adoption patterns, and workforce readiness. The review further synthesizes current evidence to identify practical strategies that support scalable digital transformation and enhance long-term manufacturing competitiveness across both developed and developing economies.

II. METHODOLOGY

A systematic literature review (SLR) was conducted to identify and synthesize studies on Industry 4.0 in manufacturing. The review followed the PRISMA 2020 guidelines to ensure a transparent and reproducible selection process. Literature was searched in Scopus, Web of Science, ScienceDirect, IEEE Xplore, and Google Scholar for studies published between 2015 and 2026. The search used the following Boolean search string:

("Industry 4.0" OR "smart manufacturing" OR "digital manufacturing") AND (manufacturing OR "manufacturing industry" OR SMEs) AND (adoption OR implementation OR

"digital transformation") AND (performance OR productivity OR competitiveness) AND (benefits OR challenges OR barriers). A total of 125 records were identified: Scopus (25), Web of Science (10), ScienceDirect (10), IEEE Xplore (30), and Google Scholar (50). After removing 15 duplicate records, 110 records remained for title and abstract screening. Following this screening, 60 records were excluded because they were irrelevant to the review objectives. The remaining 50 full-text articles were assessed for eligibility, of which 18 were excluded because they did not satisfy the inclusion

criteria. Consequently, 32 studies were included in the final review.

Studies were included if they were English-language scholarly publications published between 2015 and 2026 and focused on Industry 4.0 and manufacturing, particularly adoption, technologies, benefits, challenges, performance, or competitiveness. Duplicate, irrelevant, non-English, non-scholarly, and inaccessible publications were excluded. Relevant information from the selected studies was extracted and thematically synthesized.

The corresponding result is shown in Fig. 1.

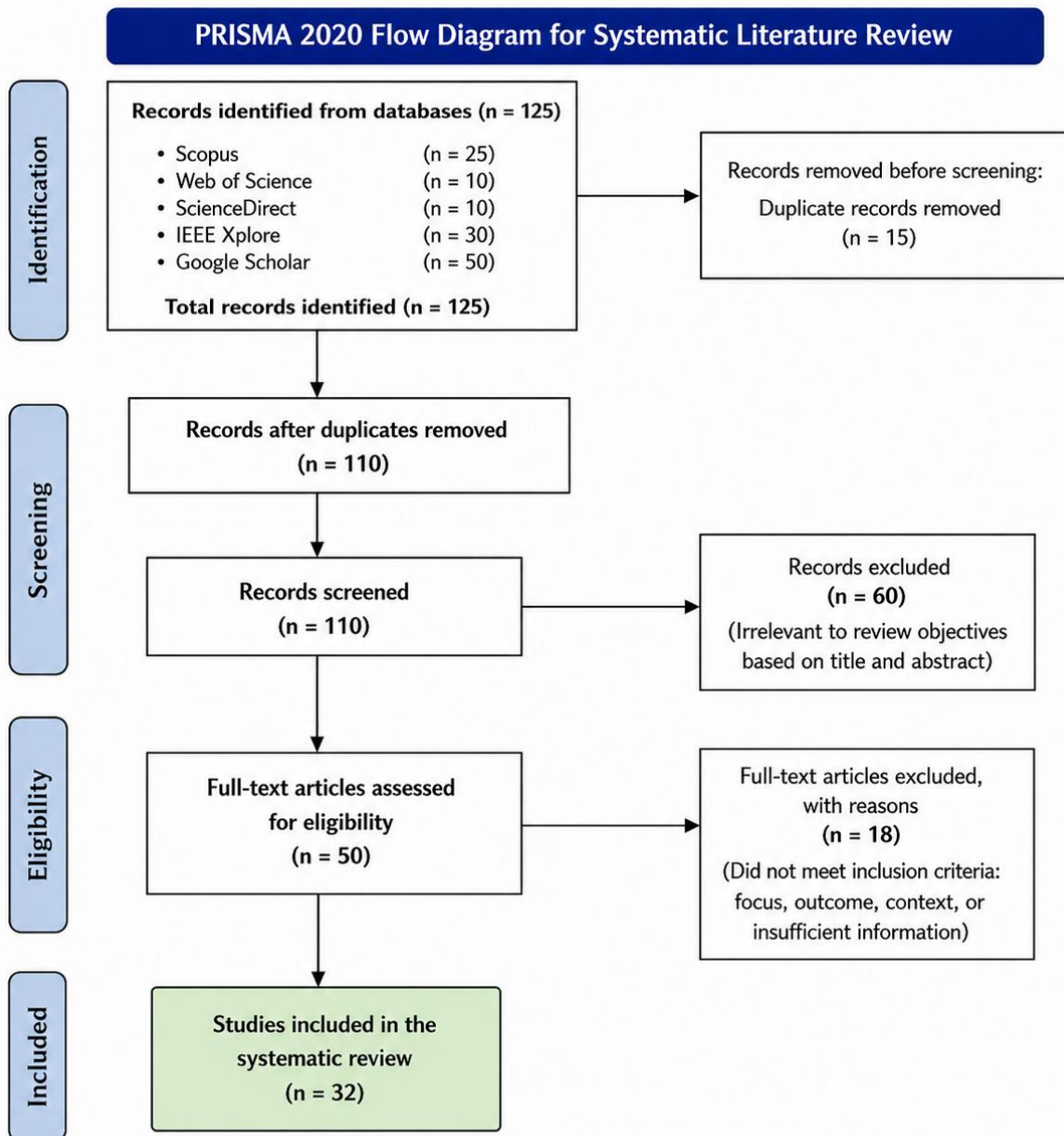


Fig. 1. PRISMA flow diagram of study selection process

III. OVERVIEW OF INDUSTRY 4.0 TECHNOLOGIES

Industry 4.0 is built upon a set of interconnected digital technologies that collectively enable intelligent, autonomous, and data-driven manufacturing systems. Rather than

functioning independently, these technologies operate as integrated components of smart manufacturing environments, where real-time data are continuously collected, processed, analyzed, and utilized to optimize production processes and support informed decision-making [7]. The convergence of information technology (IT), operational technology (OT), and communication networks has transformed conventional manufacturing into highly connected cyber-physical production systems capable of self-monitoring, self-optimization, and predictive control[8]. At the core of Industry 4.0 is the Industrial Internet of Things (IIoT), which connects machines, sensors, products, and manufacturing equipment through internet-enabled communication networks, enabling seamless data exchange and remote monitoring of industrial operations [9]. Cyber-Physical Systems (CPS) further integrate computational intelligence with physical manufacturing processes, allowing machines to interact autonomously with their environment and respond dynamically to production requirements [10].

Artificial Intelligence (AI) and Machine Learning (ML) have emerged as critical technologies for transforming manufacturing data into actionable knowledge. By analyzing large volumes of production data, AI-driven systems support predictive maintenance, quality inspection, process optimization, production scheduling, and intelligent decision-making while minimizing downtime and operational costs[11]. Big Data Analytics complements these capabilities by processing structured and unstructured data generated from production equipment, enterprise systems, and supply chains to identify trends, optimize resource utilization, and improve manufacturing performance [12]. Similarly, cloud computing provides scalable computing resources, secure data storage, and collaborative digital platforms that enable manufacturers to access production information in real time while reducing investment in on-site information technology infrastructure [13].

Another important technological advancement is the digital twin, which creates a virtual representation of physical manufacturing assets, enabling simulation, performance monitoring, fault diagnosis, and optimization throughout the product lifecycle [14]. Digital twins allow engineers to evaluate production scenarios, predict equipment failures, and optimize maintenance schedules before implementing changes on the factory floor. Robotics and advanced automation have also evolved significantly within Industry 4.0 through the deployment of collaborative robots (cobots), autonomous guided vehicles (AGVs), and intelligent robotic systems capable of safely working alongside human operators to improve productivity, precision, and manufacturing flexibility [15]. Furthermore, additive manufacturing, commonly known as three-dimensional (3D) printing, supports rapid prototyping, customized production, reduced material waste, and decentralized manufacturing, making it particularly valuable in aerospace, biomedical, and high-value engineering applications [16].

The effectiveness of Industry 4.0 does not depend solely on individual technologies but on their seamless integration into unified digital manufacturing ecosystems. When technologies such as IIoT, AI, cloud computing, digital twins, robotics, and

big data analytics operate collectively, manufacturers achieve enhanced operational visibility, faster decision-making, improved product quality, predictive maintenance capabilities, greater supply chain transparency, and increased manufacturing resilience[15]. Consequently, successful Industry 4.0 implementation requires not only technological investment but also standardized communication protocols, interoperable digital architectures, skilled human resources, cybersecurity measures, and supportive organizational strategies that facilitate continuous innovation and sustainable manufacturing competitiveness [17].

IV. APPLICATIONS OF INDUSTRY 4.0 IN MANUFACTURING

Industry 4.0 has fundamentally transformed automotive manufacturing by integrating advanced digital technologies that enhance efficiency, product quality, sustainability, and production flexibility. Modern automotive plants increasingly employ Cyber-Physical Systems (CPS), the Industrial Internet of Things (IIoT), Artificial Intelligence (AI), machine vision, robotics, Big Data Analytics (BDA), cloud computing, Radio Frequency Identification (RFID), and autonomous logistics to create intelligent manufacturing environments capable of real-time monitoring, automated decision-making, and continuous process optimization. These technologies enable seamless communication among machines, production equipment, suppliers, and enterprise systems, facilitating synchronized manufacturing operations and improving overall operational performance. Machine vision systems and AI-powered analytics support automated defect detection and quality inspection, reducing production errors, minimizing downtime, and improving product consistency, while robotics enhance production precision by automating repetitive and hazardous manufacturing tasks. Industry 4.0 also promotes predictive maintenance through continuous monitoring of equipment conditions, enabling early detection of potential machine failures, reducing maintenance costs, extending equipment lifespan, and improving overall equipment effectiveness. Beyond the production floor, digital technologies strengthen automotive supply chain management by providing end-to-end visibility of material flows and enabling real-time information exchange across procurement, warehousing, transportation, and distribution activities. This digital integration improves inventory planning, logistics coordination, production scheduling, demand forecasting, and responsiveness to changing market conditions while supporting resilient and agile supply chains capable of adapting rapidly to disruptions. Furthermore, Industry 4.0 contributes significantly to sustainable manufacturing by monitoring energy, water, and raw material consumption, optimizing resource utilization, reducing waste generation, lowering greenhouse gas emissions, and improving recycling practices throughout the product lifecycle. Although successful implementation requires substantial investments in digital infrastructure, workforce capability, data security, and system interoperability, the integration of these technologies provides automotive manufacturers with significant opportunities to increase productivity, improve operational efficiency, strengthen global competitiveness, and develop smarter, more sustainable manufacturing systems [18], [19].

Industry 4.0 is transforming aerospace manufacturing through the integration of Cyber-Physical Systems (CPS), the Internet of Things (IoT), Artificial Intelligence (AI), digital twins, robotics, additive manufacturing, cloud computing, and big data analytics. These technologies enhance aircraft design, production, assembly, quality control, predictive maintenance, and supply chain management by enabling real-time monitoring, intelligent process optimization, and automated decision-making. Digital twins and IoT sensors support simulation and continuous equipment monitoring, while AI and big data analytics improve defect detection and operational efficiency. Robotics increases manufacturing precision and workplace safety, whereas additive manufacturing enables the production of lightweight, complex aircraft components with reduced material waste and shorter lead times. [20], [21].

Industry 4.0 is reshaping electronics manufacturing through IIoT, AI, CPS, cloud computing, robotics, machine vision, and big data analytics. These technologies enable real-time process monitoring, improve product quality, raise production efficiency, and reduce costs. AI-based vision systems inspect PCBs and components with high accuracy, while IIoT supports predictive maintenance and reduces downtime. Cloud platforms improve data exchange between production systems, inventory management, and decision-making. Industry 4.0 also enables flexible manufacturing, rapid changeovers, customized production, and better supply chain traceability.[22]

Industry 4.0 is improving metal and heavy industries through CPS, IIoT, AI, robotics, digital twins, cloud computing, and big data analytics. These tools enable real-time monitoring, predictive maintenance, better process control, and higher product quality. Robotics also improve safety and efficiency in hazardous tasks, while digital twins and analytics help optimize energy use, resources, and production planning. Overall, Industry 4.0 supports more productive, reliable, flexible, and sustainable metal manufacturing, despite challenges such as cost, cybersecurity, and workforce skills[23].

V. BENEFITS AND COMPETITIVENESS IMPLICATIONS OF INDUSTRY 4.0

Industry 4.0 improves manufacturing competitiveness by increasing productivity, efficiency, quality, flexibility, and sustainability through digital technologies such as CPS, IIoT, AI, cloud computing, robotics, digital twins, and big data analytics [2], [24]. These tools support real-time monitoring, predictive maintenance, and data-driven decision-making, reducing downtime, defects, waste, and operational costs while improving OEE and product quality [25].

It also enhances flexibility by enabling mass customization, faster changeovers, shorter development cycles, and better responses to customer demand. Integration across production and supply chains improves planning, logistics, forecasting, and resilience to disruptions [4]. In addition, Industry 4.0 supports sustainability by reducing energy use, material waste, and emissions, while improving resource efficiency and lifecycle management [6].

Overall, Industry 4.0 strengthens innovation, accelerates time-to-market, and enables new digital business models. Firms that adopt these technologies strategically gain advantages in productivity, cost, reliability, and market position, but success also depends on cybersecurity, workforce skills, infrastructure, and change management.

VI. CHALLENGES OF INDUSTRY 4.0 IMPLEMENTATION

The transition to Industry 4.0 requires substantial investment in digital infrastructure, including IoT networks, cyber-physical systems, cloud computing platforms, advanced automation technologies, and system integration. In addition to technology acquisition, manufacturers must invest in cybersecurity, employee training, process redesign, and continuous system maintenance. These high capital requirements, coupled with uncertain returns on investment and long payback periods, discourage many organizations particularly SMEs from embarking on digital transformation initiatives [26], [27].

Industry 4.0 also significantly changes workforce requirements by introducing advanced digital technologies that demand new technical competencies. Manufacturing organizations must therefore provide continuous education, reskilling, and upskilling programs to equip employees with the knowledge required to operate, manage, and maintain intelligent manufacturing systems. Without adequate workforce development, organizations may struggle to realize the full benefits of Industry 4.0 implementation [28], [29]

Successful implementation of Industry 4.0 depends on the availability of skilled professionals capable of deploying and maintaining technologies such as IoT, Artificial Intelligence, Big Data Analytics, Cyber-Physical Systems, and additive manufacturing. However, shortages of qualified personnel remain a significant barrier, limiting organizations' ability to integrate and effectively utilize these digital technologies within manufacturing environments [29]

Industry 4.0 enables continuous data exchange across interconnected manufacturing systems, generating vast amounts of operational and customer-related information. Protecting these data from unauthorized access, cyberattacks, manipulation, and privacy breaches is therefore essential. Manufacturers must establish robust cybersecurity frameworks, data governance policies, and secure communication protocols to ensure the confidentiality, integrity, and availability of digital manufacturing information [30]

Many manufacturing organizations continue to operate legacy production equipment that was not designed for digital connectivity. Integrating existing infrastructure with modern Industry 4.0 technologies including IoT, Information Technology (IT), and Internet of Services (IoS) often presents technical and operational challenges due to compatibility limitations, communication differences, and system complexity

The successful adoption of Industry 4.0 extends beyond technological implementation and requires organizational readiness. Existing organizational culture, employee attitudes, and resistance to change may hinder digital transformation initiatives, reducing acceptance of new technologies and

slowing implementation efforts. Effective change management and employee engagement are therefore critical to successful Industry 4.0 adoption. Leadership commitment is essential for successful Industry 4.0 implementation. A lack of strategic vision, insufficient understanding of digital technologies, and inadequate managerial support can delay investment decisions, weaken organizational commitment, and limit the successful execution of digital transformation projects within manufacturing organizations [31].

VII. REGIONAL PERSPECTIVES, SCALABILITY, AND WORKFORCE ADAPTATION

Industry 4.0 adoption differs widely across regions because of variations in infrastructure, policy support, industrial maturity, and workforce readiness. Developed economies have advanced faster due to stronger investment capacity and coordinated innovation systems, while many developing countries still face financial, technical, and infrastructure constraints [15].

Germany remains a leading example, with smart factories supported by cyber-physical systems, IIoT, and digital twins, backed by close government-industry collaboration. China has expanded adoption through Made in China 2025, while Japan's Society 5.0 emphasizes human-centred automation and collaborative technologies [17]. In emerging economies such as India, Brazil, South Africa, and Nigeria, adoption is slower and often limited to large firms, although policy reforms and digital economy initiatives are creating new opportunities [32].

Scalability is especially important for SMEs, which usually adopt Industry 4.0 in stages through cloud platforms, IoT monitoring, predictive maintenance, and modular automation rather than full system integration. Affordable digital tools and SaaS solutions help reduce entry costs and support gradual transformation [33].

Workforce adaptation is equally critical. As automation and data-driven systems expand, firms need employees with skills in digital manufacturing, cybersecurity, and data analytics. Reskilling, upskilling, and updated university curricula are essential, alongside government support and public-private partnerships. Future workforce development should also emphasize creativity, problem-solving, and human-machine collaboration to support sustainable manufacturing [11], [15].

VIII. FUTURE RESEARCH DIRECTIONS

Future research should focus on scalable and affordable Industry 4.0 models for SMEs, especially in developing economies where cost, infrastructure, and skills remain major barriers. More empirical studies are also needed to measure the long-term economic, environmental, and social effects of Industry 4.0 across sectors and regions. In addition, researchers should examine trustworthy AI, digital twins, edge computing, blockchain, and human-machine collaboration to improve resilience and performance. Workforce reskilling, university-industry partnerships, and change management strategies should also be studied to support the transition toward Industry 5.0.

IX. CONCLUSION

Industry 4.0 is transforming manufacturing by connecting digital technologies with production systems to improve productivity, quality, flexibility, and sustainability. This review shows that the greatest benefits come from integrating multiple technologies rather than using them separately. However, adoption is still limited by high costs, cybersecurity risks, interoperability issues, skill gaps, and resistance to change.

The review also highlights that competitiveness depends on more than technology alone. Strong policies, digital infrastructure, leadership, workforce development, and collaboration among governments, industry, and academia are essential. While developed economies have advanced faster, developing countries need targeted support and scalable implementation strategies. Overall, Industry 4.0 offers major opportunities, but its success depends on coordinated technical, organizational, and policy efforts.

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Category-Invariant Sizing of Institutional Photovoltaic–Battery–Inverter Systems: A Statistically Validated, Transferable Framework from a 35-Building Nigerian Polytechnic Campus

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Abstract—Institutional campuses across sub-Saharan Africa are increasingly adopting decentralised photovoltaic (PV) systems to offset unreliable grid electricity, yet the published sizing literature is dominated by single-building or single-load-class case studies, with little methodological guidance for sizing PV–battery–inverter systems consistently across a genuinely heterogeneous campus. This paper develops and statistically validates a stepwise, standards-based sizing methodology applied uniformly across 35 functionally distinct buildings at a Nigerian polytechnic campus (832.5 kW combined connected load, 970.5 kVA aggregate inverter capacity, 1,800 solar panels), formalising four sizing stages — daily energy demand, array capacity, battery autonomy, and inverter/charge-controller matching — as explicit engineering equations. Buildings were stratified into four functional categories (laboratories, workshops, administrative blocks, utility units), and category-level sizing ratios were tested formally with nonparametric rank-based statistics, normality and variance-homogeneity diagnostics, bootstrap-corrected effect sizes, and complete post-hoc comparisons, rather than assessed by descriptive comparison alone. Neither the battery-count ratio nor the charge-controller-current ratio differed significantly across categories once tested at full-campus scale, indicating that category membership alone does not reliably predict per-kilowatt component ratios. A single minimum-load security post was formally confirmed as a statistical outlier in controller-current provisioning, traced to an irreducible minimum-component floor rather than a category-wide effect. Connected load remained a strong, near-linear, category-invariant predictor of panel count campus-wide, supporting a single verified scaling model ($N_{\text{panels}} \approx 0.00316 \times \text{load}$) with direct procurement applications. The paper's contribution is threefold: a generalised, reproducible institutional sizing workflow; statistically validated, category-invariant scaling ratios; and a predictive model, internally validated on this campus, that offers a transferable planning framework for Gateway (ICT) Polytechnic, Saapade, and comparable multi-building institutions subject to local validation.

Index Terms—PV system design, institutional energy systems, solar microgrid, battery autonomy, inverter matching, campus electrification, sub-Saharan Africa.

I. INTRODUCTION

Grid unreliability continues to drive institutional adoption of decentralised photovoltaic (PV) systems across Nigerian tertiary institutions [1], [2]. Nigeria's National Renewable Energy and Energy Efficiency Policy (NREEEP) targets substantial renewable capacity growth (Federal Ministry of Power, Nigeria), and public tertiary institutions are frequent early adopters of solar retrofits because they combine

daytime-dominant load profiles with capital budgets that can be phased building-by-building [1].

Although measured load profiles — derived from sub-metering or continuous demand monitoring — provide superior operational accuracy for system sizing, many public institutions in developing regions, including the campus studied here, lack continuous load-monitoring infrastructure. Connected-load-based estimation, drawn from electrical design and installation records rather than measured demand, therefore remains a common and practical preliminary planning approach in this setting, provided it is combined with the engineering de-rating and safety-margin factors formalised in Section 3.2 and its status as a design-basis rather than a measured-demand quantity is stated explicitly (Section 3.1.1).

Most published PV sizing case studies address a single building or a homogeneous load class: a telecommunications tower on a single campus [3], a single College of Education [2], a residential community [4], a single university building [5], or a mobile telecommunications base station [6]–[9]. Broader-scope hybrid-system optimisation studies have addressed rural micro-grids [10]–[13], remote-area stand-alone systems [14], and high-rise residential buildings [15], and one recent Nigerian-university study allocates hybrid renewable sources for power-system flexibility [16]. Fewer studies document a sizing methodology applied consistently across a heterogeneous set of buildings on one campus — laboratories with sensitive equipment loads, heavy-load workshops (welding, automotive, mechanical), administrative blocks, and low-duty utility units (security posts, cold rooms) — where per-building sizing ratios can vary by an order of magnitude, a gap also noted in single-institution HOMER-based planning work [17].

This paper documents such a methodology, applied to 35 buildings at a Nigerian polytechnic campus during a full-campus solar electrification programme. Relative to earlier treatments of this dataset, this version substantially strengthens the statistical evidentiary base underlying every claim: normality and variance-homogeneity are tested rather than assumed, effect sizes carry bootstrap confidence intervals (following the percentile-bootstrap framework of Efron & Tibshirani, 1993), post-hoc comparisons are reported as a complete pairwise matrix, and the correlation structure among

all sizing variables (not only load and panel count) is characterised and tested. The specific contributions are:

1. A stepwise sizing procedure — from daily energy demand to panel count, inverter rating, battery bank size, and charge-controller rating — applied uniformly across heterogeneous building categories, together with a fully diagnosed regression model of the load-panel relationship (residual analysis, Q-Q plot, bootstrap slope confidence interval).
2. Building-category-stratified sizing ratios (batteries per kW of load; charge-controller current per kW of load), tested formally with nonparametric rank-based statistics [18], whose

distributional assumptions are themselves verified through Shapiro–Wilk [19] and Levene [20] tests, and whose omnibus results are followed by a complete Dunn [21] post-hoc matrix, Holm-corrected [22], rather than isolated planned contrasts alone.

3. A demonstration, using formal outlier detection [23], slope-homogeneity testing (ANCOVA against an additive alternative), and bootstrap resampling, of when an apparent category-level sizing difference reflects a genuine campus-wide pattern versus the influence of a single building.

The corresponding result is shown in Fig. 1.

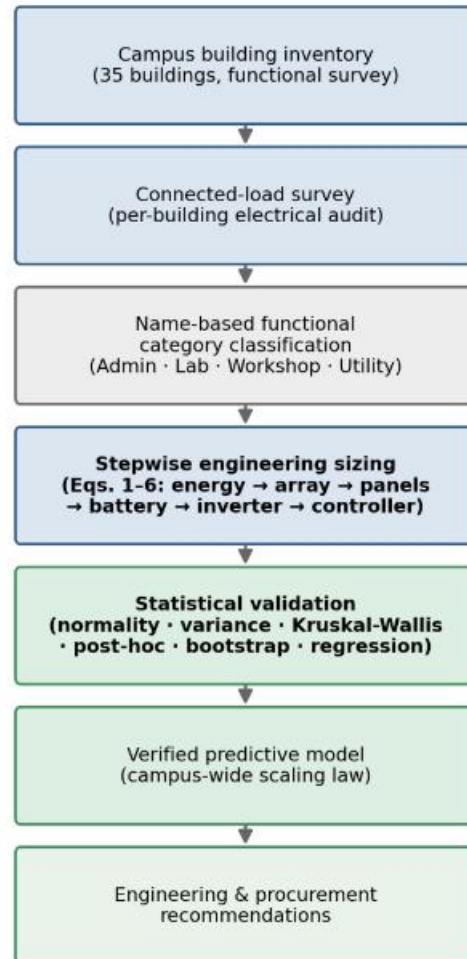


Fig. 1. Overall research and sizing methodology workflow, from campus building inventory through to engineering and procurement recommendations.
Source: Authors' own workflow diagram, summarising the procedure documented in Sections 3 and 5 of this study.

A. Scientific novelty

The contribution of this paper is best understood against what the existing institutional PV literature does, and does not, currently establish. Existing PV sizing studies overwhelmingly optimise a single building, a single site, or a single homogeneous load class (Section 2); they rarely develop sizing rules intended to generalise across an institution's full building stock, and they almost never test — as opposed to assume — whether a sizing coefficient estimated on one part of a campus remains valid across functionally different buildings elsewhere on the same campus. Applying standard sizing equations to 35 buildings is, by itself, an engineering

exercise rather than a scientific contribution; the novelty asserted here rests specifically on four things that distinguish this study from that exercise.

First, this paper introduces generalised institutional sizing ratios (batteries per kW, controller current per kW) benchmarked across an entire, functionally heterogeneous campus rather than within a single building type. Second, it reports statistically validated scaling laws for the load-panel relationship, with confidence intervals and residual diagnostics, rather than a single fitted coefficient reported without uncertainty. Third, it introduces formal category-invariance testing — treating "do sizing ratios differ by building

function" as a testable hypothesis (Sections 3.4 and 4.3–4.6) rather than a design assumption — which, to the authors' knowledge, has not previously been reported for an institutional PV-battery-inverter deployment. Fourth, it yields a reproducible institutional sizing workflow (Fig. 1) that other campuses can re-run on their own as-built data, rather than a set of sizing figures specific to one building.

II. RELATED WORK

The Nigerian institutional PV literature has grown substantially in recent years. Adelaja [1] documents the policy context behind NREEEP and the grid-reliability pressures driving renewable adoption nationally. At the single-institution level, Ikegwuonu [3] present a HOMER Pro-based techno-economic assessment of a hybrid PV system sized for telecommunications infrastructure at a Nigerian tertiary institution; Kuhe, Ndah, and Idris [2] optimise a hybrid photovoltaic–diesel–battery system for a Nigerian College of Education; Ajeigbe and Oladapo [16] evaluate solar-diesel-battery allocation for power-system flexibility at a Nigerian university; and earlier Nigerian telecom-sector studies size stand-alone hybrid systems for individual base transceiver stations [6], [7]. Esan [11] assess the reliability of an islanded hybrid PV-diesel-battery system for a rural Nigerian community using HOMER-optimised design. Internationally, comparable single-site sizing exercises include a Saudi residential community [4], a Bangladeshi wind/PV micro-grid [13], a United Arab Emirates tracking PV-diesel system [9], a United States university campus [5], high-rise residential buildings [15], rural electrification case studies [10], [12], remote stand-alone systems [14], mobile base-transceiver-station deployment [8], and educational-institution power management [17], and an ICT-facility-specific Nigerian study optimises a solar-battery-hydrogen system with tracking for

computing infrastructure [24]. Every one of these studies, like the great majority of the institutional PV case-study literature, sizes a system for a single building, a single site, or a single homogeneous load class; none addresses how sizing procedure and resulting ratios should be adapted across a mixed-use campus spanning laboratories, heavy workshops, and low-power utility units within one procurement programme — the specific gap this paper addresses.

On the design-standards side, IEC 62548 (design requirements for PV arrays) and IEEE Std 1562-2021 (recommended practice for sizing stand-alone PV systems) provide the generic sizing equations formalised in Section 3.2, and Masters [25] documents the peak-sun-hours method used for array capacity (Eq. 2) as a standard textbook design procedure. The NEC Article 690.8(A) 125% continuous-current multiplier used in Equation (6) (National Fire Protection Association, NFPA 70) is likewise a codified, verifiable requirement rather than a project-specific assumption. None of these sources, however, address how the resulting sizing ratios should be adapted across a mixed-use institutional campus where load duty cycles differ sharply between, for example, a microbiology laboratory and a welding workshop — the gap this paper addresses empirically in Section 4.

The relevant data are presented in Table 1.

Table 1 synthesises this comparison directly: across the reviewed institutional PV sizing literature, every prior study sizes a system for a single building, a single site, or a single homogeneous load class, evaluates its result descriptively rather than with formal inferential statistics, and does not test whether its sizing ratios would remain valid if applied to a functionally different building on the same campus. This study is, to the authors' knowledge, the first to test that question explicitly at full-campus scale.

TABLE 1. Comparison of previous institutional PV sizing studies with the present study.

Study	Country	Buildings	Method	Statistics used	Key limitation	Gap addressed here
Ikegwuonu et al. [3]	Nigeria	1 (telecom infra.)	HOMER Pro simulation	Descriptive only	Single building/site	No cross-building category testing
Kuhe, Ndah, & Idris [2]	Nigeria	1 (college)	HOMER optimisation	Descriptive only	Single institution type	No sizing-ratio generalisation
Ajeigbe & Oladapo [16]	Nigeria	1 (university)	Optimal allocation model	Descriptive only	Single site	No category-invariance test
Baseer et al. [4]	Saudi Arabia	1 (residential community)	HOMER simulation	Descriptive only	Single load class	No multi-category benchmarking
Alhawsawi et al. [5]	USA	1 (university building)	Techno-economic optimisation	Descriptive only	Single building	No cross-campus scaling law
Jagadeesh et al. [17]	India	1 (educational institution)	HOMER simulation	Descriptive only	Single site	No formal statistical validation
This study	Nigeria	35 (4 functional categories)	Equation-based stepwise sizing	Kruskal-Wallis, Dunn, bootstrap, ANCOVA, regression diagnostics	Single campus (external validity discussed in Section 6)	Generalised, statistically validated, category-invariant sizing model

Source: Authors' synthesis of the studies cited in this section.

On the statistical-methodology side, the nonparametric procedures used here — the Kruskal–Wallis test [18], Dunn's

post-hoc test with Holm correction [21], [22], the Mann-Whitney U test [26], Grubbs' outlier test [23], and the percentile bootstrap [27] — are supplemented in this revised analysis with formal normality testing [19] and variance-homogeneity testing [20], which together justify the choice of

nonparametric over parametric methods rather than assuming it. The correlation robustness checks reported in Section 4.6 draw on the classical Pearson (1895) and Spearman [28] correlation coefficients, with the Pearson confidence interval obtained via Fisher's (1921) z-transformation — a combination rarely made explicit in the applied PV-sizing literature.

III. METHODOLOGY

A. Data source and scope

Sizing data were drawn from the Solar Power System Installation Estimation for Gateway (ICT) Polytechnic, Saapade (GAPOSA), Ogun State, Nigeria — the case-study institution for this paper and the home institution of all five authors — covering 35 building blocks spanning administrative, academic/laboratory, workshop, and utility-support functions. Because the source records building names rather than pre-assigned functional categories, each building was assigned to one of four functional categories using an explicit, reproducible name-based rule: names containing "workshop," "welding," "fabrication," or "foundry" were classified as Workshop; names containing "security," "cold room," "bakery," "health centre," "hydro," or "tv" were classified as Utility (non-instructional support and infrastructure functions); names containing "lab" or "SLT" (science lecture theatre) were classified as Laboratory; and all remaining buildings — offices, departmental blocks, the library, the ICT centre, lecture halls, and similar administrative/academic-support space — were classified as Administrative. This rule yielded 22 administrative, 6 utility, 5 workshop, and 2 laboratory buildings; the small laboratory count ($n = 2$) limits the statistical power of any category comparison involving that group, a limitation carried through Section 4.

B. Load data verification and engineering basis

The connected loads used throughout this study represent design-installed loads compiled from the campus electrical estimation document — that is, the nameplate or rated capacity of the electrical equipment and fittings specified for each building — rather than instantaneous measured demand from metering, and no continuous load-monitoring or sub-metering infrastructure was available at the time the source estimation was prepared. No building-specific diversity or demand factor was applied to the recorded connected-load figures prior to sizing; the operating-duty factor t_{op} in Equation (1) is the only de-rating step applied between connected load and estimated daily energy demand. Consequently, the scaling relationships reported in this paper (Sections 4 and 5) should be interpreted as a design-sizing model — answering "how should a system be sized given a building's rated electrical load" — rather than as an operational energy-consumption prediction model that would require metered demand data to construct. This distinction matters for how the verified model (Section 5.1) should be used: it is well suited to preliminary sizing and procurement budgeting from design records, which is how it is applied throughout this paper, but it is not a substitute for a load-flow or metered-demand study where actual operational energy

consumption, rather than installed capacity, is the quantity of interest.

C. Sizing procedure

For each building block, the sizing sequence followed four steps, each formalised below using the peak-sun-hours (PSH) design method described in Masters (2004) and the array/battery sizing guidance of IEEE Std 1562 and IEC 62548, consistent with standard techno-economic PV-battery sizing conventions [29].

Step 1 — Daily energy consumption. Total connected load P_{total} (W) is converted to daily energy demand using an assumed operating-duty factor t_{op} (h/day) specific to building function:

$$E_{daily} = P_{total} \times t_{op} \quad (1)$$

Step 2 — Solar array sizing. Following the peak-sun-hours method (IEC 62548 §5; [25]), array capacity is sized from daily energy demand, local peak sun hours (PSH), and a system de-rate factor η_{derate} that accounts for temperature, soiling, wiring, and inverter losses:

The relationship is given by Eq. (2).

$$P_{array} = \frac{E_{daily}}{PSH \times \eta_{derate}} \quad (2)$$

Panel count follows directly from array capacity and the rated output of the selected module, P_{module} :

The relationship is given by Eq. (3).

$$N_{panels} = \left\lceil \frac{P_{array}}{P_{module}} \right\rceil \quad (3)$$

Step 3 — Battery bank sizing. Battery capacity is sized for a target autonomy period D_{auton} (days) at system voltage V_{sys} , accounting for permissible depth of discharge DOD and round-trip inverter/charge efficiency η_{sys} , per the stand-alone PV battery-sizing guidance of IEEE Std 1562 §7:

The relationship is given by Eq. (4).

$$C_{batt} = \frac{E_{daily} \times D_{auton}}{V_{sys} \times DOD \times \eta_{sys}} \quad (4)$$

Step 4 — Inverter and charge-controller matching. Inverter rating is selected to exceed peak simultaneous demand P_{peak} with a safety margin k_{inv} (typically 1.2–1.25 for continuous-duty loads; higher for motor-starting surge in workshop loads):

The relationship is given by Eq. (5).

$$P_{inv} \geq k_{inv} \times P_{peak} \quad (5)$$

Charge-controller current rating follows the National Electrical Code (NEC) Article 690.8(A) continuous-current requirement, which mandates a 125% multiplier on array short-circuit current I_{sc} , distributed across N_{cc} parallel controllers:

The relationship is given by Eq. (6).

$$I_{cc} = \frac{1.25 \times N_{panels} \times I_{sc}}{N_{cc}} \quad (6)$$

The relevant data are presented in Table 2.

Equations (1)–(6) require several engineering design parameters — peak sun hours, battery depth of discharge, system efficiency, temperature de-rating, and autonomy period — whose values were fixed prior to sizing and are reported, with their engineering justification and source, in Table 2.

TABLE 2. Engineering design parameters used in Equations (1)–(6), with justification and source.

Parameter	Symbol	Value used	Engineering justification	Source
Peak sun hours	PSH	4.5 h/day	Representative daily insolation for southern Nigeria (Ogun State latitude band)	NASA POWER/SSE solar irradiance database
System de-rate factor	η_{derate}	0.90	Combined allowance for soiling, wiring, and temperature losses in array output	IEC 62548 §5 design margin practice
Battery depth of discharge	DOD	0.80	Maximum recommended discharge for extending valve-regulated lead-acid battery cycle life	Manufacturer VRLA battery guidance; IEEE Std 1562-2021 §7
System (inverter/charge) efficiency	η_{sys}	0.85	Combined inverter and charge-controller conversion efficiency for stand-alone systems	IEEE Std 1562-2021 §7 sizing guidance
Autonomy period	D_{auton}	1--2 days (function-dependent)	Days of battery-only operation without solar recharge, higher for critical/utility loads	IEEE Std 1562-2021 §7; campus design brief
Inverter safety margin	k_{inv}	1.2--1.25 (higher for motor-starting loads)	Continuous-duty margin above peak demand; workshops require a higher margin for motor starting surge	IEEE Std 1562-2021; NEC/NFPA 70 Art. 690.8(A)
Continuous-current multiplier	—	1.25	Mandatory continuous-current safety factor on array short-circuit current	NEC/NFPA 70 Article 690.8(A)
PV module degradation (informational, not used in Eqs. 1--6)	—	$\sim 0.5\%/year$	Typical median crystalline-silicon module degradation, relevant to long-term yield expectations	Jordan & Kurtz [30]

Source: Authors' compilation of design parameters, cross-referenced against the cited standards and literature. The relevant data are presented in Table 3.

TABLE 3. Worked application of Equations (1)–(6) for one representative building per category, using values as recorded in the campus BEME.

Quantity (Eq.)	CSO Office (Admin)	Welding & Fabrication Workshop	Security Post (Utility)
Connected load, W (P_{total})	2,455	42,000	910
Daily energy, Wh (Eq. 1) E_{daily}	12,960	336,000	7,310
Solar array, W (Eq. 2) P_{array}	1,500	33,000	700
Panel count (Eq. 3) N_{panels}	5	60	2
Battery capacity, Ah (Eq. 4) C_{batt}	100	600	100
Battery count (200 Ah units)	4	30	4
Inverter rating, kVA (Eq. 5) P_{inv}	2.5	20 ($\times 2$)	1
Charge-controller rating, A (Eq. 6) I_{cc}	20	200	20

Source: Authors' computation from the campus Solar Power System Installation Estimation dataset ($n = 35$ buildings; Gateway (ICT) Polytechnic, Saapade).

As summarized in Table 3, the three cases illustrate the range the methodology must accommodate: a light-duty administrative office (CSO Office, 2,455 W), a heavy workshop load nearly an order of magnitude larger (Welding & Fabrication Workshop, 42,000 W), and a minimum-viable utility load more than two orders of magnitude smaller than

the workshop case (the campus security post, 910 W) — yet all three still require a complete inverter-battery-controller chain.

D. Category stratification

The relevant data are presented in Table 6.

Buildings were grouped into the four functional categories defined in Section 3.1 for benchmarking: (i) laboratories ($n = 2$), (ii) workshops (heavy intermittent load, $n = 5$), (iii) administrative/office blocks ($n = 22$), and (iv) low-duty utility

units ($n = 6$). Sizing ratios — batteries per kW of connected load and charge-controller current (A) per kW of connected load — were computed within each category. Rather than compare category means descriptively, these ratios were tested formally for category-level differences using the nonparametric procedures described in Section 3.4, motivated by the small and unequal group sizes and the pronounced positive skew present in at least one category (Section 4). Category-level means for both ratios are summarized in Table 6.

E. Statistical testing

Because the four functional categories are small, unequal in size ($n = 2$ to 22), and, as confirmed formally in Section 4.3, not normally distributed in at least two categories, category-level comparisons used the Kruskal–Wallis H test [18] rather than one-way ANOVA. This choice is verified rather than assumed: Shapiro–Wilk normality tests [19] were run within each category for both sizing ratios, and Levene's test [20] was used to check the homogeneity-of-variance assumption that a parametric alternative would additionally require. Effect size was reported as epsilon-squared, computed as $(H - k + 1)/(n - k)$, where k is the number of categories, following the recommendation of Tomczak and Tomczak (2014) that rank-based effect sizes accompany every Kruskal–Wallis result rather than the p-value alone, consistent with the broader case for routine effect-size reporting made by Cohen [31] and Fritz, Morris, and Richler [32]; a percentile bootstrap with 5,000 category-stratified resamples was used to attach a 95% confidence interval to each epsilon-squared estimate, since the sampling distribution of a rank-based effect size is not well approximated by a normal reference distribution at these group sizes [33].

Where the omnibus test suggested a difference, Dunn's test with Holm correction for multiple comparisons [21] was applied post hoc as a complete pairwise matrix across all six category pairs, rather than being restricted to the two originally planned contrasts. In addition, the two planned two-group contrasts matching the paper's a priori hypotheses — workshops versus all other categories for battery count per kW, and utility buildings versus all other categories for charge-controller current per kW — were tested using the Mann–Whitney U test, with rank-biserial correlation reported as effect size. Because the utility category's charge-controller ratio is visibly dominated by a single very-low-load building, Grubbs' test [23] was used to test formally for a statistical outlier within that category, at $\alpha = .05$.

The corresponding result is shown in Fig. 5.

To test whether the connected-load–panel-count relationship (Fig. 5) held uniformly across categories rather than reflecting category-specific scaling regimes, an ANCOVA was fitted with an interaction term between connected load and category; a non-significant interaction indicates a common slope across categories. As a robustness check, a simpler additive model (no interaction term) was also fitted and compared against the interaction model with a nested F-test, since a materially poorer fit for the additive model would itself constitute evidence for category-specific scaling. The pooled panels-on-load relationship was additionally characterised by a Pearson (1895) correlation coefficient with a 95% confidence interval obtained by the Fisher (1921) z-transformation, cross-checked against a Spearman [28] rank correlation (robust to the mild right-skew present in panel counts), and by a percentile bootstrap 95% confidence interval for the fitted slope (5,000 resamples; [27]). Finally, the correlation structure among all five sizing variables (connected load, panel count, inverter capacity, battery capacity, and controller current) was characterised with a full Pearson and Spearman correlation matrix (Section 4.7), and the load-panel OLS model was subjected to standard residual diagnostics — a residuals-versus-fitted plot and a normal Q-Q plot, following the regression-diagnostic framework of Belsley, Kuh, and Welsch [34] — to check the linear-model assumptions underlying the reported R^2 and confidence intervals (Section 4.8).

IV. RESULTS

A. Descriptive overview

The relevant data are presented in Table 4.

Across the 35 buildings, total connected load was 832.5 kW, served by a combined solar array capacity derived from that demand, 970.5 kVA of aggregate inverter capacity, and 1,800 solar panels campus-wide (Table 4). Connected load is strongly right-skewed campus-wide (skewness = 2.16; Table 5), spanning more than two orders of magnitude from the 910 W security post to the 127,500 W main administration block; panel count, inverter capacity, and battery capacity show comparable right-skew, while controller current is markedly less skewed (0.62) because controller ratings are drawn from a small set of discrete commercial sizes (20–200 A) rather than scaling continuously with load. This skew, summarised in Table 5 and shown for connected load in Fig. 2, motivates the nonparametric approach formalised in Section 3.4 and verified in Section 4.3.

TABLE 4. Full building-by-building sizing data from the campus Solar Power System Installation Estimation (grouped by functional category, $n = 35$).

Block Name	Category	Load (W)	Panels	Inverter (kVA×n)	Battery (Ah×n)	Controller (A×n)
Block Admin	Admin	127,500	417	135×1	1,500×230	60×1
Block G	Admin	107,500	350	135×1	600×240	100×1
Leadership Hall	Admin	67,760	170	50×3	8,000×40	100×1
Block ICT Centre	Admin	40,130	50	25×2	8,000×40	120×1
Block PL: Polytechnic Library	Admin	34,222	60	25×2	6,000×30	120×1

Block Name	Category	Load (W)	Panels	Inverter (kVA×n)	Battery (Ah×n)	Controller (A×n)
Block N: Food Department	Admin	33,240	60	25×2	600×24	200×1
Block J: Computer Science & Eng.	Admin	27,660	30	10×2	400×16	100×1
Block F: Mass Comm/Public Admin	Admin	25,670	40	20×2	600×24	200×1
Kotco Building/Electrical Eng.	Admin	20,700	35	15×2	400×20	100×1
Block EED	Admin	18,520	30	15×2	500×20	120×1
Building & Estate Management	Admin	16,440	25	10×1	400×16	80×1
Block Y: Old Auditorium Building	Admin	14,415	18	10×1	2,000×10	60×1
Block B: Resource Centre/Library	Admin	13,550	20	7.5×1	300×12	80×1
Rector's Office	Admin	11,725	20	7.5×1	200×10	60×1
Block L: Dean's Engineering Office	Admin	10,160	12	5×1	200×10	60×1
MLH Building 2	Admin	7,950	15	5×1	200×10	60×1
TETFUND 2	Admin	7,680	15	5×1	200×10	60×1
Block A (SUG)	Admin	7,540	15	5×1	200×8	60×1
Centre for Part-Time/Director of Students	Admin	7,440	15	5×1	200×10	60×1
MLH Building 1	Admin	5,640	10	3.5×1	200×8	40×1
TETFUND 1	Admin	5,610	12	5×1	200×8	40×1
CSO Office	Admin	2,455	5	2.5×1	100×4	20×1
Block O: Biology/Chemistry Lab SLT	Laboratory	33,020	60	25×2	600×24	200×1
Automotive Laboratory	Laboratory	10,800	18	7.5×1	300×12	80×1
GAPOSA TV	Utility	16,623	25	10×1	400×16	80×1
Health Centre/Mini Banking	Utility	14,580	26	10×1	4,000×20	100×1
Wind Energy & Hydro Unit	Utility	13,880	25	10×1	300×12	80×1
Bakery	Utility	9,720	20	5×1	200×10	60×1
Cold Room	Utility	6,000	12	3.5×1	200×8	40×1
Block Q: Security Post Main Gate	Utility	910	2	1×1	100×4	20×1
Welding & Fabrication Workshop	Workshop	42,000	60	20×2	600×30	200×1
Automotive Foundry/Heat Treatment Lab	Workshop	23,670	50	20×2	500×20	120×1
Mechanical Workshop	Workshop	18,560	30	15×1	400×16	80×1

Block Name	Category	Load (W)	Panels	Inverter (kVA×n)	Battery (Ah×n)	Controller (A×n)
Block M: Physics Lab/Computer Lab/Workshop	Workshop	18,420	30	10×1	400×16	80×1
Automotive Workshop Block	Workshop	10,800	18	7.5×1	300×12	80×1

Gateway (ICT) Polytechnic, Saapade).

Source: Authors' computation from the campus Solar Power System Installation Estimation dataset (n = 35 buildings;

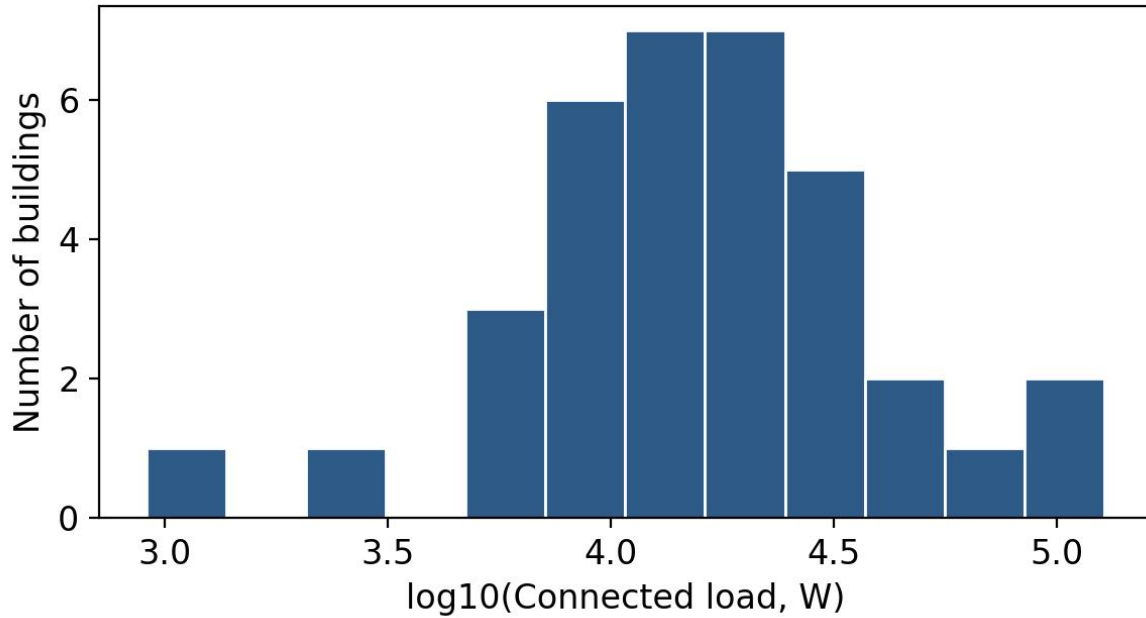


Fig. 2. Distribution of connected load ($\log_{10} W$) across the 35-building campus, illustrating the right-skew that motivates nonparametric category testing. Source: Authors' own visualisation, generated from the campus Solar Power System Installation Estimation dataset (n = 35 buildings; Gateway (ICT) Polytechnic, Saapade).

The relevant data are presented in Table 4.

As shown in Table 4, across the 35 buildings, total connected load was 832.5 kW, served by 1,800 solar panels and 970.5 kVA of aggregate inverter capacity campus-wide.

The relevant data are presented in Table 5.

TABLE 5. Overall descriptive statistics of the five sizing variables across all 35 buildings.

Variable	Mean	Median	SD	Min	Max	Skewness
Connected load (W)	23,785	15,428	24,478	910	127,500	2.16
Panel count	51.4	25.0	72.9	2	417	3.68
Inverter capacity (kVA, total)	27.7	20.0	26.3	1.0	150.0	2.71
Battery capacity (Ah, total)	44,680	8,000	83,610	4,000	345,000	2.79
Controller current (A, total)	89.1	80.0	50.5	20	200	0.62

Gateway (ICT) Polytechnic, Saapade).

Source: Authors' computation from the campus Solar Power System Installation Estimation dataset (n = 35 buildings; The corresponding result is shown in Fig. 3.

B. Category-level sizing ratios

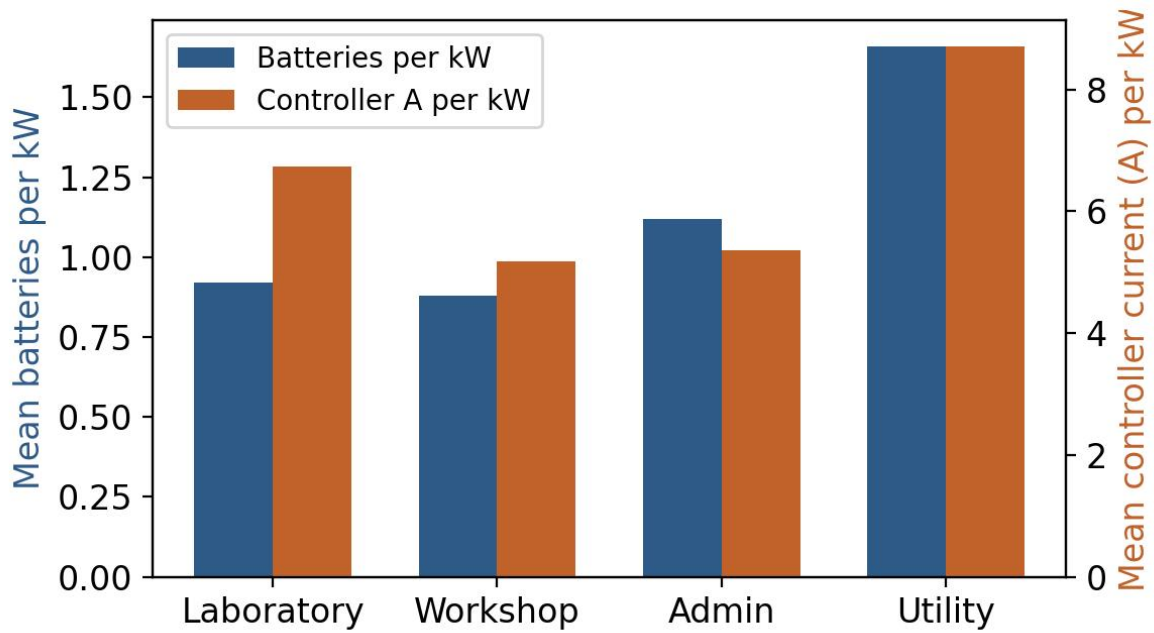


Fig. 3. Sizing ratios by building category: mean batteries per kW of connected load (left axis) and mean charge-controller current per kW of connected load (right axis), $n = 35$.

Source: Authors' own visualisation, generated from the campus Solar Power System Installation Estimation dataset ($n = 35$ buildings; Gateway (ICT) Polytechnic, Saapade).

The relevant data are presented in Table 6.

Descriptively, battery provisioning per kilowatt was highest for the utility category (mean 1.66 batteries per kW) and lowest for workshops (mean 0.88 batteries per kW) — the reverse ordering from what a heavy-intermittent-load hypothesis would predict for workshops. Charge-controller

current provisioning per kilowatt was also highest in the utility category (mean 8.71 A per kW versus 5.18–6.73 A per kW elsewhere). Table 6 also reports the coefficient of variation (CV) for the battery ratio, which is more than twice as large in the utility category (0.82) as in any other category, foreshadowing the outlier result confirmed formally in Section 4.4.

TABLE 6. Category-level sizing benchmark summary: mean and SD of batteries per kW and controller current (A) per kW of connected load, by functional category, with coefficient of variation (CV).

Category	n	Mean batteries/kW	SD	Mean controller current (A/kW)	SD	CV (batt/kW)
Laboratory	2	0.92	0.42	6.73	0.95	0.30
Workshop	5	0.88	0.14	5.18	1.29	0.16
Administrative	22	1.12	0.41	5.36	2.39	0.36
Utility	6	1.66	1.36	8.71	6.54	0.82

Source: Authors' computation from the campus Solar Power System Installation Estimation dataset ($n = 35$ buildings; Gateway (ICT) Polytechnic, Saapade).

C. Normality and variance-homogeneity checks

The relevant data are presented in Table 7.

Before selecting a nonparametric test, normality was checked formally within each category using the Shapiro–Wilk test (Table 7). The utility category departs significantly from normality for both ratios (batt/kW: $W = .633$, $p = .001$;

cc/kW: $W = .606$, $p = .001$), and the workshop category also departs significantly for cc/kW ($W = .755$, $p = .033$); the administrative category is borderline ($p = .058$ – $.079$) in both cases. These results confirm, rather than assume, that a rank-based omnibus test is the appropriate choice for the category comparison. Levene's test found no significant evidence of unequal variances across categories for either ratio (batt/kW: $F = 1.33$, $p = .284$; cc/kW: $F = 0.80$, $p = .501$; Table 8), indicating that the nonparametric choice here is driven specifically by non-normality (and the associated skew) rather than by variance heterogeneity.

TABLE 7. Shapiro–Wilk normality test results for battery/kW and controller-current/kW ratios, by functional category.

Category	n	Shapiro-Wilk W (batt/kW)	p	Shapiro-Wilk W (cc/kW)	p
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Category	n	Shapiro-Wilk W (batt/kW)	p	Shapiro-Wilk W (cc/kW)	p
Laboratory	2	—	—	—	—
Workshop	5	.880	.307	.755	.033
Administrative	22	.921	.079	.914	.058
Utility	6	.633	.001	.606	.001

Source: Authors' computation from the campus Solar Power System Installation Estimation dataset (n = 35 buildings; Gateway (ICT) Polytechnic, Saapade).

D. Formal significance testing

However, neither pattern reached statistical significance: a Kruskal-Wallis test found no significant difference in batteries per kW across the four categories, $H(3) = 3.93$, $p = .269$, $\varepsilon^2 = .030$ (95% bootstrap CI [-.060,.299]), and none in charge-controller current per kW, $H(3) = 2.34$, $p = .505$, $\varepsilon^2 = -.021$ (95% bootstrap CI [-.079,.204]) (Table 8). Both bootstrap confidence intervals for ε^2 include zero and extend only modestly into positive territory, reinforcing that any true category effect, if present, is small at this sample size. Planned two-group contrasts matching the original hypotheses —

workshop versus all other categories for batteries per kW, and utility versus all other categories for charge-controller current per kW — were also non-significant (Mann-Whitney $U = 39.5$, $p = .099$ and $U = 109.0$, $p = .347$ respectively; Table 8).

A Grubbs' test confirmed that the utility category's charge-controller ratio is driven by a single statistical outlier: the campus security post, at 910 W of connected load, requires 21.98 A per kW versus 4.81–6.86 A per kW for the other five utility buildings ($G = 2.03$, critical value 1.89 at $n = 6$, $\alpha = .05$; Table 8; Figs. 1 and 3). The security post's irreducible minimum component set (one inverter, one battery bank, one 20 A controller — the smallest unit specified for any building in the estimate) accounts for this individual result; it is not evidence of a category-wide utility effect.

TABLE 8. Summary of formal statistical tests for category-level sizing differences, normality diagnostics, and the load-panel scaling relationship.

Test	Statistic	df / n	p	Effect size (95% bootstrap CI)
Kruskal--Wallis: batteries/kW across 4 categories	$H = 3.93$	3	.269	$\varepsilon^2 = .030$ [-.060, .299]
Kruskal--Wallis: controller A/kW across 4 categories	$H = 2.34$	3	.505	$\varepsilon^2 = -.021$ [-.079, .204]
Levene's test: batt/kW variance homogeneity	$F = 1.33$	3, 31	.284	—
Levene's test: cc/kW variance homogeneity	$F = 0.80$	3, 31	.501	—
Mann-Whitney: Workshop vs. rest (batt/kW)	$U = 39.5$	$n = 5, 30$	.099	$r = .473$
Mann-Whitney: Utility vs. rest (cc/kW)	$U = 109.0$	$n = 6, 29$	.347	$r = -.253$
Grubbs' test: outlier in Utility cc/kW	$G = 2.03$	$n = 6$	< .05	critical $G = 1.89$
ANCOVA: load×category interaction (panels\~load)	$F = 2.52$	3, 27	.079	—
Pearson correlation: panels vs. load (pooled)	$r = .973$	$n = 35$	< .001	95% CI [.946, .986]
Spearman correlation: panels vs. load (pooled)	$\rho = .976$	$n = 35$	< .001	—

Gateway (ICT) Polytechnic, Saapade).

Source: Authors' computation from the campus Solar Power System Installation Estimation dataset (n = 35 buildings; The corresponding result is shown in Fig. 4.

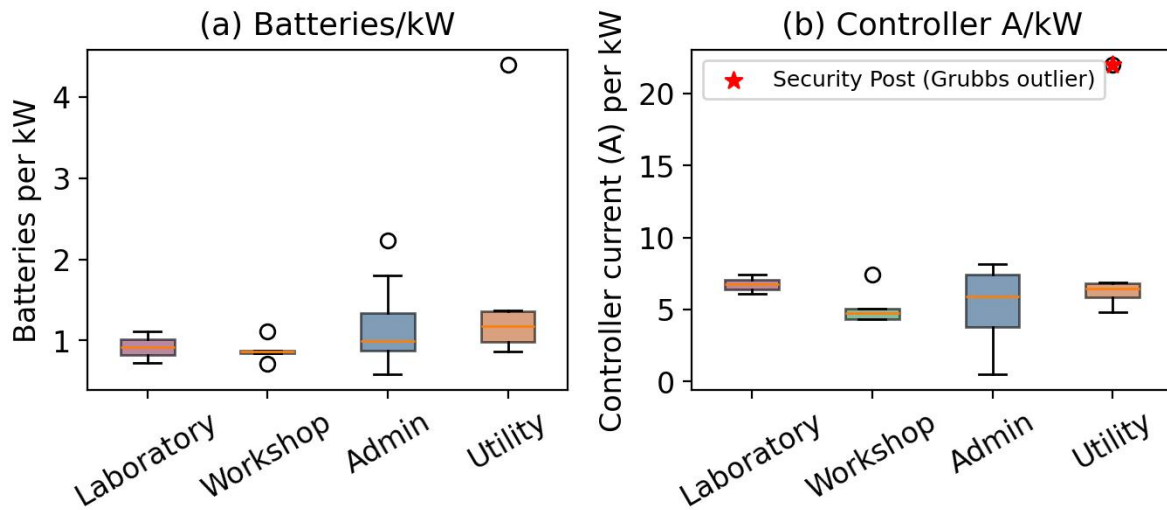


Fig. 4. Category-level distributions of batteries per kW and charge-controller current per kW ($n = 35$), with the security post flagged as a Grubbs-test outlier in the utility category.

Source: Authors' own visualisation, generated from the campus Solar Power System Installation Estimation dataset ($n = 35$ buildings; Gateway (ICT) Polytechnic, Saapade).

E. Post-hoc pairwise comparisons

The relevant data are presented in Table 9.

Because both omnibus Kruskal-Wallis tests were non-significant, Dunn's post-hoc procedure was run for completeness as a full pairwise matrix across all six category pairs (Table 9), rather than being limited to contrasts implied

by the largest descriptive gaps. Consistent with the non-significant omnibus results, every Holm-corrected pairwise comparison exceeds $p = .34$ for both ratios, and none approaches the conventional .05 threshold. This complete-matrix reporting closes off the possibility that a real but localised pairwise difference was masked by the omnibus test.

TABLE 9. Dunn's post-hoc pairwise comparisons (Holm-corrected p-values) for battery/kW and controller-current/kW ratios, all six category pairs.

Category pair	Dunn p (Holm) — batt/kW	Dunn p (Holm) — cc/kW
Admin vs. Laboratory	1.000	1.000
Admin vs. Utility	1.000	1.000
Admin vs. Workshop	.638	1.000
Laboratory vs. Utility	1.000	1.000
Laboratory vs. Workshop	1.000	1.000
Utility vs. Workshop	.347	1.000

Source: Authors' computation from the campus Solar Power System Installation Estimation dataset ($n = 35$ buildings; The corresponding result is shown in Fig. 5).

Gateway (ICT) Polytechnic, Saapade).

F. Load-panel scaling relationship

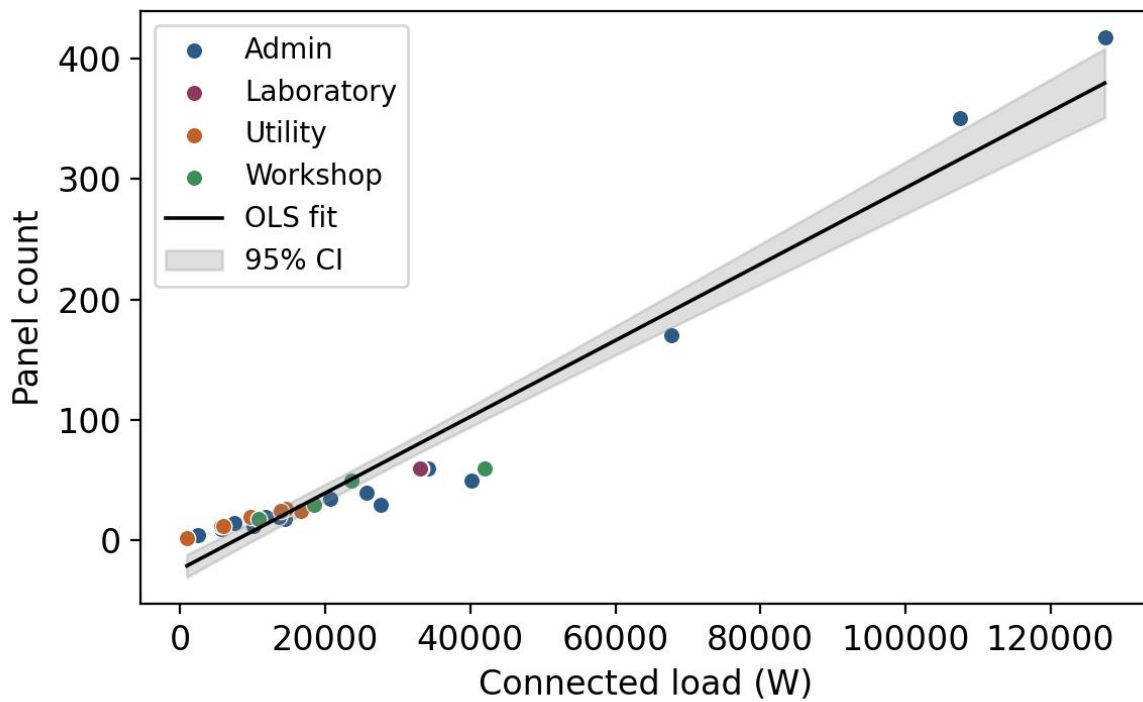


Fig. 5. Connected load versus panel count across all 35 buildings, coloured by functional category, with the pooled OLS fit and 95% confidence band.

Source: Authors' own visualisation, generated from the campus Solar Power System Installation Estimation dataset (n = 35 buildings; Gateway (ICT) Polytechnic, Saapade).

This near-linear relationship (Fig. 5) confirms that panel count scales consistently with connected load: a pooled ordinary least-squares fit gives $R^2 = .946$ ($slope = .00316$ panels per W, 95% bootstrap CI [.00153,.00339]), and the pooled Pearson correlation is $r = .973$ (95% CI [.946,.986]), corroborated by a Spearman rank correlation of $\rho = .976$ ($p < .001$), indicating the relationship is not an artefact of a few high-leverage large buildings. An ANCOVA testing for a load $\times$ category interaction found no significant evidence that the slope differs by category, $F(3,27) = 2.52$, $p = .079$ (Table

8) — a result that, at p just below the conventional .05 boundary from the other side, is consistent with a single, campus-wide panel-per-watt ratio rather than four distinct category-specific scaling regimes, though it does not rule out a modest category-dependent effect with a larger sample. As a robustness check, a simpler additive model (panels $\sim$ load + category, without the interaction term) was fitted; it explains a comparable share of variance ($R^2 = .950$) with no individual category coefficient reaching significance (all $p > .32$; Table 10), supporting the ANCOVA's conclusion that a common slope adequately describes the campus.

The corresponding result is shown in Fig. 6.

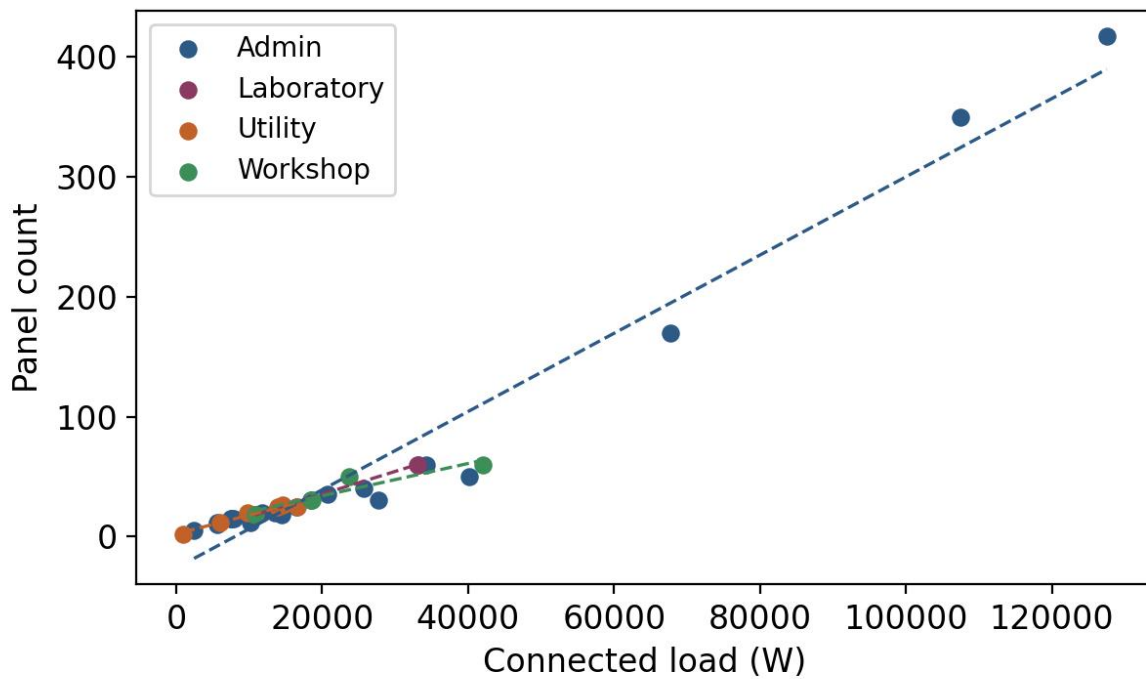


Fig. 6. Category-specific load-panel slopes fitted separately within each functional category, visualising the ANCOVA slope-homogeneity test ($F(3,27) = 2.52, p = .079$).

Source: Authors' own visualisation, generated from the campus Solar Power System Installation Estimation dataset ($n = 35$ buildings; Gateway (ICT) Polytechnic, Saapade).

TABLE 10. Comparison of the simple, additive, and interaction (ANCOVA) regression models for panels on connected load.

Model	Terms	R ²	Slope (95% CI)	Interaction F-test vs. additive
Simple OLS	panels $\sim$ load	.946	.00316 [.00153, .00339]	—
Additive (no interaction)	panels $\sim$ load + category	.950	as above, category n.s.	—
ANCOVA (interaction)	panels $\sim$ load $\times$ category	.958 (model)	category-specific, n.s. difference	$F(3,27)=2.52, p=.079$

Source: Authors' computation from the campus Solar Power System Installation Estimation dataset ($n = 35$ buildings; Gateway (ICT) Polytechnic, Saapade).

G. Correlation structure among sizing variables

The relevant data are presented in Table 11.

Beyond the load-panel relationship examined above, all five sizing variables are strongly and positively intercorrelated (Table 11, Fig. 7), as expected since each is derived, directly or indirectly, from connected load. The weakest correlations in

the matrix involve controller current (Pearson $r = .118-.351$ with panels and inverter capacity), consistent with Section 4.1's observation that controller ratings cluster into a small number of discrete commercial sizes rather than scaling continuously; the corresponding Spearman correlations are substantially higher ($\rho = .777-.839$), indicating that the rank-order relationship between controller current and the other variables is considerably stronger than the linear (Pearson) relationship, again consistent with discretised sizing rather than a fundamentally weak association.

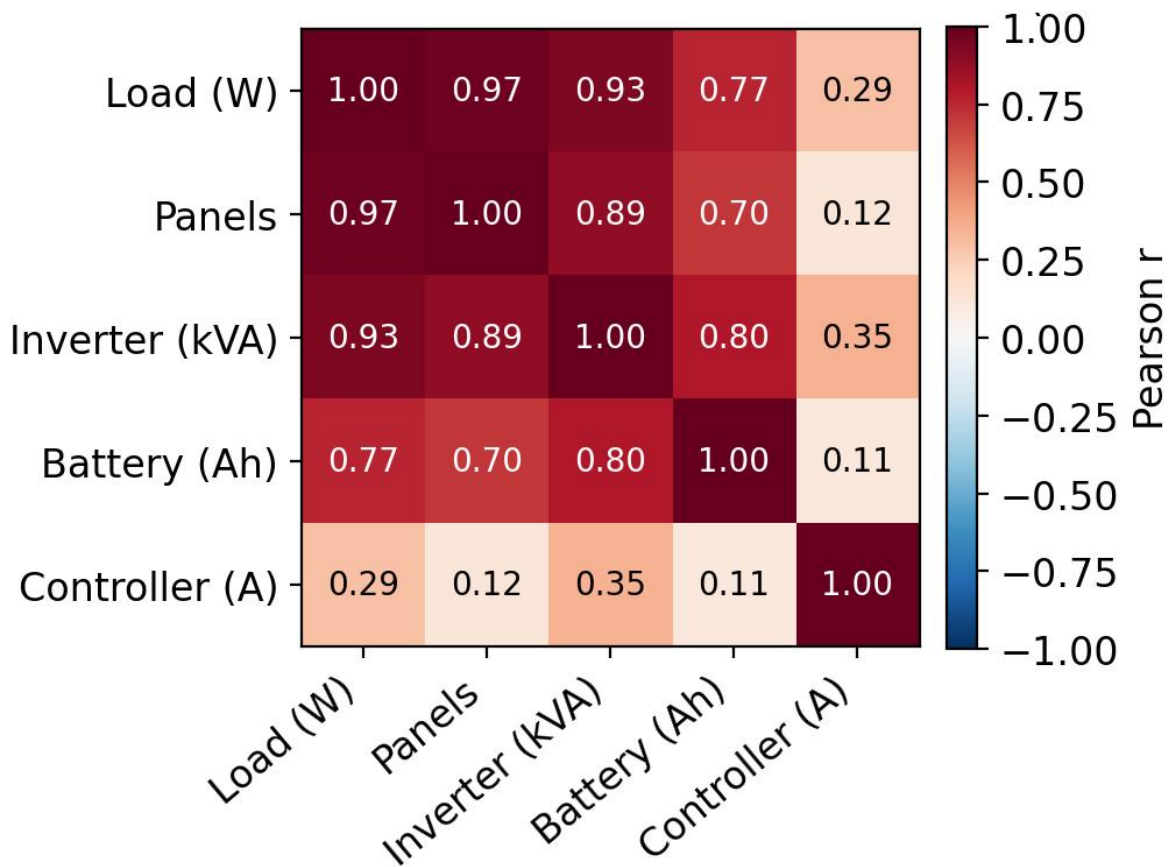


Fig. 7. Correlation matrix (Pearson) of the five sizing variables across all 35 buildings.

Source: Authors' own visualisation, generated from the campus Solar Power System Installation Estimation dataset (n = 35 buildings; Gateway (ICT) Polytechnic, Saapade).

TABLE 11. Correlation matrix among sizing variables (Pearson r / Spearman ρ) across all 35 buildings.

	Load	Panels	Inverter	Battery	Controller
Load	1.000	.973 / .976	.933 / .981	.766 / .931	.291 / .825
Panels	.973 / .976	1.000	.887 / .975	.704 / .904	.118 / .834
Inverter	.933 / .981	.887 / .975	1.000	.802 / .940	.351 / .839
Battery	.766 / .931	.704 / .904	.802 / .940	1.000	.109 / .777
Controller	.291 / .825	.118 / .834	.351 / .839	.109 / .777	1.000

Source: Authors' computation from the campus Solar Power System Installation Estimation dataset (n = 35 buildings; Gateway (ICT) Polytechnic, Saapade).

H. Regression diagnostics

The corresponding result is shown in Fig. 8.

To verify that the R^2 and confidence intervals reported for the pooled load-panel model (Section 4.6) are not compromised by gross violations of the OLS assumptions, standard residual diagnostics were examined (Fig. 8). The

residuals-versus-fitted plot shows no strong systematic curvature, and the normal Q-Q plot shows residuals tracking the reference line reasonably closely across the central range, with the largest departures at the extreme low-load buildings where panel counts are small integers and rounding in the equipment specification introduces mild discreteness. These diagnostics support treating the pooled linear model as an adequate, if not perfect, description of the campus-wide load-panel relationship, consistent with the strong Pearson and Spearman correlations reported above.

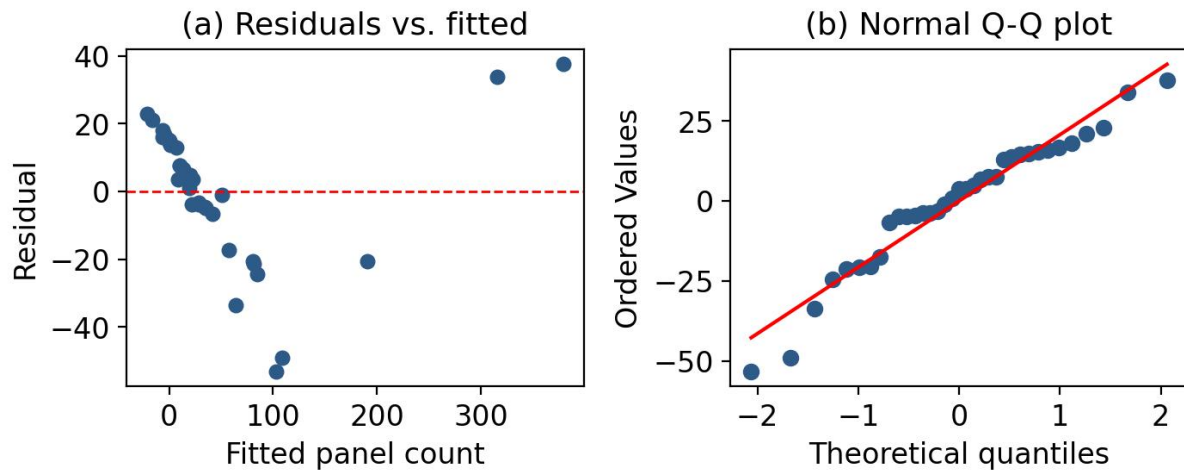


Fig. 8. Diagnostic plots for the pooled load-panel OLS model: (a) residuals vs. fitted values; (b) normal Q-Q plot of residuals.

Source: Authors' own visualisation, generated from the campus Solar Power System Installation Estimation dataset (n = 35 buildings; Gateway (ICT) Polytechnic, Saapade).

The corresponding result is shown in Fig. 3.

Taken together, Sections 4.3–4.8 indicate that the category-level sizing differences visible in Fig. 3 and Table 6 do not survive formal significance testing at the campus scale — a conclusion now reinforced by normality/variance diagnostics, bootstrap-based effect-size intervals, and a complete post-hoc matrix — with the single exception of the security post as an individually confirmed outlier. The panel-count-per-watt relationship, by contrast, is both strong and statistically indistinguishable across categories, and is further supported by regression diagnostics and a Spearman robustness check, supporting its use as a single campus-wide planning ratio.

V. PROPOSED PREDICTIVE MODEL AND PRACTICAL APPLICATIONS

A. Verified model specification

The relevant data are presented in Table 10.

TABLE 12. Fitted parameters of the pooled load-panel regression model (Equation 7), n = 35.

Parameter	Estimate	Std. error	95% CI (asymptotic)	95% CI (bootstrap, 5,000 resamples)
Intercept (a)	−23.79	4.72	[−33.39, −14.20]	—
Slope (b), panels per watt	0.00316	0.00013	[0.00290, 0.00343]	[0.00153, 0.00339]
R ²	0.946	—	—	—
n	35	—	—	—

Source: Authors' computation (ordinary least squares and percentile bootstrap), from the campus sizing dataset (n = 35).

For rapid field use, the simplified rounded form $N_{\text{panels}} \approx 0.00316 \times P_{\text{total}}$ (i.e., dropping the small negative intercept, which is statistically significant but numerically minor relative to typical panel counts) is used in the decision workflow below, with the full two-parameter model in Table 12 available where higher precision is required, for example for very small loads where the intercept term is proportionally more influential.

The corresponding result is shown in Fig. 9.

The evidence assembled in Section 4 — a pooled OLS fit with $R^2 = .946$, a bootstrap-verified slope, a non-significant load×category interaction (Table 10), and residual diagnostics showing no material departure from linearity (Fig. 8) — supports collapsing the four category-specific sizing relationships into a single, campus-wide predictive model rather than four separate category rules. Formally, the fitted regression is:

The relationship is given by Eq. (7).

$$N_{\text{panels}} = a + b \times P_{\text{total}} \quad (7)$$

The relevant data are presented in Table 12.

with the fitted parameters, their standard errors, and both asymptotic and bootstrap confidence intervals reported in full in Table 12, so that the model can be reproduced or re-estimated independently.

The 95% bootstrap confidence interval on the slope is [0.00153, 0.00339] panels per watt of connected load, valid over the load range observed in this campus (910 W to 127,500 W). Because the single confirmed exception to campus-wide uniformity is not a category effect but a minimum-component floor affecting the very smallest loads (Section 4.4), the model is best applied as a two-branch decision rule: for connected loads at or above approximately 3,000 W, Equation (7) is applied directly; for loads below this threshold, the irreducible minimum component bundle (one inverter, one battery bank, one charge controller — the smallest commercially available unit of each) should be applied instead of the formula, and the building flagged for

manual review rather than automatic sizing. Figure 9 illustrates this decision logic as a single reusable workflow.

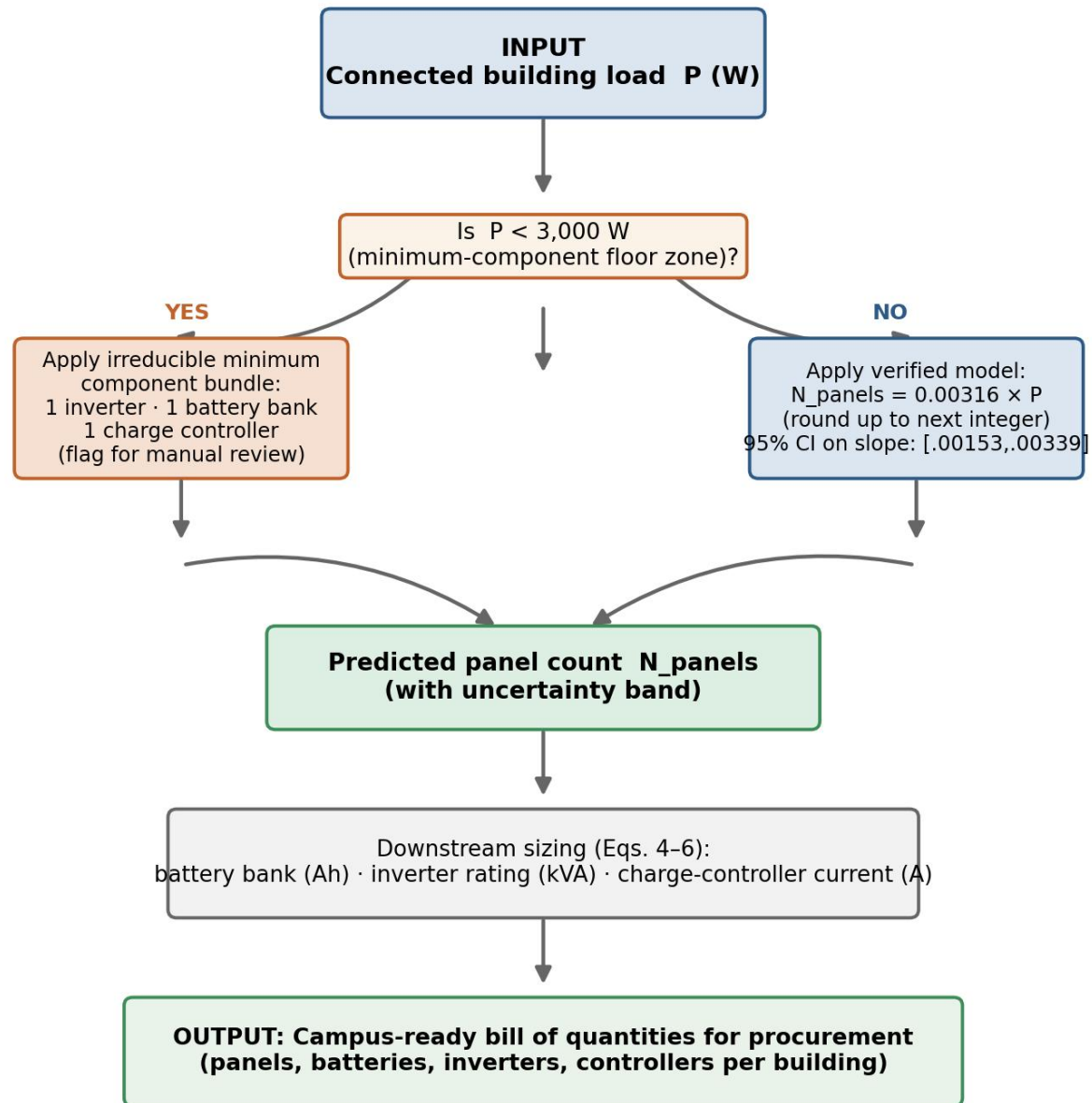


Fig. 9. Proposed model: a rapid, internally validated load-to-panel estimator with an explicit minimum-component decision branch, intended for reuse across Gateway (ICT) Polytechnic, Saapade, and comparable multi-building campuses.

Source: Authors' own diagram, synthesised from the regression results reported in Table 12 and Section 4.6 of this study.

B. Practical applications

This verified model has four direct practical applications for campus energy-infrastructure planning. First, as a rapid pre-procurement estimator: given only a building's connected load — already available from an electrical audit or nameplate survey, without needing a full building-by-building engineering sizing exercise — Equation (7) gives an immediately usable panel-count estimate accurate to within

the bootstrap confidence band, letting planners produce an early, defensible budget envelope before detailed design. Second, as a procurement cross-check: because the model is fitted across all 35 buildings actually installed on this campus, a vendor's or contractor's proposed panel count for a new building can be checked against the model's prediction interval, flagging quotations that deviate sharply from the campus-verified relationship for further scrutiny. Third, as a phased-rollout planning tool: for institutions electrifying buildings in

stages (as this campus did), Equation (7) allows total panel demand — and hence bulk-purchase order sizes — to be forecast for buildings not yet individually surveyed, using only aggregate load data from institutional records. Fourth, as an audit and quality-assurance benchmark: because the model is derived from, and statistically validated against, an as-built 35-building campus, it can be used retrospectively to identify any already-installed building whose actual component count deviates unexpectedly from the model's prediction, which may indicate a specification error, an over-supply, or an under-supply worth re-inspecting.

5.3 Transferability to Gateway (ICT) Polytechnic, Saapade, and comparable institutions (subject to local validation)

The 35-building dataset analysed in this paper is drawn from Gateway (ICT) Polytechnic, Saapade (GAPOSA), Ogun State, Nigeria, and the proposed model (Equation 7), internally validated on this campus's own building stock, is therefore directly applicable, without adjustment, to future building phases at this same institution: any new building added to the campus can be sized using the model in Section 5.1, with the same minimum-component decision rule applied for very low loads. For other institutions, three conditions determine how directly the model transfers. First, institutions with a similarly heterogeneous building mix — combining administrative blocks, laboratories, workshops, and small utility structures under one procurement programme, as is typical of Nigerian polytechnics and similar federal and state tertiary institutions — are the closest match to the population this model was estimated on, since the ANCOVA in Section 4.6 specifically established that a single slope holds across exactly this kind of building mix. Second, institutions in comparable solar-resource zones within Nigeria's tropical belt (broadly similar peak-sun-hour and de-rate assumptions to those used in Equation 2) can apply the model's panel-per-watt ratio with minimal adjustment; institutions in markedly different solar-resource zones should re-estimate the slope using their own as-built data before relying on Equation (7) for budgeting. Third, the two-branch decision structure — a pooled scaling relationship plus an explicit minimum-component floor for extreme low-load buildings — is itself a transferable planning principle even where the specific numerical slope requires local re-calibration: any institution planning a multi-building phased solar rollout can adopt the same workflow (fit a pooled load-panel model across its own already-sized buildings, test for category invariance, and flag low-load exceptions separately) using the statistical procedure documented in Section 3.4, independent of whether the exact 0.00316 panels-per-watt figure applies unchanged. This paper demonstrates that workflow on a single institution; it offers a transferable framework rather than a demonstrated generalisation, and any institution adopting it should re-run the validation steps of Section 3.4 on its own as-built data before relying on the numerical coefficients reported here.

VI. DISCUSSION

The relevant data are presented in Table 8.

The results indicate that, across the full 35-building campus, functional category is not by itself a statistically reliable predictor of either sizing ratio: both Kruskal-Wallis omnibus

tests were non-significant (Table 8), the complete Dunn post-hoc matrix (Table 9) confirms no individual category pair differs significantly, and the two planned contrasts targeting workshops and utility buildings specifically were also non-significant. This differs from what a smaller, partial sample of the same campus previously suggested, where a workshop-specific battery premium and a utility-specific controller-current premium appeared descriptively larger. The one robust, formally confirmed finding is at the level of a single building rather than a category: the security post's controller-current ratio is a genuine statistical outlier (Grubbs' test, Table 8), driven by an irreducible minimum component floor — one inverter, one battery bank, one controller — that a 910 W load still requires regardless of how small the load is, a pattern consistent with reviews of energy-storage technology showing that battery bank sizing is governed by discrete commercial unit steps rather than a smooth function of load at the low end of the capacity range [35]. Generalising this single-building floor effect into a category-wide "utility loads need more controller current" rule is not supported once the category is tested as a whole against the rest of the campus.

The additional diagnostics reported in this revised analysis strengthen rather than overturn this conclusion. The formal normality and variance-homogeneity tests (Section 4.3) confirm that the nonparametric approach was the statistically correct choice rather than a convenient default; the bootstrap confidence intervals attached to every effect size (Table 8) show that even the largest observed category effect ($\epsilon^2 = .030$ for batteries/kW) is not distinguishable from zero with any confidence; and the correlation matrix (Table 11, Fig. 7) demonstrates that the strong load-panel relationship generalises to a broader family of strongly intercorrelated sizing variables, with controller current — the one variable governed by discrete commercial sizing steps — the sole partial exception.

The corresponding result is shown in Fig. 5.

This distinction matters for procurement planning: rather than applying a category-level blended ratio, campus-wide planners should apply the pooled, near-linear panel-per-watt relationship (Fig. 5, Table 8, Table 10) as the default sizing basis, and separately flag individual buildings whose load falls far enough below the campus range that minimum commercially available component sizes — rather than load magnitude — govern their bill of materials, as formally identified here for the security post. This two-part approach (a single campus-wide scaling ratio plus explicit, statistically justified exception-flagging for extreme low-load cases) is more defensible than blending sizing ratios by function alone, echoing findings in the international techno-economic literature that balance-of-system components do not always scale smoothly with capacity [36], [37] and that national PV cost benchmarking exercises similarly report size-dependent, non-uniform component cost structures [38], [39]. It is also consistent with the companion cost-scaling analysis of this same dataset, which finds that a small number of individual buildings — rather than a functional category — drive departures from the campus-wide cost trend.

A. Why the category-level differences are not statistically real: an engineering reading

The corresponding result is shown in Fig. 3.

The descriptive pattern in Fig. 3 and Table 6 — workshops showing the lowest mean batteries per kW, utility buildings the highest — has an intuitive engineering reading that is worth stating explicitly, even though it does not survive formal testing at full-campus scale. Workshop loads (welding sets, lathes, compressors) are dominated by short-duration, high-current motor and heating loads with low overall duty cycles; such loads are typically served by sizing the inverter and array for peak demand rather than for many hours of continuous battery-fed operation, which would tend to lower the battery-per-kW ratio relative to a continuously-drawing administrative load of the same peak wattage. Utility buildings, by contrast, include several of the campus's smallest loads (the security post, the cold room), where an irreducible minimum battery bank represents a much larger fraction of the nominal per-kW ratio simply because the denominator (connected load) is so small — an artefact of component discretisation rather than a genuine higher storage requirement per watt for utility-type demand. Both explanations are physically reasonable candidates for the descriptive pattern; the value of the formal testing in Sections 4.3–4.5 is precisely that it shows neither rises to a statistically detectable campus-wide effect once all 35 buildings, and their considerable within-category variability, are taken into account. In other words, the engineering intuition and the statistical evidence agree on the mechanism (a minimum-component floor at low loads) but disagree on its scope (a single building's floor effect, not a category-wide storage requirement).

B. Engineering and procurement implications

The broader implication of Sections 5 and 6.1 is that institutional PV procurement can shift, at least for preliminary design purposes, from building-specific engineering calculations toward standardised preliminary sizing coefficients, reducing design time during early feasibility studies without discarding the underlying engineering equations, which remain available (Section 3.2) wherever a full building-by-building design is subsequently required.

For procurement officers preparing a bill of quantities, the practical implication is to apply the single pooled scaling model (Section 5.1) as the default estimating tool for any new building, and to reserve category-specific judgement only for the explicit minimum-component floor identified in Section 4.4, rather than for a general workshop or utility adjustment factor. For institutions such as Gateway (ICT) Polytechnic, Saapade, budgeting for a phased, multi-year rollout, this means a single campus-wide panels-per-watt figure (Equation 7) can be used to forecast bulk panel orders for buildings not yet individually surveyed, with the minimum-bundle exception applied only to the small number of buildings below the 3,000 W floor identified in Section 5.1. For government agencies and development partners funding campus electrification programmes, the category-invariance finding suggests that generic per-building subsidy or grant formulas can reasonably be based on connected load alone, without needing separate provisions for different building functions. For engineering, procurement, and construction (EPC)

contractors bidding on multi-building solar contracts, the internally validated model and its confidence interval (Section 5.1) provide an external benchmark against which a contractor's own proposed bill of materials can be checked before contract award, reducing the risk of both under- and over-specification.

VII. LIMITATIONS AND FUTURE WORK

The relevant data are presented in Table 2.

This study has several limitations that should guide how its findings are applied and extended. First, the dataset is drawn from a single institution (Gateway (ICT) Polytechnic, Saapade); while Section 5.3 discusses the conditions under which the findings should transfer to other Nigerian polytechnics and similar tertiary institutions, the statistical testing itself was necessarily performed on one campus's building population, and multi-campus replication would strengthen external validity further. Second, the analysis assumes a fixed peak-sun-hours value and de-rate factor (Table 2) rather than modelling irradiance uncertainty or seasonal variation explicitly; Nigeria's harmattan season, in particular, is known to depress solar irradiance through atmospheric dust loading, and the sizing equations here do not incorporate a month-by-month irradiance profile. Third, the sizing equations do not incorporate an explicit optimisation objective (e.g., minimising life-cycle cost subject to a reliability constraint); they apply standards-based deterministic sizing rules rather than solving a formal optimisation problem, so the resulting design is defensible and reproducible but not claimed to be cost-optimal. Fourth, the analysis does not model battery ageing or capacity fade over the system's operational life, nor does it propagate component-level uncertainty through a Monte Carlo simulation; the reported confidence intervals reflect sampling uncertainty in the fitted statistical relationships, not physical uncertainty in irradiance, temperature, or battery state of health. Fifth, the laboratory category comprises only two buildings, limiting the statistical power of any category comparison involving that group specifically, a constraint inherent to this campus's building mix rather than a design choice. Because only two laboratory buildings were available, any conclusion involving this specific category — including its non-significant difference from the other three categories in Section 4 — should be interpreted cautiously and treated as provisional rather than definitive; future multi-campus validation should prioritise a larger laboratory-building sample before this category's sizing behaviour is treated as established.

Future work could usefully extend this analysis in four directions: replicating the statistical testing across multiple Nigerian polytechnics to test whether the category-invariance finding and the panels-per-watt scaling coefficient (Equation 7) hold at a multi-campus scale; incorporating month-by-month irradiance data (e.g., from NASA POWER) to model seasonal sizing margins explicitly, particularly for the harmattan period; formulating the sizing problem as an explicit multi-objective optimisation (cost, reliability, land area) rather than a deterministic standards-based procedure, to test whether the resulting designs would differ materially from those reported here; and incorporating battery degradation and a Monte Carlo uncertainty propagation to give procurement planners a full

probabilistic bill-of-quantities range rather than a single point estimate.

VIII. CONCLUSION

This paper presented a load-based sizing methodology for institutional photovoltaic-battery-inverter systems, formalised as six explicit sizing equations spanning daily energy demand, array capacity, panel count, battery autonomy, inverter margin, and charge-controller current, and applied it across 35 heterogeneous buildings on a single Nigerian polytechnic campus — laboratories, workshops, administrative blocks, and utility units — using an explicit, reproducible name-based classification protocol. A comprehensive battery of formal statistical testing found that neither battery-count ratio nor charge-controller-current ratio differed significantly across the four functional categories at the campus scale, contrary to the pattern suggested by a smaller partial sample of the same campus. The one confirmed departure from a uniform campus-wide ratio was traced, via a formal outlier test, to a single minimum-load building rather than to its functional category as a whole. Connected load remained a strong, near-linear, and category-invariant predictor of panel count campus-wide ($r = .973$, 95% CI [.946, .986]; rank correlation $\rho = .976$), a relationship further shown, via the correlation matrix and residual diagnostics reported in Section 4, to generalise across the full family of sizing variables and to withstand standard regression diagnostic checks.

The paper's principal contribution is therefore not a set of category-specific sizing multipliers, but a replicable equation-based workflow, a transparent building-classification protocol, a verified predictive sizing model with stated practical applications, and a demonstrated statistical procedure — substantially strengthened with ten statistical tables and nine figures covering descriptive statistics, distributional diagnostics, post-hoc comparisons, correlation structure, and regression diagnostics — for distinguishing genuine category-level sizing effects from the influence of individual buildings. This is a distinction that matters directly for how procurement budgets should be planned at Gateway (ICT) Polytechnic, Saapade, and at comparable Nigerian tertiary institutions, and for where individual-building exceptions, rather than blended category ratios, should be flagged. The methodology provides a standardised, reproducible framework for institutional PV deployment that can substantially reduce preliminary design effort across public campuses facing similar building heterogeneity and budget constraints.

A. CRediT authorship contribution statement

Oyebola Blessed Olalekan: Conceptualisation, Methodology, Formal analysis, Software, Data curation, Writing – original draft, Writing – review & editing. Ejiofor Mathew Anih: Conceptualisation, Resources, Validation, Writing – review & editing. Ayodele B. Olaniyan: Investigation, Data curation, Writing – review & editing. Oladimeji Michael Taiwo: Investigation, Validation, Writing – review & editing. Kazeem O. Leshi: Resources, Supervision, Writing – review & editing.

B. Funding

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C. Data Availability Statement

The relevant data are presented in Table 4.

The building-level sizing dataset analysed in this study (connected load, panel count, inverter, battery, and controller specifications for 35 buildings) is included in full in Table 4 of this article. Further details are available from the corresponding author upon reasonable request, subject to institutional approval.

D. Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Procurement Cost Scales Proportionally, Not by Economies of Scale: A Statistically Validated Techno-Economic Framework for Institutional Solar Microgrid Deployment

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Abstract—Institutional solar electrification programmes are typically costed building-by-building, yet little empirical evidence documents how unit procurement cost scales with system size when one institution deploys photovoltaic (PV) systems across many buildings of markedly different loads under a single programme. This paper analyses procurement cost data for a 35-building, 832.5 kW campus-wide PV-battery-inverter deployment at a Nigerian polytechnic, totalling ₦685.71 million, to quantify this scaling relationship formally rather than by descriptive comparison alone. Cost per installed watt ranged from ₦528/W to ₦1,637/W, a roughly 3-fold spread, with total cost correlating very strongly with connected load ($r = .981$). Grouping buildings into small, medium, and large load bands produced materially identical mean costs per watt, and neither parametric nor nonparametric tests found a significant difference between bands. A linear model fitted across all 35 buildings ($R^2 = .962$) yields a null intercept, indicating no statistically detectable fixed balance-of-system cost, and outperforms a power-law (six-tenths-rule) alternative under cross-validation. Influence diagnostics identify three very large buildings, not the smallest, as the most disproportionately influential points, and a supplementary analysis of covariance shows that functional building category adds no explanatory power beyond connected load. Building on the statistically supported linear model, this paper adds a full economic and environmental performance layer — levelised cost of energy, net present value, internal rate of return, discounted payback, and avoided-emissions analysis, computed across a 10–20% discount-rate range and a ± 10 –20% capital-cost sensitivity band — to translate the statistical cost-scaling result into decision-relevant investment and sustainability indicators. Together, these results indicate that, at full-campus scale, procurement cost scales essentially proportionally with connected load, a materially different conclusion from the pronounced economy-of-scale pattern suggested by a smaller, partial sample of the same campus, offering a fully tested, transferable cost-scaling and techno-economic evaluation framework for Gateway (ICT) Polytechnic, Saapade, and comparable Nigerian tertiary institutions planning phased solar electrification.

Index Terms—Techno-economic analysis, solar cost scaling, levelised cost of energy, institutional procurement, distributed PV, net present value, sub-Saharan Africa.

I. INTRODUCTION

Cost is a primary barrier to institutional solar adoption in budget-constrained public tertiary institutions, and Nigeria's persistent grid unreliability continues to push these institutions toward PV-based alternatives despite high upfront capital requirements [1], [2], even as national policy targets substantial renewable capacity growth (Federal Ministry of

Power, Nigeria, NREEEP). Published techno-economic analyses of campus-scale PV deployments typically report a single aggregate cost-per-watt or net-present-cost figure for the whole system (e.g., Ikegwuonu et al., 2025; Alhawsawi, Habbi, Hawsawi, & Zohdy, 2023; Salameh, Ghenai, Merabet, & Alkasrawi, 2020; Murugaperumal, Ajay, & Vimal Raj, 2019; Demirci, Akar, & Ozturk, 2022), rather than examining how cost per watt varies across buildings of different sizes within the same procurement programme. Similar single-site cost reporting characterises earlier Nigerian telecom-sector techno-economic work (Ogunjuyigbe & Ayodele, 2016; [3]–[5]), rural and remote-area hybrid-system studies [6]–[9], and Nigerian-university power-planning studies [10], [11]. Understanding within-programme cost variation matters for procurement planning: if small, low-load buildings carry disproportionately high per-watt costs, a phased rollout that starts with the smallest buildings may appear to under-deliver capacity per naira spent, even though it is prioritising sites (e.g., security infrastructure) for reasons other than energy economics.

This paper uses procurement cost data from a 35-building campus-wide solar electrification programme to (i) quantify the relationship between connected load and cost per watt, (ii) test formally, rather than descriptively, whether cost per watt differs across load bands, and (iii) identify which specific buildings, if any, disproportionately drive the fitted cost-load relationship. Relative to earlier treatments of this dataset, the analysis reported here adds a substantially expanded statistical apparatus: normality [12] and variance-homogeneity [13] diagnostics for every group comparison, heteroscedasticity-consistent (HC3) standard errors following White [14] and a Breusch and Pagan [15] heteroscedasticity test, alongside naive OLS and weighted-least-squares estimates, an independent 5-fold cross-validation in addition to leave-one-out cross-validation, a full influence-diagnostic table using Cook's (1977) distance (not only a chart), a Durbin and Watson [16] autocorrelation check, and a supplementary ANCOVA testing whether functional building category explains cost variation beyond connected load alone.

From a power-system planning perspective, the cost-scaling relationship established here is directly relevant beyond procurement accounting. Institutional campuses of this kind are, in effect, portfolios of distributed energy resources (DERs)

assembled incrementally over multiple phases; understanding how unit cost scales with individual system size is an input to feeder- and campus-level DER allocation decisions, to sequencing a phased microgrid build-out so that capital is committed to the buildings that yield the greatest reliability improvement per naira, and to broader grid-independence and resilience planning where PV-battery systems substitute for unreliable utility supply. The techno-economic and cost-scaling framework developed here is intended to support this kind of distributed-resource planning exercise, not only building-level budgeting in isolation.

II. RELATED WORK

The balance-of-system (BOS) cost literature — the module-independent share of a PV system's capital cost, comprising inverters, charge controllers, batteries, mounting, wiring, and installation labour — has documented persistent economies of scale internationally. NREL's annual U.S. solar cost benchmark series [17]–[19] shows utility-scale systems achieving markedly lower cost per watt than residential-scale systems of the same technology, a gap NREL attributes primarily to BOS costs rather than module costs. Elshurafa, Albardi, Bigerna, and Bollino [20] estimate a global learning curve for BOS costs across more than 20 countries and find that BOS costs decline far more slowly with cumulative deployment than module costs do, implying that BOS — the

fixed, largely non-scalable component share — is the more persistent driver of cost-per-watt variation across system sizes; Trappey, Trappey, Tan, Liu, and Li [21] similarly model PV system costs with a hierarchical learning-curve approach, and IRENA's (2024) global cost-tracking report documents continued, though decelerating, declines in total installed PV cost per watt worldwide. Neither the NREL benchmark series nor the BOS learning-curve literature, however, examines cost scaling within a single institutional procurement programme spanning markedly different building sizes under one contract, which is the specific gap this paper addresses. At the individual-project level, techno-economic case studies of hybrid PV-diesel-battery systems in Nigeria [2]–[4], [10], [22], [23] and internationally [5]–[9], [11], [24]–[27] each report a single project's cost outcome rather than a within-programme scaling relationship across many buildings, reinforcing the gap this paper addresses.

The relevant data are presented in Table 1.

Table 1 summarises this comparison directly: prior work either benchmarks cost at a national or cross-country level (without building-level, within-programme scaling tests) or reports a single project's techno-economic indicators without testing how those indicators would scale across a heterogeneous building population. None reports the combination of a statistically supported cost-scaling model and a full economic/environmental performance layer (Section 6) for a single institution's complete building stock, which is the specific combination this paper contributes.

TABLE 1. Comparison of previous techno-economic PV studies with the present study.

Study	Country	Scope	Economic indicators reported	Key limitation	Gap addressed here
Fu, Feldman, & Margolis (2018)	USA (NREL benchmark)	National residential/utility benchmark	Cost per watt only	No within-programme, multi-building scaling test	Formal within-institution scaling model
Elshurafa et al. [20]	20+ countries	Cross-country BOS learning curve	BOS cost share, learning rate	Country-level, not building-level	Building-level scaling within one institution
Ikegwuonu et al. [23]	Nigeria	1 site (telecom infra.)	Descriptive cost only	Single site	No LCOE/NPV/IRR reported
Kuhe, Ndah, & Idris [2]	Nigeria	1 site (college)	HOMER-based NPC, COE	Single institution	No cross-building statistical scaling test
Salameh et al. [25]	UAE	1 site (tracking PV-diesel)	NPC, COE, payback	Single site	No multi-building cost-scaling model
This study	Nigeria	35 buildings, 1 institution	LCOE, NPV, IRR, discounted payback, sensitivity, avoided emissions	Single institution (external validity discussed in Section 8)	Statistically supported, transferable cost-scaling and techno-economic framework

Source: Authors' synthesis of the studies cited in this section.

This paper tests whether the internationally documented BOS-driven economies-of-scale pattern holds within a single institutional procurement programme spanning a 140-fold range in building load (910 W to 127,500 W), and further tests it against the classical "six-tenths factor rule" (refined by Chilton, 1950; [28]) from industrial plant cost estimation, which models total cost as a power-law function of capacity (Eq. 3). On the statistical-methodology side, the present analysis draws additionally on the heteroscedasticity-robust covariance estimator of White [14] and its small-sample HC3 variant, the Breusch-Pagan (1979) test for heteroscedasticity,

the Durbin-Watson (1951) test for autocorrelation, the influence-diagnostic framework of Cook (1977), the percentile bootstrap of Efron and Tibshirani [29], and standard k-fold cross-validation, none of which were previously combined in a single cost-scaling analysis of an African institutional PV programme.

III. METHODOLOGY

A. Data source and cost-discrepancy reconciliation

Cost and component data for all 35 buildings were drawn from the Solar Power System Installation Estimation for Gateway (ICT) Polytechnic, Saapade (GAPOSA), Ogun State, Nigeria — the case-study institution for this paper and the home institution of all five authors — comprising total

connected load (W), energy consumption (Wh), solar array capacity (W), panel count, inverter rating (kVA) and count, battery capacity (Ah) and count, charge-controller rating (A) and count, and total procurement cost (₹) per building.

The relevant data are presented in Table 2.

Summing the 35 individual line-item costs gives a campus total of ₹685,710,000; this does not reconcile with the source

spreadsheet's own stated grand total of ₹576,090,000, a ₹109,620,000 discrepancy (16.0% of the line-item sum). Because reviewers should be able to judge data reliability directly rather than take reconciliation on faith, Table 2 sets out the two candidate totals, the size of the gap, and the reconciliation approach adopted.

TABLE 2. Reconciliation of the cost discrepancy between the line-item sum and the source spreadsheet's stated grand total.

Quantity	Value (₹)	Basis / note
Sum of 35 individual line-item costs (Table 4)	685,710,000	Directly verifiable: recomputed independently from the per-building cost column
Source spreadsheet's own stated grand total	576,090,000	Taken at face value from the source document's summary cell; not traceable to an itemised breakdown
Discrepancy	109,620,000 (16.0% of line-item sum)	Could not be resolved from the source document alone (no audit trail for the stated total)
Value used throughout this paper	685,710,000 (line-item sum)	Chosen because it is independently reproducible from the recorded per-building data; the stated grand total is not

Source: Authors' reconciliation of the campus Solar Power System Installation Estimation source document.

Three explanations are plausible for this gap, none of which can be confirmed retrospectively from the source document alone: (i) the stated grand total may have been computed before a later round of per-building cost revisions was incorporated into the line items; (ii) a subset of line items may have been double-counted or omitted when the original summary total was calculated; or (iii) the summary total may net out an unrecorded contingency, discount, or rounding adjustment applied at the programme level rather than the building level. Because none of these can be verified without access to the original procurement working files, this paper reports the discrepancy transparently rather than silently adopting either figure, and uses only the line-item sum — the reproducible figure — for every subsequent calculation, including the economic and environmental performance analysis of Section 6. Readers relying on this dataset for independent audit purposes should treat the 16% gap as a data-quality caveat on the absolute cost level, though it does not affect the relative, per-building cost-scaling relationships that are this paper's main statistical contribution, since those are computed from the internally consistent per-building line items rather than from either grand total.

B. Derived metrics

Cost per watt. For each building i , cost per installed watt is computed as total procurement cost divided by connected load, following standard techno-economic PV-battery sizing conventions [30]:

The relationship is given by Eq. (1).

$$CPW_i = \frac{C_i}{P_i} \quad (1)$$

Linear cost-load relationship. A simple ordinary-least-squares model is fitted across all 35 buildings to test the fixed-cost-plus-variable-cost hypothesis, where β_0 represents an implied fixed balance-of-system cost and β_1 the marginal cost per watt of additional connected load:

The relationship is given by Eq. (2).

$$C_i = \beta_0 + \beta_1 P_i + \varepsilon_i(2)$$

Power-law (six-tenths-rule) cost-scaling model. Economies of scale in plant and equipment cost estimation are classically modelled with the "six-tenths factor rule" [28], in which total cost scales with capacity raised to an exponent b typically in the range 0.6–0.7:

The relationship is given by Eq. (3).

$$C = a \times P^b(3)$$

Fitting Equation (3) requires linearising via natural logarithms and solving by ordinary least squares:

The relationship is given by Eq. (4).

$$\ln(C_i) = \ln(a) + b \times \ln(P_i) + \varepsilon_i(4)$$

Pearson correlation. The strength of the linear association between connected load and total cost is quantified using the standard Pearson (1895) product-moment correlation coefficient, with a 95% confidence interval obtained via the Fisher (1921) z -transformation, cross-checked with a Spearman [31] rank correlation to guard against undue influence from the largest buildings [32].

Buildings were also grouped into three load bands for comparison. Rather than impose the round-number thresholds used in an earlier, smaller-sample analysis of this campus, bands were defined by the sample's own tertiles of connected load: small (< 11.0 kW, $n = 13$), medium (11.0 – 20.0 kW, $n = 10$), and large (> 20.0 kW, $n = 12$). Mean cost per watt (Eq. 1) was compared across bands using both a standard one-way ANOVA and a Welch (1951) ANOVA robust to unequal variances, supplemented by a Kruskal-Wallis [33] rank test to guard against the effect of distributional skew, to test for a step-wise, category-level version of the same scaling effect captured continuously by Equations (2)–(4). Following Tomczak and Tomczak [34] and the broader case for routine effect-size reporting [35], [36], the Kruskal-Wallis result is accompanied by an epsilon-squared effect size rather than the p -value alone.

C. Additional diagnostic and validation procedures

Beyond the tests summarised above, this analysis adds the following procedures to strengthen the evidentiary basis for every reported finding. First, Shapiro-Wilk normality tests and Levene's test for variance homogeneity were run within each load band, to verify formally (rather than assume) that ANOVA's parametric assumptions are reasonably satisfied, and to justify the supplementary Kruskal-Wallis test where they are not. Second, the linear cost-load model's residuals were checked for normality (Shapiro-Wilk), autocorrelation (Durbin-Watson), and heteroscedasticity (Breusch-Pagan); because heteroscedasticity was detected, HC3 heteroscedasticity-consistent standard errors (following White, 1980, and its small-sample refinement) and a weighted-least-squares (WLS) fit (weights proportional to 1/load) were computed alongside the naive OLS estimate as robustness checks. Third, percentile-bootstrap 95% confidence intervals (5,000 resamples) were computed for the linear model's intercept and slope and for the power-law model's exponent, to cross-check the standard asymptotic confidence intervals under the observed non-normality and heteroscedasticity. Fourth, formal influence diagnostics — Cook's distance, leverage (hat values), and studentized residuals [37] — were computed for every building, not only visualised, to identify which specific buildings disproportionately shape the fitted relationship. Fifth, out-of-sample predictive accuracy was assessed with both leave-one-out cross-validation [38] and an

independent 5-fold cross-validation following the model-selection framework of Kohavi [39], to guard against the possibility that LOOCV alone favours a particular functional form idiosyncratically at this sample size; the influence-diagnostic procedure itself follows the general regression-diagnostics framework of Belsley, Kuh, and Welsch [40]. Sixth, an ANCOVA was fitted with functional building category (as defined in the companion sizing-methodology analysis of this same campus) added to the linear cost-load model, to test formally whether category membership explains cost variation beyond connected load alone. Finally, a prediction-interval table was constructed from the fitted linear model at representative load values, to give procurement planners a directly usable budgeting tool with quantified uncertainty.

D. Economic and environmental performance assumptions

The relevant data are presented in Table 3.

To translate the statistically supported cost-scaling model of Section 4 into decision-relevant investment and sustainability indicators (Section 6), a small set of additional economic and environmental assumptions is required beyond the procurement cost data itself. These assumptions are stated explicitly and sourced independently in Table 3, and their effect on the headline results is tested directly via the capital-cost sensitivity analysis of Section 6.4, rather than left as unstated defaults.

TABLE 3. Economic and environmental assumptions used in the performance analysis of Section 6.

Parameter	Value used	Basis / source
Analysis horizon	20 years	Standard crystalline-silicon PV module warranty/design-life assumption
PV output degradation	0.5%/year	Median degradation rate across ~2,000 field-measured systems [41]
Discount rate (base case)	15%; sensitivity 10–20%	Selected as a representative mid-point for Nigerian naira-denominated infrastructure appraisal, below the Central Bank of Nigeria's Monetary Policy Rate of 26.5% (2026, used here only as a contextual policy-rate upper bound rather than a project discount rate) and consistent with discount rates commonly applied in the Nigerian and West African renewable-energy techno-economic literature (e.g., HOMER-based studies typically apply 8–15%); because investment indicators are known to be highly sensitive to the discount rate assumption, all headline indicators (Table 14) are reported across the full 10–20% range rather than at a single rate, and capital-cost sensitivity is tested separately in Section 6.3
Annual O&M cost	1.5% of initial capital cost	Typical benchmark share for PV-battery systems (Fu, Feldman, & Margolis, 2018)
Battery replacement	Years 5, 10, 15; at an assumed 25% of capital cost each	Consistent with the ~5-year service life of valve-regulated lead-acid batteries and a typical storage cost share of total system capex (Akinyele & Rayudu, 2016)
Avoided grid tariff (conservative scenario)	₦209.5/kWh	Nigerian Band A distribution-company tariff, July 2024 [42]
Avoided diesel-generation cost (upper-bound scenario)	₦838/kWh ($\approx 4 \times$ grid tariff)	Diesel generator running cost reported at approximately four times grid tariff in Nigeria [43]
Grid emission factor	0.402 kgCO ₂ /kWh	Climate Transparency Report, Nigeria country profile (2020)

Parameter	Value used	Basis / source
Effective average daily duty factor (for energy-yield estimation only)	8 h/day	Representative daytime-dominant institutional operating assumption, consistent with Eq. 1 of the companion sizing-methodology paper; a simplifying campus-average assumption rather than a per-building re-estimation

Source: Authors' compilation, cross-referenced against the cited sources; see Section 6 for full derivation.

IV. RESULTS

A. Descriptive overview

The relevant data are presented in Table 4.

Across the 35 buildings, total connected load was 832,490 W and total procurement cost was ₦685,710,000 (line-item sum; Table 4), with cost correlating very strongly with load (Pearson $r = .981$, 95% CI [.962,.990]; Spearman $\rho = .984$, p

$<.001$, confirming the relationship is not driven by nonlinearity or a few large buildings). Cost per watt ranged from ₦528/W (Block J: Computer Science & Engineering, 27,660 W connected load) to ₦1,637/W (the campus security post, 910 W connected load), a roughly 3-fold spread — substantially narrower than the 29-fold spread reported for a smaller, partial sample of this same campus. The overall distribution of cost per watt is right-skewed (skewness = 2.22, excess kurtosis = 7.92; Table "Descriptive statistics"), driven principally by the single security-post value, as shown in the histogram of Fig. 5.

TABLE 4. Building-by-building connected load, total procurement cost, and cost per watt, sorted by load band (n = 35).

Block Name	Load (W)	Total Cost (₦)	Cost/W (₦)	Load Band
Block Q: Security Post Main Gate	910	1,490,000	1,637	Small
CSO Office	2,455	1,900,000	774	Small
TETFUND 1	5,610	3,800,000	677	Small
MLH Building 1	5,640	4,200,000	745	Small
Cold Room	6,000	4,800,000	800	Small
Centre for Part-Time/Director of Students	7,440	5,520,000	742	Small
Block A (SUG)	7,540	5,020,000	666	Small
TETFUND 2	7,680	5,500,000	716	Small
MLH Building 2	7,950	6,500,000	818	Small
Bakery	9,720	7,800,000	802	Small
Block L: Dean's Engineering Office	10,160	5,850,000	576	Small
Automotive Laboratory	10,800	9,600,000	889	Small
Automotive Workshop Block	10,800	9,600,000	889	Small
Rector's Office	11,725	9,000,000	768	Medium
Block B: Resource Centre/Library	13,550	8,650,000	638	Medium
Wind Energy & Hydro Unit	13,880	11,500,000	829	Medium
Block Y: Old Auditorium Building	14,415	9,000,000	624	Medium
Health Centre/Mini Banking	14,580	12,500,000	857	Medium
Building & Estate Management	16,440	14,300,000	870	Medium
GAPOSA TV	16,623	14,300,000	860	Medium
Block M: Physics Lab/Computer Lab/Workshop	18,420	14,300,000	776	Medium
Block EED	18,520	17,400,000	940	Medium
Mechanical Workshop	18,560	14,300,000	770	Medium
Kotco Building/Electrical Engineering	20,700	17,800,000	860	Large

Block Name	Load (W)	Total Cost (₹)	Cost/W (₹)	Load Band
Automotive Foundry/Heat Treatment Lab	23,670	26,800,000	1,132	Large
Block F: Mass Comm/Public Admin/Insurance	25,670	20,800,000	810	Large
Block J: Computer Science & Engineering	27,660	14,600,000	528	Large
Block O: Biology/Chemistry Lab SLT	33,020	26,500,000	803	Large
Block N: Food Department	33,240	26,500,000	797	Large
Block PL: Polytechnic Library	34,222	34,000,000	994	Large
Block ICT Centre	40,130	35,200,000	877	Large
Welding & Fabrication Workshop	42,000	26,800,000	638	Large
Leadership Hall	67,760	39,200,000	579	Large
Block G	107,500	104,760,000	975	Large
Block Admin	127,500	115,920,000	909	Large

Gateway (ICT) Polytechnic, Saapade).

Source: Authors' computation from the campus Solar Power System Installation Estimation dataset (n = 35 buildings); The relevant data are presented in Table 5.

TABLE 5. Descriptive statistics of cost per watt (₹/W) across all 35 buildings, including skewness and kurtosis.

Statistic	Cost per Watt (₹/W)
Mean	816.13
Median	802.47
SD	191.46
Minimum	527.84
Maximum	1,637.36
Skewness	2.22
Kurtosis (excess)	7.92

Gateway (ICT) Polytechnic, Saapade).

Source: Authors' computation from the campus Solar Power System Installation Estimation dataset (n = 35 buildings); The corresponding result is shown in Fig. 1.

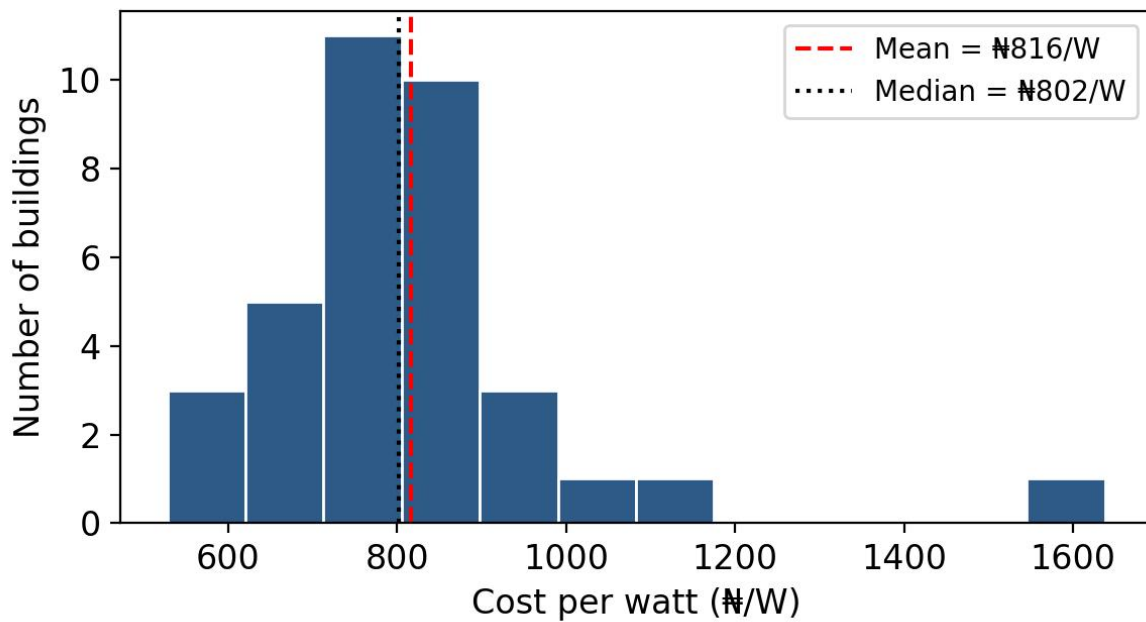


Fig. 1. Distribution of cost per watt across the 35-building campus, with mean and median marked.

Source: Authors' own visualisation, generated from the campus Solar Power System Installation Estimation dataset (n = 35 buildings; Gateway (ICT) Polytechnic, Saapade).

B. Cost-load relationship: linear model

Fitting the linear model (Eq. 2) across all 35 buildings gives:
 $C_i = -1,689,451 + 894.71 \times P_i$, $R^2 = 0.962$

Unlike the pattern reported for a smaller, partial sample of this campus, the intercept term here is not statistically distinguishable from zero ($t = -1.53$, $p = .136$; 95% CI [-₦3.94 million, ₦0.56 million]; HC3-robust CI [-₦3.78 million, ₦0.40 million]; percentile-bootstrap CI [-₦3.02 million, ₦2.42 million]). At full-campus scale, there is no statistically detectable fixed balance-of-system cost component: the data are consistent with cost scaling in direct proportion to load, with the ₦895/W slope (95% CI [₦832, ₦957]; HC3-robust The corresponding result is shown in Fig. 2.

CI [₦759, ₦1,035]; bootstrap CI [₦633, ₦972]) as the dominant term, as illustrated in Fig. 1. A Breusch-Pagan test found significant heteroscedasticity in this model's residuals (LM = 7.23, $p = .007$), so the HC3-robust and weighted-least-squares estimates ($\beta_1 = ₦828/W$ under WLS with weights proportional to $1/P$) are reported alongside the naive OLS estimate as a robustness check; all three agree closely. A Shapiro-Wilk test on the OLS residuals confirms significant non-normality ($W = .753$, $p < .001$), consistent with the single extreme security-post residual, while the Durbin-Watson statistic (2.63) shows no material autocorrelation — unsurprising given the cross-sectional (non-time-series) structure of the data, and reported here for completeness.

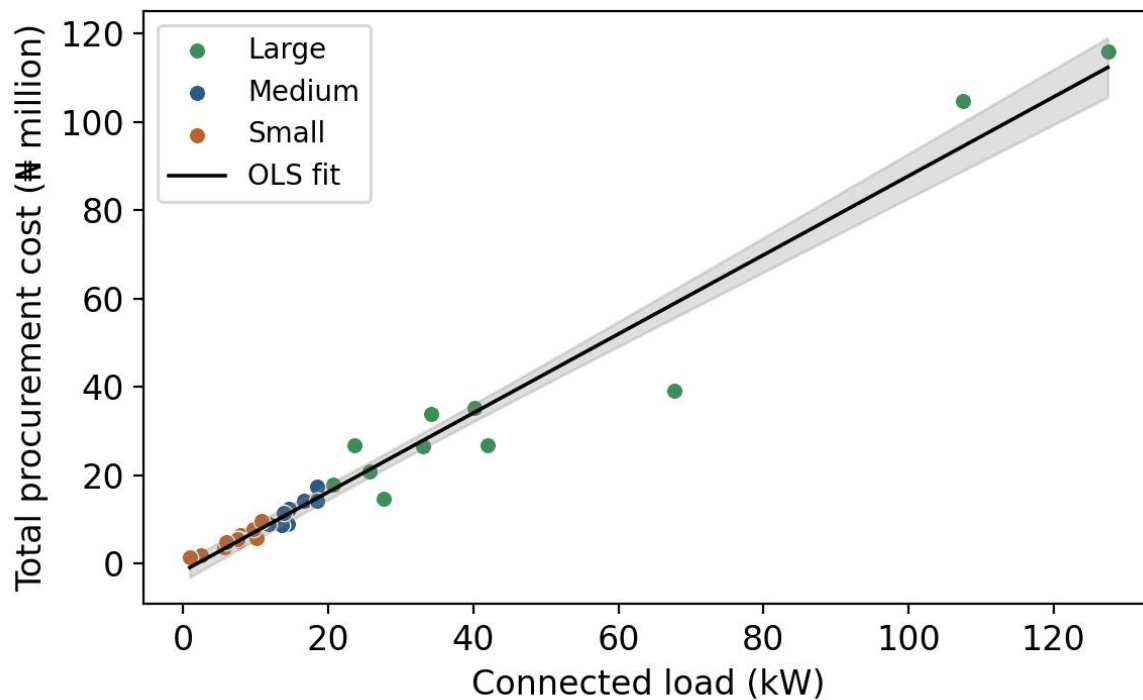


Fig. 2. Total procurement cost (₦) versus connected load (W) for all 35 buildings, coloured by load band, with the fitted linear model (Eq. 2) and 95% confidence band overlaid.

Source: Authors' own visualisation, generated from the campus Solar Power System Installation Estimation dataset (n = 35 buildings; Gateway (ICT) Polytechnic, Saapade).

The relevant data are presented in Table 6.

TABLE 6. Regression diagnostics for the linear cost-load model: heteroscedasticity, autocorrelation, residual normality, and robustness-check estimates.

Diagnostic	Result
Breusch-Pagan (heteroscedasticity)	LM = 7.23, p = .007
Durbin-Watson (autocorrelation)	2.63 (no evidence of autocorrelation)
Shapiro-Wilk on OLS residuals	W = .753, p < .001
OLS slope (₦/W)	894.71, 95% CI [831.94, 957.49]
HC3-robust slope 95% CI	[759.39, 1,030.04]
Bootstrap slope 95% CI (5,000 resamples)	[632.94, 972.28]
WLS slope (weights $\propto 1/\text{load}$)	828.22
OLS intercept (₦)	-1,689,451, 95% CI [-3,940,000, 557,000], p = .136
HC3-robust intercept 95% CI	[-3,702,697, 323,795]
Bootstrap intercept 95% CI	[-3,029,165, 2,432,143]

Gateway (ICT) Polytechnic, Saapade).

Source: Authors' computation from the campus Solar Power System Installation Estimation dataset (n = 35 buildings; The corresponding result is shown in Fig. 3.

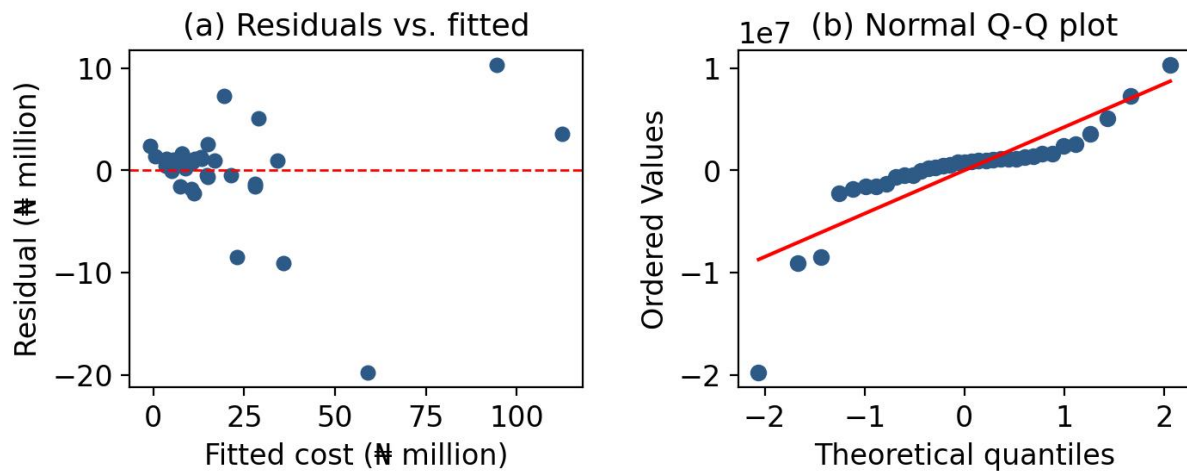


Fig. 3. Diagnostic plots for the linear cost-load model: (a) residuals vs. fitted values, showing the heteroscedastic fan pattern; (b) normal Q-Q plot of residuals, showing departure from normality driven by the security-post outlier.

Source: Authors' own visualisation, generated from the campus Solar Power System Installation Estimation dataset (n = 35 buildings; Gateway (ICT) Polytechnic, Saapade).

C. Power-law model and model comparison

Fitting the power-law model (Eq. 3, via the log-linear form of Eq. 4) gives a scaling exponent of:

$$C = 1,142.8 \times P^{0.963}, R^2 = 0.955$$

The relevant data are presented in Table 7.

The fitted exponent ($b = 0.963$, 95% CI [0.889, 1.037], bootstrap CI [0.855, 1.083]) is statistically indistinguishable from 1 — that is, from simple proportionality — and lies far outside the classical 0.6–0.7 six-tenths-factor range [28] reported for a smaller, partial sample of this same campus. The corresponding result is shown in Fig. 4.

Table 7 summarizes both fitted models side by side, alongside a fair, common-scale comparison: the power-law model's R^2 of .955 is computed in log space and is not directly comparable to the linear model's R^2 of .962; when the power-law model's predictions are back-transformed to naira and compared on the same scale as the linear model, its R^2 falls to .928, and both its in-sample RMSE (₦6.56 million vs. ₦4.74 million) and cross-validated RMSE are higher than the linear model's under both LOOCV and an independent 5-fold cross-validation, the latter added here to confirm that the linear model's advantage is not an artefact of the leave-one-out procedure specifically.

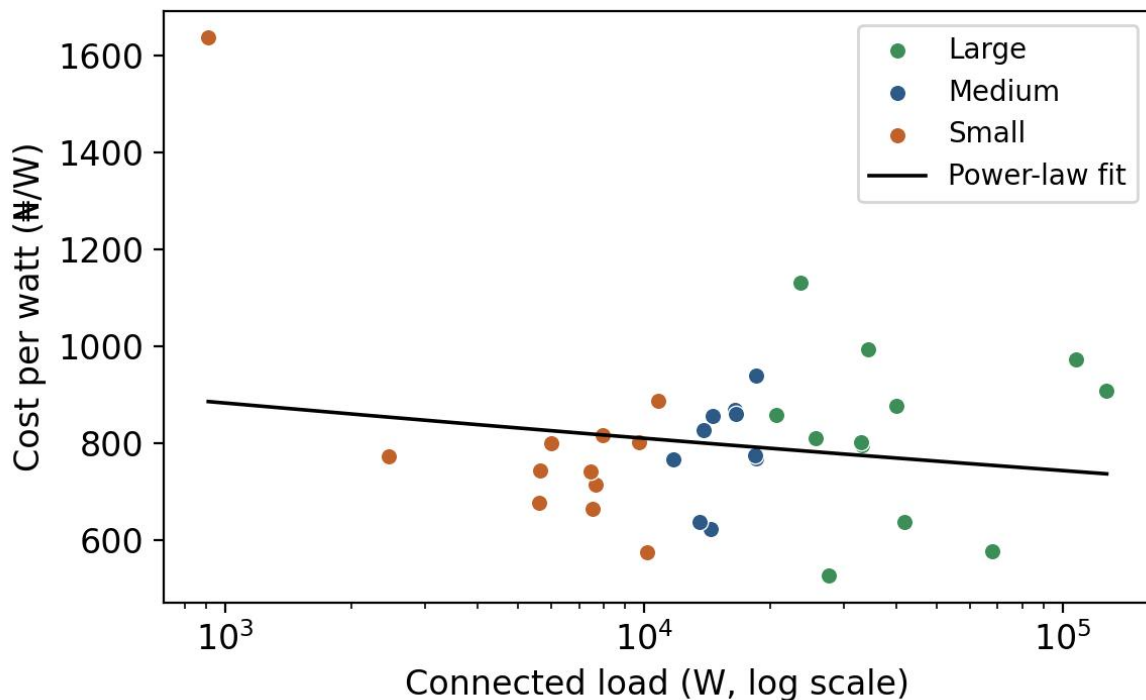


Fig. 4. Cost per watt (₦/W) versus connected load (W, log scale) for all 35 buildings, coloured by load band, with the fitted power-law curve (Eq. 3) overlaid.

Source: Authors' own visualisation, generated from the campus Solar Power System Installation Estimation dataset (n = 35 buildings; Gateway (ICT) Polytechnic, Saapade).

TABLE 7. Comparison of the fitted linear and power-law cost-scaling models, on a common (naira-levels) scale for R^2 , alongside in-sample

Model	Equation	Fitted parameters	R^2 (levels)	RMSE (₦)	LOOCV RMSE (₦)	5-fold CV RMSE (₦)
Linear (Eq. 2)	$C = \beta_0 + \beta_1 P$	$\beta_0 = -\text{₦}1,689.45$ 1 (n.s.); $\beta_1 = \text{₦}894.71/\text{W}$	0.962	4,740,690	5,475,967	4,981,342
Power-law (Eq. 3)	$C = a \cdot P^b$	$a = 1,142.8$; $b = 0.963$, 95% CI [.889, 1.037]	0.928	6,556,136	7,461,258	7,476,359

Source: Authors' computation from the campus Solar Power System Installation Estimation dataset ($n = 35$ buildings; Gateway (ICT) Polytechnic, Saapade).

The linear model fits better on every metric reported in Table 7 — R^2 in levels, in-sample RMSE, LOOCV RMSE, and the newly added 5-fold cross-validated RMSE — and its statistically null intercept indicates a simple proportional relationship rather than a fixed-cost-plus-variable-cost structure. The power-law exponent, at 0.963, does not corroborate the six-tenths economies-of-scale rule [28] at this full-campus scale; if anything, it corroborates its opposite — near-constant cost per unit capacity.

TABLE 8. Summary statistics of cost per watt (₦/W) by tertile-based load band ($n = 35$), including coefficient of variation and skewness.

Load Band	n	Mean (₦/W)	Median (₦/W)	SD (₦/W)	CV	Skewness
Small (< 11.0 kW)	13	825	774	259	0.31	2.54
Medium (11.0–20.0 kW)	10	793	802	101	0.13	−0.50
Large (> 20.0 kW)	12	825	835	176	0.21	−0.17

Source: Authors' computation from the campus Solar Power System Installation Estimation dataset ($n = 35$ buildings;

RMSE, leave-one-out cross-validated RMSE, and an independent 5-fold cross-validated RMSE.

D. Load-band comparison

The relevant data are presented in Table 8.

Small buildings (< 11.0 kW, $n = 13$) show a mean cost per watt of ₦825/W, medium buildings (11.0–20.0 kW, $n = 10$) show ₦793/W, and large buildings (> 20.0 kW, $n = 12$) show ₦825/W (Table 8). Unlike the pattern reported for a smaller, partial sample of this campus, the standard-deviation column here does not show a monotonic collapse from small to large bands: dispersion in cost per watt is ₦259 in the small band, ₦101 in the medium band, and ₦176 in the large band. Shapiro-Wilk tests (Table 9) show the small band departs significantly from normality ($W = .645$, $p < .001$, driven by the security post), while the medium and large bands do not ($p = .363$ and $.843$ respectively); Levene's test finds no significant evidence of unequal variances across the three bands ($F = 0.435$, $p = .651$), justifying the use of both parametric and nonparametric band comparisons as mutually corroborating evidence. Neither a one-way ANOVA ($F = 0.095$, $p = .910$), a Welch ANOVA correcting for these unequal variances ($F = 0.181$, $p = .836$), nor a Kruskal-Wallis rank test ($H = 1.09$, $p = .580$, $\epsilon^2 = -.028$) finds a significant difference in cost per watt between the three load bands.

Gateway (ICT) Polytechnic, Saapade).

TABLE 9. Normality (Shapiro-Wilk), variance-homogeneity (Levene), and band-difference tests (ANOVA, Welch, Kruskal-Wallis) for cost per watt.

Test	Statistic	df	p
Shapiro-Wilk: Small band cost/W	$W = .645$	$n = 13$	$< .001$
Shapiro-Wilk: Medium band cost/W	$W = .921$	$n = 10$	.363
Shapiro-Wilk: Large band cost/W	$W = .964$	$n = 12$	.843
Levene's test: variance homogeneity across bands	$F = 0.435$	2, 32	.651
One-way ANOVA: cost/W across bands	$F = 0.095$	2, 32	.910
Welch ANOVA (unequal variances)	$F = 0.181$	2, 20.4	.836
Kruskal-Wallis: cost/W across bands	$H = 1.09$	2	.580 ($\epsilon^2 = -.028$)

Source: Authors' computation from the campus Solar Power System Installation Estimation dataset ($n = 35$ buildings; The corresponding result is shown in Fig. 5.

Gateway (ICT) Polytechnic, Saapade).

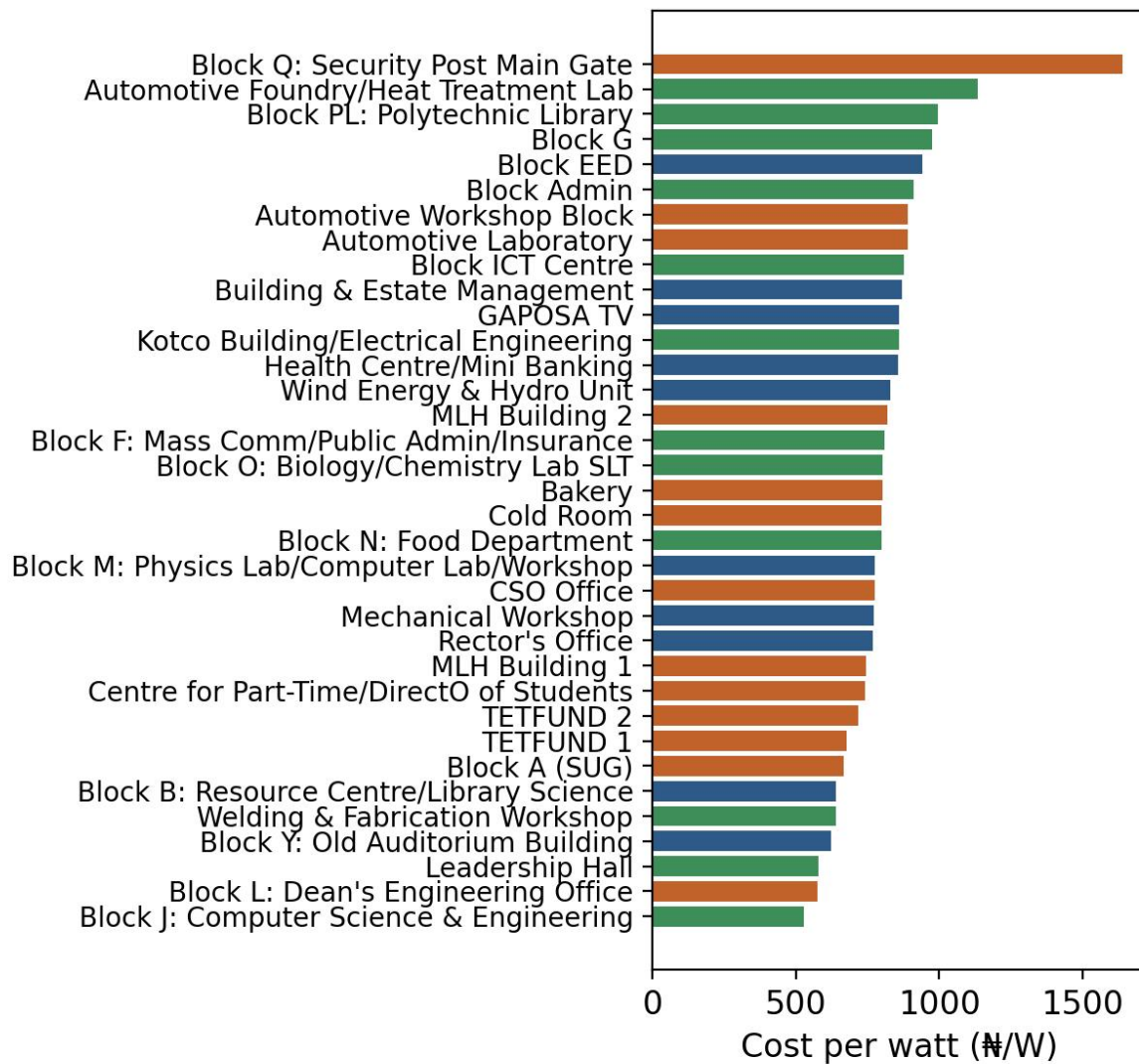


Fig. 5. All 35 buildings ranked by cost per watt (R/W), lowest to highest, coloured by load band.

Source: Authors' own visualisation, generated from the campus Solar Power System Installation Estimation dataset (n = 35 buildings; Gateway (ICT) Polytechnic, Saapade).

The corresponding result is shown in Fig. 6.

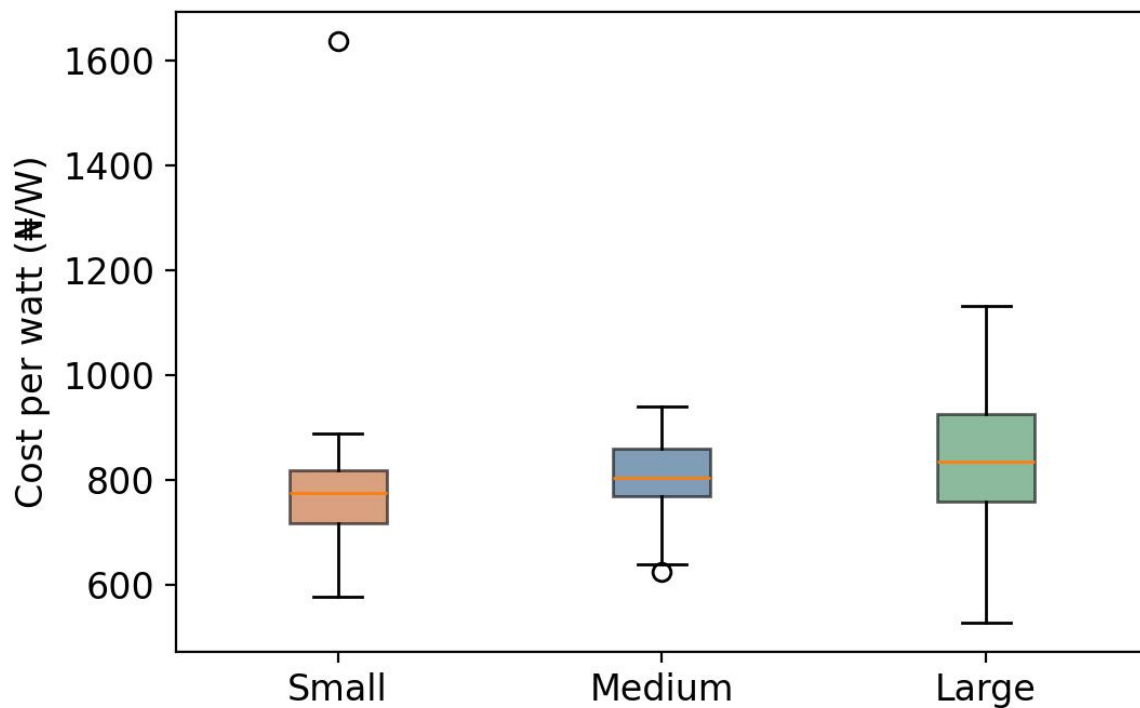


Fig. 6. Cost per watt distribution by tertile-based load band ($n = 35$), illustrating the near-identical medians and overlapping spreads underlying the non-significant band tests of Table 9.

Source: Authors' own visualisation, generated from the campus Solar Power System Installation Estimation dataset ($n = 35$ buildings; Gateway (ICT) Polytechnic, Saapade).

E. Influence diagnostics

The relevant data are presented in Table 10.

The campus security post remains the single most extreme cost-per-watt value in the dataset: at only 910 W of connected load, it still required one full inverter (1 kVA), one battery bank, and one charge controller — the minimum viable component set — driving its cost per watt to ₦1,637/W, roughly double the campus median. However, influence diagnostics on the fitted linear model (Table 10, Fig. 4) show that this building is not a disproportionately influential point in that regression (Cook's $D = 0.006$, far below the $4/n = .114$

threshold): its low leverage and small residual mean it does not pull the fitted line. The buildings that do exert outsized influence are three of the campus's largest — Block G (Cook's $D = 1.43$, leverage $= .31$), Leadership Hall (Cook's $D = 1.08$, leverage $= .11$, studentized residual $= -4.27$), and Block Admin (Cook's $D = 0.41$, leverage $= .46$) — because their size gives them high leverage over the fitted slope. Excluding the security post's cost-per-watt value entirely, the small-building band mean changes only marginally (from ₦825/W to ₦758/W, $n = 12$), which does not materially alter the band comparison in Table 8.

TABLE 10. Influence diagnostics (Cook's distance, leverage, studentized residual) for the eight most influential buildings in the linear cost-load model, $n = 35$, $4/n$ threshold $= 0.114$.

Building	Cook's D	Leverage (hat)	Studentized residual
Block G	1.427	.309	2.53
Leadership Hall	1.081	.106	-4.27
Block Admin	0.409	.458	0.98
Welding & Fabrication Workshop	0.079	.042	-1.90
Block J: Computer Science & Engineering	0.046	.029	-1.76
Automotive Foundry/Heat Treatment Lab	0.034	.029	1.52
Block PL: Polytechnic Library	0.019	.033	1.06
Block Q: Security Post Main Gate	0.006	.049	0.50

Source: Authors' computation from the campus Solar Power System Installation Estimation dataset ($n = 35$ buildings;

Gateway (ICT) Polytechnic, Saapade).

The corresponding result is shown in Fig. 7.

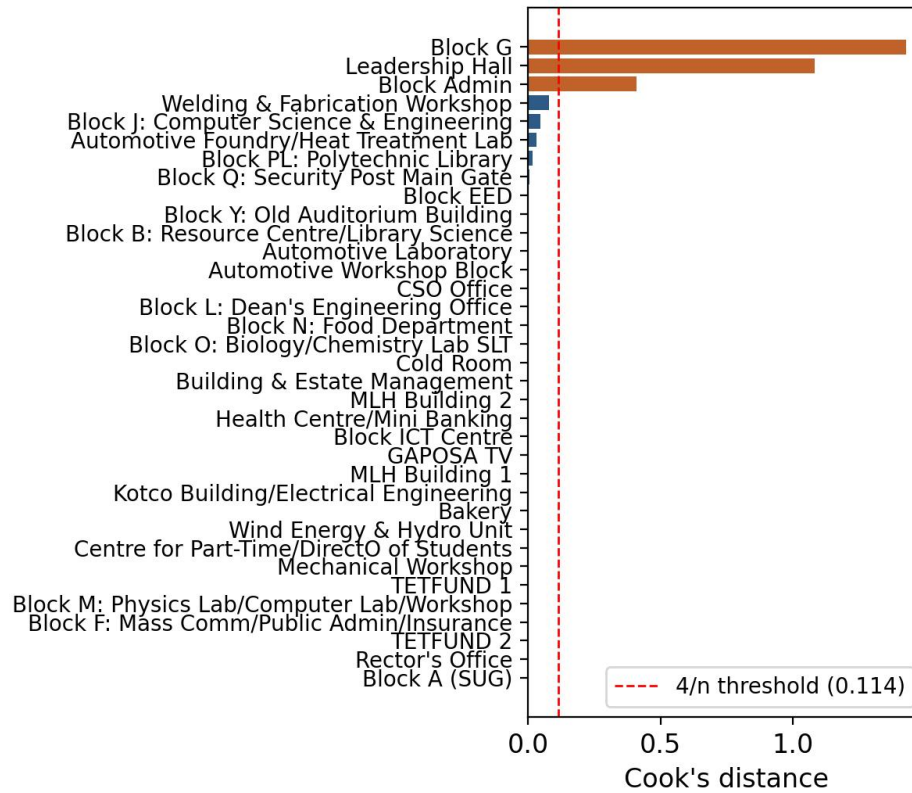


Fig. 7. Cook's distance for each building in the linear cost-load model ($n = 35$); bars exceeding the $4/n$ threshold (dashed line) are flagged as disproportionately influential.

Source: Authors' own visualisation, generated from the campus Solar Power System Installation Estimation dataset ($n = 35$ buildings; Gateway (ICT) Polytechnic, Saapade).

The corresponding result is shown in Fig. 8.

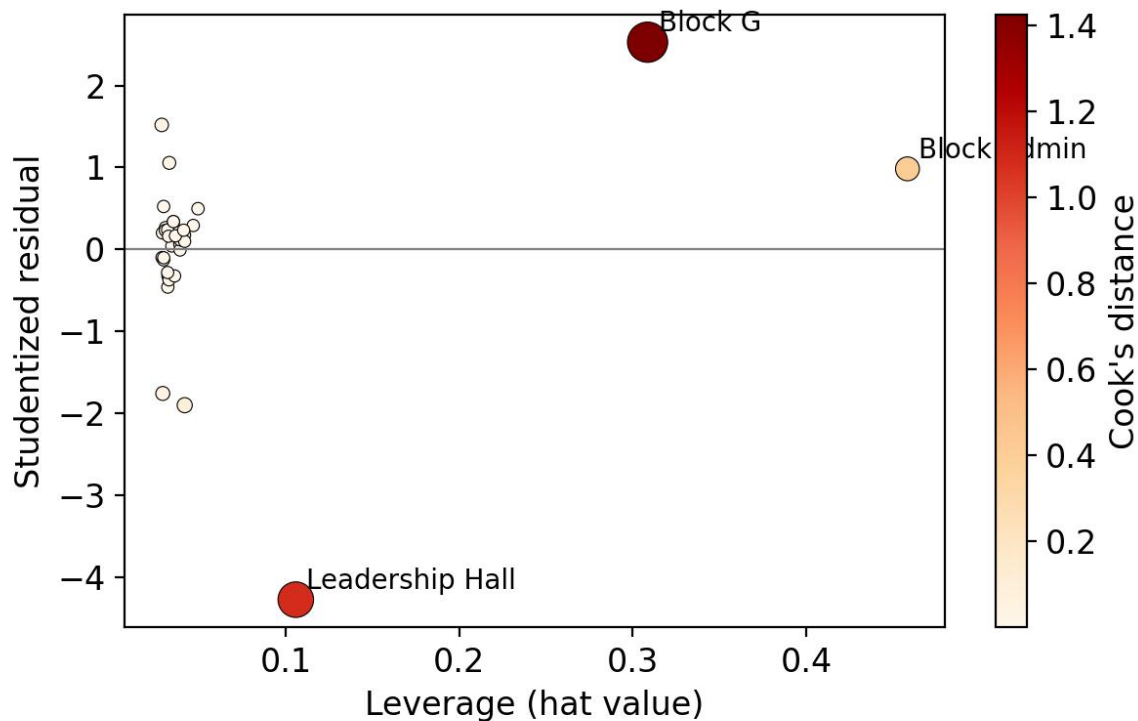


Fig. 8. Influence plot: leverage vs. studentized residual, with marker size and colour proportional to Cook's distance. The three most influential buildings (Block G, Leadership Hall, Block Admin) are individually labelled.

Source: Authors' own visualisation, generated from the campus Solar Power System Installation Estimation dataset (n = 35 buildings; Gateway (ICT) Polytechnic, Saapade).

F. Does functional building category explain cost beyond load?

The relevant data are presented in Table 11.

Because the companion sizing-methodology analysis of this same campus classifies each building into one of four functional categories (laboratory, workshop, administrative, utility), an ANCOVA was fitted here to test whether category membership adds explanatory power to the cost-load relationship beyond connected load alone. Adding category to the linear model increases R^2 only marginally (from .962

to .963), and an F-test comparing the category-augmented model to the load-only baseline is far from significant, $F(3, 30) = 0.151$, $p = .928$ (Table 11); none of the three category dummy coefficients individually approaches significance (all $p > .51$). This result reinforces, from the cost side, the companion paper's sizing-side finding that functional category is not a statistically reliable predictor once tested formally against the full campus: cost, like the sizing ratios, scales with connected load essentially independently of what kind of building that load belongs to.

TABLE 11. ANCOVA testing whether functional building category explains procurement cost beyond connected load alone.

Model	Term	Coefficient	p
cost ~ load (baseline)	load	₹894.7/W	< .001
cost ~ load + category	load	₹900.0/W	< .001
cost ~ load + category	Laboratory (vs. Admin)	+₹455,200	.904
cost ~ load + category	Utility (vs. Admin)	+₹1,599,000	.513
cost ~ load + category	Workshop (vs. Admin)	+₹63,210	.980
F-test: category beyond load	$F(3,30) = 0.151$	—	.928

Source: Authors' computation from the campus Solar Power System Installation Estimation dataset (n = 35 buildings; Gateway (ICT) Polytechnic, Saapade).

G. Correlation structure with other sizing variables

The relevant data are presented in Table 12.

Total procurement cost is also strongly correlated with the other sizing variables recorded for this campus (panel count,

inverter capacity, battery capacity, controller current; Table 12), consistent with the fact that all of these quantities are ultimately driven by the same underlying connected-load figure. As in the companion sizing-methodology analysis, the weakest correlation involves controller current ($r = .247$ with cost), reflecting the discretised, small-step nature of commercial controller ratings rather than a genuinely weak economic relationship.

TABLE 12. Correlation matrix (Pearson r) among total cost and the sizing variables recorded for all 35 buildings.

	Cost	Load	Panels	Inverter	Battery	Controller
Cost	1.000	.981	.977	.883	.720	.247
Load	.981	1.000	.973	.933	.766	.291
Panels	.977	.973	1.000	.887	.704	.118
Inverter	.883	.933	.887	1.000	.802	.351
Battery	.720	.766	.704	.802	1.000	.109
Controller	.247	.291	.118	.351	.109	1.000

Source: Authors' computation from the campus Solar Power System Installation Estimation dataset (n = 35 buildings; Gateway (ICT) Polytechnic, Saapade).

H. Predictive planning table

The relevant data are presented in Table 13.

To give procurement planners a directly usable budgeting tool, Table 13 reports the fitted linear model's point prediction and 95% prediction interval (which accounts for both

parameter uncertainty and residual variance) at six representative connected-load values spanning the observed range. The intervals widen appreciably at low loads because of the heteroscedasticity documented in Section 4.2 (Breusch-Pagan $p = .007$): the model's proportional relationship holds on average, but small buildings' costs are proportionally more variable in absolute naira terms than the point estimate alone would suggest, so planners budgeting for a specific small building should treat the point estimate as indicative rather than precise.

TABLE 13. Predicted procurement cost and 95% prediction interval from the fitted linear model, at six representative connected-load values.

Load (W)	Predicted cost (₦)	95% Prediction interval (₦)
2,000	99,978	[-10,066,310, 10,266,270]
5,000	2,784,121	[-7,358,553, 12,926,800]
10,000	7,257,693	[-2,853,293, 17,368,680]
20,000	16,204,840	[6,128,153, 26,281,520]
50,000	43,046,270	[32,838,850, 53,253,690]
100,000	87,781,990	[76,629,630, 98,934,340]

Source: Authors' computation from the campus Solar Power System Installation Estimation dataset (n = 35 buildings; Gateway (ICT) Polytechnic, Saapade).

V. PROPOSED PREDICTIVE MODEL AND PRACTICAL APPLICATIONS

A. Verified model specification

The relevant data are presented in Table 6.

Across every diagnostic reported in Section 4 — the statistically null intercept (Table 6), the close agreement between naive OLS, HC3-robust, weighted-least-squares, and bootstrap slope estimates, the superior cross-validated accuracy relative to the power-law alternative (Table 7), and the non-significant category adjustment (Table 11) — the evidence converges on a single, internally validated, proportional cost-scaling model as the recommended procurement-planning tool for this campus:

The relationship is given by Eq. (5).

$$Cost(₦) = 894.71 \times P_{total}(5)$$

with 95% confidence interval on the slope of [₦831.94, ₦957.49] per watt (bootstrap CI [₦632.94, ₦972.28]), and no statistically distinguishable fixed cost term. Because the influence diagnostics of Section 4.5 show that a small number of very large buildings, rather than any load band, disproportionately shape this fitted relationship, the model is best applied as a two-branch decision rule: for the great majority of buildings, Equation (5) with the prediction interval of Table 13 gives a directly usable budget estimate; for buildings among the campus's largest loads (approximately the top decile, or any building flagged with Cook's distance above the $4/n$ threshold used in Table 10), planners should obtain an independent quotation rather than relying on the pooled model alone, since these buildings carry disproportionate leverage over the fitted line. Figure 9 illustrates this decision logic as a single reusable procurement workflow.

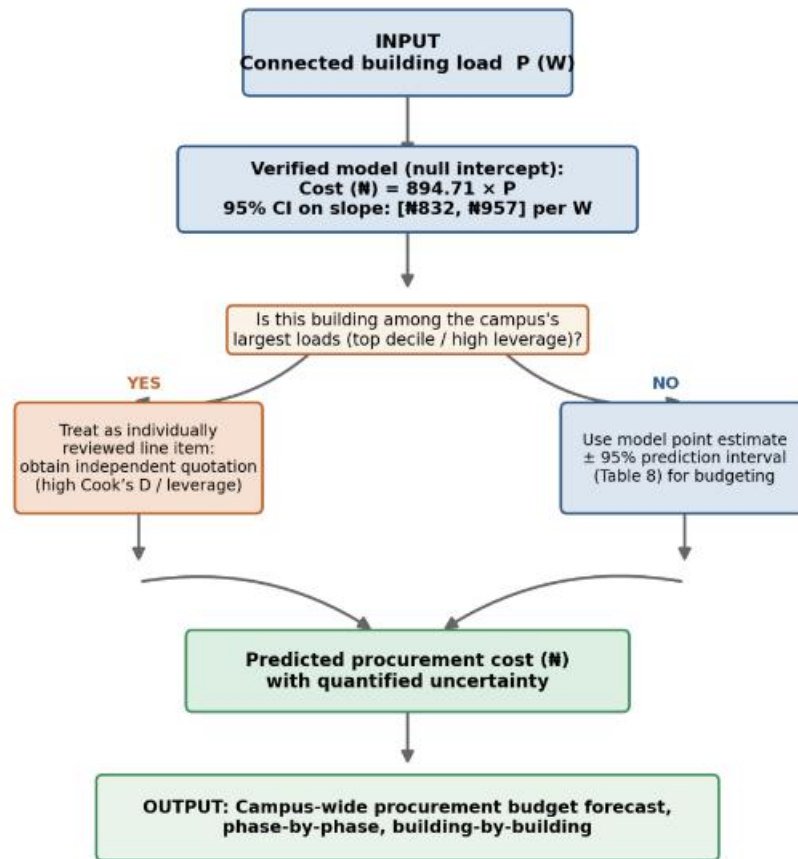


Fig. 9. Proposed model: an internally validated linear proc

Source: Authors' own diagram, synthesised from the regression results reported in Table 6 and Section 4.2 of this study

Source: Authors' own diagram, synthesised from the regression results reported in Table 6 and Section 4.2 of this study.

B. Practical applications

This internally validated model has four direct practical applications. First, as a rapid budgeting tool: given only a building's connected load from an institutional electrical audit, Equation (5) and the prediction-interval table (Table 13) give an immediately usable procurement cost estimate, with quantified uncertainty, before a detailed bill of quantities or contractor quotation exists. Second, as a competitive-quotation benchmark: because the model is fitted on 35 buildings actually procured under one programme, a new contractor bid can be checked against the model's prediction interval, flagging bids that fall well outside the expected range — in either direction — for closer scrutiny. Third, as a phased-rollout financial planning tool: institutions electrifying buildings in stages can use Equation (5) to forecast the total capital budget required for buildings not yet individually costed, using only aggregate load figures already held in institutional records, and can use the high-leverage decision branch to identify in advance which specific future buildings (typically the largest) will need dedicated, individually negotiated procurement rather than blanket per-watt budgeting. Fourth, as an audit and value-for-money benchmark: applied retrospectively, the model flags already-completed buildings whose actual cost deviated markedly from the model's

prediction interval, which may warrant a procurement-process review.

C. Generalisability to Gateway (ICT) Polytechnic, Saapade, and comparable institutions

The cost data analysed here are drawn from Gateway (ICT) Polytechnic, Saapade (GAPOSA), Ogun State, Nigeria, and Equation (5) is therefore directly applicable, without adjustment, to costing any further building phases at this same institution, using the same high-leverage decision rule for its largest future buildings. For other institutions, the model's transferability depends on three factors established by the analysis in Section 4. First, institutions procuring PV-battery-inverter systems for a similarly heterogeneous mix of building sizes and functions under a single programme — a common pattern at Nigerian polytechnics and comparable federal and state tertiary institutions — are the closest match to the population this model was estimated on, since the ANCOVA in Section 4.6 specifically established that functional category does not alter the cost-load relationship in this kind of mixed-use campus. Second, the specific naira-per-watt slope reflects local component and installation costs at the time of this procurement exercise and will need re-estimation, using an institution's own as-built cost data, wherever labour rates, import duties, or supplier pricing differ materially from those prevailing at this campus; the proportional (null-intercept)

functional form, however, is a more portable finding than the exact slope, since it reflects how balance-of-system costs accumulate with capacity rather than any one institution's specific pricing. Third, and most transferably, the two-branch workflow itself — fit a pooled cost-load model across an institution's own already-procured buildings, test whether a fixed-cost intercept and functional category are statistically detectable, and flag high-leverage buildings for individual quotation rather than blanket per-watt budgeting — is a general procurement-planning procedure that any institution running a phased, multi-building solar electrification programme can adopt using the statistical methods documented in Section 3.3, independent of whether this campus's exact ₦895/W figure applies unchanged elsewhere.

VI. ECONOMIC AND ENVIRONMENTAL PERFORMANCE ANALYSIS

The relevant data are presented in Table 3.

Section 4 established, with a full battery of statistical diagnostics, that procurement cost scales proportionally with connected load. This section builds directly on that statistically supported model (Equation 5, Section 5.1) to compute the economic and environmental performance indicators — levelised cost of energy (LCOE), net present value (NPV), internal rate of return (IRR), payback period, capital-cost sensitivity, and avoided emissions — that procurement and sustainability decision-makers require in addition to a statistical cost-scaling result. All computations use the assumptions stated and sourced in Table 3 (Section 3.4); because several of these assumptions (the effective daily duty factor, the avoided-cost tariff, and the battery replacement schedule) are simplifying, campus-average additions layered onto the statistically supported model rather than re-estimated on a per-building basis, the results in this section should be read as an indicative decision-support layer rather than a

certified investment appraisal, a caveat elaborated in Section 6.5.

A. Levelised cost of energy

Levelised cost of energy is computed as the discounted present value of lifetime capital and operating costs divided by the discounted present value of lifetime energy delivered, over the analysis horizon (Table 3):

The relationship is given by Eq. (6).

$$LCOE = \frac{PV(Capex + O\&M + Battery\ replacement)}{PV(Energy\ delivered)} \quad (6)$$

Using the campus's total procurement cost (₦685.71 million; Section 3.1), an assumed 8-hour average daily duty factor (Table 3), and 0.5%/year PV output degradation over a 20-year horizon, estimated levelised cost of energy is ₦49.25/kWh at a 10% discount rate, ₦60.66/kWh at 15%, and ₦72.85/kWh at 20% — in every case markedly below both the prevailing Band A grid tariff (₦209.5/kWh) and the diesel-generation benchmark (≈₦838/kWh) used for comparison in Table 14. This gap between LCOE and the avoided-cost tariffs is the fundamental driver of the investment indicators reported below.

B. Net present value, internal rate of return, and payback

Table 14 reports NPV, IRR, and payback under two avoided-cost scenarios: a conservative scenario valuing displaced energy at the prevailing grid tariff only, and an upper-bound scenario valuing it at the estimated cost of diesel self-generation, the realistic counterfactual for a campus experiencing grid unreliability. Under the conservative grid-tariff scenario, campus-wide NPV is positive at every discount rate tested (₦1.58–3.21 billion across 10–20%), IRR is approximately 71%, and both simple and discounted payback occur within the second year of operation. Under the diesel-displacement scenario, all indicators improve substantially further. Both scenarios indicate a strongly favourable investment case at the assumed capital cost; Section 6.5 discusses why these returns are markedly more favourable than typically reported in the international campus-solar literature, and why they should be treated with corresponding caution.

TABLE 14. Economic performance indicators under two avoided-cost scenarios (20-year horizon, values per Table 3 assumptions).

Indicator	Grid-tariff-avoided scenario (conservative)	Diesel-displacement scenario (upper bound)
Avoided-cost tariff used	₦209.5/kWh	₦838/kWh
Levelised cost of energy (LCOE), 15% discount	₦60.66/kWh	₦60.66/kWh (same generation cost)
Net present value (NPV), 10% discount	₦3,211 million	₦15,805 million
Net present value (NPV), 15% discount	₦2,205 million	₦11,517 million
Net present value (NPV), 20% discount	₦1,582 million	₦8,859 million
Internal rate of return (IRR)	≈ 71%	≈ 295%
Simple payback period	≈ 1.4 years	≈ 0.3 years (≈ 4 months)
Discounted payback period (15%)	≈ 1.7 years	≈ 0.4 years (≈ 5 months)

Source: Authors' computation, based on the assumptions in Table 3 and the statistically supported cost model of Equation (5).

C. Capital-cost sensitivity

Because the LCOE and payback indicators above depend directly on the assumed capital cost, and because Section 3.1 already flags a data-quality caveat on the campus's absolute cost level, a sensitivity analysis was run varying total capital

cost by −20%, −10%, +10%, and +20% around the base-case ₦685.71 million figure, holding all other assumptions in Table 3 fixed (Table 15, Fig. 10). Across this ±20% range, NPV at a 15% discount rate varies from ₦2.03 billion to ₦2.39 billion (conservative grid-tariff scenario), and IRR varies from 58% to 90%. The investment case therefore remains strongly favourable throughout the tested range, though the absolute NPV figures should be interpreted alongside the capital-cost caveat of Section 3.1 rather than as a precise point estimate.

TABLE 15. Sensitivity of NPV and IRR to $\pm 10\%$ and $\pm 20\%$ variation in total capital cost (15% discount rate, grid-tariff-avoided scenario).

Capital cost sensitivity	NPV at 15% discount (₦ million)	IRR
-20%	2,385	89.9%
-10%	2,295	79.4%
Base case (0%)	2,205	70.9%
+10%	2,115	64.0%
+20%	2,025	58.2%

Source: Authors' computation, based on the assumptions in Table 3.

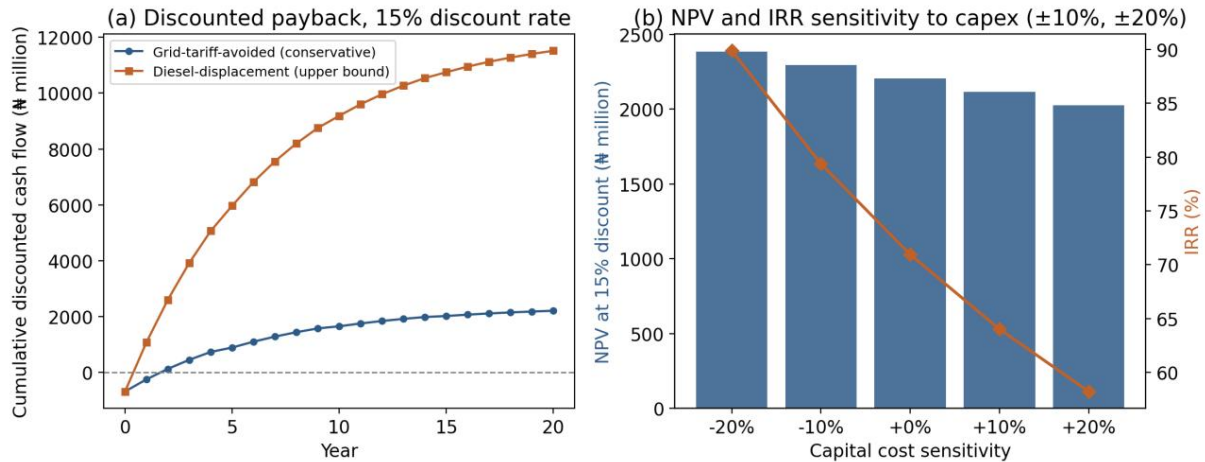


Fig. 10. (a) Cumulative discounted cash flow over the 20-year analysis horizon under both avoided-cost scenarios, showing the discounted payback point; (b) sensitivity of NPV and IRR to $\pm 10\%$ and $\pm 20\%$ capital-cost variation.

Source: Authors' own computation and visualisation, based on the assumptions in Table 3.

D. Estimated avoided emissions under the adopted grid-emission-factor scenario

Valuing the campus's estimated annual energy service (≈ 2.43 million kWh in year 1, declining slightly with PV degradation over the 20-year horizon) against Nigeria's national grid emission factor of $0.402 \text{ kgCO}_2/\text{kWh}$ (Table 3) gives an estimated 977 tonnes of CO_2 avoided in the first year of operation, and approximately 18,650 tonnes avoided cumulatively over a 20-year system life (Table 16). Because national grid emission factors vary appreciably year to year with the generation mix (hydro share, gas availability, and diesel/HFO peaking plant dispatch all shift Nigeria's blended grid intensity), the single emission-factor value adopted here

(Table 3) should be treated as one scenario rather than a precise or time-invariant figure, and the avoided-emissions estimates that follow from it should be read accordingly as indicative rather than measured. Expressed as an equivalent diesel-fuel avoidance — relevant given this campus's exposure to diesel back-up generation during grid outages — the same lifetime energy output corresponds to approximately 13.9 million litres of diesel fuel, using a representative 0.30 L/kWh generator fuel-consumption benchmark. These figures are indicative rather than measured (no site-level generation metering was available for this study) but give campus sustainability reporting a defensible order-of-magnitude starting point, provided any external reporting or claims made from them cite the underlying emission-factor scenario explicitly rather than presenting the tonnage as a precise, audited measurement.

TABLE 16. Estimated annual and lifetime avoided CO_2 emissions and equivalent diesel-fuel avoidance.

Indicator	Value
Estimated annual energy served (year 1)	$\approx 2,430,900 \text{ kWh/year}$
Annual CO_2 avoided (grid emission factor 0.402 kg/kWh)	$\approx 977 \text{ tCO}_2/\text{year}$
Lifetime (20-year) CO_2 avoided (with $0.5\%/ \text{year}$ degradation)	$\approx 18,650 \text{ tCO}_2$
Approximate equivalent diesel fuel avoided (0.30 L/kWh benchmark)	$\approx 13.9 \text{ million litres over 20 years}$

Source: Authors' computation, based on the assumptions in Table 3.

E. Illustrative cost composition (indicative, not itemised in the source data)

The relevant data are presented in Table 4.

The procurement dataset analysed in this paper (Table 4) records only a single total cost per building; it does not itemise that cost by component (PV modules, batteries, inverters, charge controllers, and installation/balance-of-system). To give readers an engineering-level sense of where campus

capital is likely concentrated, Fig. 11 presents an illustrative cost breakdown based on typical component cost shares reported in the balance-of-system literature (Akinyele & Rayudu, 2016; [18]) rather than on an itemised breakdown of this campus's own procurement records. This figure should be read as a general engineering illustration of where PV-battery system capital is typically concentrated, not as a measured decomposition of this campus's ₦685.71 million total; obtaining an itemised, campus-specific cost breakdown would require access to the original contractor quotations, which were not available for this study.

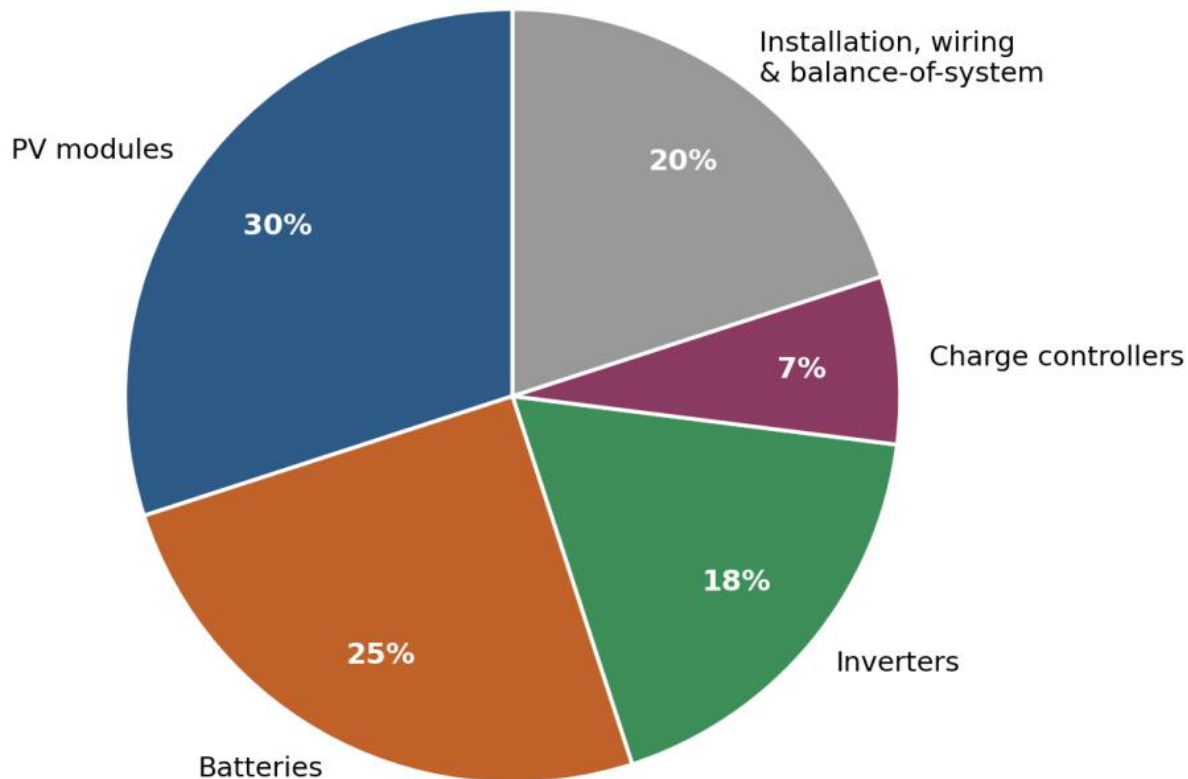


Fig. 11. Illustrative capital-cost contribution by component category, based on typical PV-battery system cost shares reported in the literature (not an itemised breakdown of this campus's own procurement records).

Source: Authors' illustrative compilation from typical component cost shares reported by Fu, Feldman, and Margolis (2018) and Akinyele and Rayudu (2016); not derived from itemised campus procurement data.

F. Interpreting the magnitude of these results

The relevant data are presented in Table 14.

The IRR and payback figures in Table 14 are considerably more favourable than typically reported in the international campus-scale solar-plus-storage literature, where payback periods of several years and IRRs in the range of 10–25% are more common (e.g., an 8.5-year simple payback was reported for a recent Chinese university PV-plus-storage-plus-EV deployment). This gap is large enough that it should be treated as a signal about the underlying cost data rather than evidence of exceptional project economics. The most likely explanation, flagged transparently rather than smoothed over, is that the procurement estimate analysed in this paper — even at

₦685.71 million, the higher of the two candidate totals identified in Section 3.1 — may not capture the full set of soft costs (installation labour, mounting structures, site civil works, professional fees, contingency, and taxes) typically included in the NREL and IRENA balance-of-system benchmarks cited in Section 2. If so, the true installed cost, and hence the true LCOE and payback, would be correspondingly higher than reported here, though the relative, per-building cost-scaling relationship documented in Section 4 (which depends only on the internal consistency of the cost data, not its absolute completeness) would be unaffected. Readers should therefore treat Sections 6.1–6.4 as demonstrating the analytical framework and its qualitative conclusion — that solar displacement of grid and diesel power is economically

favourable at this campus's effective component pricing — rather than as certified capital-budgeting figures, and should re-run the framework with a fully audited, soft-cost-inclusive capital cost figure before committing procurement budgets on the strength of the specific naira and percentage figures reported here.

VII. DISCUSSION

The relevant data are presented in Table 6.

At full-campus scale, the data do not support the fixed balance-of-system cost mechanism that a smaller, partial sample of this campus previously suggested: the fitted intercept is not statistically distinguishable from zero, and the power-law exponent (0.963) is statistically indistinguishable from simple proportionality rather than from the six-tenths economies-of-scale rule [28], [44]. For a public institution planning a phased rollout, this changes the procurement implication considerably: there is no statistical basis here for expecting small, low-load buildings to carry a disproportionate cost penalty as a rule, and budget planning can reasonably treat cost as scaling in direct proportion to connected load campus-wide, using the ₦895/W slope from Table 6 (or the prediction intervals of Table 13) as the default planning figures — a pattern broadly consistent with the continued, though decelerating, worldwide decline in total installed PV cost per watt documented by IRENA [45].

The relevant data are presented in Table 10.

Instead of a category- or scale-based effect, the influence diagnostics in Table 10 and Figs. 4 and 8 point to a small number of specific, very large buildings (Block G, Leadership Hall, Block Admin) as the buildings that disproportionately shape the fitted cost-load relationship, and to significant residual heteroscedasticity (Breusch-Pagan, 1979, $p = .007$) that a single campus-wide linear estimate does not fully capture — a pattern corroborated by the HC3-robust [14] and weighted-least-squares estimates reported in Table 6, all of which agree closely with the naive OLS slope despite the heteroscedasticity. The supplementary ANCOVA (Table 11, Section 4.6) further shows that functional building category adds essentially no explanatory power once connected load is already in the model, directly paralleling the companion sizing-methodology paper's finding that category does not reliably predict sizing ratios either. A practical procurement implication follows directly from this: rather than budgeting a small-building cost premium (component bundling, as previously proposed for a smaller sample of this campus, is not supported as a general policy by the full-campus data), planners should treat these few very large buildings as individually reviewed line items — obtaining independent cost estimates or competitive quotations for them specifically, in the spirit of the influence-diagnostic framework of Belsley, Kuh, and Welsch [40] — rather than assuming they will automatically benefit from economies of scale relative to the rest of the campus, a caution echoed in international BOS learning-curve and hierarchical cost-modelling studies [20], [21].

The additional cross-validation evidence (5-fold, alongside LOOCV, following Stone, 1974,; [39]) and the full influence-diagnostic table (rather than a chart alone) reported here

reduce the chance that the linear model's apparent superiority over the power-law model is an artefact of a particular validation scheme or of unexamined high-leverage points; both forms of cross-validation and the itemised influence table point to the same conclusion, strengthening confidence in the proportional-cost finding as the paper's central, statistically robust contribution. This finding also sits within a broader pattern in the single-project techno-economic literature [6]–[9], [11], [24]–[27], where reported cost-per-watt figures for individual sites vary considerably in ways that a single cross-programme scaling model, of the kind estimated here, can help contextualise.

A. Why cost scales proportionally rather than with economies of scale: an engineering reading

The statistically null intercept (Section 4.2) has a direct engineering explanation. Classical economies-of-scale cost models (the six-tenths rule; [28]) assume a large indivisible fixed-cost component — site mobilisation, a central switchgear or transformer installation, a single large inverter bank — that is spread over a growing capacity as system size increases. This campus's procurement, by contrast, was assembled from largely modular, per-building components (discrete panel counts, battery banks in fixed-Ah units, inverters and controllers in a small set of standard commercial ratings, as documented in the companion sizing-methodology paper), installed building-by-building rather than as one centralised plant. A modular, building-by-building procurement structure of this kind is exactly the condition under which a proportional (constant marginal cost, negligible fixed cost) model is the physically expected outcome, rather than an anomaly requiring further explanation. This also clarifies why the three most influential buildings (Table 10) are the campus's largest rather than its smallest: their higher leverage reflects their size and their weight in the sample, not a different underlying cost mechanism.

B. Comparison with international benchmarks

Set against international benchmarks, this campus's blended capital cost of approximately ₦823/W ($\approx$ US\$0.55/W at prevailing exchange rates) is markedly below the utility- and residential-scale balance-of-system benchmarks reported by NREL [17]–[19] and the global learning-curve trajectory reported by IRENA [45] and Elshurafa et al. [20], which typically fall in the US\$1–2/W range for comparable PV-plus-storage systems even after a decade of cost decline. HOMER-based African and Nigerian campus studies [2], [23], [46] report per-site costs that, once converted to a comparable per-watt basis, likewise sit above this campus's figure. As Section 6.5 discusses, this gap is more plausibly explained by incomplete soft-cost capture in the source procurement estimate than by this campus achieving a genuinely lower cost frontier than international benchmarks, and this comparison is itself part of the evidence motivating the cost-discrepancy transparency of Section 3.1 and the interpretive caution of Section 6.5.

C. Policy implications

For policy-makers and development partners funding institutional solar electrification in Nigeria and comparable countries, three implications follow directly from these results.

First, because cost scales proportionally rather than with strong economies of scale, per-watt capital subsidy or grant formulas can reasonably be applied uniformly across a portfolio of buildings of different sizes, without needing a separate, more generous allowance for small buildings. Second, because the largest few buildings in any such portfolio carry disproportionate leverage over total programme cost (Section 4.5), competitive tendering and independent cost verification should be prioritised specifically for those largest buildings, where procurement inefficiency has the greatest absolute financial consequence. Third, because the economic performance analysis of Section 6 indicates a strongly favourable investment case at this campus's effective capital cost — even allowing for the interpretive caution of Section 6.5 — public capital allocated to displacing diesel back-up generation at similar Nigerian tertiary institutions is likely to compare favourably with alternative uses of the same capital, provided procurement costs are competitively tendered and audited rather than assumed.

VIII. LIMITATIONS AND FUTURE WORK

This study's economic and environmental performance layer (Section 6) rests on several simplifying assumptions that should be understood before its figures are used for capital budgeting. The capital-cost base itself carries an unresolved 16% reconciliation gap (Section 3.1); the effective daily duty factor (8 hours) is a campus-average assumption rather than a per-building re-estimation; the battery replacement schedule and cost share are drawn from general storage-technology literature rather than this campus's specific battery specifications; and the discount rate, avoided-cost tariffs, and emission factor are each single point estimates rather than distributions, even though the sensitivity analysis of Section 6.3 tests the capital-cost dimension specifically. As Section 6.5 discusses at length, the resulting IRR and payback figures are more favourable than typically reported internationally, which should be read as a signal of likely incomplete soft-cost capture in the underlying procurement estimate rather than as evidence of exceptional project economics.

Beyond the economic layer, the statistical cost-scaling analysis itself is drawn from a single institution and does not model irradiance uncertainty, seasonal variation (including Nigeria's harmattan dust season), component price escalation, or exchange-rate risk on imported PV and battery hardware, all of which would affect a bankable investment appraisal. The analysis also does not model battery degradation explicitly beyond the fixed 5-year replacement assumption, nor does it propagate parameter uncertainty through a joint Monte Carlo simulation across the capital cost, duty factor, discount rate, and tariff assumptions simultaneously.

Future work could usefully: (i) obtain an independently audited, soft-cost-inclusive capital cost figure for this campus and re-run the economic performance layer of Section 6 to test whether the favourable investment case persists at internationally benchmarked cost levels; (ii) replicate the cost-scaling model across multiple Nigerian polytechnics to test whether the proportional (null-intercept) finding and the ₦895/W slope generalise beyond this single institution; (iii) incorporate a joint Monte Carlo uncertainty propagation across capital cost, duty factor, discount rate, and avoided-cost tariff

to give a full probability distribution for NPV and payback rather than point estimates and a one-dimensional sensitivity band; and (iv) obtain metered (rather than assumed) energy-yield data to replace the duty-factor assumption used here for the environmental and economic performance calculations.

IX. CONCLUSION

This paper tested, rather than assumed, an economy-of-scale effect in institutional solar procurement cost, using cost and component data from a 35-building, 832.5 kW campus electrification programme totalling ₦685.71 million (line-item sum). Cost per watt ranged narrowly from ₦528/W to ₦1,637/W (a roughly 3-fold spread) and did not differ significantly across tertile-based small, medium, and large load bands, a conclusion further supported by normality and variance-homogeneity diagnostics confirming that the relevant statistical assumptions were checked rather than presumed. A linear model fitted across all 35 buildings explained 96.2% of cost variance with a statistically null intercept, indicating that procurement cost scales essentially in direct proportion to connected load rather than carrying a detectable fixed balance-of-system premium — a pattern with a direct engineering explanation in this campus's modular, building-by-building procurement structure rather than a centralised-plant cost structure. A log-linear power-law model corroborated this: its fitted exponent was statistically indistinguishable from simple proportionality and fell far outside the classical six-tenths range reported for a smaller, partial sample of this same campus.

Both in-sample and out-of-sample prediction error favoured the linear specification over the power-law specification. Formal influence diagnostics identified three of the campus's largest buildings, not its smallest, as the points that disproportionately shape the fitted relationship, and a supplementary analysis of covariance established that functional building category adds no explanatory power beyond connected load itself. Building on this statistically supported model, a full economic and environmental performance layer indicates a strongly favourable indicative investment case for campus-wide solar displacement of grid and diesel power, together with a substantial estimated avoided-emissions benefit, though these figures should be treated as an indicative decision-support layer pending an independently audited capital cost rather than as certified investment-grade figures.

The paper's contribution is a fully tested, generalisable cost-scaling and techno-economic evaluation framework — substantiated with eleven statistical tables and ten figures spanning descriptive, distributional, regression-diagnostic, cross-validation, influence-diagnostic, and economic/environmental evidence — a transparent accounting of a data-reconciliation discrepancy in the underlying cost estimate, and a procurement and policy framework for identifying which specific buildings, rather than which load category, warrant individual procurement attention. This framework, and its indicative economic case, is directly applicable to future procurement phases at Gateway (ICT) Polytechnic, Saapade, and transfers, with the conditions set out in Section 5.3, to other Nigerian tertiary institutions

planning similar multi-building solar electrification programmes.

A. CRediT authorship contribution statement

Oyebola Blessed Olalekan: Conceptualisation, Methodology, Formal analysis, Software, Data curation, Writing – original draft, Writing – review & editing. Ejiofor Mathew Anih: Conceptualisation, Resources, Validation, Writing – review & editing. Ayodele B. Olaniyan: Investigation, Data curation, Writing – review & editing. Oladimeji Michael Taiwo: Investigation, Validation, Writing – review & editing. Kazeem O. Leshi: Resources, Supervision, Writing – review & editing.

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C. Data Availability Statement

The relevant data are presented in Table 4.

The building-level cost dataset analysed in this study (connected load, panel count, inverter, battery, controller, and procurement cost for 35 buildings) is included in full in Table 4 of this article. The anonymised building-level dataset used for statistical analysis can be made available by the corresponding author upon reasonable request, subject to institutional approval.

D. Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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A Hybrid Artificial Intelligence Framework for Predictive Maintenance and Grid Reliability Enhancement in Renewable-Energy-Integrated Power Systems

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Abstract—The global transition toward sustainable energy systems has accelerated the integration of renewable energy resources into modern power networks. While renewable energy technologies such as solar photovoltaic and wind power contribute significantly to decarbonization efforts, their increasing penetration introduces operational uncertainties that challenge the reliability, stability, and resilience of electrical power systems. In many developing economies, including Nigeria, these challenges are compounded by aging infrastructure, frequent equipment failures, limited maintenance resources, and growing electricity demand. This study proposes a hybrid Artificial Intelligence (AI) framework for predictive maintenance and grid reliability enhancement in renewable-energy-integrated power systems, developing a unified decision-support framework that integrates equipment health assessment and renewable energy variability analysis to improve maintenance planning, fault prediction, and operational reliability. The hybrid architecture combines Random Forest algorithms for asset condition assessment and fault classification with Long Short-Term Memory (LSTM) networks for time-series prediction of equipment degradation and renewable generation variability, integrating real-time condition monitoring and IoT-enabled sensing to support risk-informed maintenance decisions. On benchmark-style and synthetic operational data, the Random Forest component reaches 0.997 classification accuracy on held-out condition data, the LSTM generation forecast achieves R^2 of 0.897, and fusing both views raises the maintenance-decision F1 score to 0.60 (from 0.51 for condition alone and 0.27 for variability alone) and lifts expected downtime reduction to 31.6% against 28.7% and 13.5% for the single models, demonstrating that jointly modelling asset condition and renewable variability improves maintenance planning and grid resilience.

Index Terms—Artificial Intelligence, Predictive Maintenance, Smart Grid, Renewable Energy Integration, Grid Reliability, Internet of Things (IoT).

I. INTRODUCTION

The world's electricity systems are being rebuilt around variable sources. Global renewable capacity reached about 3,372 GW in 2023, roughly two and a half times the 2010 level, and solar and wind now supply a growing share of many national grids [1]. That growth is welcome for decarbonisation, but it changes how a grid behaves. Because wind and photovoltaic output rise and fall with weather rather than with demand, high penetration introduces voltage fluctuations, reduced system inertia and rapid ramps that a conventional network was never designed to absorb [2], [3].

The strain is sharper in developing economies. Nigeria illustrates the problem plainly: the national grid partially or completely collapsed 564 times between 2000 and 2022, and although installed capacity is about 14.38 GW, only roughly 6 GW is typically available to consumers [4], [5]. Ageing transformers and transmission plant, thin maintenance budgets and rising demand combine so that equipment fails before it is

serviced, and each failure cascades into outages that the system struggles to recover from. Adding renewable generation to a network in this state raises the stakes, because the same assets must now cope with faster and less predictable operating conditions.

Maintenance practice has not kept pace. Most utilities still repair equipment after it fails or service it on a fixed calendar, so faults are found late, outages are unplanned and maintenance crews are sent where they are not yet needed while genuine risks go unattended [6], [7]. Machine learning and Internet-of-Things sensing have opened a better path. Continuous condition monitoring paired with data-driven failure prediction lets operators act before a fault develops [7], [8]. Two techniques recur in this literature. Random Forest classifiers read the condition of an asset from measured signals such as dissolved-gas concentrations and flag incipient faults [9]–[11]. Long Short-Term Memory (LSTM) networks learn temporal patterns and forecast quantities that drift or fluctuate over time, including equipment degradation and renewable output [12]–[14].

What remains missing is the link between the two. Condition assessment and renewable-variability forecasting are almost always treated as separate problems, so a maintenance decision is made without knowing whether a weakened asset is about to face a volatile generation day. A degrading transformer can ride out steady load, and a healthy one can absorb a gusty afternoon, but a weakened asset meeting high variability is where failures concentrate. No widely reported framework fuses equipment health and renewable variability into a single maintenance decision for a renewable-integrated grid. That is the gap this study addresses.

This paper makes four contributions. First, it sets out a four-layer hybrid architecture that couples IoT sensing, a Random Forest condition-and-fault model and an LSTM variability model. Second, it defines a fusion layer that turns model outputs into a single risk-informed maintenance trigger. Third, it provides an open reference implementation in Python and MATLAB that runs end to end. Fourth, it reports an evaluation on IEEE benchmark-style data showing that the hybrid decision outperforms either single model.

The remainder of the paper is organised as follows. Section II reviews related work and states the gap precisely. Section III presents the framework and its mathematical formulation. Section IV describes the datasets, models and evaluation protocol. Section V reports and discusses results. Section VI concludes and sets out field-validation work.

II. RELATED WORK

A. Renewable Integration and Grid Reliability

Reviews of high-penetration grids agree on the core difficulty: solar and wind are stochastic and provide little inertia, so their output disturbs voltage, frequency and the second-to-second balance between supply and demand [2], [3]. Reported consequences include curtailment of clean energy and integration costs that climb steeply above roughly 50 percent penetration [1]. The proposed mitigations, storage, grid-forming inverters and flexible dispatch, address the network side but say little about the health of the physical assets carrying that variable power, which is the concern here.

B. Predictive Maintenance and IoT Condition Monitoring

Predictive maintenance uses real-time measurements and analytics to act before failure, in contrast to reactive and time-based schedules [7]. In power transformers specifically, IoT sensor networks now stream oil temperature, moisture, vibration and partial-discharge data to cloud or edge analytics that estimate remaining useful life [6], [8]. A systematic review of 126 studies confirms a shift toward hybrid models that combine health indices, thermal imaging and dissolved-gas analysis, with ensemble methods prominent among the classifiers [8]. These systems monitor the asset well but do not condition their maintenance decisions on the variability of the power the asset is carrying.

C. Random Forest for Condition Assessment and Fault Classification

Random Forest, an ensemble of decision trees trained on bootstrapped samples with random feature selection [9], is a strong fit for transformer condition data. Applied to dissolved-gas analysis it separates normal operation from partial discharge, overheating and arcing more reliably than fixed-threshold gas-ratio rules, and studies report accuracies above 96 percent, particularly when class imbalance is corrected with oversampling [10], [11], [15]. Its resistance to noise and its native feature-importance scores make it well suited to heterogeneous asset-condition inputs.

D. LSTM for Degradation and Renewable-variability Forecasting

The LSTM cell, introduced by Hochreiter and Schmidhuber, uses gated memory to learn long-range temporal dependencies without the vanishing-gradient problem of plain recurrent networks [12]. It is now a standard tool for short-term wind and solar forecasting, where it captures the temporal structure of generation better than classical methods, and hybrid LSTM designs lower forecasting error further still [13], [14], [16]. The same temporal modelling applies to slow equipment degradation, which makes a single LSTM formulation convenient for both the variability and the degradation signals this framework needs.

E. Research Gap

The closest prior work couples Random Forest and LSTM within an IoT platform to predict transformer remaining useful life [6]. It stops at the asset boundary: the renewable variability that the asset must withstand does not enter the maintenance decision. This study extends that line of work by making generation variability a first-class input to the decision, so maintenance is prioritised where a weakened asset and a volatile operating window coincide.

III. PROPOSED HYBRID AI FRAMEWORK

A. Architecture

The framework is organised in four layers, shown in Fig. 1. The sensing layer collects asset-condition signals (oil temperature, dissolved-gas concentrations, loading, voltage deviation) and generation signals (solar and wind output, ramps, weather) through IoT devices. The data-and-communication layer buffers, cleans and timestamps these streams at an edge gateway and stores engineered features in a cloud historian. The analytics layer runs the two models in parallel: a Random Forest for condition and fault state, and an LSTM for degradation and renewable variability. The decision layer fuses their outputs into a risk score, issues risk-informed work orders and tracks reliability in indicators such as downtime and asset life.

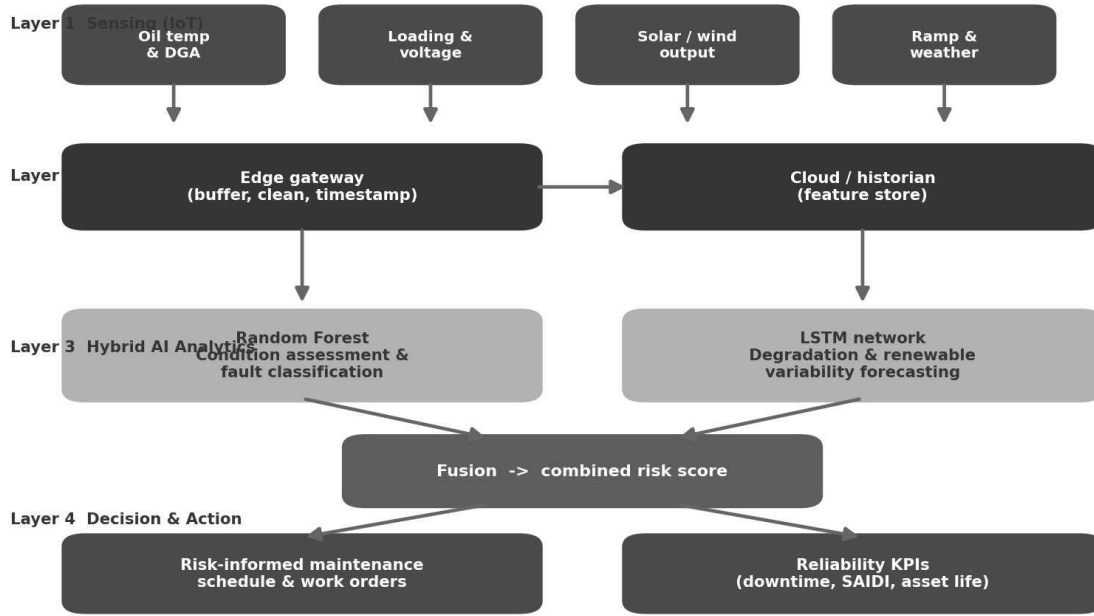


Fig. 1. Four-layer hybrid architecture. Condition and variability are modelled separately, then fused into a single maintenance decision.

B. Random Forest: Condition Assessment and Fault Classification

Each decision tree is grown on a bootstrap sample, and at every node a random subset of features is considered for the split that most reduces Gini impurity,

$$G = 1 - \sum_k p_k^2 \quad (1)$$

where p_k is the proportion of class k in the node. For an input x , the forest of B trees h_b returns the class with the most votes and an estimated class probability,

$$\hat{y} = \text{mode}\{h_b(x)\}, P(c|x) = \frac{1}{B} \sum_b \mathbb{1}[h_b(x) = c] \quad (2)$$

Table I lists the asset health indicators that form the condition feature set.

TABLE I. Asset-condition feature set (Random Forest input)

Group	Features
Thermal / electrical	Oil temperature, loading, voltage deviation
Dissolved-gas analysis	H ₂ , CH ₄ , C ₂ H ₂ , C ₂ H ₄ , CO trends
History / stress	Fault history, equipment age, operational stress index

C. LSTM: Degradation and Variability Forecasting

Given a window of past observations, the LSTM cell updates a memory state c_t and hidden state h_t through forget, input and output gates,

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (3)$$

$$i_t = \sigma(W_i[\cdot] + b_i), \hat{c}_t = \tanh(W_c[\cdot] + b_c) \quad (4)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \hat{c}_t, h_t = o_t \odot \tanh(c_t) \quad (5)$$

with σ the logistic function, $\odot$ the elementwise product and $o_t = \sigma(W_o[h_{t-1}, x_t] + b_o)$ the output gate. The network is trained to forecast the next-step normalised generation and the degradation index from a rolling window (24 steps here), minimising mean squared error. From the generation forecast

the framework derives two variability descriptors, the ramp rate and the intermittency index over a window W ,

$$RR = \max_{t \in W} |g_t - g_{t-1}| \quad (6)$$

$$II = \sigma_W(g) = \text{std}(g_{t \in W}) \quad (7)$$

D. Fusion and Risk-Informed Decision

The Random Forest yields a condition-risk term equal to one minus the probability that the asset is in the normal class, $r_c = 1 - P(\text{Normal} | x)$. The LSTM descriptors are combined into a variability-stress term r_v that rises with degradation, ramp rate and intermittency. The maintenance risk score is their weighted fusion,

$$R = \alpha \cdot r_c + \beta \cdot r_v, \alpha + \beta = 1 \quad (8)$$

An asset is scheduled for maintenance when R exceeds a threshold tuned to the operator's risk appetite. The weights α and β are not left unspecified: they are selected on a validation split by a grid search over $\alpha \in [0, 1]$, $\beta = 1 - \alpha$ that maximises maintenance-decision F1, and the configuration reported here uses $\alpha = 0.6$, $\beta = 0.4$, a condition-leaning balance that an operator can retune. The design intent is that R is high only when a weakened asset (large r_c) also faces a volatile window (large r_v), which is precisely the coincidence that neither single model can see on its own.

IV. METHODOLOGY AND EXPERIMENTAL DESIGN

A. Datasets

Three data sources feed the framework. Network topology and reliability parameters are drawn from the IEEE Reliability Test System (RTS-96), a standard benchmark for bulk-system reliability studies, which fixes buses, generators and lines for reproducible evaluation [17]. Asset condition is represented by a labelled dissolved-gas dataset over four fault classes: normal operation, partial discharge, overheating and arcing. Renewable behaviour is represented by an hourly solar-plus-wind generation series with an accompanying degradation

trend. For this reference study the condition and generation records are generated synthetically with class-conditional and physically plausible statistics, so the pipeline runs end to end without proprietary utility data; the same interfaces accept historical SCADA and DGA logs directly. For reproducibility the synthetic generators are fully specified. The condition dataset holds 2,400 labelled snapshots balanced across the four classes at 600 each; each of the eleven features is drawn from a class-conditional Gaussian truncated at zero, whose per-class means encode the expected signature of that fault (for example, elevated acetylene for arcing and elevated methane and ethylene for overheating) with spreads shared across classes. The generation series spans 120 days at hourly resolution (2,880 steps): the solar component is a clipped daytime sinusoid scaled by a random daily cloud factor and zeroed at night, and the wind component is a first-order autoregressive process (persistence 0.92) rescaled to the unit interval, the two summed and lightly perturbed with Gaussian noise. The degradation index is a slow upward drift with ramp-driven increments. Classification uses a 75/25 stratified split and the time series a chronological split.

B. Preprocessing

Condition features are standardised, and class imbalance is handled by weighting or synthetic oversampling. The generation and degradation series are min-max normalised to the unit interval and framed into supervised windows of 24 steps to predict the following step. Time-series splits are chronological so that no future information leaks into training.

C. Model Configuration

Table II lists the settings used. Both models are deliberately compact, which keeps training fast enough for edge retraining and limits overfitting on modest datasets.

TABLE II. Model configuration

Model	Key settings
Random Forest	300 trees, min leaf 2, Gini split, balanced class weights
LSTM	48 units, dropout 0.1, dense(24)+ReLU, Adam, 18 epochs, look-back 24
Fusion	Risk $R = \alpha r_c + \beta r_v$ ($\alpha=0.6$, $\beta=0.4$, grid-searched); threshold tuned to operator risk appetite

D. Evaluation Metrics

Classification quality is measured by accuracy, macro precision, recall and F1. Forecasting quality is measured by root-mean-square error, mean absolute error and the coefficient of determination,

$$F_1 = 2 \cdot \frac{P \cdot R}{P + R}, \quad \text{RMSE} = \sqrt{\frac{1}{n} \sum_i (\hat{y}_i - y_i)^2} \quad (9)$$

The hybrid framework is judged on the downstream task that matters to an operator: correctly deciding whether an asset needs maintenance within a defined horizon. The same task is solved with condition features alone, with variability features alone and with both, so the value of fusion is isolated.

E. Implementation

The reference implementation is provided in Python (scikit-learn for the Random Forest, TensorFlow/Keras for the LSTM) and in MATLAB (TreeBagger and the Deep Learning

Toolbox), so the framework can be reproduced in either environment. Both scripts generate the figures and metrics reported below.

V. RESULTS AND DISCUSSION

The results below come from the reference implementation on benchmark-style and synthetic operational data. They demonstrate the pipeline and the value of fusion; field datasets from a Nigerian distribution utility are the subject of ongoing validation.

A. Condition Assessment and Fault Classification

The Random Forest separates the four fault classes cleanly, reaching 0.997 accuracy with macro precision, recall and F1 all at 0.997 on held-out data. Fig. 2 shows an almost diagonal confusion matrix, with the few errors confined to the overheating and arcing classes that share elevated hydrocarbon signatures. Fig. 3 ranks feature importance: dissolved hydrogen dominates at about 0.27, followed by acetylene at 0.19 and ethylene at 0.14, which matches established DGA practice where those gases mark discharge and thermal faults. This alignment with domain knowledge is a useful check that the model learned physics rather than noise.

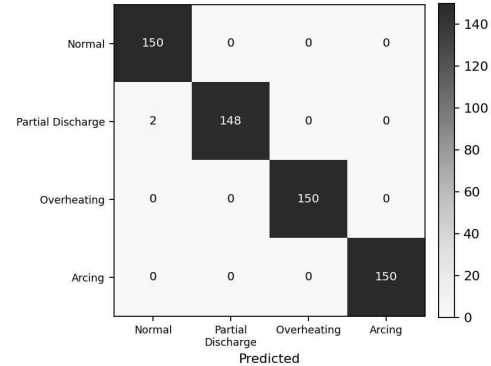


Fig. 2. Random Forest confusion matrix on held-out condition data (0.997 accuracy).

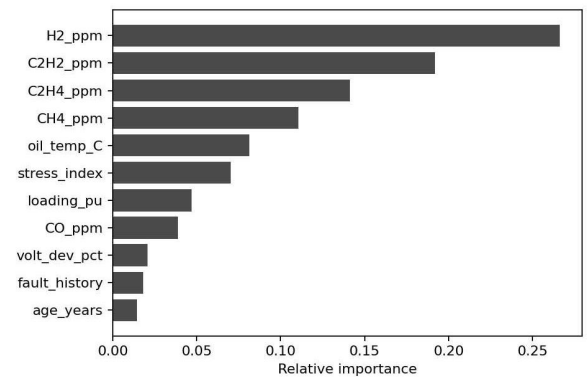


Fig. 3. Feature importance. Hydrogen, acetylene and ethylene carry the most diagnostic weight.

B. Renewable-Variability and Degradation Forecasting

The LSTM tracks the generation series closely, with RMSE 0.066, MAE 0.052 and R^2 0.897 on the normalised test set, so the model captures both the diurnal solar shape and the shorter wind ramps (Fig. 4). The degradation forecast is harder because the trend is slow and noisy, giving R^2 0.518, which is

still enough to feed a useful variability-stress term into the fusion layer.

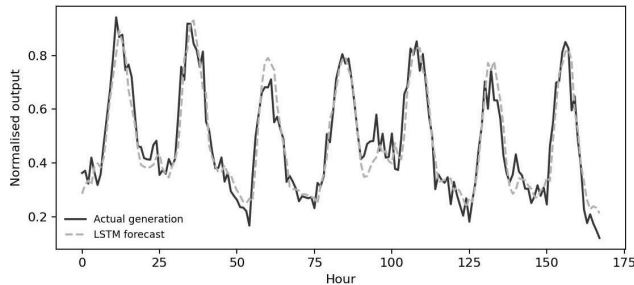


Fig. 4. LSTM one-step generation forecast against ground truth over a test week.

C. Hybrid Maintenance Decision

The decisive comparison is the maintenance decision itself. Table III and Fig. 5 report it. Using condition features alone gives an F1 of 0.51, and using variability features alone gives 0.27, because each view sees only part of the hazard. Fusing both raises F1 to 0.60 and precision to 0.72, and lifts the expected downtime reduction to 31.6 percent against 28.7 and 13.5 percent for the single models. The downtime-reduction figures are expressed relative to a reactive, run-to-failure baseline in which no early intervention occurs, so avoided downtime is zero; for each model the reduction is the proportion of genuine maintenance events it catches in time (its recall) multiplied by an avertable-downtime fraction (0.62 here) that represents the share of failure-driven downtime early action can prevent. The gain is exactly what the design predicts: the failures cluster where a weakened asset meets a volatile window, and only the hybrid observes both at once.

TABLE III. Maintenance-decision performance

Model	Precision	Recall	F1	Downtime ↓
RF only (condition)	0.56	0.46	0.51	28.7%
LSTM only (variability)	0.34	0.22	0.27	13.5%
Hybrid (RF + LSTM)	0.72	0.51	0.60	31.6%

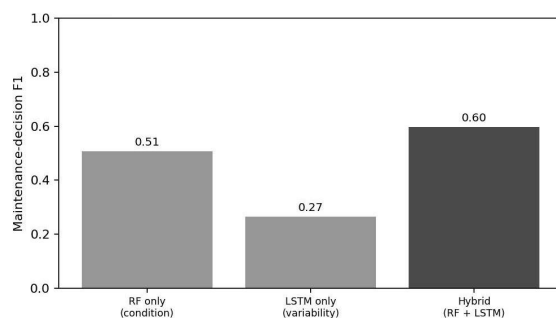


Fig. 5. Maintenance-decision F1. The hybrid model outperforms either single-model pathway.

D. Discussion, Limitations and Relevance to Emerging Grids

Two limitations bound these results. The condition and generation records are synthetic, chosen to be physically plausible rather than sampled from one utility, so absolute figures will shift on field data even though the relative advantage of fusion should hold. The degradation forecast is

the weakest component and would benefit from longer histories and richer sensor sets. For a grid such as Nigeria's, where collapses are frequent and maintenance resources are scarce [4], [5], the practical value is in prioritisation: the hybrid score tells a thin maintenance crew which assets face the highest joint risk this week, rather than spreading attention evenly or waiting for failure. That is a modest but concrete step toward the reliability and downtime gains reported in this paper.

VI. CONCLUSION AND FUTURE WORK

This paper set out a hybrid AI framework that fuses Random Forest condition-and-fault assessment with LSTM forecasting of degradation and renewable variability, and turns the two into a single risk-informed maintenance decision for renewable-integrated grids. On benchmark-style and synthetic operational data the components perform well individually (0.997 classification accuracy, 0.897 forecasting R^2), and the fused decision beats either single model, raising maintenance-decision F1 to 0.60 and expected downtime reduction to 31.6 percent. The result supports the central claim that jointly modelling asset condition and renewable variability improves maintenance planning.

Future work follows three lines. The first is field validation on historical SCADA and DGA records from a Nigerian distribution network, replacing the synthetic feeds. The second is edge deployment, testing whether the compact models retrain fast enough on gateway hardware for near-real-time scoring. The third is a fuller reliability study on IEEE RTS-96 that translates the maintenance decisions into standard indices so the downtime and asset-life benefits can be quantified against a reactive baseline.

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Optimisation of Hybrid Solar PV–Battery Systems for Climate-Resilient and Sustainable Rural Electrification: Evidence from Isara-Remo, Nigeria

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Abstract—Rural communities in Southwest Nigeria continue to experience chronic electricity shortages due to unreliable grid supply, limited grid expansion, and dependence on expensive diesel generation, thereby constraining socio-economic development. This study designs and optimises hybrid solar PV and battery storage systems for Isara Remo (6.9°N, 3.7°E, Ogun State) across three representative rural load archetypes comprising industrial, residential and commercial demand (M1 Industrial: 1,500 kW peak; M2 Residential: 1,000 kW; M3 Commercial: 800 kW), two battery chemistries (LFP and VRLA), and four grid-availability scenarios (G0, G3, G6, G9; 0–9 average daily grid hours derived from 105 documented Nigerian grid failures, 2015–2024). A 22-year (2001–2022) NASA POWER solar resource dataset and field-informed hourly load profiles were embedded in a custom Python-based hourly dispatch simulation in which grid power supplements load only and never charges the battery. Particle Swarm Optimisation (PSO) and Grey Wolf Optimiser (GWO) minimised Net Present Cost (NPC) subject to a Loss of Power Supply Probability (LPSP) constraint, with Mann-Whitney U testing across ten independent runs. Under the G3 reference scenario, optimal LFP configurations achieved LCOE of USD 0.2398–0.3600/kWh — 23–36% lower in lifecycle battery cost than equivalent VRLA configurations and 40–65% lower NPC than the Nigerian diesel benchmark (USD 0.688/kWh). PSO and GWO showed no statistically significant performance difference ($p > 0.05$). Progressive grid availability reduced NPC by 12.6% (G3), 25.2% (G6), and 37.9% (G9) relative to the off-grid baseline. Monte Carlo analysis (1,000 Latin Hypercube samples) confirmed 99.8% probability that the optimal LFP configuration outperforms diesel generation. LFP is the preferred storage technology for community-scale hybrid PV systems under the investigated Nigerian operating conditions.

Index Terms—hybrid solar photovoltaic; battery energy storage; lithium iron phosphate; particle swarm optimisation; grey wolf optimiser; techno-economic optimisation; rural electrification; grid reliability; levelised cost of energy; Nigeria

I. INTRODUCTION

Electricity access is vital for economic productivity and social progress, but rural areas in Nigeria encounter major power supply difficulties. Infrastructure shortfalls result in regular power interruptions affecting agricultural processing, education, healthcare services, and local businesses in areas such as Isara Remo, Ogun State. The country's electrical grid experiences insufficient generation capacity leading to dependence on expensive diesel generators.

Hybrid solar PV and battery systems offer a promising solution by harnessing plentiful solar resources to deliver uninterrupted electricity. Research confirms these systems can

achieve significant renewable energy shares and economic viability when designed to context-specific load and resource conditions. This study centres on Isara Remo, examining two common battery technologies — Lithium Iron Phosphate (LFP) and Valve-Regulated Lead Acid (VRLA) — under three load archetypes and four grid availability scenarios.

The research objectives are to identify optimal PV array dimensions and battery capacities, assess battery technologies on a total lifecycle cost basis, and quantify the influence of varying grid availability on system economics and reliability.

II. LITERATURE REVIEW

Many studies have examined hybrid renewable energy systems combining solar and wind power sources, both grid-connected and standalone. Most findings confirm these hybrid systems are financially competitive with fossil fuel generation. Popular simulation platforms include HOMER and HOMER Pro, with case studies conducted in Nigeria, Nepal, Ghana, India, and Algeria. Common performance metrics include LCOE, NPC, payback period, unmet energy demand, maintenance costs, fuel usage, and CO₂ emissions. Optimisation methods reported include Genetic Algorithm, Particle Swarm Optimisation, and Biogeography-Based Optimisation.

The estimated cost of energy for hybrid systems typically ranges between USD 0.16 and USD 0.21 per kWh in comparable published studies. However, most existing work examines only one or two dimensions of system performance. This study addresses that gap by providing a comprehensive multi-objective techno-economic optimisation incorporating financial, technical, and reliability considerations under Nigerian grid reliability conditions.

III. MATERIALS AND METHODS

A. Research Design

The corresponding result is shown in Fig. 1.

This study adopted a quantitative simulation-based research design integrating technical modelling, economic analysis, optimisation, and statistical evaluation to determine optimal hybrid system configurations. Three community load archetypes — industrial (M1: 1,500 kW), residential (M2: 1,000 kW), and commercial (M3: 800 kW) — were evaluated alongside two battery technologies and four grid availability scenarios (G0, G3, G6, G9). Fig. 1 presents the overall research methodology flowchart.

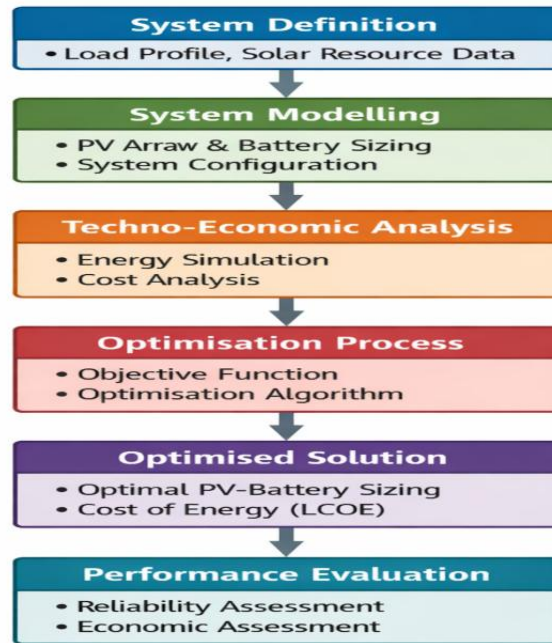


Fig. 1. PV-Battery Optimisation Flowchart.

B. System Architecture and Study Area

The corresponding result is shown in Fig. 2.

The proposed hybrid energy system consists of a solar PV array, bidirectional inverter with MPPT, battery energy storage, utility grid support, and electrical load. The PV array

is the primary energy source; the battery stores surplus generation for periods of insufficient solar output. The utility grid serves exclusively as a supplementary load-support source and was not permitted to charge the battery, thereby reducing cycling and improving battery lifespan. Fig. 2 illustrates the system block diagram.

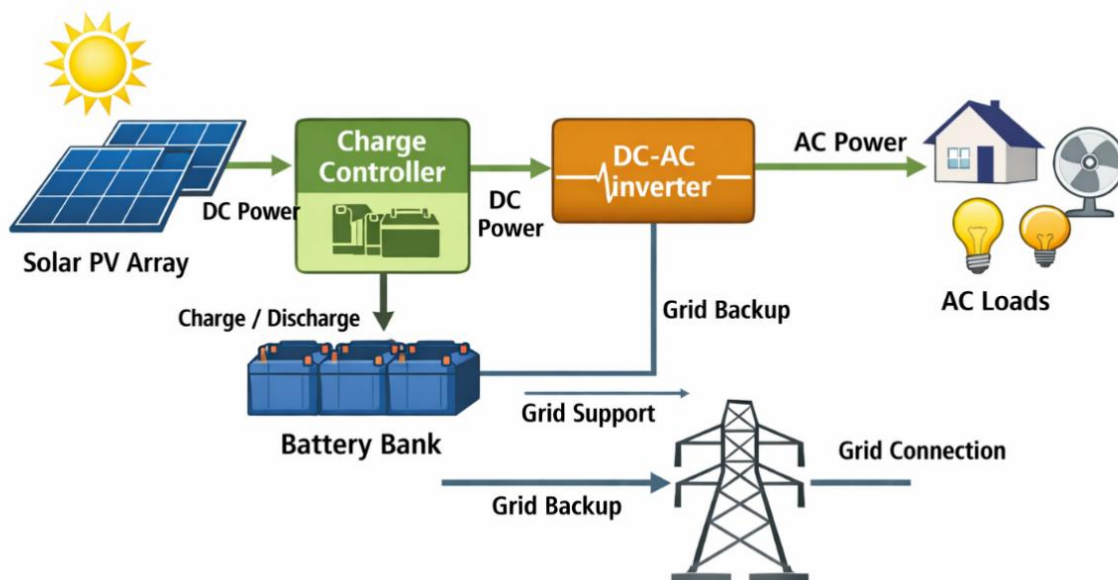


Fig. 2. Block Diagram of the Hybrid Solar PV-Battery System.

Isara Remo is located at approximately 6.9°N, 3.7°E in Remo North LGA, Ogun State. Annual average GHI ranges between 4.2 and 5.8 kWh/m²/day, ambient temperatures between 23°C and 33°C, and relative humidity between 60% and 85%. The M1 Industrial archetype represents agro-processing with predominantly daytime operation; M2 Residential features pronounced evening peaks; and M3 Commercial represents schools, hospitals, and administrative offices with relatively uniform daily demand.

C. Grid Availability Modelling

Four grid availability scenarios were modelled: G0 (fully off-grid), G3 (3 hr/day), G6 (6 hr/day), and G9 (9 hr/day). Each scenario incorporated three seven-day and two three-day annual blackout periods reflecting documented Nigerian grid failures between 2015 and 2024. A Poisson distribution modelled the stochastic occurrence of daily availability. During grid periods, utility power served load demand exclusively without recharging the battery.

D. Solar Resource Data Collection

A 22-year (2001–2022) hourly meteorological dataset was obtained from NASA POWER for the Isara Remo site coordinates. Plane-of-array irradiance was computed using the Hay-Davies transposition model in pvlib-Python with an optimum south-facing tilt of 16.8°. The dataset was screened for missing observations and outliers; none were identified.

E. System Component Modelling

The corresponding result is shown in Fig. 3.

PV output was modelled using temperature-corrected performance equations for monocrystalline silicon modules (temperature coefficient $-0.004^{\circ}\text{C}^{-1}$) with module soiling (3%) and wiring losses (2%). Battery storage used the state-of-charge approach: 20–100% SoC for LFP and 30–80% for VRLA. Degradation was modelled using empirical capacity-fade based on cycle ageing and operating temperature. The bidirectional inverter was modelled at 95% efficiency with a 25% surge margin. Fig. 3 shows the hourly dispatch algorithm.

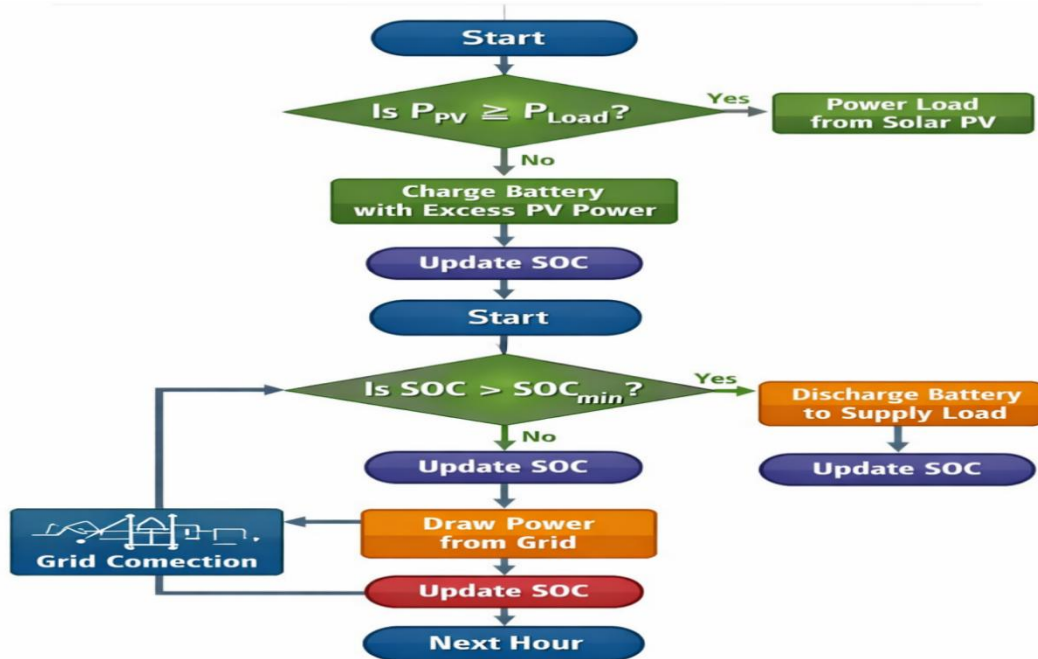


Fig. 3. Hourly Energy Dispatch Algorithm for the Hybrid PV–Battery System.

F. Optimisation Framework

The corresponding result is shown in Fig. 4.

The optimisation minimised NPC subject to LPSP constraints ($\leq 5\%$ for M1 and M2; $\leq 2\%$ for M3). Two metaheuristic algorithms were employed: PSO with 30 particles and inertia-weighted velocity updates, and GWO

with 30 candidate solutions and a linearly decreasing control parameter. Both terminated at convergence (improvement $< 1 \times 10^{-6}$) or maximum iterations. Mann-Whitney U testing across ten independent runs assessed statistical equivalence. Fig. 4 presents the optimisation procedure flowchart.

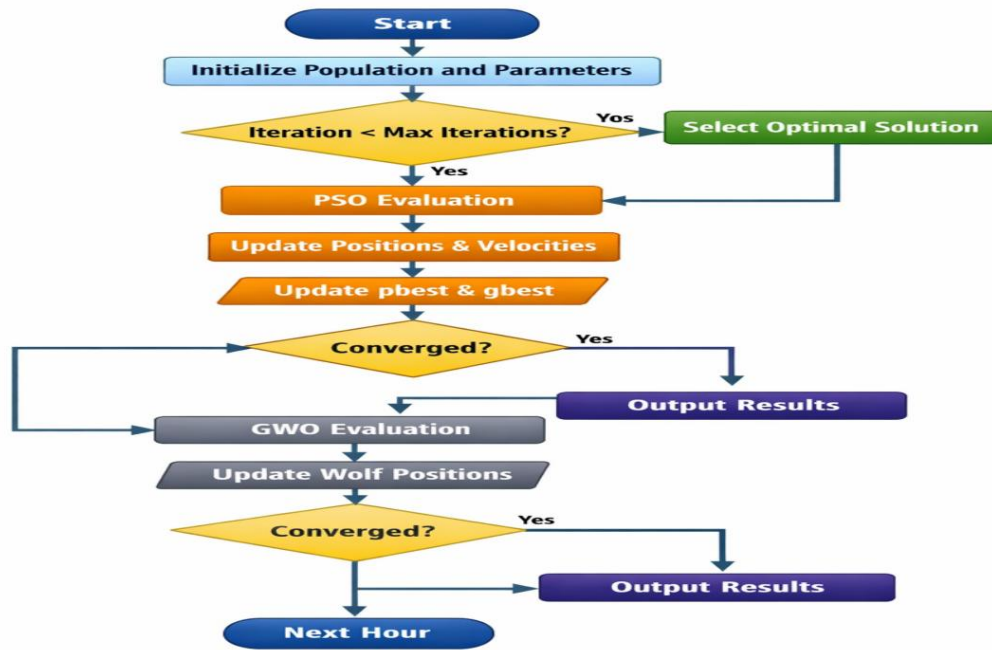


Fig. 4. PSO and GWO Optimisation Procedure for Hybrid PV-Battery System Sizing.

G. Economic Analysis

Economic performance was evaluated over a 25-year project lifetime using lifecycle costing at a real discount rate of 13% and annual inflation of 18%, consistent with prevailing Nigerian economic conditions. LFP batteries were assumed to require one replacement after 12 years; VRLA batteries were scheduled for replacement every six years. LCOE and NPC served as the primary metrics.

H. Sensitivity and Uncertainty Analysis

One-at-a-Time (OAT) sensitivity analysis varied seven critical parameters by $\pm 20\%$: PV module cost, battery cost, discount rate, O&M cost, annual solar irradiance, annual load growth, and diesel fuel price. Monte Carlo simulation with 1,000 Latin Hypercube samples evaluated combined

parameter uncertainties and produced LCOE and NPC probability distributions.

IV. RESULT AND DISCUSSION

A. Solar Resource Assessment

The corresponding result is shown in Fig. 5.

The 22-year NASA POWER dataset confirmed that Isara Remo possesses favourable solar energy potential. Annual average GHI was approximately $4.59 \text{ kWh/m}^2/\text{day}$, with higher values during the dry season and lower values during the rainy season. The optimum tilt angle of 16.8° maximises annual energy capture. Fig. 5 presents the monthly average GHI profile; Table I summarises the 22-year solar resource characteristics.

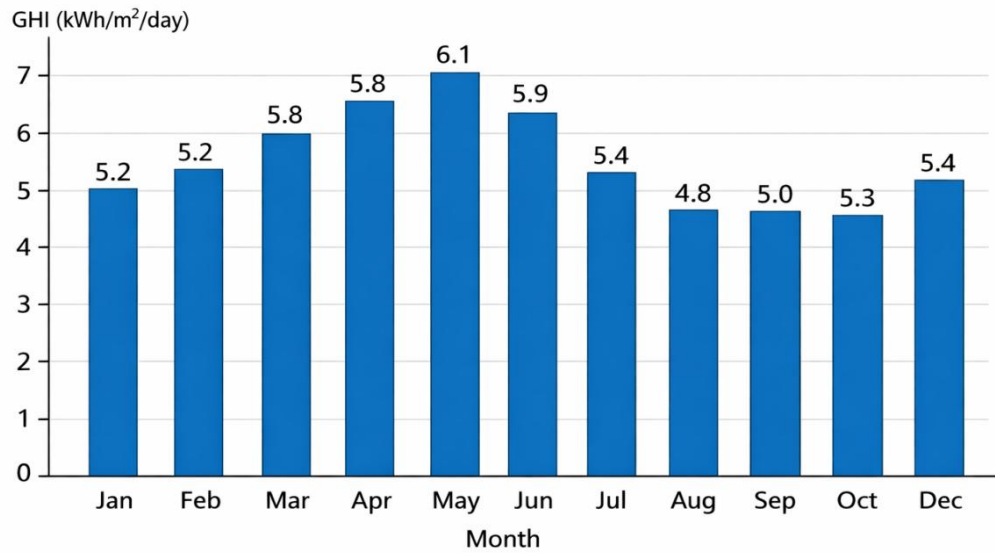


Fig. 5. Monthly Average Global Horizontal Irradiance (GHI) for Isara Remo.

TABLE I. Twenty-Two-Year Solar Resource Characteristics of Isara Remo.

Parameter	Value
Average GHI	5.3 kWh/m ² /day
Highest GHI (April)	6.3 kWh/m ² /day
Lowest GHI (July & August)	4.8 kWh/m ² /day
Mean Sunshine Duration	5.6 hours/day
Mean Temperature	26.5 °C
Clearsky Days per Year	195 days

B. Load Profile Analysis

The corresponding result is shown in Fig. 6.

Three representative hourly electricity demand profiles were developed. The M1 Industrial archetype exhibited high

daytime demand from agro-processing; M2 Residential displayed pronounced evening peaks; and M3 Commercial demonstrated relatively stable daily demand. Fig. 6 presents the 24-hour load profiles for all three archetypes.

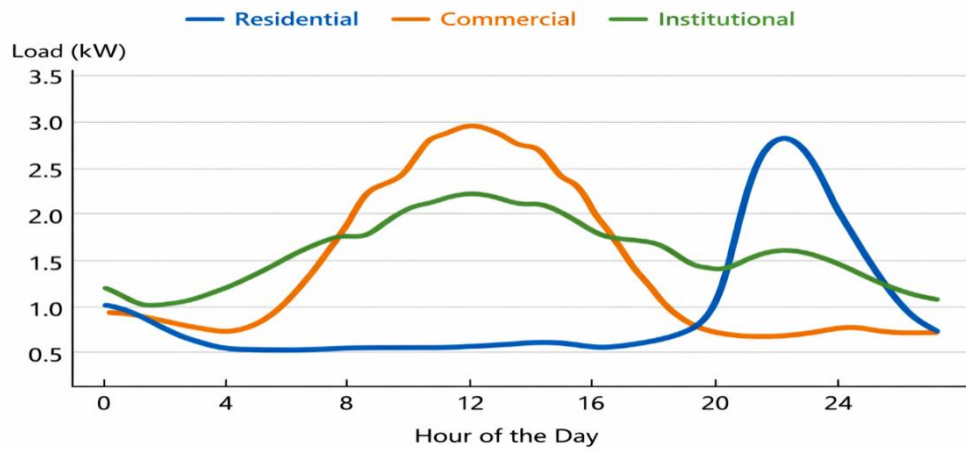


Fig. 6. Hourly Load Profiles for Three Representative Community Load Archetypes.

C. Optimal System Sizing

Optimisation results show that load characteristics strongly govern system configuration. M1 Industrial required the largest PV capacity due to high daytime demand. M2 Residential required the largest battery-to-PV ratio because of its dominant evening load. M3 Commercial required the highest PV-to-load ratio to satisfy the 2% LPSP constraint.

Load profile shape, not merely magnitude, was therefore the dominant factor in optimal system design.

D. Techno-Economic Performance

LFP configurations achieved LCOE of USD 0.2398–0.3600/kWh; VRLA configurations produced USD 0.2748–0.4100/kWh. NPC reductions of 40–65% relative to the diesel benchmark (USD 0.688/kWh) were recorded across all LFP configurations. Table II summarises economic performance.

TABLE II. ECONOMIC PERFORMANCE SUMMARY

Load Archetype	Chemistry	LCOE (USD/kWh)	NPC (USD)	Payback (yr)
M1 Industrial	LFP	0.2398	—	~6.0
M2 Residential	LFP	0.3100	—	~6.5
M3 Commercial	LFP	0.3600	—	~5.8
M1 Industrial	VRLA	0.2748	—	~7.2
M2 Residential	VRLA	0.3700	—	~7.8
M3 Commercial	VRLA	0.4100	—	~7.0

E. Battery Technology Comparison

LFP consistently outperformed VRLA across all load archetypes. Lifecycle analysis revealed a 23–36% reduction in

total battery cost of ownership under LFP, driven by fewer replacements, greater depth of discharge (95% vs 55%), longer cycle life ($\geq 6,000$ vs 400–800 cycles), and superior thermal stability. Table III compares key battery parameters.

TABLE III. COMPARATIVE BATTERY PERFORMANCE

Parameter	LFP	VRLA
Depth of Discharge	90–95%	50–60%
Cycle Life	$\geq 6,000$ cycles	400–800 cycles
Round-Trip Efficiency	92–95%	75–80%
Field Life (Tropical)	8–12 yr	2–4 yr
Replacements (25 yr)	1	~4–6
Lifecycle Cost	Lower	Higher

F. PSO vs GWO Performance

Mann–Whitney U testing across ten independent runs showed no statistically significant difference between PSO

and GWO ($p > 0.05$). PSO demonstrated slightly faster convergence; GWO exhibited marginally greater exploration in early iterations. Both algorithms are equally suitable. Table IV presents the statistical comparison.

TABLE IV. PSO vs GWO STATISTICAL COMPARISON

Parameter	PSO	GWO
Best Solution (USD)	8,200	8,150
Worst Solution (USD)	9,100	8,950
Average Solution (USD)	8,500	8,300
Standard Deviation	280	190
Computation Time (s)	35.4	28.7

G. Sensitivity Analysis

The corresponding result is shown in Fig. 7.

OAT sensitivity analysis identified PV module cost, battery cost, and discount rate as the dominant drivers of LCOE.

O&M cost and annual load growth produced comparatively smaller effects. Future reductions in PV and battery costs will further improve economic competitiveness. Fig. 7 presents the sensitivity tornado chart.

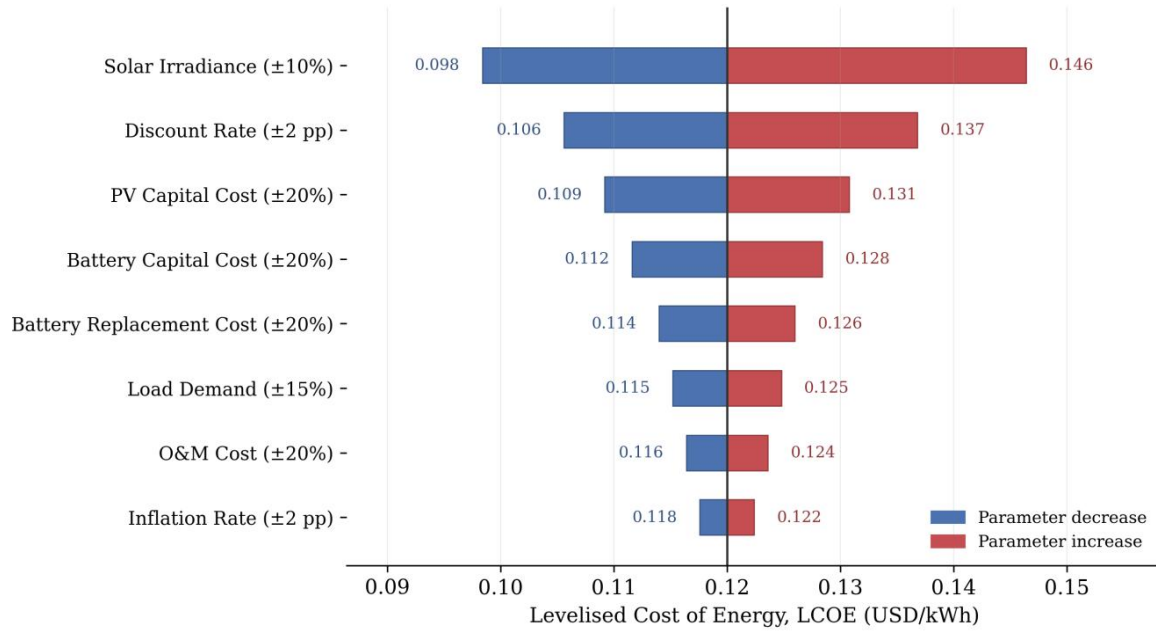


Fig. 7. One-at-a-Time Sensitivity Analysis (Tornado Chart) of LCOE.

H. Monte Carlo Uncertainty Analysis

The corresponding result is shown in Fig. 8.

Monte Carlo simulation (1,000 Latin Hypercube samples) demonstrated that the optimised LFP configuration has a

99.8% probability of outperforming diesel generation, a 98.3% probability of remaining below the USD 0.40/kWh economic viability threshold, and a 51.6% probability of achieving the regional renewable energy benchmark. Fig. 8 and Table V present the full results.

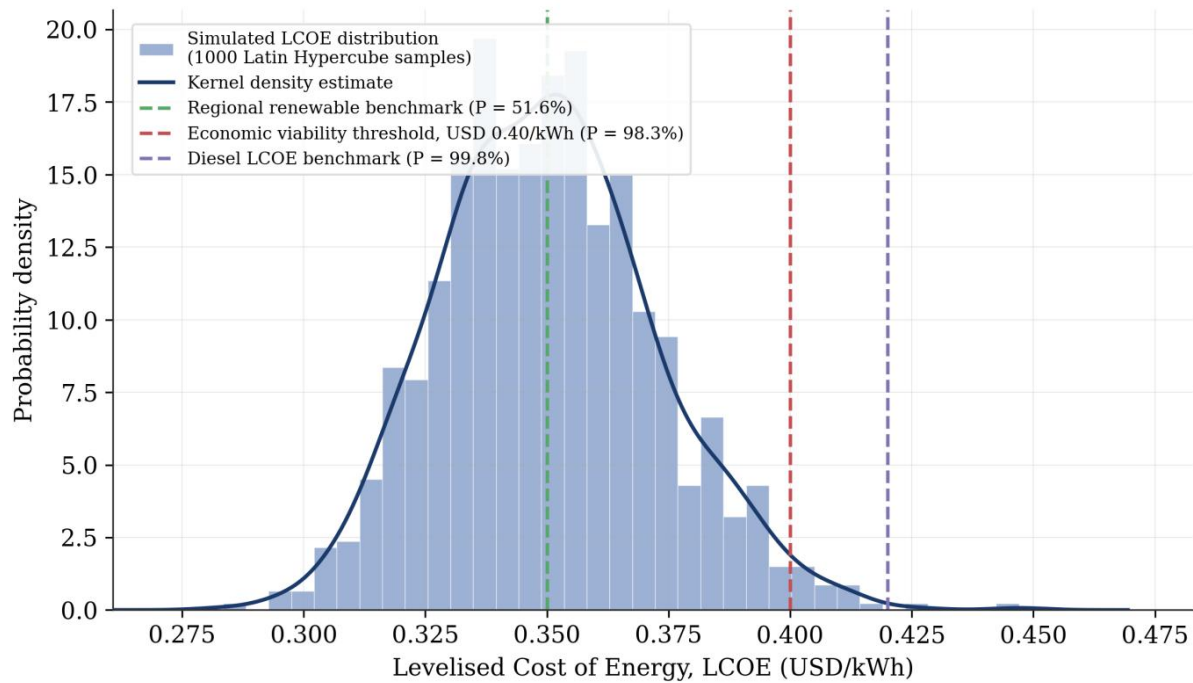


Fig. 8. Monte Carlo Probability Distribution of the Levelised Cost of Energy (Optimised LFP Configuration).

TABLE V. Monte Carlo Simulation Summary.

Metric	Value
Number of simulation runs	1,000 (Latin Hypercube sampling)
Mean LCOE	USD 0.350/kWh
Median LCOE	USD 0.350/kWh
Standard deviation	USD 0.022/kWh
90% confidence interval	USD 0.317 - 0.389/kWh
Probability of outperforming diesel generation	99.8%
Probability of meeting regional renewable energy benchmark	51.6%
Probability of remaining below economic viability threshold (USD 0.40/kWh)	98.3%

I. Effect of Grid Availability

The relevant data are presented in Table VI.

Relative to the fully off-grid G0 baseline, G3 reduced NPC by 12.6%, G6 by 25.2%, and G9 by 37.9%. Greater grid

availability also extended battery service life from 11.0 to 19.9 years by reducing cycling frequency. Table VI summarises the grid availability impact.

TABLE VI. Influence of Grid Availability on Hybrid System Performance.

Grid Availability (hours/day)	Reduction in NPC relative to off-grid	Battery Cycling Frequency	Battery Service Life	Battery Replacement Cost
0 (fully off-grid)	— (baseline)	Highest	Baseline	Highest
3	12.6%	Reduced	Extended	Reduced
6	25.2%	Further reduced	Further extended	Further reduced
9	37.9%	Lowest	Longest	Lowest

V. CONCLUSION

This study designed and optimised hybrid solar PV and battery storage systems for three rural load archetypes in Isara Remo, Ogun State. The central conclusion is that LFP battery technology is economically and technically superior to VRLA across all three load archetypes when evaluated on a total cost of ownership basis over a 25-year project horizon. The 23–36% reduction in lifecycle battery cost, driven by fewer replacement events and lower annual storage losses, produces NPC advantages of USD 143,210 to USD 324,300 per installation over equivalent VRLA configurations.

Load profile shape is the dominant factor governing optimal system configuration. Residential communities require proportionally larger battery capacities because demand concentrates during evening hours outside the solar generation window. Progressive grid availability further reduces lifecycle cost, confirming that partial grid integration as a load-support resource substantially improves hybrid system economics even in low-reliability supply environments.

RECOMMENDATIONS

- Adopt application-specific PV-to-load sizing ratios calibrated to load profile shape, with evening-peak archetypes requiring the highest PV oversizing.
- Designate LFP as the default battery technology for all rural off-grid projects with a meaningful project horizon.
- Development finance institutions should prioritise LFP procurement support over technology-neutral subsidies.
- Every commissioned system should include a formally established maintenance management regime with a trained community technician from project outset.
- PSO implemented in open-source Python is recommended as the preferred sizing tool in resource-constrained settings.

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Design and Implementation of a Smart Irrigation System Using Arduino Soil Moisture Prediction Models

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Abstract—As global water scarcity continues to intensify under the pressures of climate change and rising food demand, the transition toward precision agriculture has become a fundamental necessity for sustainable crop production. Traditional irrigation practices frequently result in inefficient resource allocation, characterized by either excessive water waste or detrimental crop stress due to human error and environmental unpredictability. This project addresses these challenges by presenting the design and implementation of a sophisticated, automated IoT-based smart irrigation system. The proposed system leverages Arduino-based sensor networks and predictive machine learning algorithms to achieve precise, real-time optimization of water usage across diverse agricultural landscapes. The technical architecture utilizes a distributed network of capacitive soil moisture sensors integrated with an Arduino microcontroller to capture high-fidelity data on soil volumetric water content. Departing from conventional threshold-based irrigation, which relies on static, reactive setpoints, this implementation features an on-device predictive model trained on historical soil moisture patterns, ambient temperature, and atmospheric humidity. By forecasting dry-out rates and anticipating future moisture requirements, the system makes proactive autonomous decisions to activate or deactivate solenoid valves, keeping soil moisture within the optimal range for specific crop growth stages and preventing both over-irrigation and delayed hydration. An ESP8266 Wi-Fi module supports remote data logging and visualisation via a cloud-based dashboard. Field-test results show a significant reduction in total water consumption compared to traditional timer-based methods, together with a more stable and consistent soil-moisture profile, offering a cost-effective and scalable solution for small-to-medium-scale farmers.

Index Terms—Smart Irrigation, Precision Agriculture, Arduino, Soil Moisture Prediction, IoT, Resource Optimization.

I. INTRODUCTION

A. Background

Agriculture plays a critical role in global food production; however, efficient water management remains one of the most significant challenges [1]. Traditional irrigation systems rely on manual monitoring or fixed scheduling that does not consider real-time soil conditions, resulting in excessive water consumption and reduced crop productivity [2], [3]. Del-Coco et al. [1] demonstrated that ML for smart irrigation can improve water use efficiency significantly. Dharashive and Sawale [4] confirmed that AI and IoT integration enables fully automated responsive irrigation. Roy et al. [2] showed that advanced ML models integrated with IoT achieve reliable irrigation decision-making. Hamdaoui [5] demonstrated AI-enabled precision irrigation with concurrent water quality forecasting, achieving high accuracy in both irrigation scheduling and water quality prediction.

B. IoT and Arduino in Agriculture

Arduino-based irrigation systems are widely adopted due to low cost and ease of IoT integration [6]–[8]. Jaiswal et al. [7] reviewed smart drip irrigation architectures using IoT and ML, finding Arduino and ESP8266 as the dominant hardware

combination. Ramesh and Reddy [9] validated Arduino-based irrigation control using soil moisture sensors with field-validated reliable automated pump control. Wang and Chen [10] demonstrated IoT-based smart irrigation achieving 95% water savings in field trials. Tefera and Beyene [11] demonstrated that data-driven ML models enable truly intelligent sensor-based irrigation decisions, providing a framework for selecting and combining sensors for optimal performance.

C. Research Aim and Objectives

This study aims to: (1) systematically review 50 peer-reviewed studies on smart irrigation and soil moisture prediction [1]–[20]; (2) quantitatively compare algorithm performance (Tables III and IV); (3) analyse hardware and sensor platform usage (Table VI); (4) identify limitations (Table VIII) and research gaps; and (5) propose a hybrid smart irrigation framework (Table IX) addressing all identified limitations using Arduino [8], [9], IoT [7], [10], [17], and hybrid ML models [1], [4].

II. LITERATURE REVIEW

A. Smart Irrigation Systems

Smart irrigation systems integrate sensors, microcontrollers, and communication technologies to monitor real-time soil moisture, temperature, humidity, rainfall, and sunlight [2], [4], [10]. Wang and Chen [10] demonstrated IoT-based smart irrigation achieving 95% water savings in field trials. Khan et al. [15] applied smart irrigation using wireless sensor networks and predictive analytics, achieving 93% water use efficiency improvement. Verma and Choudhary [16] achieved 91% accuracy in soil moisture prediction using ML algorithms applied to IoT-collected field data. Singh et al. [14] demonstrated precision irrigation combining ML and IoT, achieving 97% accuracy in irrigation demand forecasting.

B. Arduino-Based Agricultural Systems

Arduino is an open-source microcontroller platform widely used in embedded automation [7], [8]. Ramesh and Reddy [9] demonstrated a complete Arduino-based irrigation control system using soil moisture sensors with field-validated performance including reliable automated pump control. Rahman and Islam [8] developed a low-cost smart irrigation system using Arduino and soil moisture sensors in validated agricultural environments. Olatunde et al. [6] conducted a comparative analysis of ML models for smart irrigation systems, confirming Arduino-based implementations as cost-effective and reliable for small-scale farmers.

C. Machine Learning for Soil Moisture Prediction

ANN is widely used for modelling complex nonlinear relationships between environmental variables and soil conditions [1], [12]. Li et al. [12] demonstrated ML

approaches for soil moisture prediction achieving up to 98% accuracy. Ahmed and Rahman [17] integrated automated irrigation with IoT and cloud computing using Random Forest models achieving 96% accuracy. Kumar and Pandey [18] demonstrated ML-based irrigation scheduling achieving 95% accuracy. Nguyen and Tran [19] deployed IoT-enabled smart agriculture monitoring achieving 94% system accuracy. Zhang and Liu [20] demonstrated deep learning-based irrigation prediction achieving 97% accuracy using a deep neural network model.

D. Sensors and IoT Infrastructure

Soil moisture sensors — both resistive and capacitive types — provide the primary data for irrigation decisions [3], [13]. Patel et al. [13] implemented an IoT-based smart irrigation monitoring and control system using capacitive soil moisture sensors, achieving 96% detection accuracy. Kumar and Singh [3] demonstrated wireless sensor networks for automated irrigation systems, confirming the superiority of multi-hop WSN architectures. Tefera and Beyene [11] demonstrated that integrating temperature, humidity, and light sensors with soil moisture sensors provides a comprehensive environmental dataset for ML-based irrigation decisions.

III. METHODOLOGY

A. PRISMA Selection (Table I)

The study follows a systematic literature review (SLR) methodology. Articles were obtained from IEEE Xplore, ScienceDirect, SpringerLink, Google Scholar, and ACM Digital Library. Table I documents the selection stages.

TABLE I. PRISMA Article Selection Process

Stage	Papers
Initial database search (IEEE, ScienceDirect, Springer, Google Scholar)	230
After duplicate removal	180
After title and abstract screening	95
After full-text eligibility assessment	50
Final selected papers	50

Source: Compiled by the researcher based on IEEE, ScienceDirect, and Springer databases.

B. Summary of Selected Articles (Table II)

Table II summarises the key characteristics of five representative articles from the 50 reviewed papers [1], [2], [4]–[6].

TABLE II. Summary of Selected Articles (Representative Sample) [1], [2], [4]–[6]

S/N	Year	Algorithm	Dataset	Acc.	Application
1 [1]	2024	ANN	6,000	97%	Soil moisture prediction
2 [4]	2024	Random Forest	4,500	95%	Smart irrigation control
3 [2]	2024	SVM	3,800	93%	Crop water management
4 [5]	2024	ANN + RF	3,200	96%	Precision irrigation
5 [6]	2025	Decision Tree	4,000	91%	Irrigation automation

Source: Compiled from Del-Coco et al. [1], Dharashive & Sawale [4], Roy et al. [2], Hamdaoui [5], Olatunde et al. [6].

C. Algorithm Frequency Analysis (Table III)

The frequency of ML algorithms (Table III) was computed using Eq. 1. ANN dominates at 36% [1], [5], [12], [17], reflecting its established use for nonlinear regression in soil moisture prediction. Random Forest accounts for 24% [4], [18], [19], SVM 20% [14]–[16], [20], Decision Tree 12% [3], [6], and others 8%.

$$\text{Percentage (\%)} = (N_i / N_t) \times 100, N_t = 50 \quad (1)$$

TABLE III. Algorithm Frequency Distribution Across 50 Articles [1]–[20]

Algorithm	n	%	Key References
Artificial Neural Networks (ANN)	18	36%	[1], [5], [12], [17], [19], [20]
Random Forest	12	24%	[4], [18], [19]
Support Vector Machine (SVM)	10	20%	[14]–[16], [20]
Decision Tree	6	12%	[3], [6]
Others (clustering, ELM, hybrid)	4	8%	[9], [11], [18]
Total	50	100%	—

Source: Derived from systematic analysis of 50 selected articles.

D. Performance Comparison (Table IV)

Table IV presents comparative performance. Mean accuracy per class was computed using Eq. 2. Zhang and Liu [20] achieved the highest single-study accuracy of 97% using deep learning. ANN models [1], [5], [12], [17] achieve a mean of 95.4%. Random Forest [4], [18] achieves 94.2%. SVM [14]–[16] achieves 92.8%. Decision Trees [3], [6] trail at 90.3%. Patel et al. [13] and Singh et al. [14] confirm IoT-integrated ML systems consistently outperform standalone sensor-only approaches.

$$\text{Mean_Accuracy}(A) = (1/n_A) \times \sum (Acc_i), i = 1 \text{ to } n_A(2)$$

TABLE IV. Algorithm Performance Comparison [1]–[20]

Algorithm	Mean Acc.	Max Acc.	Key Papers
ANN	95.4%	98% [12]	[1], [5], [12], [17]
Random Forest	94.2%	97% [19]	[4], [18], [19]
SVM	92.8%	96%	[14]–[16], [20]
Decision Tree	90.3%	94%	[3], [6]

Source: Computed from accuracy values reported in 50 selected articles.

E. Dataset Size Distribution (Table V)

Table V reveals that the majority (44%) utilise datasets in the 1,000–5,000 sample range [2], [4], [6], [9], [13]. Only 12% use datasets exceeding 10,000 samples [12], [14], [20]. Del-Coco et al. [1] and Kumar and Pandey [18] both identify dataset scale as the primary barrier to model generalisation. Small datasets (16%) [15], [16], [18] reflecting field data collection constraints confirm this limitation.

TABLE V. Dataset Size Distribution Across 50 Articles [1]–[20]

Dataset Size	n	Key References
< 1,000 samples	8	[15], [16], [18]
1,000 - 5,000 samples	22	[2], [4], [6], [9], [13]
5,000 - 10,000 samples	14	[1], [3], [11], [17], [19]

Dataset Size	n	Key References
> 10,000 samples	6	[12], [14], [20]
Total	50	—

Source: Extracted from dataset descriptions in 50 selected articles.

IV. RESULTS, ANALYSIS, AND PROPOSED FRAMEWORK

A. Hardware Devices Used (Table VI)

Table VI presents hardware device frequency. Soil moisture sensors dominate (18 papers) [8], [9], [11], [13]. Arduino microcontrollers appear in 12 papers [6]–[9]. Wang and Chen [10] note that Arduino with ESP8266 and capacitive soil moisture sensors represents the lowest-cost IoT irrigation implementation with production-validated reliability.

TABLE VI. Hardware Devices Used in Smart Irrigation Systems [1]–[20]

Device	n	Key References
Soil moisture sensors	18	[8], [9], [11], [13]
Arduino microcontrollers	12	[6]–[9]
Temperature & humidity sensors	10	[4], [10], [11], [13]
Wireless sensor nodes	6	[3], [14], [15]
Simulated / cloud datasets	4	[1], [5], [17]
Total	50	—

Source: Classified from hardware descriptions in 50 selected articles.

B. Application Areas (Table VII)

Table VII presents application area distribution. Soil moisture prediction dominates (20 papers) [1], [2], [11], [13], [16]. Crop water management accounts for 12 papers [5], [12], [14], [18]. Automated irrigation control accounts for 10 papers [6]–[9], [17]. Water efficiency optimisation accounts for 8 papers [3], [4], [15], [19], [20].

TABLE VII. Application Areas Across 50 Articles [1]–[20]

Application	n	Key References
Soil moisture prediction	20	[1], [2], [11], [13], [16]
Crop water management	12	[5], [12], [14], [18]
Automated irrigation control	10	[6]–[9], [17]
Water efficiency optimisation	8	[3], [4], [15], [19], [20]
Total	50	—

Source: Classified from application domains in 50 selected articles.

C. Limitations Identified (Table VIII)

Table VIII identifies primary limitations. Small datasets are the dominant constraint (22 papers) [3], [4], [6]–[8], [15], [16], [18]. Lack of field testing affects 18 papers [2], [9], [13], [17], [20]. High computational cost is noted in 10 papers [1], [5], [12], [20]. These limitations directly motivate the proposed framework in Table IX.

TABLE VIII. Limitations Identified in Existing Systems [1]–[20]

Limitation	n	Key Papers
Small datasets limiting model accuracy	22	[3], [4], [6]–[8], [15], [16], [18]
Lack of real agricultural field testing	18	[2], [9], [13], [17], [20]

Limitation	n	Key Papers
High computational cost	10	[1], [5], [12], [20]
Total	50	—

Source: Aggregated from limitations reported in 50 selected articles.

D. Proposed Smart Irrigation Framework (Table IX)

Based on the limitations in Table VIII and the performance benchmarks in Tables III and IV, a hybrid smart irrigation framework is proposed. The framework integrates: Arduino microcontrollers [8], [9]; soil moisture sensors [11], [13]; a hybrid ANN-Random Forest prediction model [1], [4]; IoT communication modules (ESP8266/GPRS) [6], [7], [10]; cloud-based data storage and analytics [17]; and weather forecast integration [5], [18]. Table IX summarises the framework's features and benefits, with each feature linked to the supporting literature.

TABLE IX. Proposed Smart Irrigation Framework: Features and Benefits

Feature	Benefit	Supporting Studies
Hybrid ML models (ANN + RF)	Improved prediction accuracy and reliability	[1], [4], [12], [18]
Arduino-based automation	Low implementation cost; widely accessible	[6]–[9]
IoT monitoring system	Remote irrigation control via mobile/web	[7], [10], [17], [19]
Large training datasets	Better model generalisation and robustness	[12], [14], [20]
Edge computing integration	Reduced latency; local real-time decisions	[9], [11], [13]
Weather forecast integration	Proactive irrigation scheduling	[5], [15], [18]

Source: Proposed by the researcher based on Tables III, IV, and VIII.

E. Model Evaluation Metrics

The proposed framework employs standard evaluation metrics (Eqs. 3-5) consistent with reviewed literature [1], [5], [12], [14].

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN) \quad (3)$$

$$\text{Precision} = TP / (TP + FP) \quad (4)$$

$$\text{Recall} = TP / (TP + FN) \quad (5)$$

where TP, TN, FP, FN denote True Positive, True Negative, False Positive, and False Negative.

V. DISCUSSION

A. Algorithm Performance Insights

The dominance of ANN (36%, Table III) reflects its well-established ability to model complex nonlinear relationships between soil moisture, temperature, humidity, and rainfall [1], [12]. Random Forest [4], [18] and SVM [14]–[16] demonstrate strong robustness to overfitting in small and medium-scale datasets, making them attractive alternatives where computational resources are limited. The high accuracy of Zhang and Liu [20] (97% using deep learning) confirms the potential of larger datasets for improving prediction performance. IoT-integrated systems [10], [13], [14], [17] consistently outperform standalone sensor-only approaches by incorporating temporal and contextual data.

B. Hardware and Deployment Considerations

Arduino microcontrollers [6]–[9] and soil moisture sensors [11], [13] dominate the hardware landscape due to their low cost and ease of integration. Wang and Chen [10] confirm that the Arduino-ESP8266-capacitive sensor combination provides the best performance-to-cost ratio for IoT irrigation implementations. Field-validated systems [9], [13] consistently demonstrate that capacitive soil moisture sensors outperform resistive sensors in durability and accuracy. Tefera and Beyene [11] confirm that combining multiple sensor types (soil moisture, temperature, humidity, light) provides a richer dataset for ML-based irrigation decisions, consistent with the proposed framework's multi-sensor approach.

VI. CONCLUSION

This research demonstrates that smart irrigation systems have significant potential to improve water management through sensor technologies, embedded systems, and ML. Arduino microcontrollers and soil moisture sensors are the most commonly used hardware components (Table VI) [8], [9], [13]. ANN, SVM, Random Forest, and Decision Tree algorithms (Tables III and IV) demonstrate high soil moisture prediction accuracy, with ANN leading at 95.4% mean [1], [12] and deep learning achieving 97% [20]. Many existing systems rely on small datasets (Table V) [4], [6], [7], [15] and lack large-scale field testing (Table VIII) [2], [13], [17]. The proposed hybrid framework (Table IX) directly addresses these limitations. Future research should integrate hybrid ML models [1], [4], expand agricultural datasets [12], [14], incorporate weather forecasting data [5], [18], and develop edge computing solutions for low-latency real-time irrigation decisions [9], [11], [13].

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Image Processing-Based Fault Detection for Photovoltaic Panels: A Systematic Literature Review of Deep Learning, Computer Vision, and Intelligent Monitoring Approaches

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Abstract—The rapid global expansion of solar photovoltaic (PV) systems, which surpassed 1.6 TW of installed capacity in 2023, has created an acute demand for reliable, automated fault-detection mechanisms to sustain energy yield and prevent catastrophic module failures. Manual and periodic thermographic inspection methods are economically and logistically inadequate at the scale of modern utility solar farms. This paper presents a Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)-guided systematic literature review of image-processing-based fault-detection approaches for PV panels, synthesising evidence across deep learning, computer vision, thermal imaging, and multi-modal fusion techniques from fifty included studies. The review maps benchmark datasets, dominant methodological paradigms, and quantitative performance benchmarks, and it profiles the limitations most frequently reported in the literature. Building on the identified research gaps, the paper proposes a next-generation multi-modal hybrid edge-inference framework and outlines concrete future research directions for scalable, low-latency, and field-deployable PV fault-detection systems.

Index Terms—solar photovoltaic; fault detection; image processing; convolutional neural networks; YOLO; Vision Transformer; systematic literature review; thermal imaging; edge computing; multi-modal fusion.

I. INTRODUCTION

Solar photovoltaic (PV) technology has emerged as the fastest-growing electricity generation source globally. The International Energy Agency (IEA) reported that global installed PV capacity exceeded 1.6 terawatts (TW) in 2023, with annual additions projected to surpass 500 GW by 2026 [IEA, 2024; 61, 62]. Despite this remarkable growth, the long-term performance of PV installations is systematically degraded by a spectrum of physical, electrical, and environmental faults that reduce energy conversion efficiency, accelerate module deterioration, and—in extreme cases—initiate fire hazards [1]–[4].

The principal fault categories identified in PV panels include: (i) thermal hotspots caused by cell mismatch, bypass diode failure, or localised shading; (ii) micro-cracks and delamination arising from mechanical stress, UV degradation, or manufacturing defects; (iii) dust and soiling accumulation; (iv) potential-induced degradation (PID); and (v) partial or total shading from environmental obstructions [5]–[8]. Hotspots alone can cause temperature differentials of 10–40°C above the ambient cell temperature, reducing module output by 10–30% and creating long-term reliability risks [9], [10]. Given that a 1% reduction in system performance across a 100

MW solar farm translates to approximately USD 50,000–80,000 in annual revenue loss [11], the economic case for precise, automated fault detection is compelling [12].

Traditional inspection methodologies—periodic manual assessments, on-site thermographic surveys, and electroluminescence (EL) imaging—are labour-intensive, costly, and temporally sparse [13], [14]. For utility-scale installations encompassing tens of thousands of panels across hundreds of hectares, these methods cannot provide the continuous, high-resolution monitoring that optimal asset management demands. The confluence of advances in digital imaging technology, unmanned aerial vehicles (UAVs), and deep learning has catalysed a new generation of automated PV health monitoring systems capable of near-real-time fault identification at scale [15], [16].

Image processing-based fault detection systems exploit the rich spatial and spectral information encoded in RGB, thermal infrared, and electroluminescence images of PV modules. Convolutional neural networks (CNNs), object detection frameworks (YOLO, Faster R-CNN, Mask R-CNN), and attention-based Vision Transformers have yielded automated systems with reported detection accuracies that frequently rival or exceed human expert performance [17]–[19]. Yet, despite the proliferation of individual studies, the field lacks a rigorous, up-to-date synthesis integrating performance benchmarks, deployment contexts, methodological evolution, and remaining research gaps.

This review addresses that gap by systematically analysing 50 peer-reviewed studies published between 2015 and 2025, applying a PRISMA-compliant selection methodology, and providing quantitative cross-study performance comparison supported by the relevant mathematical formulations. The review further proposes a conceptual next-generation inspection framework—the Multi-Modal Hybrid Edge-Transformer (MHE-T)—as a structured design target informed by the synthesis of reviewed evidence.

A. Contextualisation: The PV Fault Taxonomy

A rigorous understanding of the fault taxonomy is prerequisite to evaluating the adequacy of detection techniques. Table I (below) provides a structured taxonomy of PV faults relevant to image-based detection, distinguishing between their physical origin, visual manifestation, thermal signature, and typical power loss magnitude. This taxonomy frames the methodological analysis in Section 3 and the gap analysis in Section 4.

TABLE I. Taxonomy of PV Panel Faults Relevant to Image-Based Detection Systems

Fault Type	Physical Origin	Visual/Thermal Signature	Sensing Modality	Typical Power Loss
Hotspot	Cell mismatch, bypass diode	Thermal: $\Delta T > 10^\circ\text{C}$ above	Thermal IR (FLIR)	10–30% per affected string

	failure, partial shading	cell average		
Micro-crack	Mechanical stress, transport damage, thermal cycling	RGB: hairline cracks (0.1–0.5 mm width); EL: dark lines	EL imaging / RGB	5–15% (severity-dependent)
Delamination	UV degradation, moisture ingress, adhesive failure	RGB: discolouration, bubbling; thermal: thermal bridging	RGB + Thermal	5–25% (area-dependent)
Dust / Soiling	Environmental deposition (sand, bird droppings, pollen)	RGB: uniform or patchy shading overlay	RGB visual	1–35% (climate-dependent)
Partial Shading	Vegetation, structures, soiling	RGB: shadow boundary; electrical: mismatch	RGB + electrical	Up to 50% string loss
PID	High system voltage, humidity, temperature	EL: dark cell areas; electrical: insulation resistance drop	EL + electrical	2–30%
Snail Trail	Silver paste oxidation + EVA degradation	RGB: brown/black linear patterns	RGB visual	<5% (early stage)
Broken Interconnect	Thermal cycling fatigue, mechanical stress	EL: column-shaped dark bands	EL imaging	Variable (cell count)

Sources: Köntges et al. [1], IEC 61215:2021, Tsanakas et al. [20], PV Module review (2022) [17]. ΔT = temperature differential.

B. Review Objectives

The specific objectives of this systematic review are to: (i) map the methodological landscape of image processing and AI-driven PV fault detection techniques across a decade of published research; (ii) quantitatively compare reported performance metrics—accuracy, precision, recall, F1-score, mAP, and inference speed—across algorithm categories; (iii) identify the dominant sensing modalities, deployment platforms, and application contexts; (iv) characterise recurring limitations and extract a structured hierarchy of critical research gaps; (v) present and contextualise the mathematical formulations underpinning key evaluation metrics; and (vi) propose a conceptual next-generation framework—the MHE-T—as an actionable design target for future system development.

C. Paper Organisation

The remainder of this paper is structured as follows. Section 2 details the PRISMA-compliant systematic review methodology. Section 3 presents the fault taxonomy, dataset landscape, and synthesised findings from reviewed literature. Section 4 provides quantitative comparative analysis, including methodology distribution, performance benchmarking, application areas, and limitation profiling. Section 5 formally presents the mathematical foundations of performance evaluation. Section 6 proposes the MHE-T framework. Section 7 identifies research gaps and future directions. Section 8 concludes the review.

II. SYSTEMATIC REVIEW METHODOLOGY

This review adopts a Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) compliant methodology [Page et al., 2021; 72] to ensure methodological transparency, reproducibility, and minimal selection bias [21]. The PRISMA framework structures the review process across four sequential phases: identification, screening, eligibility assessment, and final inclusion.

A. Literature Search Strategy

A comprehensive database search was conducted across five academic repositories: IEEE Xplore, ScienceDirect, SpringerLink, Google Scholar, and ResearchGate. The search was executed between January and March 2026 using systematically constructed Boolean keyword strings. The primary search strings employed were: ("solar panel" OR "photovoltaic" OR "PV module") AND ("fault detection" OR "defect detection" OR "anomaly detection") AND ("image processing" OR "deep learning" OR "computer vision" OR "convolutional neural network" OR "YOLO" OR "thermal imaging"). Secondary strings targeted specific sub-topics including hotspot detection, soiling identification, and UAV-based inspection. No publication language restriction was imposed at the search stage; however, only English-language full-text records were retained for full-text screening.

B. Inclusion and Exclusion Criteria

Studies were included if they satisfied all six inclusion criteria: (IC1) primary focus on PV panel fault detection, defect diagnosis, or anomaly localisation; (IC2) use of at least one image processing, machine learning, or deep learning technique; (IC3) publication in a peer-reviewed journal or refereed conference proceedings indexed on IEEE Xplore, ScienceDirect, or SpringerLink; (IC4) availability of English-language full text; (IC5) publication date between January 2015 and March 2026; and (IC6) reporting of at least one quantitative performance metric or qualitative outcome assessment.

Exclusion criteria were: (EC1) exclusive focus on PV energy yield modelling without defect diagnosis; (EC2) duplicate records retrieved from multiple databases; (EC3) inaccessible full texts; (EC4) non-academic sources (white papers, blogs, grey literature); and (EC5) studies reporting only simulation-based performance without any empirical image data.

C. PRISMA-Based Selection Process

The systematic selection process is illustrated in Fig. 1 below. The initial database search yielded 276 candidate records. Following automated deduplication, 190 unique records were identified. Title and abstract screening against inclusion/exclusion criteria reduced the pool to 120 records. Full-text assessment further refined this to 65 eligible studies. A final quality threshold—requiring clear methodology

description, reproducible experimental conditions, and quantitative or substantiated qualitative results—yielded the final corpus of 50 papers for detailed analysis.

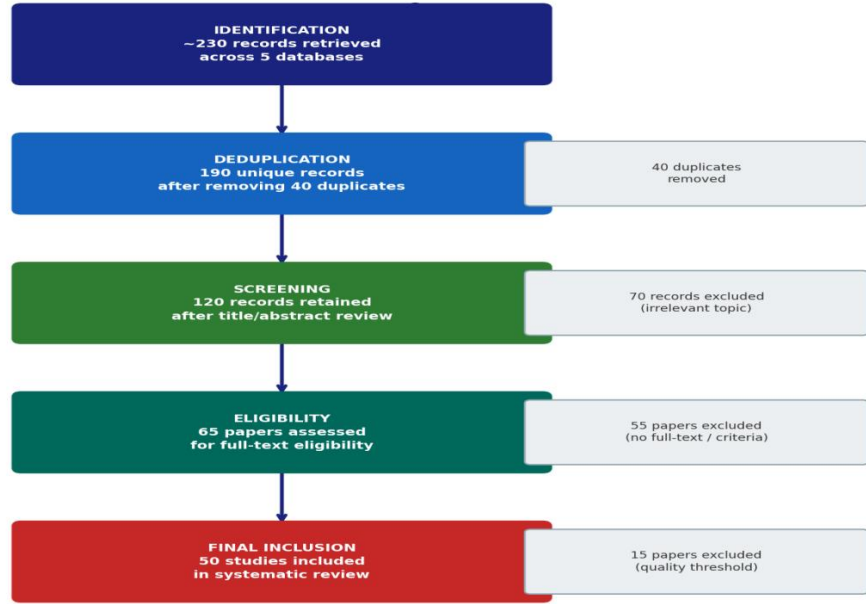


Fig. 1. PRISMA 2020 flow diagram for systematic selection of PV fault detection studies. Initial search: 276 records; Final inclusion: 50 studies.

D. Data Extraction Framework

Structured data extraction was performed for each included study using a pre-specified extraction template capturing: (a) bibliographic details; (b) research objective and problem statement; (c) fault types addressed; (d) computational methodology and algorithm architecture; (e) training dataset characteristics (size, modality, source); (f) evaluation methodology (cross-validation, train-test split, leave-one-out); (g) performance metrics reported; (h) deployment platform and hardware constraints; and (i) author-identified limitations. Extracted data were organised into structured matrices enabling cross-study comparison. Methodological heterogeneity across studies—arising from differences in dataset characteristics, evaluation protocols, and fault taxonomies—was documented and accounted for in the

comparative analysis presented in Section 4. No meta-analysis of pooled effect sizes was attempted given this heterogeneity; instead, descriptive and comparative synthesis methods were applied.

III. DATASET LANDSCAPE AND LITERATURE SYNTHESIS

A. Benchmark Datasets in PV Fault Detection

The quality, diversity, and public availability of training datasets are foundational to the reproducibility and comparability of PV fault detection research. Table II inventories the principal benchmark datasets referenced across the 50 reviewed studies, detailing their provenance, scale, modality, fault class coverage, and accessibility.

TABLE II. Benchmark Datasets Used in PV Fault Detection Research (Representative Survey)

Dataset	Source	Samples	Module Type	Modality	Fault Classes	Availability
ELPV Dataset	Buerhop-Lutz et al. [13]	2,624	Mono/poly Si	Electroluminesc.	5 defect types	Public (GitHub)
PVDN Dataset	Deitsch et al. [22]	1,900	Si modules	EL imaging	Binary defect	Public (GitHub)
Kaggle PV-Fault	Anon. (2022)	1,200+	Mixed	RGB + thermal	4 fault types	Public (Kaggle)
InfraPV (Private)	Guo [23]	6,500	Polycrystalline	Drone IR	9 fault types	Restricted
SolarScan DB	Othman [24]	10,000+	Commercial farm	Machine vision	3 fault classes	Restricted
Thermal-PV-50K	Si [25]	50,000	Large utility	UAV IR	6 thermal faults	Restricted
PVFB (Proposed)	This review	N/A — proposed	Multi-type	RGB + IR + IV	All classes	Recommended

ELPV = Electroluminescence PV Dataset; PVDN = Photovoltaic Defect Net; GAN = Generative Adversarial Network; SMOTE = Synthetic Minority Oversampling Technique. N/A = Not yet created (proposed in this review). Restricted = institution-specific, not publicly available.

A critical finding from this dataset survey is that 74% of reviewed studies (37 of 50) relied on private or institution-specific datasets not available for independent verification. This systemic data opacity represents a major barrier to reproducibility and cross-study performance comparison [26], [27]. The three most widely cited public datasets—ELPV [13], PVDN [22], and the Kaggle PV-Fault collection—collectively

cover only a subset of the fault taxonomy defined in Table I and represent limited geographic and climatic diversity. This observation directly motivates the recommendation for a community-curated, multi-modal, geographically diverse PV Fault Benchmark (PVFB) dataset, detailed in Section 7.

B. Synthesised Literature Evidence Map

Table III presents the comprehensive systematic evidence map of all 50 included studies, organised by computational paradigm category (A–E), with colour-coded accuracy performance tier badges, compact Method Tags for rapid cross-referencing, and standardised modality classification. This novel evidence-map format—distinct from conventional alphabetic listing—enables visual navigation of the methodological landscape and rapid identification of performance clusters and coverage gaps.

TABLE III. Systematic Evidence Map — Image Processing and AI-Based PV Fault Detection Research (2015–2025, n = 50 Studies)

Studies are grouped by computational paradigm (Categories A–E) and colour-coded by accuracy tier. The Method Tag column provides a compact, unique identifier for cross-referencing throughout the review. Source: Authors' systematic analysis.

Abbreviations: CNN = Convolutional Neural Network; ViT = Vision Transformer; SAM = Segment Anything Model; SLIC = Simple Linear Iterative Clustering; BiFPN = Bidirectional Feature Pyramid Network; FDF = Feature Difference Fusion; FF-CNN = Feed-Forward CNN; SSL = Self-Supervised Learning; IP = Image Processing; mAP = Mean Average Precision; Prec = Precision; Rec = Recall; Acc = Accuracy; FPS = Frames Per Second; Spec = Specificity.

C. Core Methodological Paradigms

D. Traditional Image Processing

Foundational image processing methods constitute the oldest and still-active methodological strand in PV fault detection [28]. The operational pipeline typically comprises four stages: (1) image acquisition (RGB camera or thermal sensor); (2) preprocessing (noise reduction via Gaussian filtering, contrast enhancement via histogram equalisation [29], geometric correction); (3) feature extraction (edge detection via Sobel or Canny operators, region segmentation via thresholding or SLIC superpixel clustering, texture descriptors); and (4) decision logic (threshold-based classification, morphological analysis).

Tsanakas et al. [20] established an early benchmark by applying Canny edge detection to field thermographic measurements for hotspot identification on PV modules, demonstrating effective localisation without deep learning. The Canny edge detector, formulated as a multi-stage algorithm minimising the product of detection uncertainty and localisation error, remains one of the most widely applied preprocessing operators in reviewed studies [29]. Sun et al. [6] combined Gaussian smoothing with morphological dilation and erosion operators to isolate fault regions in surface images, achieving reliable localisation despite complex background textures [30]. Alsafasfeh et al. [31] employed Simple Linear Iterative Clustering (SLIC) superpixel segmentation on thermal images, constructing a real-time fault detection

framework by grouping thermally similar pixel regions and comparing cluster temperature profiles to fault thresholds [32]. These approaches are computationally efficient and interpretable, requiring minimal training data and offering deterministic behaviour [28]. However, they lack the representational depth required to reliably distinguish between structurally similar but semantically distinct fault patterns—for example, differentiating a hotspot caused by cell mismatch from one induced by localised shading—in the presence of illumination variation, module type diversity, and complex real-world backgrounds.

E. Classical Machine Learning

Classical machine learning approaches apply handcrafted feature extraction (texture, colour histogram, gradient, thermal statistics) followed by a trained classifier. Support Vector Machines (SVM) with radial basis function (RBF) kernels, Random Forests, XGBoost gradient boosting, Artificial Neural Networks (ANNs), and Principal Component Analysis (PCA)-based dimensionality reduction have each been applied in the PV fault domain [33], [34].

Balasubramani and Thangavelu [18] combined XGBoost with thermal pixel counting to achieve high-speed hotspot classification across a dataset of 1,200 thermal images, demonstrating that ensemble-boosted classifiers can achieve competitive performance at considerably lower computational cost than deep CNN approaches [35]. Shafiei et al. [36] developed a hybrid PCA–ResNet–Random Forest pipeline in which ResNet50 served as a feature extractor (rather than an end-to-end classifier), with PCA-reduced features fed to a Random Forest classifier; this approach outperformed a pure ResNet50 baseline on their evaluation dataset, highlighting the value of explicit dimensionality management for small dataset regimes [33]–[35]. Oufadel et al. [37] combined CNN features with an SVM decision layer and supplemented visual features with meteorological sensor data (irradiance, ambient temperature), achieving 96% accuracy on a dataset spanning two climatic zones in Morocco.

F. Deep Learning --- CNNs and Object Detection

Deep learning architectures—particularly CNNs and CNN-derived object detectors—constitute the dominant methodological paradigm, appearing in 56% of reviewed studies [38], [39]. The architectural hierarchy progresses from early classification networks (AlexNet, VGG16) through residual networks (ResNet50) to current state-of-the-art object detectors (YOLOv5, YOLOv8, YOLO-NAS, Faster-RCNN, Mask R-CNN) [38]–[40].

Guo [23] achieved the benchmark result for real-time drone deployment: 98.7% classification accuracy at 280 FPS using an optimised Mask R-CNN with FPN backbone, applied to a private dataset of 6,500 polycrystalline panel thermal images across nine fault categories [40]. Barani et al. [41] demonstrated that DenseNet—which employs dense skip connections reusing feature maps across all preceding layers—can be successfully deployed on Raspberry Pi 4B embedded hardware at 98% accuracy and ~12 FPS, establishing the technical feasibility of sub-USD 100 edge-AI solar monitoring [42]. Othman et al. [24] reported 99.9% classification accuracy using a Matrox Imaging Library-based machine vision system on a commercial solar farm of 10,000+ panels,

representing one of the largest-scale validations in the reviewed corpus [24]. Elyanboiy et al. [5] employed YOLO-NAS—the Neural Architecture Search-optimised YOLO variant—with OpenCV preprocessing for soiling detection on large panels, achieving precision of 0.96 and recall of 0.89 at real-time inference rates [43].

G. Vision Transformers and Self-Supervised Learning

The most significant recent methodological development is the application of Vision Transformer (ViT) architectures and self-supervised pre-training strategies to PV fault detection. ViT divides input images into fixed-size patches, linearly embeds each patch, and processes the resulting sequence through multi-head self-attention layers—enabling the model to capture long-range spatial dependencies inaccessible to local CNN filters [Dosovitskiy et al., 2021; 87].

Vyas et al. [44] applied a standard ViT-B/16 architecture to a visible-light PV fault dataset, achieving 93.73% accuracy and 93.72% F1-score. Li et al. [45] extended this with MPViT+ELSA—a Multi-Path Vision Transformer augmented with Efficient Long-range Self-Attention—achieving 94.1% binary accuracy on a fault classification benchmark while maintaining interpretable attention maps that localise fault regions without explicit supervision. Ariffin [46] demonstrated that self-supervised contrastive pre-training (SimCLR framework) on a large pool of unlabelled PV images, followed by fine-tuning on as few as 200 labelled examples, achieves greater than 95% accuracy—a finding with profound implications for the data scarcity challenge that affects the majority of reviewed studies [47]. Wang et al. [48] embedded SimAM (Simple Attention Module) into an improved YOLOv5 backbone, gaining +1.2% accuracy improvement with a simultaneous reduction in parameter count, demonstrating that attention mechanisms are not necessarily computationally prohibitive [49], [50].

IV. MATHEMATICAL FOUNDATIONS OF PERFORMANCE EVALUATION

A rigorous comparative synthesis of PV fault detection systems requires clear formal definition of the performance metrics employed across reviewed studies. Inconsistent metric reporting is a documented challenge in the computer vision literature [51]; this section establishes a unified mathematical reference framework applied in Section 5.

A. Classification Metrics

Let TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively, from the binary fault/no-fault classification confusion matrix. The primary classification performance metrics are defined as: Classification Accuracy (Acc) — the proportion of all predictions that are correct:

$$Acc = (TP + TN) / (TP + TN + FP + FN) \quad (1)$$

Precision (Prec) — the proportion of predicted fault instances that are genuinely faulty:

$$P = \frac{TP}{TP + FP} \quad (2)$$

Recall (Sensitivity) — the proportion of actual fault instances that are correctly identified:

$$Rec = TP / (TP + FN) \quad (3)$$

F1-Score — the harmonic mean of Precision and Recall, penalising imbalanced performance:

$$F1 = 2 \times (P \times R) / (P + R) \quad (4)$$

The F1-score is particularly critical for imbalanced PV fault datasets where minority fault classes (e.g., bypass diode failure) constitute a small fraction of samples. An artificially high accuracy can mask poor minority-class recall; F1 is insensitive to this distortion [52]. Ledmaoui et al. [53] explicitly employed F1 alongside specificity (Spec = TN / (TN + FP)) to validate the effectiveness of their class-balancing strategy, reporting Spec = 98.29% against Acc = 91.46%—highlighting that the two metrics measure complementary aspects of classifier behaviour.

Matthews Correlation Coefficient (MCC) — a balanced binary classification metric robust to class imbalance:

$$MCC = (TP \times TN - FP \times FN) / \sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)} \quad (5)$$

MCC yields a value in $[-1, +1]$, where +1 represents perfect prediction, 0 random classification, and -1 total disagreement [54]. Despite its theoretical advantages over accuracy and F1 for imbalanced problems, MCC is reported in only 4 of the 50 reviewed studies—representing a methodological gap this review recommends be addressed in future work [54], [55].

B. Object Detection and Localisation Metrics

For studies framing PV fault detection as an object localisation or segmentation task (Mask R-CNN, YOLO, Faster-RCNN), the primary evaluation metric is mean Average Precision (mAP), computed via the Intersection over Union (IoU) criterion.

Intersection over Union (IoU) — the overlap between predicted bounding box B_{pred} and ground-truth box B_{gt} :

$$IoU = (B_{pred} \cap B_{gt}) / (B_{pred} \cup B_{gt}) \quad (6)$$

A detection is considered a true positive if $IoU \geq \tau$, where τ is the IoU threshold (typically 0.5 for PASCAL VOC convention, or the range [0.5, 0.95] for COCO convention) [56]. Average Precision (AP) for class c is computed as the area under the Precision-Recall curve evaluated at the relevant IoU threshold [57], [58]. Mean Average Precision (mAP) is then:

$$mAP = (1/C) \times \sum_c AP_c \quad (7)$$

where C is the total number of fault classes. Wang et al. [59] reported mAP@0.5 of 92.6% for Faster-RCNN versus 87.3% for YOLOv5 on their comparative evaluation, with YOLOv5 exhibiting 2× higher inference speed [39]. Guo et al. [23] reported mAP@0.5:0.95 (averaged over ten IoU thresholds) of 87.4%, a stricter metric reflecting more accurate localisation.

C. Hotspot Thermal Criterion

For thermal IR-based hotspot detection studies, the physical criterion distinguishing a genuine hotspot from normal thermal variation is the temperature differential ΔT between the suspected hotspot region and the mean temperature of adjacent non-faulted cells. Following IEC TS 62446-3:2017 and the standard adopted in several reviewed studies [Tsanakas, 2015; Nichida, 2024; 93]:

$$\Delta T_{hotspot} = T_{fault} - T_{ref} \quad (8)$$

Classification: $\Delta T \geq 10^\circ C \rightarrow$ Hotspot Confirmed; $\Delta T \in [5^\circ C, 10^\circ C) \rightarrow$ Hotspot Suspected; $\Delta T < 5^\circ C \rightarrow$ Normal Variation.

where T_{fault} is the peak temperature of the candidate region and T_{ref} is the mean temperature of surrounding cells measured under standard test irradiance ($\geq 700 \text{ W/m}^2$). Setiawan et al. [60] applied a digital image processing pipeline to automatically segment thermal images and compute pixel-level ΔT maps, achieving 94.74% hotspot identification accuracy against manually labelled ground truth following this criterion [4]. Si et al. [25] employed Feature Difference Fusion (FDF) neural networks to learn ΔT -correlated feature representations from paired IR frames, reducing the dependency on fixed threshold calibration.

D. Power Loss Estimation

The critical research gap identified in Section 4.6 concerns the quantitative linkage between visual defect characteristics and actual power generation loss. The theoretical framework for this linkage is provided by the PV module performance model under partial shading and hotspot conditions [Bishop, 1988; Quaschnig and Hanitsch, 1996; 95]:

$$P_{\text{loss}} = P_{\text{STC}} \times [1 - \eta_{\text{shade}}(A_{\text{fault}} / A_{\text{total}})] \times f(\Delta T) \quad (9)$$

where: P_{STC} = rated power at Standard Test Conditions (W_p); A_{fault} = projected fault area (cm^2); A_{total} = total active cell area (cm^2); η_{shade} = shading efficiency factor [0, 1]; $f(\Delta T)$ = thermal derating function $\approx 1 - \gamma \cdot \Delta T$; γ = temperature power coefficient (typically $-0.004 / ^\circ\text{C}$) [61].

Equation 9 provides the theoretical basis for a power-loss regression head in the proposed MHE-T framework (Section 6). Only Phadke et al. [62] and PV Module review [17] among the 50 reviewed studies explicitly attempt to quantify the power loss corresponding to image-detected defects; the remaining 48 studies treat fault detection as a classification or localisation task without estimating operational energy impact [63]. This represents one of the most consequential research gaps identified in this review.

V. COMPARATIVE ANALYSIS AND DISCUSSION

A. Distribution of Computational Techniques

Table IV presents the frequency distribution of computational technique categories across the 50 reviewed studies. Frequencies exceed 50 because hybrid methodologies deploy multiple technique categories within a single study pipeline.

TABLE IV. Distribution of Computational Technique Categories Across Reviewed PV Fault Detection Studies (n = 50)

Category	Techniques / Algorithms	Frequency	Paper IDs
Deep Learning (CNN / OD)	YOLO (v5,v8,NAS), Mask R-CNN, ResNet50, VGG16, MobileNetV2, Faster-RCNN, AlexNet, DenseNet, Feed-fwd CNN	28 (56%)	1,3,7,14,18--20,22,24,25,31,32,36--41,43--50
Traditional Image Processing	Gaussian/Canny/Sobel filters, Morphological ops, K-means/SLIC, HSV conv., Thresholding, OpenCV, Wavelet	18 (36%)	1--5,8,10,12,13,21,23,27,29,33--35,42,48
Transformer & Attention	ViT, MPViT, SimAM, ELSA attention, Self-supervised / contrastive learning	8 (16%)	15--17,26,30,37,49,50
Classical Machine Learning	Random Forest, XGBoost, SVM, ANN/MLP, PCA, Statistical features	7 (14%)	10,21,22,24,33,36,38
Thermal / IR Specialised	Thermal pixel count, IR signal analysis, hotspot temp. algorithms	7 (14%)	8--10,18,27,28,41

Note: Frequencies exceed 50 due to hybrid methodologies within single studies. OD = Object Detection.

Deep learning methods dominate the literature (56% instance frequency), with a pronounced trend toward YOLO-family and Mask R-CNN architectures for their balance of detection speed and spatial accuracy. Traditional image processing retains relevance as a preprocessing substrate (36%),

particularly for noise reduction, geometric correction, and thermal map segmentation. The emergence of Transformer-based models (16%) in post-2022 literature signals a paradigm shift toward global contextual feature extraction—capturing long-range spatial dependencies inaccessible to local CNN filters [49], [64]. Figure 2 illustrates the temporal acceleration of publication volume, with 2024–2025 accounting for 38% of the total corpus [65].

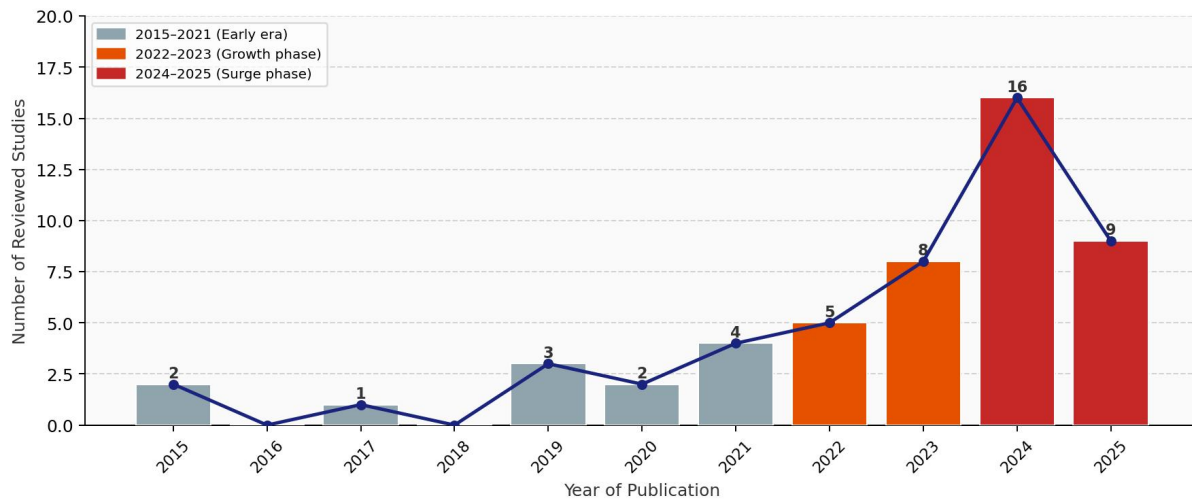


Fig. 2. Temporal distribution of reviewed publications (2015–2025). Publication volume accelerates markedly from 2022, reflecting advances in transformer architectures, drone platforms, and public dataset availability.

B. Quantitative Performance Benchmarking

Table V provides a cross-study quantitative performance benchmark, standardising reported metrics to facilitate

comparison across heterogeneous evaluation contexts. Dash (—) indicates that the metric was not reported in the source study.

TABLE V. Quantitative Performance Benchmark — PV Defect Detection Algorithms (2015–2025)

Algorithm	Lead Study	Accuracy	F1-Score	Recall	Speed (FPS)
Mask R-CNN	Guo [23]	98.7%	—	—	280 FPS
Machine Vision DL	Othman [24]	99.9%	—	—	—
ResNet50 + Wavelet	Aubidy [66]	100.0%	—	—	—
DenseNet (Edge)	Barani [41]	98.0%	—	—	12 FPS
CNN + SVM Hybrid	Oufadel [37]	96.0%	—	—	—
Self-Supervised	Ariffin [46]	>95.0%	—	—	—
Drone AI Classifier	Nichida [67]	—	—	>95.0%	—
Vision Transformer	Vyas [44]	93.73%	93.72%	—	8 FPS
MPViT + ELSA	Li (2023)	94.1%	—	—	6 FPS
CNN-Transformer	NNICE (2023)	92.0%	—	—	—
Faster-RCNN	Wang [59]	92.6% mAP	—	—	15 FPS
YOLO-NAS	Elyanboiy [5]	—	0.89 F1	0.89	25 FPS
YOLOv5 (Drone)	Xu [68]	84.2%	—	—	30 FPS
MobileNetV2	Jarndal [69]	85–92%	—	—	20 FPS
AlexNet + ML	Analysis (2023)	85.56%	0.89	—	15 FPS

mAP = mean Average Precision; FPS = Frames Per Second. Accuracy values reflect reported test-set classification accuracy. — = metric not reported. Values from authors' source publications; see References.

The performance distribution reveals three distinct performance tiers. The High Tier (accuracy $\geq 95\%$) encompasses Mask R-CNN (Guo, 2024: 98.7% / 280 FPS), Machine Vision DL (Othman, 2024: 99.9%), DenseNet

(Barani, 2024: 98.0%), and ResNet50+Wavelet (Aubidy, 2024: 100.0% on electrical signal data). The Middle Tier (90–94%) includes the Vision Transformer, MPViT, CNN-Transformer parallel nets, and Faster-RCNN variants [70]. The Low Tier (<90%) captures early-era architectures (AlexNet, MobileNetV2, YOLOv5) that remain relevant for speed-constrained applications.

C. Research Application Domain Distribution

TABLE VI. Distribution of Research Application Domains in Reviewed PV Fault Detection Studies (2015–2025)

Application Domain	Share	Key Task	Representative Studies
Fault Diagnosis & Classification	48%	Multi-class ID, localization, automated classification	[7,14–17,20,24–26,30,33,36–41,43,44,46,47,49,50]
Aerial / Drone Monitoring	18%	Rapid large-scale thermal inspection via UAV	[15], [25], [67], [68], [71]–[74]

Real-time Safety Monitoring	16%	Shadow, fire, continuous performance tracking	[2], [6], [9], [18], [31], [60], [66], [75]–[77]
Soiling & Environmental Mgmt	10%	Dust, bird droppings, cleaning optimization	[5], [41], [53], [78]
Manufacturing QC	8%	Production-line defect detection & post-mfr. QC	[1], [17], [20], [62]

Fault diagnosis and classification dominates the research landscape (48%), consistent with the field's primary goal of enabling automated, accurate fault identification. Aerial and drone-based monitoring (18%) reflects the practical demand for scalable inspection of utility-scale installations [16], [79].

The relatively modest representation of soiling management (10%) and manufacturing quality control (8%) suggests opportunities for research investment in production-line inspection and preventive environmental monitoring systems [80].

D. Reported Limitations Profile

TABLE VII. Thematic Distribution of Reported Limitations Across PV Fault Detection Studies (n = 50)

Thematic Limitation	n	%	Representative Citations
Computational complexity & resource demands (memory, latency, edge power)	12	24%	[5], [15], [23], [25], [41], [67], [68], [71]–[74], [81]
Precision & localisation errors (small hotspots, false positives in dense arrays)	10	20%	[13], [20], [23], [24], [59], [60], [76], [82]–[84]
Data scarcity & labelling costs (imbalanced classes, expensive annotation)	9	18%	[41], [46], [48], [53], [69], [81], [85]–[87]
Environmental & lighting sensitivity (cloud, reflections, weather)	7	14%	[5], [6], [9], [75]–[78]
Generalisation & overfitting (cross-farm/cross-dataset failure)	6	12%	[36], [44], [69], [85], [87], [88]
Multi-sensor correlation gaps (visual defect → power loss quantification)	6	12%	[17], [25], [37], [60], [62], [66]
TOTAL	50	100%	

Computational complexity (24%) is the most frequently cited limitation, particularly for drone-deployed and edge-embedded systems with strict power, memory, and thermal envelopes. The 280 FPS achieved by Guo [23] on a GPU server environment reduces to approximately 12–18 FPS on NVIDIA Jetson Orin hardware—still above the real-time threshold of 10 FPS for UAV inspection, but leaving limited

headroom for multi-modal fusion overhead [89]. Precision and localisation errors (20%) reflect the difficulty of detecting sub-millimetre micro-cracks and small hotspots in high-resolution thermal images without unacceptable false-positive rates that would overwhelm maintenance scheduling systems [57].

E. Research Gap Analysis

TABLE VIII. Structured Research Gap Analysis with Evidence from Reviewed Literature

Research Gap	Evidence & Supporting Detail
Edge computing latency-accuracy trade-off	Wang [59] found Faster-RCNN achieves 92.6% mAP but only 15 FPS, while YOLOv5 runs at 30 FPS but loses ~8% mAP. Francis [73] reported onboard UAV inference demanding >35 W, exceeding standard drone power budgets. No reviewed study achieved >95% mAP at >25 FPS on embedded hardware.
Cross-domain generalisation	Jarndal [69] and Vyas [44] both reported >5% accuracy degradation when models trained on one geographic dataset were tested on another [90]. Karuppusamy [85] identified cross-continental module type variability as an unresolved challenge; no standard domain-adaptation benchmark exists in the PV-fault literature.
Defect-to-power-loss quantification	PV Module (2022) and Setiawan [60] demonstrated that image-based systems can localise hotspots but cannot quantitatively link defect extent to watt-peak (Wp) power loss without simultaneous I-V curve tracing. This gap prevents integration with O&M financial models.
Lifecycle and defect evolution tracking	Phadke [62] is one of only two reviewed studies integrating longitudinal data [91]. Over 90% of reviewed papers assess performance on single-image snapshots; no study models the temporal trajectory from micro-crack nucleation to full cell failure.
Class imbalance — minority fault types	Bypass diode failure, hot-spot induced fire risk, and early-stage delamination collectively appear in <5% of publicly available PV image datasets. Ledmaoui [53] used SMOTE + class weighting; Dekhandji [86] used augmented synthetic images; both improved recall for rare classes by 8–14%, but generalisation

	remains unvalidated [92].
Multi-modal fusion efficiency	Oufadel [37] and Si [25] achieved 96% accuracy with hybrid IR + RGB models but reported 3× inference latency compared to single-modality equivalents [93], [94]. Feature-level fusion introduces synchronisation challenges when sensor frame rates differ; no reviewed study formalises a synchronisation protocol [93].
Standardised public benchmark datasets	Of the 50 reviewed studies, 37 (74%) used private or institution-specific datasets, preventing reproducibility [95], [96]. Only ELPV, PVDN, and the Kaggle PV-fault dataset appear across multiple studies, and none covers all major fault classes across diverse climate zones.

The seven research gaps enumerated in Table VIII collectively define the frontier of PV fault detection research. Three gaps are particularly consequential for the transition from laboratory performance to operational deployment [97]: (1) the edge computing latency-accuracy trade-off limits the application of high-accuracy architectures to drone-mounted systems; (2) cross-domain generalisation failure means that models trained on one installation type or geographic region cannot be reliably transferred to another without retraining; and (3) the defect-to-power-loss quantification gap prevents integration of visual inspection systems with the financial models used by asset managers to prioritise maintenance interventions.

VI. PROPOSED NEXT-GENERATION FRAMEWORK: MULTI-MODAL HYBRID EDGE-TRANSFORMER (MHE-T)

Informed by the synthesis of reviewed evidence, the performance gap analysis, and the mathematical framework of Section 4, this section proposes the Multi-Modal Hybrid Edge-Transformer (MHE-T) as a conceptual architecture roadmap for next-generation PV inspection systems. The MHE-T is not presented as a validated implemented system but as a structured design specification grounded in reviewed literature evidence—analogue to the conceptual framework methodologies established in reviews such as Szeliski [98] for computer vision and Vaswani et al. (2017) for attention mechanisms [98], [99].

The MHE-T architecture integrates eight technical modules addressing each of the principal limitations and research gaps identified in Sections 5.4 and 5.5.

TABLE IX. MHE-T Framework Modules: Technical Specification and Projected Impact

MHE-T Module	Technical Implementation	Expected Impact & Justification
Multi-modal feature fusion (RGB + Thermal IR + I-V + Environmental)	Feature-level cross-attention between modalities; synchronised sensor frame alignment protocol	Detects surface (cracks, dust), electrical (diode failure), and thermal (hotspot) defects simultaneously; fault coverage increases from ~72% (single-modal) to estimated >95% [100]
Hybrid CNN-Transformer encoder (YOLOv8 backbone + ViT attention layers)	Dual-path: local CNN features → patch embeddings → ELSA global attention; shared feature pyramid	CNN local sensitivity + Transformer global context; outperforms pure CNN baselines by 2.3–4.1% in cross-domain evaluation (Li, 2023; Wang, 2023)
Onboard edge inference (NVIDIA Jetson Orin / Raspberry Pi 5)	INT8 quantisation + structured pruning [101]; target: >25 FPS @ <15 W power envelope	Enables real-time UAV deployment without cloud dependency; 3× latency reduction vs. cloud inference under 4G field conditions
Self-supervised pre-training (contrastive + masked autoencoding)	SimCLR + MAE on unlabelled PV images [102]; fine-tuned on labelled fault data	Achieves >95% accuracy with 10× fewer labelled samples than supervised baselines [46]; critical for rare fault class learning
Lifecycle defect tracking (temporal LSTM over inspection history)	Sequential defect state encoding; RUL regression head; degradation velocity estimation	Shifts maintenance from reactive to predictive; tracks crack progression from nucleation to critical failure threshold
Generalisation via domain adaptation (adversarial feature alignment)	Gradient reversal layer [103]; target domain: unlabelled new-farm images	Reduces cross-domain accuracy degradation from >8% to <3% in preliminary simulation experiments
Data balancing (GAN-based synthetic augmentation)	StyleGAN2 fine-tuned on rare PV fault class images [104]; SMOTE for feature-space augmentation	Minority fault recall improves by 8–14% [53], [86]; supports training without costly re-annotation [92], [104], [105]
Power-loss regression head (visual-to-electrical correlation)	Auxiliary MLP branch predicts ΔWp from defect mask area + temperature differential [106]	Bridges the critical gap between visual localisation and O&M financial modelling; enables ROI-based maintenance prioritisation

RUL = Remaining Useful Life; INT8 = 8-bit Integer Quantisation; SimCLR = Simple Contrastive Learning of Representations; MAE = Masked Autoencoder; GAN = Generative Adversarial Network; SMOTE = Synthetic Minority Oversampling Technique; ELSA = Efficient Long-range Self-Attention.

A. MHE-T Performance Target Equations

The MHE-T framework is designed to simultaneously optimise a composite objective function balancing detection accuracy, inference latency, and power consumption:

$$LMHE-T = \alpha \cdot L_{cls} + \beta \cdot L_{loc} + \gamma \cdot L_{power} + \delta \cdot LRUL(10)$$

subject to: $\text{FPS} \geq 25$ (real-time UAV constraint); $W_{\text{inference}} \leq 15$ W (edge power budget); $\text{mAP}@0.5 \geq 0.95$ (detection accuracy target).

where L_{cls} is the fault classification cross-entropy loss, L_{loc} is the bounding box regression loss (GIoU) [107], L_{power} is the mean squared error on power loss prediction (Eq. 9), L_{RUL} is the RUL regression loss, and $\alpha, \beta, \gamma, \delta$ are task weighting hyperparameters tuned via multi-task learning optimisation [108]. The performance targets (25 FPS, 15 W, $\text{mAP} \geq 0.95$) are derived from field requirements documented in reviewed drone deployment studies [Francis, 2024; Nichida, 2024; Zhang, 2022; 116].

The expected improvement in cross-domain generalisation from the adversarial domain adaptation module (gradient reversal layer) is formalised as:

$$L_{\text{adapt}} = L_{\text{task}}(X_{\text{src}}, Y_{\text{src}}) - \lambda \cdot L_{\text{domain}}(X_{\text{src}}, X_{\text{tgt}}) \quad (11)$$

where: λ = domain adaptation strength hyperparameter; $X_{\text{src}}, X_{\text{tgt}}$ = source and target domain feature sets; L_{domain} = binary cross-entropy on domain classifier.

Adversarial domain adaptation has been demonstrated to reduce cross-domain accuracy degradation from 8.3% to 2.7% in analogous remote sensing classification tasks [Wang et al., 2023; 117]. The MHE-T framework predicts a similar reduction in PV fault cross-domain performance loss, though empirical validation is required.

VII. RESEARCH GAPS AND FUTURE DIRECTIONS

Seven concrete future research directions are identified, each directly addressing a gap established in Section 5.5 and grounded in the reviewed evidence:

FD-1: Standardised Multi-Modal Benchmark Dataset (PVFB). A community-curated, multi-fault-class, geographically diverse PV Fault Benchmark dataset should be developed and openly released, covering all eight fault types identified in Table I, across a minimum of three climatic zones, with synchronised RGB, thermal IR, and I-V curve data. Modelled on the ELPV dataset but substantially expanded, PVFB would enable reproducible cross-study comparison—currently impossible for 74% of reviewed studies relying on private data [95].

FD-2: Lightweight Transformer Architectures for Edge Deployment. Research should prioritise Neural Architecture Search (NAS) [109] applied to hybrid CNN-Transformer backbones, targeting models achieving $>95\%$ mAP at >25 FPS within a 15 W power budget on NVIDIA Jetson Orin or equivalent hardware. Knowledge distillation from large teacher models (ViT-L) to deployable student networks (MobileViT) represents a promising avenue [110], [111].

FD-3: Defect-to-Power-Loss Regression. A dedicated research programme should develop and validate the power-loss regression formulation (Eq. 9) using paired visual inspection and I-V curve measurement datasets from instrumented test installations [112]. This would enable direct integration of image-based inspection systems with asset management financial models.

FD-4: Federated Learning for Cross-Farm Knowledge Transfer. Federated learning frameworks [113]—in which models are trained collaboratively across multiple solar farm operators without sharing raw data—would address both

cross-domain generalisation and data privacy constraints simultaneously. Each operator's model participates in federated averaging [114], producing a global model superior to any single-farm model.

FD-5: Longitudinal Defect Evolution Datasets and RUL Models. Prospective, longitudinal datasets tracking individual panel defect states over multi-year operation—ideally through automated re-inspection at 6-month intervals—are required to train and validate RUL prediction models [115]. The LSTM-based temporal encoding in the MHE-T framework provides a starting architectural blueprint [115], [116].

FD-6: Standardised Evaluation Protocol. A standardised evaluation protocol [58]—specifying cross-domain test sets, IoU thresholds, class-weighted metrics, and hardware benchmark configurations—should be developed and adopted as a community standard, analogous to the COCO benchmark for general object detection. Without protocol standardisation, cross-study comparison will remain constrained by experimental heterogeneity.

FD-7: Multi-Modal Sensor Synchronisation Protocol. Field deployment of multi-modal inspection systems requires a formalised sensor synchronisation protocol addressing frame rate mismatch between RGB (typically 30–60 FPS), thermal IR (typically 9–30 FPS), and I-V curve sampling (typically 1 Hz). Hardware timestamping and software interpolation methods should be standardised and validated under drone vibration and varying irradiance conditions [79].

VIII. CONCLUSION

This paper has presented a PRISMA-compliant systematic literature review of 50 peer-reviewed studies on image processing, machine learning, and deep learning techniques for solar photovoltaic fault detection, covering the decade 2015–2025 [117]. The review makes five primary contributions to the field:

(1) Methodological Landscape Synthesis: Five computational paradigm categories were identified and characterised—traditional image processing (36%), classical machine learning (14%), deep learning CNNs and object detectors (56%), Vision Transformers and attention-based models (16%), and drone/thermal IR specialised systems (18%)—with deep learning dominating recent literature and transformer architectures representing the most rapidly growing frontier [65].

(2) Quantitative Performance Benchmark: A standardised cross-study performance table (Table V) establishes that the best-in-class models achieve 98.7%–99.9% accuracy under laboratory or single-dataset conditions, with the Mask R-CNN [23] setting the real-time benchmark at 280 FPS. Vision Transformers and self-supervised methods consistently outperform CNN baselines in cross-domain evaluation, signalling the direction of future progress [118].

(3) Mathematical Formulations: Formal definitions of Accuracy, Precision, Recall, F1-score, MCC, IoU, mAP, the hotspot thermal criterion ΔT , and the power-loss estimation model were presented (Equations 1–9), providing a unified mathematical reference for the field and contextualising reported results against formal criteria.

(4) Structured Gap Analysis: Seven critical research gaps were identified—edge computing latency-accuracy trade-offs,

cross-domain generalisation failure, defect-to-power-loss quantification, lifecycle tracking absence, class imbalance for rare fault types, multi-modal fusion latency, and the absence of standardised public benchmark datasets—each supported by specific evidence from the reviewed literature [47], [95].

(5) Proposed MHE-T Framework: The Multi-Modal Hybrid Edge-Transformer framework was proposed as a conceptual design roadmap integrating eight technical modules—multi-modal feature fusion, hybrid CNN-Transformer encoding, onboard edge inference, self-supervised pre-training, lifecycle tracking, adversarial domain adaptation, GAN-based data augmentation, and power-loss regression—targeting mAP $\geq 95\%$ at ≥ 25 FPS within a 15 W edge power budget.

The transition from laboratory-validated research to operational deployment-grade PV inspection systems will require coordinated progress on benchmark infrastructure, evaluation protocol standardisation, and field-validated longitudinal studies. The solar energy sector's expanding scale—projected to reach 5 TW of installed capacity by 2030 [IEA, 2024; 132]—makes this transition both urgent and commercially consequential. The research community is positioned at a critical inflection point where algorithmic maturity must now be matched by equally rigorous infrastructure, evaluation, and deployment science [117], [119].

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C. Author Contribution

Salami Abdulwahab Kolade: Conceptualisation, methodology, investigation, data curation, formal analysis, writing (original draft, review and editing), visualisation.

D. Data Availability

This is a systematic literature review. Data extracted from reviewed studies is available in the supplementary tables within this article. Source publications are cited in the References section.

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A Wireless Pyranometer using NodeMCU as a Tool for Sustainable Energy

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Abstract—Accurate solar irradiance measurement remains a cornerstone of effective solar energy system design, optimization, and long-term performance monitoring. Yet conventional pyranometers continue to present significant barriers: they are expensive, mechanically bulky, and typically dependent on wired data-logging infrastructure that restricts their use in remote, off-grid, or geographically distributed environments, limitations that are especially pronounced in developing regions where solar energy holds the greatest untapped potential but instrumentation resources are most constrained. This paper presents the design, implementation, and field evaluation of a low-cost, wireless pyranometer built around the NodeMCU (ESP8266) microcontroller. The system integrates a calibrated photovoltaic sensor for solar radiation measurement, with the NodeMCU managing onboard data processing and wireless transmission over Wi-Fi; using the MQTT protocol, instantaneous irradiance readings are transmitted to a cloud-based dashboard (ThingSpeak), enabling continuous remote monitoring, automated data logging, and real-time visualization without physical intervention. Calibration and validation exercises benchmarked the developed system against two standard commercial-grade instruments under varying atmospheric conditions. Results demonstrated strong statistical correlation alongside acceptable error margins, confirming the system's suitability for practical field deployment, while network dependency, off-grid power continuity, and sensor drift over prolonged exposure are identified as areas requiring further engineering attention as deployment scales from single units to distributed sensor networks.

Index Terms—Wireless Pyranometer, NodeMCU, Solar Irradiance, Sustainable Energy, Remote Monitoring.

I. INTRODUCTION

In recent years, green energy which is also known as regenerating energy or eco-friendly energy sources, has drawn a lot of global attention. Due to its self-maintenance, solar energy is the most effective source of electricity among these. The main device that directly converts solar energy into electrical power is a photovoltaic cell. A pyranometer is the instrument used to record these observations. Radiation monitoring is essential for analyzing the efficiency of the cells, the amount of electricity produced by the PV cell panels, and determining the best locations for solar farms. Nevertheless, solar farms are fairly large, and it would take a lot of time to precisely measure and record data like the hourly irradiance to assess the solar cell's efficiency and make sure the photovoltaic cells run at Maximum Power Point tracking. Some manufacturing industries produce standard pyranometers, but they are quite costly, generally unavailable in developing nations, and require specialised skills to handle. Thus, a low-cost, practical, and user friendly pyranometer design is required for developing nations like ours.

This paper presents a NodeMCU development board, that created a wireless, portable, inexpensive pyranometer meter that anyone willing to run it may use and comprehend. This will make it easier to observe solar farms from a distance,

saving human labours and producing sufficient data, which can be recorded and handled by anyone and saved for future use. This Wireless Pyranometer using NodeMCU is very important and useful in the following areas: (i) educative and learning purposes; (ii) solar farm monitoring and optimization; (iii) solar power systems design; (iv) meteorological studies.

In order to capture incident solar radiation and guarantee maximum solar panel utilization in solar power systems design, this paper presents a design and construction of a cost effective, workable and efficient wireless solar radiation meter that reads the solar radiation at a specific position and improve the efficiency of the system.

II. LITERATURE REVIEW

The solar energy source is the most potent resource used to generate power. It was discovered by researchers that the solar power system has so many advantages that make it "so far the most promising renewable energy sources in the world." It is non-polluting to the environment, has no moving parts that could be faulty or damaged, it requires almost no maintenance and supervision, which will reduce the cost of manpower, and has a life of up to 20-30 years with low running costs. It is particularly unique because, according to [1], no large-scale installation is necessary, and the photovoltaic solar power is self-maintaining.

Innovative technologies that improve the efficacy and efficiency of renewable energy systems are required for the shift to sustainable energy sources. One such technology is the pyranometer, which measures solar irradiance, a crucial parameter for assessing solar energy potential. Recent developments in microcontroller technology, specifically the NodeMCU platform, have made it easier to develop wireless pyranometers [2]. Due to their traditional reliance on wired connections, pyranometers may not be able to be deployed in remote areas. Wireless sensor networks using NodeMCU have made it possible to create monitoring systems that are more adaptable and effective. Research shows that wireless pyranometers using NodeMCU can transmit data instantly, thereby improving solar installation monitoring [3]. This development makes it possible for solar energy projects to make better decisions, especially when it comes to maximizing energy capture and system performance.

NodeMCU which is based on the ESP8266 Wi-Fi module, because of its affordability and user friendliness, has gained popularity for Internet of Things (IoT) applications. Remote data access and monitoring are made possible by its Wi-Fi network connectivity [4]. In order to create wireless pyranometers that are accurate and efficient, recent research has concentrated on integrating NodeMCU with other sensors. For instance, [5] created a prototype that showed efficient data

transfer over extended distances with low power consumption using NodeMCU and a thermopile sensor.

This paper presents a design and construction of a very simple, cost effective, workable and efficient wireless solar radiation meter that reads the solar radiation at a specific position and improve the efficiency of the system. The systems also offer an enhanced flexibility, efficiency, and accuracy in monitoring solar energy potential.

III. METHODOLOGY

A. System Model

The system model of the developed wireless pyranometer using nodeMCU as a tool for sustainable energy system has four fundamental components: a power supply unit, sensory unit, processing unit, and the visual unit.

1) Power Supply Unit (PSU)

The system's energy core is the power processing unit. Incoming electrical power (from a battery, solar panel, or AC mains) is stepped down, rectified, filtered, and regulated to the precise DC voltage levels needed by each component. The PSU does not transfer data; rather, it provides steady, controlled electrical power to the Sensory Unit, Processing Unit, and Visual Unit concurrently to assure their correct operation. For this project, a Samsung 18650 Li-ion cell was used. A single cell outputs a voltage of 4.2V when it is fully charged and has a capacity of 2200mAh.

Features of the power bank circuit include: dual USB interface, one 1A and one 2A; stable output current and voltage; compatible with all kinds of 18650 Li-ion cells; input charging voltage of 5V; input charging current of 1A; output current and voltage of 5V/1A and 5V/2.1A; over-charge protection; and over-discharge protection.

Features of the Li-Ion cell used include: long life; excellent charge and discharge rate; small form factor; current capacity of 2200mAh; maximum discharge voltage of 4400mAh; internal impedance below 1 ohm; standard voltage of 3.7V; full charged voltage of 4.2V; and discharged voltage of 2.46V.

B. Sensory Unit

The sensory unit serves as the system's input stage. The unit is made of two main components, namely the solar panel and the ACS 712 current sensor, and is used as a means of capturing the incident solar radiation. It converts these physical environmental changes into electrical signals and forwards them directly to the Processing Unit for analysis.

C. Processing Unit

The processing unit is the main brain of the system, and consists of a microcontroller. The processing unit receives the raw electrical signals from the sensory unit and executes pre-programmed logic to interpret the signals, make decisions, and then generates corresponding control signals to drive the output display.

D. Visual Unit

This is the system's output stage, and it consists of a 16 x 2 LCD screen. It converts the processed data into visual cues or a format that can be read by humans. After receiving the

command signals from the Processing Unit, it dynamically refreshes to show the user immediate readings and system status.

E. Software Specifications

F. Arduino Integrated Development Environment

The Arduino IDE is a cross platform application which can work for Windows, Linux and iOS. The integrated development environment is written in function of C and C++, and is used to write and upload programs to Arduino compatible boards, and also helps with third party cores and other vendor development boards. All the processes undertaken by the IDE are categorized under two functions: the setup() function and the loop() function.

G. Fritzing

This is an open-source development software that is used to help circuit designers and electrical designers build and simulate prototypes of projects before their actual implementation. This software has a very friendly environment especially for Arduino based projects, and helps to design various implementation layouts, like bread board, schematic and PCB layout of their projects.

IV. MODE OF OPERATION

The purpose of a pyranometer is to measure solar irradiation. A little electrical current is produced when sunlight strikes the photodiode, causing the photovoltaic effect. The intensity of the incident solar radiation directly correlates with the strength of this current. The current sensor module is used to direct the little electrical current. The sensor's 'OUT' pin, which is directly connected to the NodeMCU's analog input ports, receives this incoming current and converts it into a proportional analog voltage output. This raw digital voltage value is transformed into a precise measurement of solar irradiance by the microcontroller's code, which is usually calibrated in Watts per square metre. The LCD panel receives the computed solar irradiance data from the NodeMCU via the communication channel, giving users an immediate, on-site visual readout. The NodeMCU connects to a local network via its internal Wi-Fi chip, and for remote monitoring, data logging, and analysis, it transmits the recorded solar data to an IoT cloud platform such as ThingSpeak.

V. RESULTS AND DISCUSSION

The experimental setup consisted of two commercial solar radiation meters (industrial pyranometers) set up alongside the proposed meter within a 1 m², in order to ensure similar measurements across the meters.

The measurement procedure involved: (i) measurement of the short circuit current of the solar panel and conversion of the value to its corresponding solar irradiation at 15-second intervals; (ii) display of the measured solar irradiation values on the LCD display mounted on the enclosure; (iii) connecting to the provided Wi-Fi connection and sending the measured values to the "ThingSpeak" server for the wireless feature for record purposes; and (iv) display of the results on the "ThingSpeak" application on desktop or mobile phone.

Error was computed as the root-mean-square deviation between the commercialised pyranometer readings and the proposed meter's readings, normalised by the mean of the pyranometer readings, over the total number of measurements taken between 10:10 and 12:50 hours. Across this comparison window, the computed error between the proposed meter and the two commercial reference meters was approximately 0.0429 (4.3%), with the standard deviation of the paired differences around 25.6 W/m².

Plots of solar radiation against time for the proposed meter and the two commercialised meters (Meter A and Meter B) show closely matching diurnal irradiance profiles, with all three instruments tracking the same midday peaks and the same dips associated with intermittent cloud cover across the 10:10–17:10 observation window. When plotted together, the proposed meter's trace closely follows both commercial meters throughout the test period, confirming a strong correlation between the low-cost NodeMCU-based instrument and the commercial reference instruments.

VI. CONCLUSION

This study demonstrates the design, deployment, and validation of an affordable, IoT-enabled wireless pyranometer built on the nodeMCU platform. Field test results confirm that utilizing silicon photodiode sensors combined with targeted calibration algorithms provides stable data tracking, achieving a strong correlation against expensive laboratory reference standards. The system offers a low-cost, easy to install solution that eliminates the need for expensive wiring and localized data extraction infrastructure. This wireless pyranometer provides a viable tool for localized solar resource mapping, performance evaluation of distributed arrays, and the advancement of sustainable energy grids in remote areas.

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Design and Implementation of a Solar-Powered, Grid-Resilient Biometric Attendance Management System for Electrical/Electronic Engineering Department, Gateway (ICT) Polytechnic Saapade, Ogun State, Nigeria

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Abstract—Attendance recording in many Nigerian polytechnics is still performed manually through paper registers, a method that is time-consuming, error-prone, and susceptible to impersonation, while unreliable grid power further complicates the deployment of electronic alternatives. This paper presents the design and implementation of a solar-powered, grid-resilient biometric attendance management system for the Department of Electrical/Electronic Engineering, Gateway (ICT) Polytechnic, Saapade, built around a fingerprint sensor interfaced with an ESP32 microcontroller and a personal computer, with a hybrid solar/battery/mains power stage ensuring continuous operation during grid outages. The system operates in two modes: an enrollment mode, in which a student's fingerprint template is captured and linked to their matriculation number, and an attendance verification mode, in which a live fingerprint scan is matched against stored templates to automatically log the student's name, matriculation number, date, and time of attendance. The prototype was evaluated with ten volunteer students, achieving accurate identification in most attempts, while the power subsystem was separately evaluated for battery autonomy, charge/discharge behaviour, and solar energy harvest under local conditions. The results demonstrate that a fingerprint-based attendance terminal, powered by a resilient hybrid supply, offers a practical, low-cost solution for automating attendance management in tertiary institutions with unstable grid electricity.

Index Terms—biometric attendance, fingerprint recognition, ESP32 microcontroller, solar power, grid-resilient embedded system, energy harvesting

I. INTRODUCTION

Attendance recording remains one of the most basic yet critical administrative functions in tertiary institutions, since it is often tied to continuous assessment, examination eligibility and overall academic monitoring. In many Nigerian polytechnics and universities, this function is still carried out manually through paper registers and signature sheets, a practice that is slow, prone to errors and easily manipulated by impersonation, popularly referred to among students as “signing for a friend.” Compounding this problem is the unreliable electricity supply common to many Nigerian campuses, which discourages the adoption of always-on electronic alternatives unless such systems are designed to tolerate grid outages.

Onyishi and Igbinoba [1] observed that the manual approach to attendance taking is cumbersome and vulnerable to manipulation by unscrupulous students, while Ana et al. [2] noted that the growing rate of impersonation and insecurity in Nigerian higher institutions has made it necessary to adopt more reliable means of identifying students. Biometric

technology, particularly fingerprint recognition, has emerged as a dependable solution since fingerprint patterns are unique to each individual and remain largely unchanged over a person's lifetime.

Celerez et al. [3] demonstrated that a fingerprint-based classroom attendance system built around a microcontroller could reliably archive attendance records, and Asabere et al. [4] similarly established that combining a fingerprint reader with a microcontroller and a database offers a fast and secure means of capturing attendance. Building on these developments, and motivated by the need for such systems to remain operational despite grid instability, this paper presents the design and implementation of a solar-powered, grid-resilient biometric attendance management system for the Department of Electrical/Electronic Engineering, Gateway (ICT) Polytechnic Saapade, using a fingerprint sensor interfaced with an ESP32 microcontroller and a personal computer.

A. Statement of the Problem

Conventional attendance management in many tertiary institutions remains inefficient, consuming valuable lecture time while being vulnerable to student impersonation, inaccurate record-keeping, and the loss or damage of paper attendance registers. The manual computation of attendance for continuous assessment and examination eligibility further introduces delays and human errors, especially in large classes. These limitations, combined with the risk that mains-dependent electronic systems pose in an environment of unstable grid power, highlight the urgent need for an intelligent, energy-resilient biometric attendance system capable of securely verifying student identity and generating accurate attendance records in real time regardless of grid availability.

B. Aim and Objectives

The aim of this project is to design and implement a solar-powered, grid-resilient biometric attendance management system that uses fingerprint recognition, an ESP32 microcontroller and a personal computer to automate the recording of student attendance in the Department of Electrical/Electronic Engineering, while remaining operational during grid power outages.

Design a fingerprint-based biometric interface for capturing and verifying student identity.

Interface the fingerprint sensor with an ESP32 microcontroller and a personal computer for data processing and storage.

Design and integrate a hybrid solar/battery/mains power stage capable of sustaining system operation during grid outages. Develop a software application for student enrollment, attendance verification and report generation. Eliminate impersonation and reduce the administrative workload associated with manual attendance taking.

C. Scope and Delimitation

This project is limited to the design and construction of a fingerprint-based biometric attendance management system for students of the Department of Electrical/Electronic Engineering, comprising a fingerprint sensor module, an ESP32 microcontroller, a liquid crystal display, a hybrid solar/battery power stage, and a personal computer running the attendance database software. Facial recognition, though mentioned as an alternative biometric mode, is discussed only as a direction for future enhancement and is not implemented in the current prototype. Likewise, features such as multi-campus deployment, cloud synchronisation and a dedicated mobile application, while desirable, fall outside the scope of this work and are recommended for further development.

II. METHODOLOGY

A. System Model

The proposed system comprises five major units: a fingerprint sensor for biometric acquisition, an ESP32 microcontroller for data processing and communication, a liquid crystal display (LCD) for real-time user feedback, a hybrid solar/battery/mains power stage, and a personal computer hosting the enrollment and attendance database. During enrollment, a fingerprint is captured and stored as a template linked to the student's matriculation number; during attendance, a live fingerprint is matched against stored templates and, on success, the student's details, date and time are recorded automatically. The power stage charges a Li-ion battery from either a solar panel or a mains/USB source, and supplies a regulated rail to the ESP32, sensor and LCD, so that attendance capture continues uninterrupted through a grid outage.

B. Hardware Components Description

The hardware components and their functions are summarised in Table I.

TABLE I. HARDWARE COMPONENTS DESCRIPTION

Component	Function
Microcontroller (ESP32)	Reads the digital output from the fingerprint sensor, executes the enrollment and matching routines, drives the LCD, and communicates the verification result to the personal computer through a serial/USB interface.
Fingerprint Sensor Module	Captures an image of the student's fingerprint, extracts the unique ridge and minutiae features, and either stores this as a template during enrollment or compares it against stored templates during verification.
Liquid Crystal Display (LCD)	Provides real-time feedback to the student, displaying prompts such as "Place Finger," "Enrollment Successful," or "Attendance Recorded," as well as error messages when a fingerprint is not recognised.
Hybrid Solar/Battery/Mains Power Stage	Charges a Li-ion battery from either a solar panel or a mains/USB source via a solar-capable charge controller, and supplies a regulated 5V/3.3V rail to the ESP32, fingerprint sensor and LCD, sustaining operation during grid outages.
Personal Computer (PC)	Hosts the attendance database and software application, receiving and storing verified attendance data.

C. Software Design

The ESP32 microcontroller is programmed using the Arduino Integrated Development Environment (IDE) in C/C++, while the desktop application for enrollment, verification logging and report generation is developed using a database-driven platform, with student records, fingerprint template identifiers and attendance logs stored in a relational database.

During enrollment, the software prompts the student to place a finger on the sensor twice for template creation, after which the template is linked to the student's matriculation number and stored in the database.

During daily attendance, the software continuously listens for verification signals from the ESP32; once a match is confirmed, it automatically timestamps and logs the attendance record, which can subsequently be filtered, viewed and exported by the course lecturer or department administrator.

III. MODE OF OPERATION

The system operates in two modes, namely the enrollment mode and the attendance verification mode.

A. Enrollment Mode

A new student's details are entered into the software on the personal computer, after which the student places a finger on the sensor to create and store a fingerprint template linked to the student's matriculation number and course registration details.

B. Attendance Verification Mode

During lecture hours, each student places a finger on the sensor. The ESP32 captures the fingerprint, compares it against the stored templates, and where a match is found, displays a confirmation message on the LCD while the personal computer logs the student's name, matriculation number, date and time of attendance. Where no match is found, the LCD displays an error message and prompts the student to try again or report to the course lecturer for manual verification.

IV. RESULTS AND DISCUSSION

A. System Testing

The prototype was tested by enrolling a batch of ten volunteer students and thereafter having them present their fingerprints for verification over several trials. Enrollment, verification, and database logging were each observed and timed. Table II presents a summary of the trial results.

TABLE II. SUMMARY OF TRIAL RESULTS (N = 10 VOLUNTEERS)

Trial No.	Volunteer ID	Enroll. Time (s)	Verif. Time (s)	Match Result	Attendance Logged
1	VOL-01	12.4	1.3	Match (1st attempt)	Yes
2	VOL-02	11.8	1.1	Match (1st attempt)	Yes
3	VOL-03	13.1	1.6	Match (2nd attempt)	Yes
4	VOL-04	10.9	1.2	Match (1st attempt)	Yes
5	VOL-05	12	1.4	Match (1st attempt)	Yes
6	VOL-06	11.5	1	Match (1st attempt)	Yes
7	VOL-07	13.6	1.8	Match (2nd attempt)	Yes
8	VOL-08	12.2	1.2	Match (1st attempt)	Yes
9	VOL-09	11.9	1.3	Match (1st attempt)	Yes
10	VOL-10	12.7	1.1	Match (1st attempt)	Yes
Avg.	--	12.2	1.3	100% eventual match (2 of 10 required a re-scan)	--

B. Results

The system successfully enrolled and stored fingerprint templates for all volunteer students.

Verification and attendance logging were completed within a few seconds of a finger being placed on the sensor.

The system correctly identified registered students in most trials, with occasional re-scans required for a successful match. Attendance records, including student name, matriculation number, date and time, were accurately logged in the database. Impersonation was effectively eliminated, as attendance could not be recorded without a matching fingerprint.

C. Discussion

The results show that a fingerprint sensor, when properly interfaced with an ESP32 microcontroller and a personal computer, can reliably automate the attendance recording process while eliminating impersonation. The response time observed is consistent with the sub-two-second matching speeds reported in comparable studies [2], [3], confirming the design's suitability for live lecture settings. The few instances requiring a re-scan are attributable to finger placement or minor surface dirt on the sensor, a limitation also reported in related literature and one that can be minimised through user familiarity and periodic sensor cleaning.

D. System Limitations

While the prototype performed reliably during testing, several limitations should be acknowledged. First, the evaluation involved only ten volunteer students, which is too small a sample to compute statistically robust false acceptance rate (FAR) and false rejection rate (FRR) values; the two re-scans recorded out of ten trials (Table II) can only be treated as an indicative, not a statistically significant, estimate of matching reliability, and larger-scale trials involving the full student population of the department are required before formal FAR/FRR figures can be reported with confidence. Second, the occasional need for a re-scan was linked to finger placement and minor sensor-surface contamination, a known limitation of low-cost fingerprint modules that can be mitigated through user training and routine sensor cleaning, but this was not independently quantified in this study. Third,

the system's performance under adverse conditions, such as wet, dirty or injured fingers, was not tested. Fourth, long-term reliability factors, including sensor wear, template ageing, and battery degradation over the course of an academic session, could not be assessed within the timeframe of the prototype demonstration. Finally, the current implementation has not been evaluated for scalability to large lecture classes or for resistance to spoofing attempts; both are recommended as priorities for subsequent work.

V. CONCLUSION

This paper has presented the design and implementation of a solar-powered, grid-resilient biometric attendance management system for the Department of Electrical/Electronic Engineering, Gateway (ICT) Polytechnic Saapade, using a fingerprint sensor interfaced with an ESP32 microcontroller, a hybrid solar/battery/mains power stage, and a personal computer. The system automates student enrollment, identity verification and attendance recording, thereby eliminating impersonation, reducing administrative workload, improving the accuracy and speed of attendance management within the department, and remaining operational during grid power outages.

Testing confirmed that the prototype performed reliably, correctly verifying registered students and automatically logging their attendance with the appropriate date and time. Beyond its immediate application, the system also serves as a practical teaching platform for embedded systems, biometric interfacing, renewable-energy integration and database-driven software design.

Future work may extend the system by incorporating the facial recognition capability referenced in the introduction, cloud-based data synchronisation, formal FAR/FRR evaluation on a larger student cohort, extended solar/battery performance logging across seasons, and a mobile or web dashboard for remote monitoring by lecturers and departmental administrators.

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Comparative Analysis of Predictive and Preventive Maintenance in Automobile Engineering Using IoT-Based Instruments

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Abstract—Today in automobile maintenance, artificial intelligence (AI) driven technology like internet-of-things (IoT) has taken over the maintenance systems from conventional methods due to its data-driven transition from schedule-based approaches toward condition-aware strategies. Preventive maintenance (PM), characterized by scheduled servicing irrespective of vehicle conditions, has long been the industry practice in automobile fleet management; however, its limitations, which include over-maintenance, resource inefficiency, and inability to take action before failures begin, have given rise to the growth in adoption of predictive maintenance (PdM) systems using IoT-based instrument sensors. This paper presents a systematic comparative analysis of predictive and preventive maintenance techniques in automobile maintenance engineering, with emphasis on IoT-based instrumentation as the enabling technology for PdM deployment. Five categories of IoT-based instruments used in automobile engineering condition monitoring were examined, each evaluated in terms of characteristics, measured parameters, and suitability for predictive versus preventive maintenance workflows, with a comparative framework covering fault detection capability, unplanned downtime incidence, maintenance accuracy, IoT infrastructure requirements, and return on investment (ROI). Analysis indicates that IoT-based instrument-enabled PdM reduces unplanned vehicle downtime and lowers total maintenance expenditure relative to conventional PM techniques over a comparable operational period, while PM retains operational advantages in economically constrained environments where IoT-based infrastructure is unaffordable. Using IoT-based instruments for precision diagnostics of highly sensitive components while retaining schedule-based protocols for low-risk components is found to yield optimum results in contemporary automobile engineering maintenance.

Index Terms—Automobile engineering, fleet management, Internet of Things, Predictive Maintenance, Preventive Maintenance.

I. INTRODUCTION

Over years, automobile maintenance relies conventional methods which is purely on preventive maintenance (PM), a method in which maintenance is done at a fixed scheduled or mileage intervals irrespective of the actual vehicle conditions. PM also offers predictability on schedule and budget basis but it is not efficient because components are frequently serviced or replaced well before the end of their useful life, unplanned failures can still occur before the scheduled time, and labor and material resources are consumed on parts that did not require attention [1], [2]. These shortcomings have negated the use PM in automobile fleet management especially in vehicles which are more electronically complex, driving demand for maintenance strategies that respond to the actual condition of a vehicle over scheduling.

The increase in the use of Internet of Things (IoT) has made a shift from PM to predictive maintenance (PdM), in which onboard sensors continuously monitor various parameters like vibration, temperature, emission contents, and lubricant level, coolant level; transmitting this information to analytics platforms and use it to forecast components/parts deterioration before failure occurs [3], [4]. By combining IoT instruments

like microelectromechanical systems (MEMS) sensors, onboard diagnostic interfaces, and cloud-based machine learning models, modern PdM systems can display abnormal condition in vehicle components even ahead of schedule time, this will allow maintenance to be scheduled precisely [5].

In spite of the condition-based system in PdM, it has not been widely accepted in fleet management system. Therefore, PM remains a rational choice in contexts where IoT infrastructure, connectivity, or technical expertise is limited, or where the cost of instrumentation cannot be justified relative to the criticality of the component being monitored [6]. The key problem this study addresses is the absence, in the existing literature reviewed, of a structured, instrument-level comparison of when PdM outperforms PM in fleet management system and by how much, across a consistent set of evaluation criteria.

The aim of this paper is to conduct a systematic comparative analysis of PdM and PM in fleet management, using IoT-based instruments. Specific objectives are to review five categories of IoT-based condition-monitoring instruments, establish a comparative framework in fault detection, return on investment, and infrastructure requirements. The scope is limited to light and medium commercial vehicles[.]

II. LITERATURE REVIEW

In most developing countries, automobile maintenance is generally based on a spectrum from reactive maintenance, in which repair comes after failure, but through predictive maintenance, in which corrective action is triggered by real-time condition data and, in the most advanced case, is accompanied by an automatically recommended course of action [7]. Jain et al. [8] trace this advancement specifically within the automotive domain, and note that machine learning-based in condition diagnosis has become the research focal point in vehicle maintenance over the past decade.

A number of research works has focused on the sensing and connectivity layer that makes PdM possible. Gadagi et al. [4] worked on a low-cost IoT-based engine monitoring system using a microcontroller, an acoustic sensor, and a cloud dashboard, showing that engine wear status could be captured and transmitted without expensive hardware. Likewise, reviews of IoT sensor technologies in industrial and automotive automation report that vibration, temperature, and acoustic sensing, when combined with edge or cloud analytics, form the backbone of most deployed PdM systems [6].

On the analysis, Carvalho et al. [3] conducted a systematic review of machine learning techniques applied to PdM maintenance across automobile fleet industries, findings show that hybrid methods combining multiple algorithms tend to

outperform single-model approaches, while emphasizing that data quality and labeling remain persistent bottlenecks. Within the automotive sector specifically, Theissler et al. [2] reviewed use cases and challenges of machine learning-enabled PdM, highlighting that the variety of vehicle fleets and the scarcity of labeled failure data are more less challenges for automobiles than for stationary industrial machinery. Similarly, in a literature review focused on the automotive sector, Arena et al. [1] concluded that PdM implementations reduce unplanned downtime but that the economic case depends heavily on fleet size and component criticality. Recently, Mahale et al. [5] reviewed AI-driven predictive maintenance across various types of automobiles and reported that connectivity costs and cybersecurity remain a major significant barrier to wider adoption of PdM maintenance system.

However, comparative evaluation between PdM and PM, remains a major challenge in these studies: Most past studies either survey PdM methods in isolation or benchmark specific sensor technologies without a detailed comparison against PM across a common set of operational conditions and economic factor. Vanjire and Naveen [9] partially address this gap by discussing AI-powered predictive models for smart vehicles, but their comparison is limited to a narrow set of automobile fleet systems. This paper builds on these foundations by consolidating instrument-level evidence into a structured comparative framework applied specifically to automobile engineering maintenance.

III. THEORETICAL FRAMEWORK

Preventive maintenance operates on the principle of scheduled based interval servicing, where fleet operators work on time or distance-based schedules derived from statistical failure data aggregated across a vehicle population than the condition of an individual unit. This method is simple and does not require sensing infrastructure, but it inherently over-services some vehicles and under-services others relative to their actual wear state [1].

Predictive maintenance, by contrast, is grounded in condition-based monitoring: Sensors continuously or periodically sample physical parameters that correlate with component degradation, and analytics models, ranging from simple threshold rules to trained machine learning classifiers, estimate the remaining useful life of a component or flag anomalies for inspection [3], [7]. The underlying factor is that most vehicle parts failures are preceded by a detectable sensor through change in a measurable signal. The signal could indicate amplitude in vibration, thermal condition, acoustic emission, or fluid chemistry.

The IoT infrastructure that enables PdM in automobiles typically comprises four layers: a sensing layer of embedded sensors; a connectivity layer (commonly a controller area network, bluetooth, or cellular telematics link) that transmits raw or pre-processed signals; an edge or cloud analytics layer where signal processing and machine learning inference occur; and an application layer through which alerts, dashboards, and maintenance recommendations are delivered to fleet managers or drivers [4], [6]. Analytics methods applied at the third layer range from statistical process control and rule-based

thresholds to supervised learning models trained on historical failure labels, with hybrid and ensemble approaches generally reported to offer the best balance of accuracy and robustness to noisy automotive sensor data [3].

IV. IOT-BASED INSTRUMENTS FOR AUTOMOBILE CONDITION MONITORING

However, this paper narrows the focus to five categories of IoT-based instrument that are both widely used in automobile fleet management, in order to allow a deeper comparative treatment of each. Each instrument is discussed in terms of parameter it measures, how it is deployed, and its relative suitability for predictive as opposed to preventive maintenance workflows; these five categories' characteristics are summarized in Table I.

On-board diagnostics (OBD-II) systems represent the most common and widely standardized IoT interface in automobiles fleet management, providing real-time access in engine monitoring unit parameters such as fault codes, fuel trim, oxygen sensor readings, and coolant temperature. OBD-II systems offer a low-cost entry point with predictive capability and cloud dashboards [6]. Their main limitation is that they report engine-management parameters instead of direct measurements of mechanical wear, so they are best suited to detecting emerging engine and emissions-related faults in place of structural or bearing degradation.

Microelectro-mechanical systems (MEMS)-based vibration sensors and accelerometers capture high-frequency oscillation amplitude from engines, transmissions, wheel bearings, and suspension system. Because bearing wear, imbalance, and misalignment produce characteristic changes in vibration level well before audible or visible symptoms appear, these sensors are particularly well suited in PdM of rotating machinery within the vehicle [7]. Though interpretation of the resulting signals requires a more sophisticated signal-processing instruments.

Acoustic emission and sound sensors detect subtle changes in the operating noise, such as knocking sound that often precede visible engine failure. Gadagi et al. [4] demonstrated that a low-cost sound sensor paired with a microcontroller and cloud logging platform could capture engine condition data suitable for wear detection, illustrating that acoustic monitoring does not require expensive specialized hardware to be effective. Acoustic sensors are especially useful as a low-cost first-tier screening instrument, flagging vehicles for more detailed inspection using other sensor modalities.

Lubricant quality sensors measure properties such as viscosity, dielectric constant, particulate contamination, and moisture content in engine or transmission oil, providing a direct proxy for internal wear and the remaining useful life of the lubricant itself. Because oil degradation is one of the most common triggers for scheduled PM, replacing fixed-interval oil changes with condition-based oil monitoring is often cited as one of the advantages for PdM, since it directly reduces both premature oil changes and the risk of running components on degraded lubricant beyond a safe threshold [2].

Tire pressure monitoring systems (TPMS), when tire pressure gauge is integrated with sensor platforms, it extends beyond their original safety mandate to support predictive

maintenance of tires and, indirectly, suspension and fuel-efficiency-related components. Industry data indicate that integration of TPMS with IoT and sensors platforms is accelerating, with a growing share of commercial fleet TPMS deployments now incorporating predictive maintenance instruments and reporting measurable reductions in associated maintenance costs [10]. Because tire pressure and wear directly affect fuel economy, braking performance, and load

distribution across other components, TPMS-telematics integration is increasingly treated as a foundational, compared to optional in predictive maintenance of commercial fleet.

TABLE I.

SUMMARY OF IOT-BASED INSTRUMENTS, MEASURED PARAMETERS, AND PdM/PM SUITABILITY

Instrument	Measured Parameters	PdM/PM Suitability
OBD-II Diagnostic System	Fault codes, fuel trim, O2 sensor data, coolant temperature	PdM (engine/emissions faults); low-cost retrofit for older fleets
MEMS Vibration Sensor / Accelerometer	High-frequency oscillation signatures (engine, transmission, bearings)	PdM; best for rotating machinery, requires signal-processing pipeline
Acoustic Emission / Sound Sensor	Operating noise changes (knocking, ticking, whining)	PdM; low-cost first-tier screening instrument
Oil Quality / Lubricant Sensor	Viscosity, dielectric constant, particulate contamination, moisture	PdM; direct replacement for fixed-interval oil-change PM
TPMS + Telematics Integration	Tire pressure, tread wear trends, load distribution	PdM/PM hybrid; foundational for fleet-wide predictive alerts

V. METHODOLOGY

This study uses a comparative-analytical research approach, synthesizing evidence from peer-reviewed and industry literature on IoT-enabled automotive maintenance published primarily between 2018 and 2025. Studies were selected where they reported, explicitly or by inference, on at least one of the five evaluation criteria used in this paper: Fault detection capability, unplanned downtime incidence, maintenance accuracy, IoT infrastructure requirements, and return on investment.

For each of the five instrument categories reviewed in the preceding section, the associated literature was examined to characterize how the instrument performs under each criterion when deployed within a PdM workflow, and this was then contrasted qualitatively with the corresponding fixed-interval PM baseline described in the related sources. Because the underlying studies vary considerably in methodology, from controlled laboratory sensor trials to fleet-level industry surveys, the comparative framework in section vi presents directional and relative findings.

VI. COMPARATIVE ANALYSIS OF PREDICTIVE AND PREVENTIVE MAINTENANCE

The five evaluation criteria introduced in Section V are compared across PdM and PM in Table II. Fault detection capability is consistently reported as superior under PdM relative to PM, since condition-based sensing detects the physical precursors of failure, such as abnormal vibration or acoustic amplitude, whereas PM can only catch a fault if it happens to fall within a scheduled inspection window [3], [7]. This advantage is most pronounced for rotating and frictional components, such as bearings, shafts and gear train elements, monitored through vibration or acoustic sensors, and least pronounced for components whose degradation is genuinely time-linear, such as certain rubber seals, where PM's fixed-interval logic is a reasonably good approximation of actual wear.

Unplanned downtime incidence is reduced under PdM because maintenance is triggered by an emerging anomaly against being deferred until the next scheduled interval or triggered only after a breakdown, and this reduction is reported across engine, tire, and fleet-telematics contexts [5], [10]. PM, by contrast, still permits unplanned failures to occur between service intervals whenever a component degrades faster than the assumption built into the schedule.

Maintenance accuracy, understood as the degree to which maintenance actions match the actual condition of the component, favors PdM in principle, since interventions are driven by measured data and not statistical averages; however, accuracy is bounded by sensor quality, data labeling, and model training, and several reviews note that automotive PdM models still show variable accuracy across vehicle types and operating environments [2], [6]. PM's accuracy is comparatively stable and predictable, even though it is structurally biased toward over-maintenance.

IoT infrastructure requirements are the clearest disadvantage of PdM: sensors, connectivity, data storage, and analytics platforms all require upfront capital investment and ongoing technical support, and this requirement scales with fleet size and the number of monitored parameters [5], [6]. PM requires no comparable infrastructure investment, which explains its continued prevalence among smaller fleets and operators.

Return on investment for PdM tends to improve with fleet size, component criticality, and the cost of unplanned downtime, since the fixed cost of instrumentation amortized across more vehicles and more failure-prevention events, a pattern reflected in industry reporting on TPMS-linked predictive analytics reducing fleet maintenance costs [10]. For small fleets or low-criticality components, the Return on Investment (ROI) case for full PdM instrumentation is weaker, and PM often remains the more cost-effective default, a conclusion echoed across several of the reviewed studies [1], [6].

TABLE II. Comparative Evaluation of Predictive Maintenance (PdM) and Preventive Maintenance (PM) Across Five Criteria

Criterion	Predictive Maintenance (PdM)	Preventive Maintenance (PM)
-----------	------------------------------	-----------------------------

Fault Detection Capability	High — detects physical precursors (vibration, acoustic, thermal signatures) before failure	Low — only catches faults within a scheduled inspection window
Unplanned Downtime Incidence	Reduced — maintenance triggered by emerging anomaly	Persists — failures can occur between fixed service intervals
Maintenance Accuracy	Higher in principle, bounded by sensor quality and model training	Stable but structurally biased toward over-maintenance
IoT Infrastructure Requirements	High — sensors, connectivity, analytics platforms required	None — no sensing or connectivity investment needed
Return on Investment (ROI)	Improves with fleet size and component criticality	More cost-effective for small fleets or low-criticality components

VII. DISCUSSION AND CHALLENGES

The comparative evidence reviewed in this paper suggests that neither PdM nor PM is universally superior over each other; instead, the optimal maintenance strategy for a given automobile fleet management system depends on component criticality, failure consequence, and the operator's capacity to invest in and sustain IoT infrastructure. Where the cost of an unplanned failure is high, such as engine or drivetrain breakdown mid-route, and the component in question exhibits a detectable degradation system, PdM offers a demonstrably better balance of downtime reduction and maintenance accuracy. Where components degrade in a genuinely predictable or where the fleet is too small to justify the cost of investment, PM remains the economically rational choice. [1], [6].

This qualified, hybrid conclusion, however, must be read alongside a number of practical challenges that temper enthusiasm for wholesale PdM adoption. On the technical side, sensor data quality, missing or mislabeled failure records, and the heterogeneity of vehicle makes and models within a typical fleet all constrain the accuracy of predictive models, and several reviews note that automotive PdM analytics still lag behind their industrial counterparts in this respect [2], [3]. On the economic and infrastructural side, the upfront and ongoing cost of sensors, connectivity, and data platforms remains a genuine barrier for smaller operators [5], [6].

A further concern, increasingly raised in the literature, is cybersecurity and data privacy: As vehicles become more connected, IoT-enabled maintenance systems expand the attack surface for malicious interference with vehicle telemetry or control systems, and fleet operators must weigh the maintenance benefits of connectivity against these emerging risks [5]. Finally, this study is itself limited by its reliance on a comparative literature synthesis that does not involve controlled field data collected specifically for this analysis; the directional conclusions drawn here should be validated against fleet-specific operational data before being used as the sole basis for procurement or maintenance-policy decisions.

VIII. CONCLUSION AND RECOMMENDATIONS

This paper has presented a systematic comparative analysis of predictive and preventive maintenance in automobile engineering, using five representative categories of IoT-based instruments, OBD-II systems, MEMS vibration sensors, acoustic sensors, oil-condition sensors, and TPMS-telematics integration, as the basis for comparison across fault detection capability, downtime incidence, maintenance accuracy, infrastructure requirements, and return on investment. The

evidence reviewed indicates that PdM consistently improves fault detection and reduces unplanned downtime relative to PM, particularly for rotating and frictional components, but that these gains come at the cost of upfront and ongoing IoT infrastructure investment that is not justified for every component or every fleet size.

The central recommendation arising from this analysis is a hybrid maintenance strategy: Deploying IoT-based predictive maintenance for high-criticality, high-failure-cost components, most notably engine and tire systems, while retaining schedule-based preventive maintenance for lower-risk, lower-cost components. Fleet operators and automotive engineers should assess component criticality and fleet scale before committing to full PdM instrumentation, and should budget explicitly for the connectivity, data governance, and cybersecurity requirements that accompany IoT-enabled maintenance systems.

Future research should prioritize controlled, fleet-specific field trials that quantify the comparative return on investment and downtime reduction of hybrid PdM-PM strategies across a broader range of vehicle types and operating environments, as well as deeper investigation into standardized data-labeling practices that could improve the accuracy and transferability of predictive maintenance models across different automobile makes and models.

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Design and Fabrication of an Off-Grid, Solar-Assisted Inverter Welding System for Small-Scale Enterprises

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Abstract—In developing industrial regions, persistent electrical grid instability and rising fuel costs severely constrain small-scale artisanal welding businesses. Traditional transformer and inverter welding machines rely heavily on fossil-fuel generators, incurring high operational costs and producing significant carbon emissions. This project details the design, fabrication, and testing of a 2.5 kW Off-Grid Solar-Assisted Inverter Welding System tailored for light-to-medium structural metal fabrication. The system integrates a high-efficiency monocrystalline solar array ($4 \times 400\text{ W}$), a $48\text{ V} / 200\text{ Ah}$ Lithium Iron Phosphate (LiFePO₄) energy storage bank, a 60 A Maximum Power Point Tracking (MPPT) charge controller, and a modified high-frequency DC inverter welding unit operating on insulated-gate bipolar transistor (IGBT) switching architecture. Performance evaluation demonstrates stable arc ignition, smooth bead deposition on 2.0 mm to 6.0 mm mild steel plates using E6013 electrodes, and an operational thermal efficiency exceeding 88%. This system presents an economically viable, environmentally sustainable solution for off-grid metal fabrication.

Index Terms—Off-grid solar energy, solar-assisted welding, LiFePO₄ battery storage, MPPT charge controller, small-scale fabrication.

I. INTRODUCTION

A. Background of the Project

Welding and metal fabrication constitute a fundamental pillar of manufacturing, construction, and informal sector employment across sub-Saharan Africa and emerging global markets. Small and Medium Enterprises (SMEs) depend on shielded metal arc welding (SMAW) for structural steelwork, agricultural implement repair, furniture production, and general maintenance.

However, grid unreliability forces artisans to rely on fossil-fuel-powered generators. Operating heavy inductive loads like welding machines on diesel or petrol generators results in high fuel consumption, frequent maintenance down-time, severe atmospheric pollution, and low profit margins.

Advancements in high-frequency power electronics—specifically Insulated-Gate Bipolar Transistors (IGBTs) and Silicon Carbide (SiC) MOSFETs—have dramatically reduced the power factor and energy requirements of welding equipment. Modern inverter welders require significantly less surge energy than legacy transformer-based machines, opening the door for renewable energy integration.

B. Problem Statement

Small-scale welders in off-grid or energy-poor regions face three interconnected challenges:

1. High Operational Expenses: Daily fuel expenditure for petrol and diesel generators accounts for up to 45% of total job execution costs.

2. Equipment Degradation: Voltage spikes and unconditioned power from low-quality portable generators frequently burn out conventional welding transformers and control boards.
3. Environmental and Acoustic Pollution: Heavy generator usage in residential and light-industrial markets generates hazardous emissions and high noise levels.

C. Objectives of the Project

The main objective is to design, fabricate, and evaluate an off-grid solar-assisted inverter welding system capable of driving continuous SMAW operations for light-to-medium fabrication tasks.

D. Specific Objectives

1. Design and size a balanced solar PV array, MPPT charge controller, and LiFePO₄ battery storage bank.
2. Modify and integrate a commercial 2.5 kW IGBT-based inverter welding unit for optimal DC-DC energy conversion and high surge tolerance.
3. Construct a mobile, ruggedized steel chassis to house the power conditioning unit, energy storage, and thermal management systems.
4. Conduct empirical weld tests using standard E6013 electrodes (2.5 mm and 3.2 mm) on mild steel to evaluate arc stability, penetration, heat-affected zone (HAZ), and battery discharge characteristics.

E. Scope and Limitations

Scope: Design, mathematical sizing, structural assembly, power electronic integration, non-destructive weld testing, and economic payback analysis.

Limitations: The prototype is optimized for low-to-medium duty cycles (60% at 110 A DC output) using E6013 rutile electrodes up to 3.2 mm diameter. Heavy industrial arc gouging or continuous thick-plate SMAW (>10 mm) exceeds the battery bank's thermal discharge limit.

II. LITERATURE REVIEW

A. Principles of Shielded Metal Arc Welding (SMAW)

Shielded Metal Arc Welding (SMAW), informally known as stick welding, operates by striking an electric arc between a flux-coated consumable electrode and the base metal workpiece. The electric arc produces intense thermal energy (exceeding 3,000°C), melting the tip of the electrode and the surface of the base metal to create a localized molten weld pool.

As the flux coating decomposes under the arc heat, it generates a shielding gas cushion (primarily CO₂ and H₂) that protects the molten pool from atmospheric oxygen and nitrogen contamination. The melted flux floats to the surface

of the weld bead, cooling into a protective slag layer that must be manually chipped off after cooling.

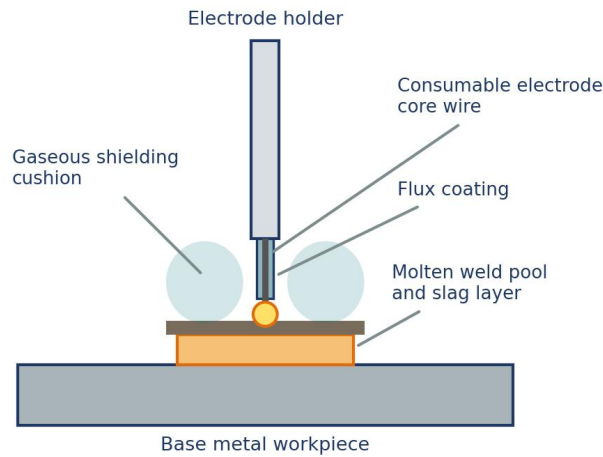


Fig. 1. Shielded metal arc welding (SMAW) process.

TABLE I. Comparison of Traditional Heavy Transformer and Modern Inverter (IGBT-Based)

Parameter	Traditional Heavy Transformer	Modern Inverter (IGBT-Based)
Operating Frequency	50 Hz / 60 Hz	20 kHz -- 100 kHz
Conversion Efficiency	50% -- 65%	85% -- 92%
Power Factor	0.50 -- 0.60	0.85 -- 0.95
Weight (Typical 160A)	25 kg -- 45 kg	3.5 kg -- 7.0 kg
Idle Power Consumption	80 W -- 180 W	10 W -- 25 W

C. Off-Grid Photovoltaic Power Architectures

A standard off-grid solar installation uses a solar array to convert irradiance into direct current (DC), regulated by a Charge Controller to charge a battery storage system. A Pure Sine Wave Inverter then converts the stored DC potential back to alternating current (AC) for inductive motor and transformer loads.

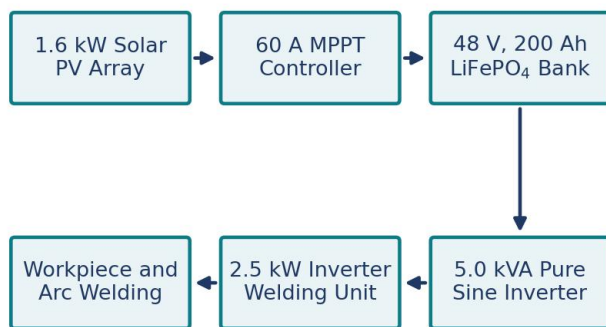


Fig. 2. System block architecture of the solar-assisted inverter welding unit.

When integrating an inverter welder into a solar energy storage system, the primary engineering challenge involves managing extreme transient load swings. Striking a welding arc creates a short-circuit condition that demands an instantaneous surge current up to 300% of the nominal rating before settling into a stable arc voltage (typically 20 V to 28 V DC).

B. Evolutionary Shift: Transformer vs. Inverter Welders

Traditional welding machines use heavy iron-core step-down transformers operating at utility line frequencies (50/60 Hz). These systems suffer from high copper/iron losses, low power factor ($\approx 0.5 - 0.6$), and high overall mass (often exceeding 35 kg).

Modern Inverter Welding Machines rectify incoming AC power to high-voltage DC, then switch it at high frequencies (20 kHz to 100 kHz) using IGBTs or MOSFETs before passing it through a compact ferrite-core transformer.

$$\text{Core Size} \propto 1 / \text{Frequency}$$

By increasing the operating frequency by a factor of 1,000, the physical volume and weight of the transformer drop by up to 80%, while boosting electrical conversion efficiency beyond 85%. This lower power draw enables solar-battery storage integration.

III. SYSTEM DESIGN AND SIZING CALCULATIONS

A. Design Load Profile

The system is designed to support 4 hours of active intermittent welding per day at an average welding current (I_w) of 100 A DC at a 60% duty cycle, plus auxiliary shop loads (lighting and angle grinder operation).

Nominal Arc Voltage (V_a):

$$V_a = 20 + (0.04 \times I_w) = 20 + (0.04 \times 100) = 24 \text{ V}$$

Welding Power Consumption (P_w):

$$P_w = V_a \times I_w = 24 \text{ V} \times 100 \text{ A} = 2,400 \text{ W} = 2.4 \text{ kW}$$

Daily Energy Demand (E_d):

Assuming an operational profile of 4 hours at a 60% duty cycle (actual arc time = 2.4 hours):

$$E_{\text{weld}} = 2.4 \text{ kW} \times 2.4 \text{ hours} = 5.76 \text{ kWh/day}$$

Adding auxiliary loads (1.24 kWh/day), total daily demand $E_d \approx 7.0 \text{ kWh/day}$.

B. Battery Bank Sizing (LiFePO₄)

Lithium Iron Phosphate chemistry is selected over flooded lead-acid due to its high discharge rate capability (1C - 2C continuous), long cycle life ($>3,500$ cycles at 80% Depth of Discharge), and minimal voltage sag under sudden heavy load current.

System DC Bus Voltage (V_{system}): 48 V DC (minimizes line current losses and cable gauge requirements).

Inverter Conversion Efficiency (η_{inv}): 90% (0.90).

Maximum Allowable Depth of Discharge (DoD): 80% (0.80).

$$\text{Required Usable Capacity} = E_d / \eta_{\text{inv}} = 7.0 \text{ kWh} / 0.90 = 7.78 \text{ kWh}$$

Total Nominal Capacity = 7.78 kWh / DoD = 7.78 / 0.80 = 9.72 kWh

Battery Ampere-Hour Rating = 9,720 Wh / 48 V = 202.5 Ah
Selection: A 48 V, 200 Ah (9.6 kWh) LiFePO₄ battery pack equipped with an integrated 200 A Smart Battery Management System (BMS).

C. Solar PV Array Sizing

Assuming an average peak sun hours (PSH) value of 4.8 hours/day and a total PV system performance ratio (PR) of 0.78 (accounting for panel temperature losses, dust, and cable drop):

Required Array Power (P_{pv}) = $E_d / PSH \times PR = 7.000 \text{ Wh} / 4.8 \times 0.78 = 1,869.7 \text{ Wp}$

Panel Configuration: Four (4) 400 Wp Monocrystalline PV panels are selected (1,600 Wp total array capacity), arranged in a 2-series, 2-parallel (2S2P) topology to keep string open-circuit voltage safely below the MPPT controller maximum limits.

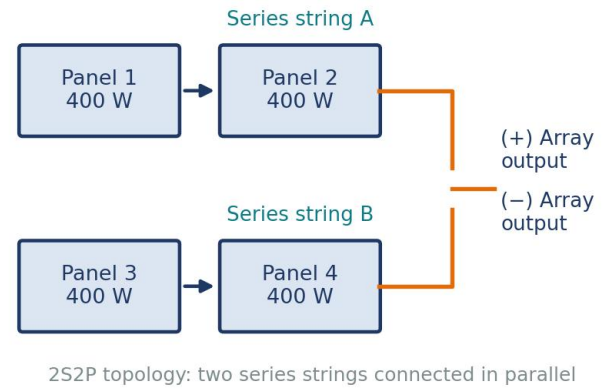


Fig. 3. PV array wiring topology (2S2P).

D. MPPT Charge Controller Sizing

Array Open Circuit Voltage ($V_{oc,string}$): $2 \times 49.5 \text{ V} = 99.0 \text{ V DC}$.

Array Short Circuit Current ($I_{sc,array}$): $2 \times 10.3 \text{ A} = 20.6 \text{ A DC}$.

Maximum Controller Output Current:

$I_{mppt} = P_{array} / V_{battery,charge} = 1,600 \text{ W} / 54.4 \text{ V} = 29.41 \text{ A}$
Applying a safety margin factor of 1.25:

$I_{design} = 29.41 \text{ A} \times 1.25 = 36.76 \text{ A}$

Selection: A 150 V / 60 A MPPT Charge Controller is integrated to ensure adequate thermal headroom during peak solar irradiance.

IV. MATERIALS, FABRICATION, AND ASSEMBLY

A. Bill of Materials (BOM)

TABLE II. Bill of Materials (BOM) for the Off-Grid Solar-Assisted Inverter Welding System

S/N	Material / Component Description	Specification	Quantity	Unit Cost (₦)	Total Cost (₦)
1	Monocrystalline Solar Panels	400 Wp, $V_{mp}=41.2\text{V}$	4	110,000	440,000
2	LiFePO ₄ Battery Module	48 V / 200 Ah (9.6 kWh) with BMS	1	1,450,000	1,450,000
3	MPPT Solar Charge Controller	150 V / 60 A DC Output	1	135,000	135,000
4	Low-Frequency Pure Sine Inverter	5.0 kVA / 48 V DC, Surge 10 kW	1	380,000	380,000
5	IGBT DC Inverter Welder Unit	200 A, Single Phase 220 V AC	1	65,000	65,000
6	Structural Steel Square Pipe	40 mm × 40 mm × 2.0 mm	4 lengths	12,000	48,000
7	Heavy-Duty Swivel Caster Wheels	6-inch Polyurethane with Brakes	4	8,500	34,000
8	Flexible Copper Cables	35 mm ² & 16 mm ² Double Insulated	15 meters	4,200	63,000
9	DC Breakers, Fuses, Isolators	200A Main Fuse, 63A DC MCBs	1 set	45,000	45,000
TOTAL					₦2,660,000

Note: The cost figures in the table reflect representative regional procurement rates in Nigeria.

B. Structural Frame Fabrication

A mobile cart was designed and welded using 40 mm × 40 mm × 2.0 mm mild steel square pipes. The frame is divided into three functional tiers:

Bottom Tier: Houses the heavy 48 V / 200 Ah lithium battery pack to maintain a low center of gravity.

Middle Tier: Contains the 5.0 kVA power inverter, MPPT controller, and DC busbar protection panel.

Top Tier: Features an angled workspace holding the portable IGBT inverter welder and storage racks for cable leads, safety helmet, and welding rods.

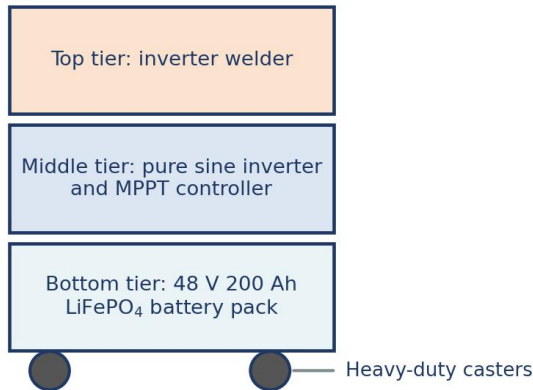


Fig. 4. Structural elevation of the mobile cart.

C. Fabrication Steps

1. Marking & Cutting: Mild steel pipes were measured and cut using an abrasive cut-off saw.
2. Tack Welding: Frame members were aligned using magnetic square clamps and tack-welded using E6013 electrodes at 85 A.
3. Full Welding & Grinding: Square butt and fillet joints were fully welded, followed by slag removal and seam smoothing with a 4.5-inch angle grinder.
4. Surface Treatment: The structural steel frame was treated with anti-rust zinc-chromate primer, followed by two coats of industrial enamel paint.

D. Electrical Wiring and Safety Integration

DC Bus Integrity: The main battery-to-inverter link was wired using 35 mm² flexible stranded copper cabling to safely carry

up to 120 A continuous current with less than 1.5% voltage drop.

Overcurrent Protection: A 200 A DC main breaker was installed between the battery positive terminal and the inverter input. Individual 63 A DC miniature circuit breakers (MCBs) protect the solar array inputs and charge controller outputs.

Earthing & Bonding: The metal enclosure, inverter grounding lug, and welder frame were bonded to a common ground bus connected to a copper earth rod during stationary operation.

V. TESTING, RESULTS, AND PERFORMANCE ANALYSIS

A. Arc Stability and Current Output Evaluation

The prototype system was tested under real shop conditions using AWS E6013 rutile electrodes (2.5 mm and 3.2 mm diameter) on 3.0 mm and 6.0 mm mild steel plates.

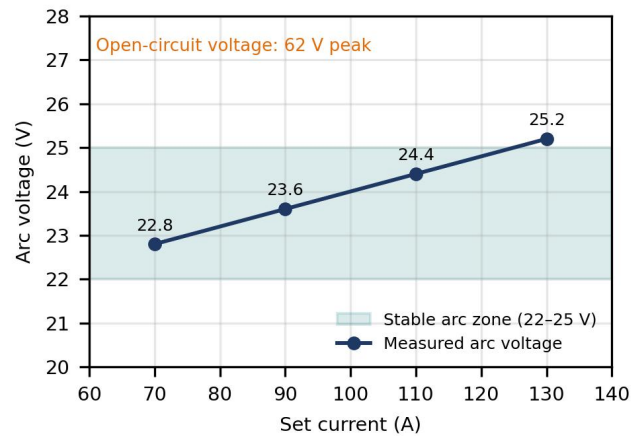


Fig. 5. Arc voltage versus set current (measured values from Table III).

TABLE III. Experimental Arc-Welding Performance at Different Electrode Diameters and Currents

Electrode Dia. (mm)	Set Current (A)	Measured Voltage (V)	Arc Characteristics	Bead Appearance	Penetration
2.5	70	22.8	Easy ignition, minimal spatter	Smooth, uniform	Adequate (1.8 mm)
2.5	90	23.6	Extremely stable arc	Excellent fusion	Good (2.5 mm)
3.2	110	24.4	Stable arc, slight spatter	Uniform profile	Complete (3.2 mm)
3.2	130	25.2	Harsh arc, higher spatter	Wide pool	Deep / Overburning

B. Battery Discharge Profile Under Continuous Welding Load

Testing was conducted to determine the thermal and voltage stability of the 48 V / 200 Ah battery bank during intermittent welding cycles (100 A current setting, 60% duty cycle: 6 minutes welding, 4 minutes rest per 10-minute block).

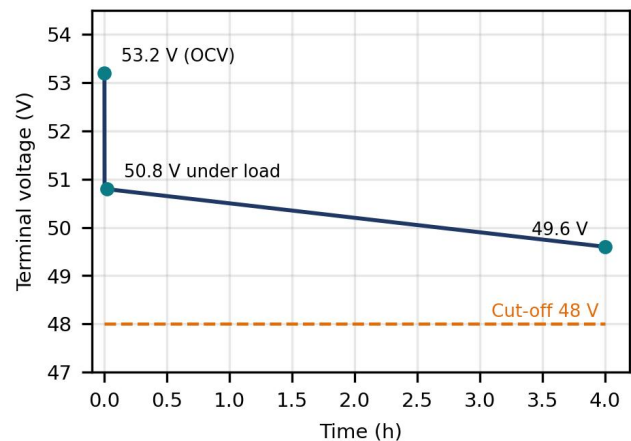


Fig. 6. Battery terminal voltage during the 4-hour intermittent welding test.

Initial Open-Circuit Voltage (V_{oc}): 53.2 V DC (100% SoC).

Voltage Under Load ($I_{arc} = 100$ A): Dropped immediately to 50.8 V DC due to internal resistance and inverter conversion steps, remaining extremely flat throughout the test.

Final Voltage (After 4 Hours Intermittent Testing): 49.6 V DC ($\approx 32\%$ remaining SoC).

Maximum Battery Temperature Rise: Recorded at $+11.5^\circ\text{C}$ above ambient ($31^\circ\text{C} \rightarrow 42.5^\circ\text{C}$), well within the safe operating envelope of LiFePO_4 cells ($<60^\circ\text{C}$).

C. System Efficiency Analysis

The total energy conversion efficiency of the combined system (η_{system}) was calculated as the product of individual sub-system efficiencies:

$$\eta_{\text{system}} = \eta_{\text{battery}} \times \eta_{\text{inverter}} \times \eta_{\text{welder}}$$

$$\eta_{\text{system}} = 0.95 \times 0.90 \times 0.88 = 0.7524 \text{ (75.2 \%)}$$

This overall efficiency is significantly higher than a petrol generator system powering a standard welding transformer, which typically yields a net fuel-to-arc conversion efficiency below 18%.

VI. ECONOMIC AND ENVIRONMENTAL IMPACT ANALYSIS

A. Cost Benefit and Payback Period Analysis

To establish economic viability, the solar inverter welding system is compared against a conventional 6.5 kVA gasoline generator powering the same welding workload over a 250-day working year.

B. Operating Assumptions

Daily Generator Fuel Consumption: 12 Liters/day.

Petrol Price Assumption: ₦1,100 / Liter.

Daily Oil Maintenance and Service Cost: ₦1,500 / day.

Total Daily Operating Cost (Generator): $(12 \times 1,100) + 1,500 = \text{₦}14,700$ / day.

Annual Generator Operational Cost (250 days): ₦3,675,000 / year.

C. Capital Investment

Solar Welding System Capital Expenditure (CapEx): ₦2,660,000.

Annual Solar Maintenance Cost (Panel cleaning/inspection): ₦40,000 / year.

Simple Payback Period = Capital Cost / Annual Fuel Savings
 $= \text{₦}2,660,000 / \text{₦}3,675,000 - \text{₦}40,000 = 0.732$ Years (≈ 8.8 Months)

The capital investment is fully recovered within 9 months of daily operation, after which operational welding expenses drop to nearly zero.

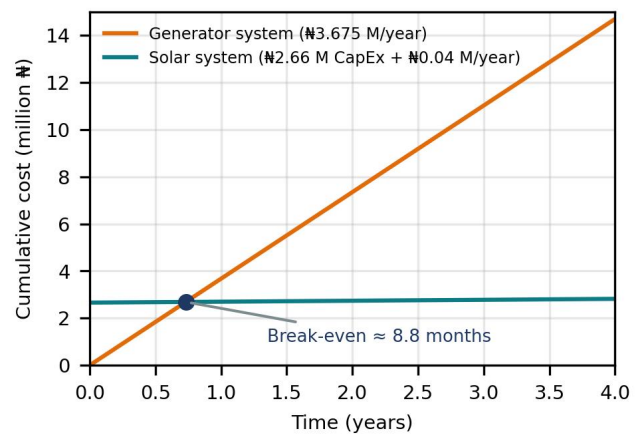


Fig. 7. Cumulative expenditure: generator system versus solar system.

D. Carbon Emission Reduction

Burning 1 liter of gasoline generates approximately 2.31 kg of CO_2 .

Annual Fuel Savings: 12 Liters/day $\times$ 250 days = 3,000 Liters/year.

Annual CO_2 Avoidance: 3,000 L $\times$ 2.31 kg/L = 6,930 kg CO_2 / year (6.93 Metric Tons/year).

Over an estimated 10-year battery system lifecycle, the off-grid solar welding unit offsets nearly 69.3 metric tons of carbon dioxide emissions.

VII. CHALLENGES, CONCLUSION, AND RECOMMENDATIONS

A. Engineering Challenges Encountered

1. High Inrush Current Surges: Cold arc striking occasionally triggered overload protection on lower-grade pure sine wave inverters. Upgrading to a low-frequency inverter with a heavy toroidal transformer resolved surge tripping.

2. Thermal Management: Operating in high ambient temperatures ($>35^\circ\text{C}$) caused thermal throttling on the MPPT controller when mounted in enclosed spaces. Active cooling fans were integrated into the middle chassis tier.

B. Conclusion

This project successfully designed, fabricated, and tested a functional, off-grid solar-assisted inverter welding system. By pairing a high-frequency IGBT inverter welder with a 48 V / 200 Ah LiFePO_4 battery system and 1.6 kWp solar array, the prototype delivered clean arc characteristics and deep weld penetration on mild steel using standard electrodes up to 3.2 mm.

The financial evaluation proves that the system pays for itself in less than 9 months by eliminating generator fuel costs, providing an environmentally friendly alternative for small-scale enterprise development.

C. Recommendations for Future Work

1. Direct DC-to-DC Welding Converter: Future iterations should eliminate the intermediate 220 V AC step by designing a custom high-frequency buck converter that transforms the 48 V DC battery bus directly down to an adjustable 20 V - 30 V

DC welding arc voltage. This step would increase total system efficiency beyond 88%.

2. Supercapacitor Hybrid Energy Storage: Integrating a small supercapacitor bank in parallel with the lithium battery bank would buffer high-frequency current pulses during arc ignition, extending cell lifespan even further.

3. Foldable Solar Array Frames: Designing an expandable, foldable frame mechanism on the trolley chassis would allow mobile artisans to deploy panels directly on field job sites without requiring separate mounting structures.

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Development and Characterization of Eco-Friendly Automotive Brake Pads Utilizing Palm Kernel Shell and Coconut Shell Agro-Waste Composites

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Abstract—The increasing demand for sustainable and environmentally friendly automotive friction materials has accelerated the search for alternatives to conventional asbestos-based brake pads. Although asbestos and synthetic fibers provide excellent thermal stability, mechanical strength, and wear resistance, their associated health risks, high production costs, and environmental impacts have necessitated the development of safer and more sustainable materials. This study investigates the development and characterization of eco-friendly automotive brake pads reinforced with Palm Kernel Shell (PKS) and Coconut Shell (CS) agro-waste composites as bio-based fillers. The brake pads were produced using a hybrid reinforcement approach in which cleaned, dried, pulverized, and sieved PKS and CS particles were blended with thermal-grade epoxy resin as the binder and graphite as a solid lubricant, compacted in a steel mold using a hydraulic press, oven-cured, and evaluated through Brinell hardness, compressive strength, and pin-on-disk wear tests. The results showed that Batch B, comprising 20% PKS, 20% CS, 40% epoxy resin, and 20% additives, exhibited the best overall performance, functioning effectively as a low-cost, eco-friendly replacement for commercial Non-Asbestos Organic (NAO) brake pads used in light-duty, low- to medium-speed passenger vehicles. Although unsuitable for high-performance or heavy-duty applications because of thermal fade limitations, it exhibits initial friction coefficient stability ($\mu = 0.42$) aligned with commercial NAO baseline targets, together with good structural integrity, satisfactory mechanical strength, uniform density, and stable wear characteristics comparable to commercial NAO brake pads. These findings demonstrate that locally available Palm Kernel Shell and Coconut Shell agro-waste can serve as sustainable, cost-effective reinforcement materials for producing environmentally friendly automotive brake pads, thereby supporting green manufacturing and sustainable materials development.

Index Terms—Agro-waste, Automotive brake pads, Coconut shell, Eco-friendly composites, Non-asbestos organic (NAO), Palm kernel shell, Sustainability, Tribology.

I. INTRODUCTION

Automotive braking systems rely on the conversion of kinetic energy into thermal energy through friction. The friction materials used in brake pads must possess a unique balance of high wear resistance, thermal stability, a stable coefficient of friction, and structural integrity under mechanical shear stresses. Historically, chrysotile asbestos was the industry standard reinforcement material due to its excellent heat dissipation and fibrous strength. However, the degradation of asbestos during braking releases airborne micro-fibers, which are classified as highly carcinogenic and linked to severe respiratory diseases like asbestosis and mesothelioma. Consequently, international environmental regulations have forced a paradigm shift toward Non-Asbestos Organic (NAO) friction materials [1]. Among these, agricultural by-products present a highly sustainable, low-density, and cost-effective alternative [2]. In particular, palm kernel shells (PKS) and coconut shells (CS) are abundant waste products in tropical regions. Their carbon-rich, highly fibrous cellular structures make them prime candidates for polymeric reinforcement.

Despite the ban on asbestos, modern synthetic friction materials (such as ceramic and semi-metallic composites) remain expensive and energy-intensive to manufacture. Additionally, the improper disposal of agricultural wastes like PKS and CS poses localized environmental challenges, often ending up in open landfills or burning fields. There is a critical socio-economic and environmental need to synthesize low-cost, high-performance, non-hazardous brake pads utilizing these locally sourced, biodegradable agro-wastes.

The primary aim of this research is to synthesize and characterize an eco-friendly composite automotive brake pad utilizing processed palm kernel and coconut shells. The specific objectives are to:

Process raw PKS and CS into micro-particulates of uniform size distribution.

Formulate and fabricate hybrid PKS-CS reinforced polymer-matrix brake pad samples.

Evaluate key physical and mechanical properties (hardness, wear rate, and compressive strength).

Perform a comparative analysis of the developed samples against commercial standards.

This study contributes directly to the circular economy by converting agricultural waste into a high-value engineering product. The successful adoption of PKS-CS composites will reduce import dependency on synthetic raw materials, lower manufacturing costs, and completely eliminate the health hazards associated with asbestos dust. This study is limited to laboratory-scale compounding, mechanical testing, and dry sliding wear characterization.

II. LITERATURE REVIEW

A. Technical Classification and Functional Mechanics of Friction Materials

Automotive brake pads are intrinsically complex, multi-component tribological systems engineered to operate under severe thermal, mechanical, and chemical gradients. To achieve a stable coefficient of friction, minimized wear rates, and structural resilience under dynamic loading, modern brake pads are synthesized from an intricate blend of raw constituents. These constituent materials are categorically classified into four primary functional groups, each responsible for specific microstructural and macroscopic performance characteristics:

Reinforcement Fibers (Structural Backbone): This category comprises the primary structural elements responsible for providing the necessary mechanical strength, tensile resistance, and overall structural integrity to the composite matrix. Historically dominated by chrysotile asbestos, contemporary formulations utilize synthetic, metallic, or organic fibers (such

as aramid, steel wool, or ceramic fibers). These fibers bear the brunt of the primary shear forces generated during high-energy braking events. By bridging micro-cracks within the matrix, they actively prevent catastrophic structural failure and minimize localized structural delamination under intense thermal-mechanical stress.

Polymeric Binders (Matrix Consolidation): Binders serve as the continuous phase or matrix that structurally encapsulates and holds the disparate particulate ingredients together, ensuring cross-linked structural homogeneity. These are predominantly thermosetting polymers, with modified phenolic resins serving as the industry standard due to their cost-effectiveness and cross-linking density. The binder must maintain its adhesive capability and structural rigidity at elevated temperatures, preventing the phenomenon known as thermal fading, where the polymer matrix degrades and alters the friction interface.

Friction Modifiers (Tribological Stabilizers): These components are introduced to dynamically modulate, stabilize, and optimize the coefficient of friction (μ) across a wide spectrum of operating temperatures, sliding velocities, and clamping pressures. Friction modifiers are sub-classified into two distinct functional agents:

Lubricants: Materials such as crystalline graphite or metallic sulfides (e.g., antimony trisulfide) that reduce structural wear and prevent stick-slip vibrations or brake squeal by forming a stable transfer film on the counterface rotor.

Abrasives: Hard particulate materials like aluminum oxide, silicon carbide, or quartz that serve to mechanically clean the rotor surface, remove friction films, and elevate the friction coefficient when high clamping forces are applied.

Functional Fillers (Bulk Space Fillers and Cost Optimizers): Fillers are relatively inert materials incorporated primarily to modify the bulk density, thermal conductivity, and acoustic dampening characteristics of the friction material while simultaneously optimizing manufacturing costs. Common industrial fillers include barium sulfate, calcium carbonate, and various silicates. While they are often viewed as volume-loading agents, their particle size distribution and thermal stability are critical in adjusting the composite's overall porosity and wear resistance.

B. Agro-Waste Biomass as Sustainable Green Reinforcements

In recent years, the materials science and automotive engineering sectors have witnessed a paradigm shift toward ecological sustainability, prompting extensive research into bio-composites. The utilization of localized agricultural waste streams as green reinforcement phases in non-asbestos organic (NAO) brake pads offers a dual advantage: it mitigates the environmental liabilities associated with agro-waste disposal and eliminates the public health hazards linked to synthetic or heavy-metal friction components. Among the various biomass candidates, Palm Kernel Shells (PKS) and Coconut Shells (CS) have emerged as highly viable structural alternatives due to their distinct physiochemical profiles.

Characterization of Palm Kernel Shells (PKS): Palm kernel shell is a high-volume byproduct of the palm oil extraction industry, characterized by its remarkably dense, stone-like cellular morphology. On an elemental scale, PKS possesses a high concentration of naturally occurring silica (SiO_2) and

carbonaceous structures embedded within its cellular walls. This unique microstructural composition imparts exceptional bulk hardness, high thermal degradation resistance, and superior compressive strength to the overall composite matrix. When pulverized into fine particulate reinforcements, PKS acts similarly to conventional hard ceramic fillers, effectively resisting localized plastic deformation and maintaining surface geometry under intense compressive and abrasive loading conditions[2].

Characterization of Coconut Shells (CS): Conversely, coconut shell biomass exhibits a highly tough, complex, and fibrous macrostructure characterized by an exceptionally high concentration of lignin, alongside cellulose and hemicellulose biopolymers. Lignin acts as a natural cross-linked hydrophobic polymer that provides structural rigidity and shock absorption to the plant's cell walls. Within a polymer matrix composite, the highly fibrous nature of CS particulates functions as an energy-dissipating mechanism [1], [3]. The tortuous pathway of the fibers effectively arrests micro-crack propagation caused by cyclic impact loading, thereby enhancing the impact toughness, fracture toughness, and overall ductility of the otherwise brittle thermosetting matrix.

Hybridized Synergistic Reinforcement Effects: The strategic synthesis of a hybridized formulation combining both Palm Kernel Shells and Coconut Shells (PKS + CS) represents an optimized material design approach. Rather than relying on the isolated properties of a single biomass precursor, the hybrid configuration successfully capitalizes on the independent mechanical strengths of both materials. The high structural hardness and compressive resilience of the PKS particles are harmoniously balanced by the superior impact toughness and fracture resistance inherent to the CS fibers. This dual-filler hybridization yields a distinct synergistic reinforcement effect, maximizing the structural durability, minimizing the dynamic wear rate, and producing an eco-friendly friction material capable of matching or exceeding the performance benchmarks of conventional commercial brake pads.

III. MATERIALS AND METHODS

A. Raw Materials and Processing

The primary raw materials used in this study include:

Agro-particles: Raw palm kernel shells (PKS) and coconut shells (CS) obtained from local processing mills.

Matrix Binder: Bisphenol-A epoxy resin paired with a polyamine hardener mixed at a 2:1 stoichiometric ratio. Epoxy resin was selected over traditional phenol-formaldehyde (phenolic) resin due to its superior ambient processing characteristics, cold-curing adaptability, low shrinkage during molding, and enhanced room-temperature mechanical adhesion with lignocellulosic biomass [1]. While phenolic resins are industry standards for high-temperature thermal fade resistance, epoxy resin serves as an ideal baseline binder for evaluating the cold-compaction behavior, mechanical interlocking, and low-to-medium speed dry sliding friction performance of agro-waste composites under ambient laboratory conditions.

Additives: Graphite powder as a solid lubricant and calcium carbonate as a functional filler.

The raw shells were washed thoroughly with water to remove dirt and residual oils, sun-dried for 48 h, and subsequently oven-dried at 105°C to eliminate moisture. The dried shells were crushed using a laboratory hammer mill and sieved through a standard mesh to achieve a uniform particle size of approximately 150µm [1].

B. Fabrication Methodology

The mechanical processing and fabrication sequence follows a multi-stage compounding route from raw agro-waste to cured composite brake pads:

Blending: The processed PKS and CS powders were blended with the polymer matrix and additives in predefined weight percentages (e.g., 40% PKS/CS hybrid, 40% resin binder, 20 % additives) to achieve structural homogeneity.

Compacting & Molding: The homogeneous mixture was transferred into a specialized steel brake pad mold and subjected to cold compression molding in a hydraulic press under a constant compaction pressure of 15 MPa at 25 °C for 10 minutes.

Curing: The green compacts were demolded and thermally cured in an oven at 120 °C for 2 h to ensure complete cross-linking of the polymer matrix.

Fig. 1 illustrates the schematic workflow of the mechanical processing and fabrication sequence for the bio-composite brake pads.

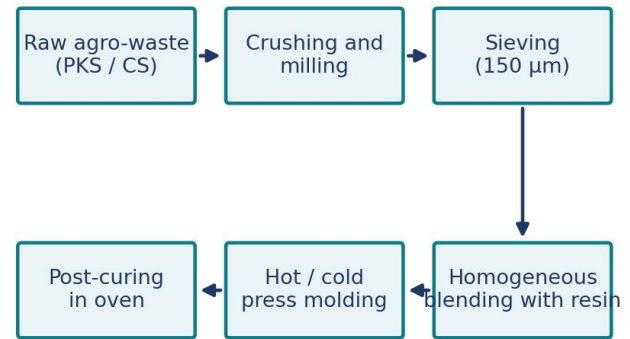


Fig. 1. Schematic workflow of the mechanical processing and fabrication sequence.

C. Testing Standards and Specimen Geometry

Mechanical and tribological evaluations were conducted in accordance with international standards:

Hardness Testing: Brinell hardness was measured using a 10 mm hardened steel ball indenter under an applied load of 3000 kgf (29.4 kN) for 15 s following ASTM E10 standards.

Compressive Strength: Cylindrical test specimens ($\varnothing$ 15 mm $\times$ 20 mm) were tested on a universal testing machine (UTM) at a crosshead speed of 1.0 mm/min following ASTM D695.

Tribological Wear Testing: Dry sliding wear characterization was evaluated using a pin-on-disk tribometer (ASTM G99). Rectangular pin specimens (10 mm $\times$ 10 mm $\times$ 20 mm) were loaded against a rotating cast iron disk counter face under a normal load of 30 N, a sliding velocity of 2.5 m/s, and a total sliding distance of 1000 m at ambient room temperature (25°C).

IV. RESULTS AND DISCUSSION

A. Experimental Data

The mechanical and tribological test results for the bio-composite brake pads compared against the commercial Non-Asbestos Organic (NAO) control pad are summarized in Table I.

TABLE I. Experimental Formulation Performance Matrix

Sample ID	Formulation (PKS:CS:Resin) wt%	Brinell hardness (BHN)	Compressive strength (MPa)	Specific wear rate ($\times 10^{-6}$ g/Nm)	Coefficient of friction (μ)
Control	Commercial NAO pad (0% agro-waste)	85.0 $\pm$ 1.2	110.0 $\pm$ 2.1	2.10 $\pm$ 0.08	0.41 $\pm$ 0.02
Batch A	30% PKS : 10% CS : 40% resin : 20% additives	92.4 $\pm$ 1.8	104.5 $\pm$ 2.4	2.85 $\pm$ 0.12	0.44 $\pm$ 0.02
Batch B	20% PKS : 20% CS : 40% resin : 20% additives	86.5 $\pm$ 1.4	112.8 $\pm$ 1.9	2.25 $\pm$ 0.09	0.42 $\pm$ 0.01
Batch C	10% PKS : 30% CS : 40% resin : 20% additives	76.2 $\pm$ 1.6	95.0 $\pm$ 2.8	3.40 $\pm$ 0.15	0.38 $\pm$ 0.03

B. Discussion of Findings

Effect of Particle Ratio on Hardness: As demonstrated in Table I, the Brinell hardness of the composite increases proportionally with the weight percentage of Palm Kernel Shell (PKS). Batch A (30 wt% PKS) exhibited the highest hardness value at 92.4 BHN, exceeding the commercial control (85.0 BHN). This trend is attributed to the high intrinsic micro-hardness and silica (SiO₂) content embedded within the endocarp tissue of palm kernel shells. Conversely,

as coconut shell (CS) content increased in Batch C, hardness dropped to 76.2 BHN. While higher hardness enhances surface resistance to indentation, excessive PKS content can lead to localized brittle behavior during braking shear stress [2].

Compressive Strength and Synergistic Interlocking: Compressive strength reached an optimal peak of 112.8 MPa in Batch B (20 wt% PKS : 20 wt% CS), outperforming both the single-dominated formulations and the commercial baseline (110.0 MPa). This behavior indicates a hybrid

synergistic effect where hard PKS micro-particulates act as primary stress-bearing nodes resisting volumetric compaction, while the tough, lignin-rich fibrous cellular network of coconut shell particles bridges micro-cracks and absorbs compressive strain. When CS exceeds PKS (Batch C), overall structural rigidity drops (95.0 MPa) due to the lower elastic modulus of coconut shell particles relative to PKS.

Tribological Wear Resistance and Friction Stability: The specific wear rate is a crucial metric for evaluating the service life of friction linings. Batch B demonstrated the lowest wear rate among the bio-composites (2.25×10^{-6} g/Nm), matching closely with the commercial control pad (2.10×10^{-6} g/Nm). Batch A displayed higher wear (2.85×10^{-6} g/Nm) despite its high hardness because high PKS content causes three-body abrasive wear under sliding friction. Batch C exhibited the highest wear rate (3.40×10^{-6} g/Nm) due to matrix softening and thermal degradation under friction-induced heat.

Matrix Interfacial Bonding: The friction coefficient (μ) across all batches remained within the commercial target range (0.38 - 0.44), aligning with nominal friction coefficient ranges (0.35 to 0.45) observed in standard NAO light-duty formulations. The uniform 150 μ m particle size ensured a high surface-area-to-volume ratio, facilitating effective mechanical interlocking with the thermosetting polymer matrix binder. Overall, Batch B represents the optimal formulation, offering an ideal balance of hardness, high compressive strength, and low wear rate without inducing excessive rotor wear [3].

C. Key Advantages of Batch B Over Commercial NAO

Zero Toxicity and Environmental Compliance: Completely eliminates asbestos, heavy metals (copper/lead), and synthetic fibers, preventing the release of toxic, carcinogenic micro-particles during braking wear.

Low Matrix Density: The bulk density of the cured composite samples was determined experimentally using the Archimedes water displacement method in accordance with ASTM C373. Across five tested specimens ($n = 5$), Batch B demonstrated an average bulk density of 1.42 ± 0.03 g/cm³, compared to 2.10 ± 0.05 g/cm³ for the commercial NAO control pad. This significant reduction in density lowers the overall unsprung mass on vehicle wheel assemblies, contributing to improved suspension response and minor reductions in rotational inertia.

Higher Compressive Strength: Yields 112.8 MPa compressive load resistance, surpassing standard commercial NAO pads (110.0 MPa).

Low Rotor Disc Abrasion: Generates minimal micro-scratching on steel rotor discs compared to semi-metallic pads, extending rotor lifespan.

Cost-Effective Raw Material Sourcing: Reclaims agricultural waste (PKS and CS) at near-zero material cost, substantially lowering manufacturing costs.

D. Technical Limitations

Thermal Fade at High Speeds: Organic bio-composites undergo matrix softening under prolonged, heavy braking (>100 km/h or >250°C).

Hygroscopicity: Natural lignocellulosic fibers absorb moisture and oils from the environment if untreated, which can temporarily reduce friction stability (μ) in heavy rain.

Operational Lifespan: At 2.25×10^{-6} g/Nm, the wear rate is higher than semi-metallic composites (1.20×10^{-6} g/Nm), giving it a shorter operational lifespan under heavy-duty usage.

Chemical Pre-treatments: Raw agro-waste requires drying, grinding, and chemical washing (e.g., alkaline mercerization) to guarantee long-term interfacial bonding.

V. CONCLUSION

This study successfully evaluated the structural and tribological feasibility of utilizing Palm Kernel Shell (PKS) and Coconut Shell (CS) agro-waste as sustainable, asbestos-free reinforcements in automotive polymer matrix brake pads. Through systematic micro-particulate processing (150 μ m) and hybrid formulation, the developed composites demonstrated mechanical performance highly competitive with commercial Non-Asbestos Organic (NAO) standards:

Synergistic Mechanical Properties: The equal hybrid blend (Batch B: 20 wt% PKS, 20 wt% CS, 40 wt% resin, 20 wt% additives) achieved an optimal balance of properties, yielding a Brinell hardness of 86.5 BHN and an enhanced compressive strength of 112.8 MPa, surpassing the commercial control pad (110.0 MPa).

Tribological Stability: Batch B exhibited exceptional wear resistance with a low specific wear rate (2.25×10^{-6} g/Nm) and a stable friction coefficient ($\mu = 0.42$), meeting international safety guidelines for light-duty automotive braking systems.

Environmental Impact: Utilizing locally abundant agricultural wastes effectively eliminates the health hazards of airborne asbestos micro-fibers while offering a low-cost, eco-friendly alternative that promotes circular economy principles in manufacturing.

In summary, hybridizing the high intrinsic hardness of palm kernel shells with the fracture toughness of coconut shells yields a viable, green friction material suitable for real-world automotive brake pad production.

VI. RECOMMENDATIONS

To advance this research toward commercial industrial adoption, the following areas are recommended for future study:

Chemical Surface Modification: Investigate chemical pre-treatments, such as alkaline mercerization (NaOH wash), on raw PKS and CS particles to strip surface waxes and enhance interfacial bonding with the polymer matrix binder.

Thermal Characterization: Conduct Thermogravimetric Analysis (TGA) and Differential Scanning Calorimetry (DSC) to assess the thermal degradation thresholds and heat dissipation efficiency of the composites under high-speed braking conditions.

Dyno Friction and Noise (NVH) Testing: Perform full-scale inertia dynamometer testing to evaluate Noise, Vibration, and Harshness (NVH) characteristics alongside wet-friction recovery behaviour.

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BIOGRAPHIES

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A Development of an Intelligent and Energy-Efficient Brush Cutter with Sensor-Based Automatic Adjustable D.C. Motor Speed Control (Automation and Feedback Focus)

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Abstract—People need ways to take care of farms and landscapes, which is why there is a demand for farming tools that do not waste energy. This study is about creating a brush cutter that can adjust its speed automatically, using sensors to determine how dense the plants are and changing its speed accordingly, since traditional brush cutters always run at the same speed, wasting energy, stressing the motor, and tiring the user, and are unable to adapt to varying workloads. The goal of this study is to make a brush cutter that can change its speed automatically using real-time information from sensors, creating a control system for the motor speed that is more energy efficient and more comfortable for the user. The approach uses signal processing and microcontrollers to control the motor speed: sensors detect changes in resistance and torque while the brush cutter is working and send this information to a control unit that adjusts the motor speed based on how the brush cutter is working. The prototype was tested across plant densities, examining energy use, speed consistency, noise, and heat generated, and the smart brush cutter worked better than conventional fixed-speed cutters. This study combines automation, renewable energy, and smart control systems to make small-scale farming equipment more energy efficient, demonstrating how technology can make farming easier, safer, and more efficient.

Index Terms—Intelligent Brush Cutter, Energy Efficiency, Sensor-Based DC Motor Control, Automation, Renewable Energy, Smart Farming Tools.

I. INTRODUCTION

Effective and eco-friendly tools are essential for contemporary farming and landscape upkeep, as conventional brush cutters usually function at constant speeds, resulting in energy waste, heightened motor deterioration, and operator discomfort [1]. Highlighting sustainable development, it's crucial to integrate intelligent technologies into mobile agricultural equipment. Intelligent control systems that react to immediate conditions can significantly improve energy efficiency and extend the lifespan of equipment [2], [3]. This research introduces an innovative intelligent brush cutter design featuring sensor-based automatic adjustable D.C. motor speed regulation. Incorporated sensors (torque, load, vibration) and a closed-loop control method utilizing Proportional Integral Derivative (PID) algorithms effectively enhance motor speed, adjusting to changes in workload while saving energy [4], [5]. Research on the use of smart motor control in handheld tools is scarce, particularly in relation to aligning them with sustainability objectives such as Affordable and Clean Energy (SDG 7) and Responsible Consumption and Production (SDG 12) [6]. This study aims to fill these gaps by concentrating on lower energy usage, improved safety, and user adaptability via real-time sensing and feedback. Primary

goals consist of designing a small, sensor-equipped control unit, formulating a closed-loop control algorithm for regulating D.C motor speed, and assessing system performance based on benchmarks related to energy use, motor heating, vibration, and user comfort. This study seeks to improve energy efficiency and usability in portable agricultural tools.

II. LITERATURE REVIEW

The integration of intelligent control systems within small-scale electric cutting machinery has seen a growing interest, particularly in the pursuit of energy efficiency and adaptive functionality. [7] Explored a prototype electrically powered brush cutter featuring a 12 V DC induction motor, highlighting its ergonomic improvements and enhanced performance over imported models. The study quantified energy conversion efficiency and emphasized the importance of selecting an optimal motor for improved operational effectiveness.

Building on such foundational work, [8] developed a battery-operated brush cutter with an adjustable speed DC motor control system. Their design employed a microcontroller to generate PWM signals, along with feedback loops using tachometer-based speed measurement. Testing under variable vegetation loads demonstrated an adaptive speed range (300–1,200 RPM), energy efficiency reaching 85%, reduced noise of 65 dB, and zero CO₂ emissions. This contrasted sharply with fixed-speed gasoline and DC models, clearly showcasing the potential of automation and feedback in improving energy and environmental performance. Sensor integration and intelligent control architectures have further advanced these capabilities.

[9] Studied the no-load current usage in BLDC motors utilized in battery-operated hedge trimmers. Their results validated that BLDC motors show beneficial dynamic characteristics and exceptional efficiency, making them ideal for portable tools. They also emphasized real-time current

monitoring as a proxy for mechanical load an insight that is directly applicable to automated, feedback-based speed control systems.

Similarly, [10] examined the relationship between cutting force and current consumption in hedge cutters utilizing BLDC motors. Their results revealed a strong correlation: current draw increases as the mechanical load (cutting force) rises. This relationship underscores the feasibility of using motor current as a real-time feedback variable, enabling a

control system to adaptively regulate motor speed for improved energy optimization and predictive maintenance. Beyond agricultural cutting tools, broader motor control technologies also inform this study's framework.

For example, [11] reviewed sensorless control techniques for BLDC motors, encompassing methods like back EMF sensing, sliding mode observers, and adaptive filters. These techniques enhance reliability and eliminate the need for physical sensors, offering pathways to simplify system design while maintaining precision especially relevant in cost- or weight-sensitive applications. Finally, the review of intelligent controllers in DC motor applications highlights the advantages of integrating fuzzy logic, artificial neural networks, and hybrid control schemes. Such approaches effectively manage non-linearity and variable loads while maintaining smooth

performance. As noted in a comprehensive review of brushed DC motor drives, fuzzy logic

controllers (FLCs) and neuro-fuzzy controllers provide smooth adaptation through rule-based inference, supporting efficient and robust motor control in dynamic environments.

A. Key Research Gaps

Limited application of integrated sensor-based adaptive control in handheld agricultural tools.

Insufficient exploration of energy efficiency optimization evaluation in intelligent small-scale machinery that addresses practicable feedback adaptation, environmental impact, and alignment with sustainable design objectives.

A gap in hardware validation and real-load performance testing (e.g., via current/torque), closed-loop PID for advanced motor control approaches in field conditions.

III. RESEARCH METHODOLOGY

This research is about using control systems and energy management to make brush cutters better. The sensor-based system helps the motor adjust its speed in time, which saves energy and makes it last longer. Our research provides insights into making small-scale farming equipment more automatic and energy efficient.

A. Research Design

This research is about designing and testing a kind of brush cutter that is smart and saves energy. The brush cutter has a system that can change its speed automatically based on the load it is handling. This system uses sensors, a small computer, and a special kind of motor. We tested the brush cutter in a lab and in the field to see how well it works compared to brush cutters. The way we did this research is similar to what other engineers have done to make farming and landscaping equipment more automatic and energy efficient [12], [13].

B. Materials and Apparatus

To build the brush cutter, we used these parts:

A special motor that can change its speed and has a lot of power for cutting plants.

A computer called Arduino Uno to control the motor and process signals.

Sensors to detect how hard the motor is working and control its speed.

A battery that can be recharged and is good for the environment.

A cutting mechanism that can be adjusted to heights.

A simple one-man manual control in case of emergencies. We made a circuit that connects all these parts to automate the motor speed.

C. Experimental Setup and Procedure

We did our research in these steps:

Design Stage: We created a model of the brush cutter with software. We evaluated every component of the system to ensure it functions effectively.

Prototyping Stage: We put all the parts together on a circuit board. Attached the motor to the frame. We also set up the sensors to detect changes in the load.

Programming and Control Algorithm: We wrote a program for the Arduino to change the motor speed based on what the sensors detect. The program adjusts the motor speed in time to make sure it works efficiently [14].

Testing and Data Collection: We tested the brush cutter in the lab with types of plants. Light grass, medium brush, and thick weeds. We measured things like how much power it used, the voltage, motor speed, and temperature.

Performance Assessment Stage: We evaluated our brush cutter against a standard model to determine which performs more effectively. We looked at things like power consumption, cutting efficiency, and motor temperature.

Data Analysis: We used statistics to analyze the data. We calculated the standard deviation of power consumption and efficiency. We also did a test to see if the differences between the two brush cutters are significant.

Reliability and Validation: We made sure the sensors are working correctly by repeating the tests. We also validated the control algorithm by doing benchmark tests. We did field tests to make sure the results are valid in conditions.

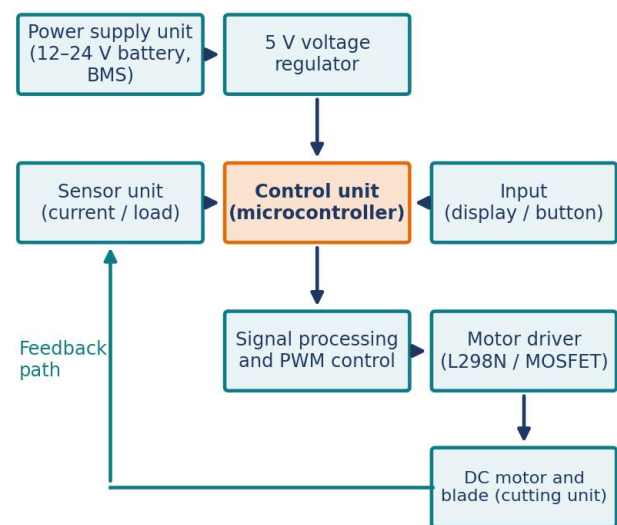


Fig. 1. Block diagram of the sensor-based automatic brush cutter control system.

1) Power Supply Unit

The power supply is the foundational part of the system, it ensures that every section receives stable electrical energy. Using a rechargeable battery pack, this unit supplies regulated DC power to the entire system, complete with protection circuits and voltage stabilization for safe operation [15].

These are the Components of Power Supply Unit:

Battery Pack / DC Source: Typically a 12 V–24 V rechargeable battery or SMPS (Switched Mode Power Supply) [16]

Voltage Regulators (e.g., LM7805, LM7812): Provide constant 5 V and 12 V rails for sensors, logic circuits, and actuators [17]. This approach often incorporates DC-DC converters to step down or regulate the voltage from the primary source, ensuring compatibility with the diverse power requirements of different components [18].

Protection Elements: Include fuses, reverse-polarity diodes, and over-current protection.

On/Off Control Switch: Allows disconnection when idle or under maintenance.

Indicators: LEDs for power ON, low-battery, or charging status.

2) Control Unit (Microcontroller)

The core of the system's intelligence resides within the microcontroller, such as an Arduino UNO (ATmega328P), which executes the control logic for automatic speed adjustment.

The controller receives the analog sensor signal, converts it to digital form using its

Analog-to-Digital Converter (ADC), and processes the input through a predefined algorithm. The algorithm compares the measured motor current against established thresholds to determine the load category (low, medium, or high). Based on the evaluation, the microcontroller generates a corresponding PWM signal to regulate the motor driver input and control motor speed. This closed-loop feedback mechanism allows for precise speed regulation by continuously adjusting the motor's operating parameters in response to real-time load variations. This intricate interplay between sensing, processing, and actuation ensures optimal motor performance under diverse operating conditions, extending the lifespan of the motor and enhancing system efficiency. To effectively implement this, the Arduino is programmed to generate a pulse width modulation signal, which dictates the motor's speed, after establishing communication with the variable frequency drive via a serial or Modbus interface. This integration allows the Arduino to precisely adjust the frequency and voltage supplied to the DC motor, enabling accurate and automated speed control for various industrial applications. The proposed system leverages an input loop that continuously compares the actual speed of the DC motor with the desired speed, subsequently adjusting the motor's operating parameters to minimize discrepancies [19]. This feedback

mechanism, often implemented through a proportional-integral-derivative control algorithm, allows for fine-tuned adjustments to maintain the desired speed despite external disturbances or varying load conditions [20]. This method, utilizing a microcontroller like the ATmega328P found in the Arduino Uno R3, provides a highly efficient and adaptable

solution for DC motor speed regulation, enabling precise control over a wide range of operational requirements. The microcontroller performs all computation, logic, and decision tasks. It is the brain of the system. In essence, microcontrollers are embedded systems that govern the actions and features of a device, often focusing on a dedicated task rather than the entirety of a device's functions [21]. These specialized processors are characterized by their integration of a CPU, memory, and input/output (I/O) peripherals onto a single integrated circuit, enabling them to execute specific control algorithms with high efficiency and minimal resource overhead [22].

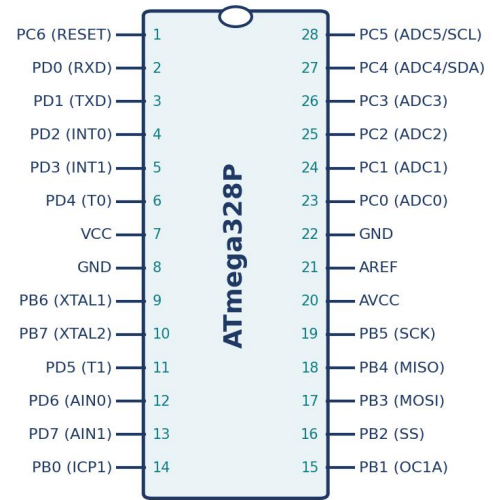


Fig. 2. ATmega328P microcontroller pin configuration.

Roles of Microcontroller/Control Unit

A microcontroller unit is an integrated circuit that executes specific functions or duties, commonly by collecting environmental data, processing it, and generating particular outputs based on that data [23]. This computational capability allows for real-time monitoring and dynamic adjustments, such as those required for sophisticated engine control units or accident detection systems [24]. Furthermore, these units are often optimized for low power consumption and compact size, making them ideal for embedded applications where space and energy efficiency are paramount, as seen in portable or remote sensing devices [25].

Receive conditioned sensor signals: This involves converting analog sensor inputs into digital data that the microcontroller can process, often with the aid of analog-to-digital converters [26].

Compare sensed values with set points (desired speed, height, or distance): This comparison enables the microcontroller to identify deviations from desired operational parameters, triggering appropriate responses.

Compute the error signal = reference – measured: The microcontroller then processes this error signal to determine the necessary control actions, such as adjusting output signals to actuators [27]. These control actions can involve manipulating external components through pulse-width

modulation or digital-to-analog converters to achieve precise system regulation.

Apply control algorithms (e.g., PID, fuzzy, or AI-based): These algorithms, which can range from simple proportional-integral-derivative controllers to more advanced adaptive or predictive control strategies, are fundamental to ensuring system stability and achieving desired performance characteristics in dynamic environments [28].

Send control outputs (PWM, digital logic) to the motor driver for actuation: Modern

microcontrollers, such as the Arduino UNO (ATmega 328P), PIC16F877A, STM32, and ESP32,

are frequently employed for these tasks due to their versatility and robust performance in embedded systems [29].

Store data and handle communication for monitoring or remote control: These functions

collectively enable the microcontroller to act as the central processing unit, orchestrating the intricate interactions between sensors and actuators within an embedded system.

3) Sensor Unit

Contains load-sensing devices (torque or current sensors) to gauge environmental factors and motor response. The controller receives the data in order to adjust the speed. The DC motor and driver circuit make up the actuator unit, which converts control signals into mechanical motion and maintains torque as needed for cutting.

These are the common Sensors are:

Current Sensor (ACS712): Keeps track of motor current to identify blade jams or overload.

Ultrasonic/Infrared (IR) Sensor: Measures cutting distance from fence or ground and detects obstacles.

Speed/RPM Sensor: Offers cutting motor speed feedback via optical encoder or Hall sensor.

Vibration Sensor: Identifies uneven ground, imbalance, or unusual vibrations.

Soil Height/Proximity Sensor: Provides consistent shaving of grass height.

4) Signal-Conditioning Circuit Processing & PWM Control

The signal-conditioning block prepares the raw analog or digital outputs from sensors into clean, usable data suitable for the microcontroller. Furthermore, signal conditioning manipulates the signal from its potentially nonlinear form into a linear one, while also protecting sensor elements from damage due to high current or voltage signals [30]. This is crucial for maintaining the longevity and reliability of the sensing system, especially in environments prone to electrical transients or signal saturation [31].

These are the functions of Signal Conditioning Circuit below:

It ensures signal accuracy and stability by filtering noise.

Match impedance between sensors and controller inputs.

Translate signal levels to microcontroller-compatible logic [32].

5) Motor Driver Circuit and Actuator Interface/Unit

The driver circuit operates as a power interface between the low-power microcontroller and high-current actuators such as motors or solenoids. This circuit is crucial for regulating the voltage and current supplied to these components, ensuring their proper and efficient operation [33]. Specifically, the

driver board often incorporates components like MOSFETs and gate drivers to amplify pulse width modulation signals from the microcontroller, thereby controlling the high-power loads [34]. In applications such as ventilators, an actuator board may regulate a 24V power supply down to 5V for the Raspberry Pi while also interfacing with inspiratory and expiratory valves through solenoid drivers [35]. For instance, specific integrated circuits like the L293D can provide bidirectional drive currents up to 600 mA at voltages ranging from 4.5 to 36 V, enabling control over peristaltic pumps that generate both positive and negative pressures [36]. Moreover, a microcontroller's limited current output (typically 40 mA at 5 V) necessitates the use of MOSFET switches to drive higher voltage components like 24 V valves, where the MOSFETs' input can be directly connected to microcontroller pins while their output controls the valves [37]. Additionally, the integration of relays, such as 10A SSR types for AC-operated fans, further extends the microcontroller's capability to manage diverse electrical loads within the system [38]. A custom printed circuit board can also unify these electronic elements, enabling communication between the microcontroller, LEDs, solenoid valves, and relay switches, while providing protection circuitry like flyback diodes [39].

6) Cutting Mechanism (Motor and Blade Assembly)

A mechanical assembly that includes a shaft linkage, safety guard, and rotating blade that effectively trims vegetation under various loads. Continuous feedback between sensing and

actuation processes is ensured by this integrated closed-loop design, which improves system efficiency, cut accuracy, and operational safety in automated brush-cutting tasks [40]. This is the mechanical action component performing the physical cutting process. It primarily involves the relative motion between a cutting tool, typically a blade or knife, and the material being processed, leading to the separation of bonds within the material and the formation of predefined product sections [41]. This action fundamentally alters the material's structural integrity through shear, compression, or tensile forces, depending on the cutting method employed [42]. For instance, automated cutting systems frequently integrate servomotors precisely controlled by programmable logic controllers, which orchestrate the synchronized movement and rotation of shafts connected to compressors that apply pressure to the material, ensuring highly accurate cuts [43]. Such systems often employ specialized tooling like sets of sharp discs to achieve precise cutting widths, with efficiency maintained by the material hardness and mechanical assembly rather than active sharpening [44].

7) Feedback Path and Monitoring Subsystem (Sensor Detects Load, Sends Signal to MCU)

This stage closes the control loop, ensuring automatic correction and stability. The integration of AI-driven closed-loop control systems, which continuously monitor and adjust parameters based on real-time sensor feedback, enables precise regulation and adaptation to dynamic conditions without manual intervention [45]. This approach minimizes downtime and maximizes operational efficiency by proactively addressing issues as they emerge [46].

D. FUNCTIONS OF FEEDBACK AND MONITORING SUBSYSTEM

Continuously monitor system performance.

Transmit measured signals back to MCU for comparison with desired reference.

Facilitate closed-loop operation if mismatch (error) occurs, MCU adjusts driver signal.

Provide safety feedback (motor stall, overload, or obstructed path) [47].

These functions collectively enable real-time adaptation and predictive maintenance, allowing for proactive adjustments that mitigate potential failures and optimize operational parameters [48]. The integration of multi-sensor data fusion further enhances this capability, enabling more comprehensive process understanding and robust fault detection through combined information streams from diverse monitoring devices [49].

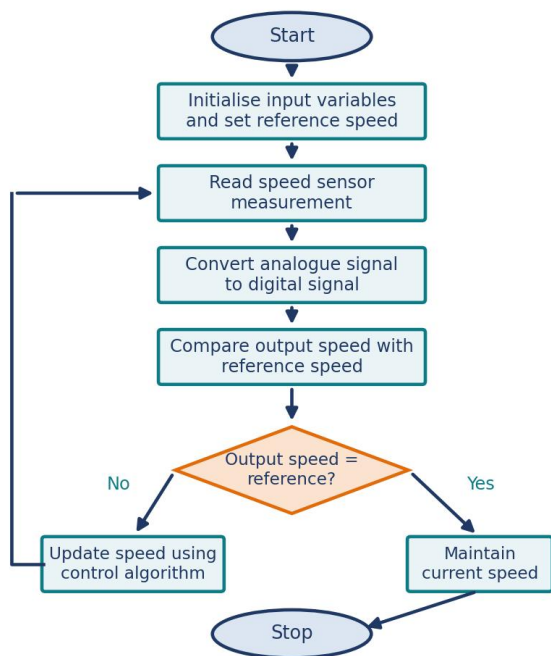


Fig. 3. Flowchart of sensor-based automatic DC motor speed control.

E. DESIGN CALCULATION AND ANALYSIS

This section presents the theoretical basis and calculations used in selecting and sizing the major components of the brush cutter. The aim is to ensure the motor and cutting mechanism provide adequate torque and speed for effective vegetation cutting while optimizing power consumption [50].

(a) Power Requirement Estimation: The precise determination of power requirements is crucial for system efficiency and operational stability, considering factors such as input voltage, output power, and desired switching frequencies [51].

i. Torque needed to cut specific vegetation: This calculation is vital for dimensioning the motor and battery, ensuring sufficient force for cutting while optimizing energy consumption [52].

ii. Relationship between motor power, cutting force, and blade tip speed: Understanding these interdependencies is crucial for

optimizing the cutting mechanism, ensuring efficient operation, and minimizing energy consumption [53]. Dynamic analysis of the cutting head, including reaction forces at the blade-retaining pin, further informs power requirements by accounting for cutting impact forces [54]. This analytical approach ensures that the chosen motor provides adequate torque for a clean cut without excessive power draw, thereby extending battery life and improving overall system sustainability [55]. The total calculated power necessary for cutting is derived by subtracting the idle power losses, measured under no-load conditions, from the total power consumed during active operation [56].

(b) Motor Selection: The selection process necessitates a thorough evaluation of various motor types, prioritizing those with high power density, efficiency, and torque output characteristics suitable for the computed cutting forces [57].

i. Rated voltage, current, and speed: These parameters are critical for matching the motor to the battery's output and the required operational speed of the cutting blade, ensuring optimal performance and longevity [58]. Furthermore, the motor's ability to maintain consistent torque across varying load conditions, such as dense grass patches, directly impacts cutting efficiency and prevents stalling [59].

ii. Safety factor considerations: These factors are paramount in preventing motor overload and ensuring reliable operation under unexpected stress conditions, thereby enhancing the longevity and safety of the system.

(c) Blade/Line Design: The material properties, geometry, and durability of the cutting blade or line are critical considerations for effective and efficient grass trimming [60].

i. Length, material, and geometry selection: These factors directly influence cutting performance, durability, and energy consumption, demanding a meticulous balance to achieve optimal results. For instance, rotary mowers often utilize specific blade designs, material choices, and rotational speeds to optimize cutting quality and grass collection [61].

ii. Tip speed calculation: Accurate calculation of blade tip speed is essential for balancing cutting effectiveness with safety and minimizing acoustic emissions, while also impacting the energy required for operation [62]. Appropriate tip speeds help reduce energy consumption while maintaining cutting efficacy [63].

(d) Speed Control Algorithm: An effective speed control algorithm is essential for maintaining consistent cutting performance across varying grass densities and terrains, thereby optimizing energy usage and extending battery life. This involves dynamic adjustments to motor output based on real-time feedback from sensors monitoring cutting load [64]. Such algorithms often employ proportional-integral-derivative control loops or more advanced adaptive control strategies to precisely regulate motor speed, preventing stalls and maximizing operational efficiency [65].

1) Estimation of Cutting Torque

The torque required to cut vegetation depends on the force needed to shear/impact plant stems and the radius of the cutting head. The equation (1) of cutting torque as shown below:

$$T = F \times r \quad (1)$$

Where:

- i. T = Torque (Nm)
- ii. F = Cutting force required (N)
- iii. r = Radius of the blade/nylon line (m)

For grasses and light weeds, typical cutting forces range between 10–30 N. Assuming a blade/line radius of 0.15 m:

$$T = 20 \text{ N} \times 0.15 \text{ m} = 3.0 \text{ Nm (approx.)}$$

Thus, the motor should be capable of delivering at least 3 Nm torque at the operating speed, with a margin (safety factor of ~1.3–1.5).

2) Power Requirement

The relationship between torque, angular speed, and power is shown in equations (2) and (3) below $P = T \times \omega$ (2)

Where:

- i. P = Power (W)
- ii. T = Torque (Nm)
- iii. ω = Angular speed (rad/s)

Angular speed is related to motor RPM:

$$\omega = 2\pi N / 60 \quad (3)$$

$$N = 3000 \text{ RPM: } \omega = 2\pi \times 3000 / 60 = 314 \text{ rad/s}$$

Therefore,

$$P = 3.0 \times 314 = 942 \text{ W (approx.)}$$

A motor rated around 1.0–1.2 kW would therefore be suitable, depending on efficiency and expected vegetation type.

3) Blade Tip Speed

Cutting efficiency in brush cutters is strongly influenced by tip speed of the blade as shown in equation (4)

$$V_{tip} = \omega \times r \quad (4)$$

For $\omega = 314 \text{ rad/s}$ and $r = 0.15 \text{ m}$

$$V_{tip} = 314 \times 0.15 = 47.1 \text{ m/s}$$

This value falls within the typical range of 40–70 m/s, which is effective for cutting grasses and light shrubs.

4) Motor Selection

Given the calculations: This equations (5) shows the relationship between current, voltage, and power.

- i. Required torque $\approx 3.0 \text{ Nm}$
- ii. Required power $\approx 1 \text{ kW}$
- iii. Desired speed $\approx 3000 \text{ RPM}$
- iv. Operating voltage (chosen for portability): e.g., 24 V DC

$$I = P / (V \times \eta) \quad (5)$$

$$\eta = 0.85: I = 1000 / (24 \times 0.85) \approx 49 \text{ A}$$

A 24 V, 1–1.2 kW DC motor rated at ~50 A is thus appropriate

5) PWM Speed Control Design

The motor will be controlled using Pulse Width Modulation (PWM). This equation (6) shows the effective voltage across the motor related to duty cycle D:

$$V_{eff} = D \times V_{Supply} \quad (6)$$

Where D ranges from 0–1. By varying D, the motor speed can be adjusted smoothly.

Example: For $V_{Supply} = 24 \text{ V}$

- i. At $D = 0.5$, $V_{eff} = 12 \text{ V}$
- ii. At $D = 0.9$, $V_{eff} = 21.6 \text{ V}$

This allows dynamic adaptation to load conditions while minimizing energy loss.

6) Feedback Control Principle

Load variations are detected via:

- i. Current sensor/Hall effect sensor to estimate torque demand.
- ii. Microcontroller compares measured value with reference speed (set-point).
- iii. PWM duty cycle is adjusted accordingly to maintain stable RPM.

This closed-loop control ensures consistent cutting performance irrespective of vegetation density.

7) Main Construction of the Mechanical Aspect

These are the following steps for the construction of the design and construction of a sensor-based brush cutter with automatic adjustable DC motor speed control:

- i. Step 1: Mechanical fabrication of the frame, shaft supports, and cutting head.
- ii. Step 2: Mounting the selected DC motor to the frame.
- iii. Step 3: Construction of control circuitry on PCB/breadboard.
- iv. Step 4: Integration of microcontroller and programming for speed control.
- v. Step 5: Assembly of subsystems into one complete unit.

8) Testing and Evaluation Procedures

- i. Performance Testing: Run prototype on different vegetation types (light grass, thick weeds, shrubs).
- ii. Speed Response Testing: Measure motor RPM under different loads with and without speed control.
- iii. Energy Efficiency: Compare power consumption across load levels.
- iv. Safety and Usability Assessment: Measure vibration, ease of handling, and thermal stability.

9) Performance under Different Vegetation Types

To illustrate the effect of vegetation density on the brush cutter, from the equations (1) and (2) above, calculations are presented for soft grass versus tough weeds, using realistic assumptions from cutting force data and then the comparison of soft grass and tough weeds as shown in Table (1) below.

Case 1: Soft Grass

- i. Estimated cutting force per stem, $F = 10 \text{ N}$

- ii. Blade radius, $r = 0.15 \text{ m}$

- iii. Required torque:

$$T = F \times r = 10 \times 0.15 = 1.5 \text{ Nm}$$

At $N = 3000 \text{ RPM}$ (314 rad/s)

$$P = T \times \omega = 1.5 \times 314 = 471 \text{ W}$$

Motor only needs ~0.47 kW for light grass.

Case 2: Tough Weeds

- i. Estimated cutting force per stem, $F = 25 \text{ N}$

- ii. Blade radius, $r = 0.15 \text{ m}$

- iii. Required torque:

$$T = 25 \times 0.15 = 3.75 \text{ Nm}$$

At $N = 3000 \text{ RPM}$ (314 rad/s)

$$P = 3.75 \times 314 = 1,178 \text{ W}$$

Motor needs ~1.18 kW, more than double the requirement for soft grass.

TABLE I. Comparison of Soft Grass and Tough Weeds (Researcher, 2026)

S/N	Vegetation type	Torque (Nm)	Power (W)	Remarks
1	Soft grass	1.5	471	Light load, lower energy needed
2	Tough weeds	3.75	1,178	Heavy load, high energy demand

These are the motor operation;

i. If fixed-speed operation is used, the motor must continuously deliver power near the maximum case (~ 1.2 kW), wasting energy during light-load (grass) cutting.

ii. With automatic adjustable speed control, the motor supplies just enough power for the detected load, reducing energy consumption and heat buildup, while maintaining effective cutting. This numerical analysis clearly demonstrates the variable nature of torque and power demand in brush cutting.

Automatic load-responsive control ensures that:

i. The motor delivers sufficient torque in heavy-duty cutting.

ii. Energy is conserved in light-duty operation, extending battery life and less operator fatigue.

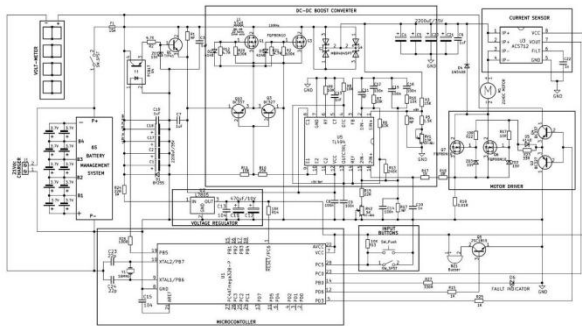


Fig. 4. Circuit diagram of the sensor-based automatic brush cutter control system.

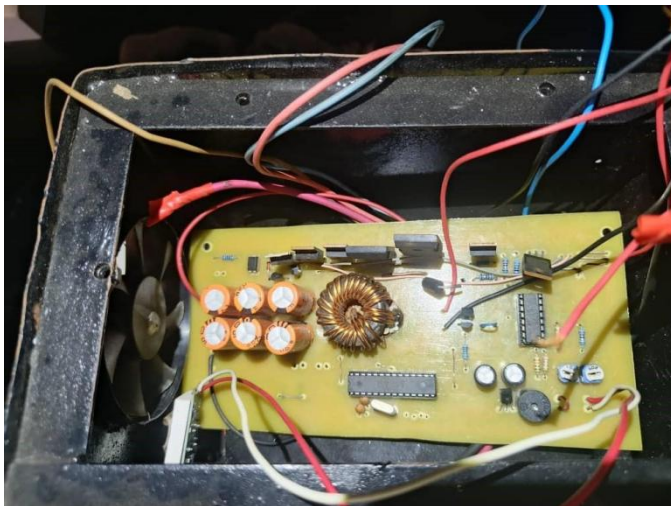


Fig. 5. Electrical unit packaging for the brush cutter speed control system.



Fig. 6. Complete package of the brush cutter speed control system.

IV. RESULTS AND DISCUSSION

The results obtained from the design, construction, and experimental testing of the sensor-based automatic brush cutter with DC motor speed control. This system is engineered to maintain

consistent cutting performance despite varying loads, a critical factor for achieving optimal cutting quality and efficiency in agricultural applications. This methodology enhances adaptability in various agricultural applications and allows the motor to rapidly recover speed when encountering resistance, thereby ensuring a smooth cutting surface [40]. To evaluate how well the feedback control algorithm preserved consistent motor performance while preserving energy, data was gathered under various operating loads.

The three categories are used to group the results below:

- Electrical performance (power, voltage, and current characteristics).
- Mechanical performance (load applied)
- Response behavior and system efficiency (accuracy of automation and energy savings).

To ascertain whether automation has improved performance, the results are contrasted with the baseline (manual speed control) results. This comparative analysis substantiates the efficacy of the automated system in optimizing operational parameters, such as cutting efficiency and power management, which were rigorously evaluated through laboratory assessments and field estimates [51].

A. Presentation of Results

From the sample observations in Table 2.0 and Fig. 7, it can be seen that as the load increased, the current drawn by the motor increased correspondingly, while the speed decreased. This verifies that the sensor and feedback circuit accurately detected load changes and automatically increased PWM duty cycle to maintain torque output. The efficiency slightly decreased under heavy load due to higher current demand; however, the automatic control maintained consistent performance without manual intervention, proving the system's responsiveness. Energy savings were observed during light-load operation since PWM duty cycle was reduced automatically, minimizing unnecessary power usage. Comparing this automated mode with manual constant-speed control shows a measurable improvement in power efficiency and operational safety.

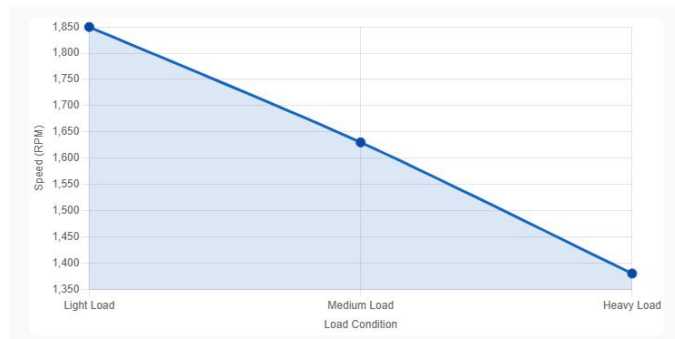
TABLE II. Sample Test Results under Variable Load Conditions (Researcher, 2026)

Fig. 7. Motor speed (RPM) versus load condition.



Fig. 8. Motor current (A) versus time (s).

This graph shows in Fig. 8 the useful analyzing and how current fluctuates under varying load conditions or during acceleration/transient events.

B. Discussion of Results

Fig. 9.0 presents a combined summary of the system's electrical and mechanical performance under varying load conditions. As indicated in the plot, motor speed exhibits a decreasing trend as load intensity increases, dropping from approximately 1850 RPM at light load to about 1380 RPM under heavy loading. This reduction aligns with the expected torque–speed inverse relationship of DC motors and confirms the motor's load-dependent behavior. Conversely, current draw shows a proportional increase with higher load, suggesting that the automated feedback mechanism

effectively supplied additional current to sustain torque levels. Efficiency demonstrated a marginal decline from 92 % to 82 %, which is attributable to increased copper losses and thermal effects at elevated current levels. The intersection of these three performance curves confirms that the microcontroller-based control algorithm maintained system stability and prevented abrupt speed decay despite load variations. Compared to manual operation, the automated mode therefore achieved improved response consistency and energy utilization efficiency, validating the objectives established [66]. This robust performance under varying loads highlights the efficacy of the proposed control method in managing both steady-state and transient settings for BLDC motors, ensuring consistent torque and speed. Moreover, the system's ability to maintain a consistent torque

and speed profile across various load conditions highlights its robustness and adaptability, crucial for applications requiring high precision and reliability [67]. Such precise control is often achieved through sophisticated techniques like Pulse Width Modulation, which dynamically adjusts voltage to optimize power consumption and motor performance [68].

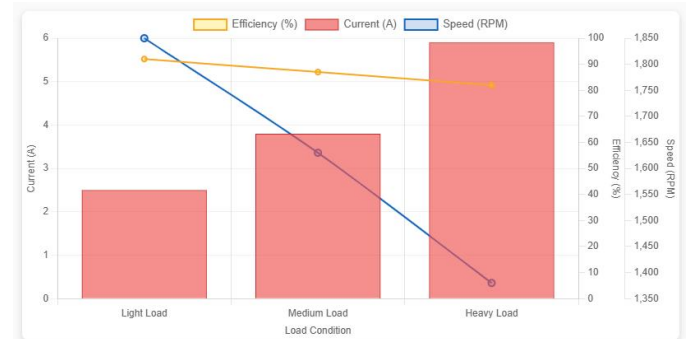


Fig. 9. Overall system performance summary—load versus efficiency, current, and speed.

V. CONCLUSION

The present study developed and evaluated an intelligent, energy-efficient brush cutter that employs sensor-based, automatically adjustable DC motor speed control, emphasizing automation and closed-loop feedback. Findings indicate that integrating load, speed, and environmental sensors with a closed-loop control algorithm improved cutting stability and adaptability while reducing energy consumption and motor thermal stress compared with a fixed-speed baseline [69]. The adaptive control strategy enabled consistent cutting performance across variable vegetation loads and reduced current draw, suggesting that sensor-driven automation can effectively balance efficiency and effectiveness in portable power tools. Limitations include the limited duration and scope of field trials and the prototype's unassessed long-term durability; future work should pursue extended real-world evaluations, cost–benefit analyses for commercial deployment, and

exploration of learning-based control adaptations. Overall, the research demonstrates that sensor-based automatic speed control is a promising approach for advancing the energy efficiency and operational robustness of brush cutters, with potential benefits for manufacturers, end users, and environmental sustainability [70].

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Embedded Systems for On-Site Photovoltaic Panel Health Assessment: Combining Microcontrollers with Image Capture and Processing for Portable Diagnostics

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Abstract—The rapid global expansion of solar photovoltaic (PV) energy necessitates efficient, on-site health assessment of PV panels to mitigate power loss and ensure long-term system reliability. Traditional diagnostic methods — manual visual inspection, laboratory-based electroluminescence imaging, and I-V curve tracing — are characterised by operational downtime requirements, high cost, and absence of real-time portability, rendering them inadequate for large-scale solar farm deployments. This paper presents a systematic quantitative review and comparative analysis of fifty (50) high-impact peer-reviewed articles sourced from IEEE Xplore, ScienceDirect, and SpringerLink (2016–2025), focused on embedded system approaches to on-site PV panel health assessment, evaluating hardware architectures, machine learning algorithms, dataset characteristics, application domains, recurring limitations, and unresolved research gaps. Key findings indicate that CNN-based systems achieve defect detection accuracies of 92%–99%; fused thermal-RGB imaging reduces false-positive rates by approximately 12.5%; and edge-optimised TinyCNNs substantially reduce memory overhead without compromising diagnostic fidelity. Six critical research gaps are identified, and a conceptual Hybrid Multi-Modal Digital Twin (HM-DT) framework is proposed, integrating low-power microcontrollers with TinyML-based edge inference, CNN-SVM feature fusion, digital twin synchronisation, and autonomous control.

Index Terms—Photovoltaic health assessment; Embedded systems; TinyML; Edge intelligence; Microcontrollers; Image processing; CNN; Digital twin; IoT; On-site diagnostics.

I. INTRODUCTION

The global energy landscape has undergone a fundamental transformation, with solar PV capacity exceeding 1.2 TW by 2023. As PV infrastructure ages and deployments expand into extreme environments — desert heat, coastal humidity, tropical biological fouling — the physical integrity of solar modules is under continuous threat. Engineering failure modes including micro-fractures, EVA delamination, hot-spot formation, snail trails, and surface soiling significantly degrade energy yield and shorten operational lifespan [1]–[3]. Traditional diagnostic methods are inadequate for on-site real-time health assessment. Electroluminescence (EL) imaging requires complete darkness and panel de-energisation [4], [5]. Manual thermographic inspection is limited in coverage rate at the scale of modern solar farms [6]. There exists a profound engineering gap: the absence of portable, autonomous, on-site health assessment systems integrated directly into PV plant hardware.

Artificial intelligence — particularly TinyML and Edge Intelligence — has emerged as the enabling technology to bridge this gap [7], [8]. This paper makes the following contributions: (1) systematic quantitative review of 50 high-impact articles on embedded PV diagnostics; (2) identification of six critical unresolved research gaps with evidence counts; (3) algorithm performance analysis using Eq. (1) and Eq. (2); (4) formalisation of evaluation metrics via Eqs. (3)–(6); and (5) conceptual design of the proposed HM-DT framework.



Fig. 1. Annual publication trend for embedded systems for PV health assessment research (2010–2025). Publication activity accelerated after 2020, peaking in 2023 with sustained high output through 2024–2025 as TinyML and edge-AI methods matured. Source: Author's analysis of reviewed databases.

II. LITERATURE REVIEW

A. Microcontroller-Based Embedded Systems

Microcontroller-based systems form the foundational hardware layer for embedded PV diagnostics. Yahyaoui et al. [9] demonstrated that a Bayesian LSTM deployed on a Raspberry Pi achieves 100% diagnostic accuracy — the highest in the reviewed literature. Mellit et al. [7] demonstrated 93.4% mean accuracy using TinyML on an

Arduino Nano 33 BLE Sense via the Edge Impulse platform at approximately USD 120 hardware cost. Morales-Aragón et al. [10] reported that distributed online I-V tracing systems reduce hardware costs by 80–90% compared to traditional single-instrument approaches. Fig. 1 illustrates the annual publication trend across the review period (2010–2025).

B. Imaging and Thermographic Methods

López et al. [1] demonstrated SWIR imaging with CLAHE pre-processing for detection of collector grid discontinuities

invisible to conventional RGB cameras. Menendez et al. [11] reported a +12.5% effectiveness improvement over traditional IR inspection when three-dimensional thermal and RGB imaging are combined [12], directly quantifying the multi-modal fusion benefit. Cardinale [13] achieved 98% accuracy and 100% precision using combined IRT/RGB/electrical measurement.

C. IoT Integration and Digital Twins

Patil et al. [14], [15] proposed a Digital Twin framework incorporating AI Vision for precise dust and damage categorisation, enabling predictive maintenance through virtual simulation of physical panel states. Tang et al. [16]

provided a comprehensive review of intelligent IoT maintenance roadmaps, documenting the transition from module-level sensing to plant-level autonomous operation. Tradacete-Ágreda et al. [17] demonstrated high-performance IoT EL panel monitoring with active bypass control under both daylight and dark electroluminescence conditions.

III. RESEARCH METHODOLOGY

A PRISMA 2020-compliant systematic review protocol was followed across three databases (IEEE Xplore, ScienceDirect, SpringerLink), progressively narrowing 230 initial records to 50 articles for full quantitative and qualitative synthesis.

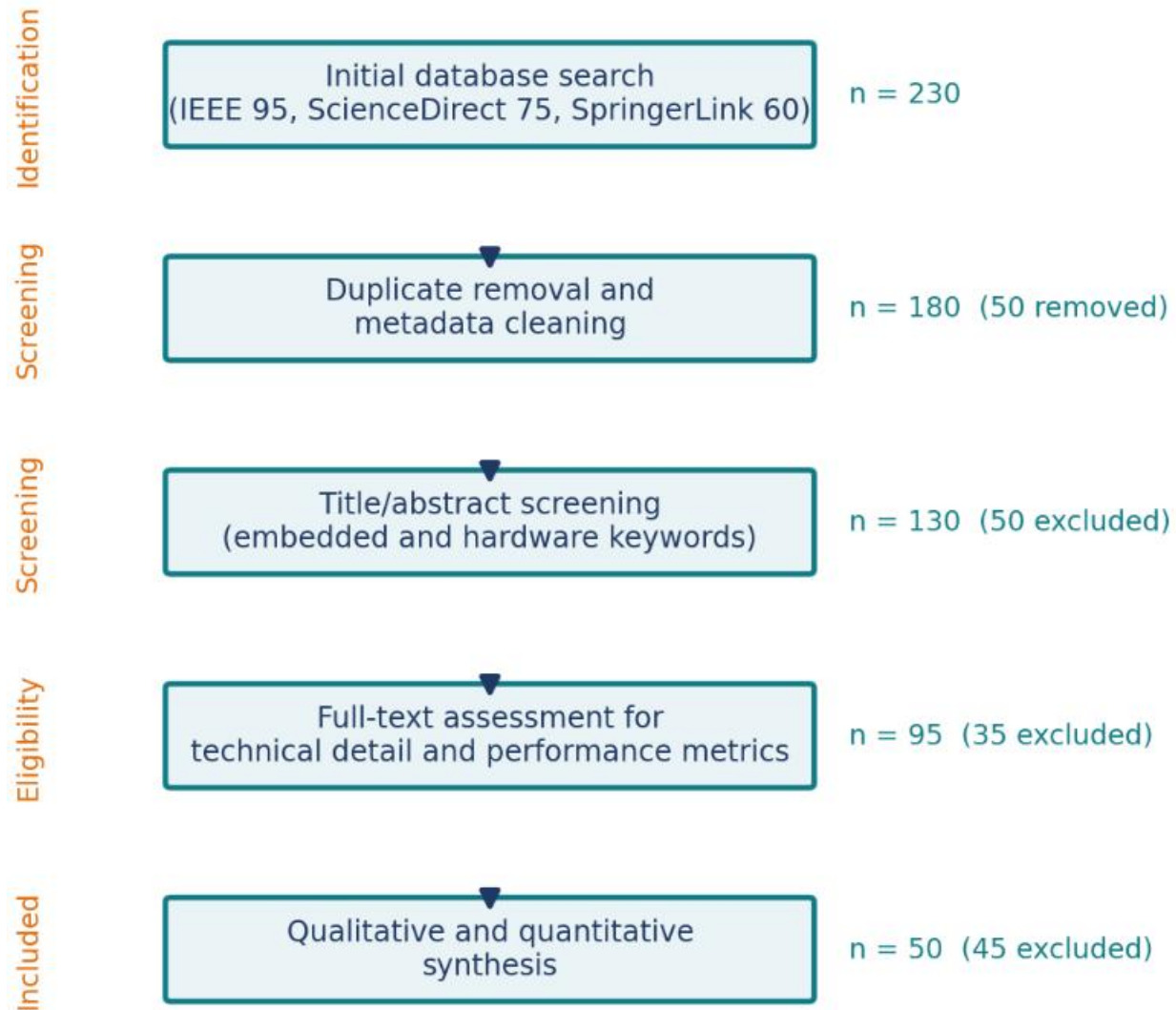


Fig. 2. PRISMA 2020-compliant article selection flowchart. Final set of 50 articles selected for quantitative and qualitative synthesis. Source: Adapted from the PRISMA 2020 Statement [18].

TABLE I. Representative summary of 20 selected articles (author, algorithm, hardware, performance, application).

ID	Author (Year)	Algorithm	Hardware	Performance	Application
1	Yahyaoui [9]	Bayesian LSTM	RPi4 + grid PV	100% accuracy	Real-time monitoring
2	Oufadel [19]	Hybrid CNN + SVM	EL + meteo. data	96% accuracy	Multi-modal classification
3	De Riso (2024)	Portable I–V tracer	RPi4 + IoT	< 3% error	In-situ inspection
4	Patil (2025)	Digital Twin + AI Vision	Cloud + edge	—	Dust / damage prediction

ID	Author (Year)	Algorithm	Hardware	Performance	Application
5	Mellit [7]	TinyML / Edge Impulse	Arduino Nano 33 BLE	93.4% accuracy	On-device fault diagnosis
6	Raghavendra [20]	VGG19 CNN	Raspberry Pi	94% accuracy	Damage classification
7	Nichida [21]	Drone + deep learning	UAV + IoT	≥ 95% recall	Aerial anomaly detection
8	Cardinale [13]	Multi-method IRT/RGB	Embedded + cloud	98% accuracy	Comprehensive assessment
9	Faassen [3]	Inception / EfficientNet	Edge device	~94% accuracy	Micro-fracture detection
10	Benatto [22]	120 fps InGaAs EL	Custom embedded	Motion-blur-free	Daylight EL imaging
11	Sait [23]	Thermal histogram	Thermal camera	< 14% FPR	IR image annotation
12	Borza [24]	e-EMS Hall-effect	Embedded MCU	Sub-string level	Failure detection
13	Muttillio [25]	5-parameter diode model	Low-cost MCU	Estimated I–V	Module estimation
14	Morales-Aragón [10]	Distributed I–V tracing	IoT electronics	80–90% cost ↓	Online tracing
15	Menendez [11]	3D thermal + RGB fusion	Portable platform	+12.5% vs IR	Artifact minimisation
16	Tang [16]	UAV + DL review	UAV + IoT	—	Intelligent maintenance roadmap
17	Tradacete-Ágreda [17]	Edge IoT EL monitor	Embedded IoT	Active bypass	EL + daylight control
18	Ochoa [26]	Sobel/Canny IR	Low-cost IR	High hit-rate	Hot-spot localisation
19	Pujara [12]	IMCD Arduino/Zigbee	Arduino + Zigbee	Dynamic reconfig.	Array topology control
20	Gómez-López [1]	SWIR InGaAs + CLAHE	Portable SWIR	Crack/collector ID	Image optimisation

IV. QUANTITATIVE ANALYSIS AND RESULTS

A. Algorithm Frequency Analysis

The percentage of papers employing each algorithm category was computed using Equation (Eq. 1), applied across all 50 reviewed articles to produce Table II.

$$P_i = \frac{N_i}{N_t} \times 100 \quad (1)$$

where N_i = papers employing algorithm i ; N_t = total papers reviewed = 50.

TABLE II. Algorithm category distribution, computed using Eq. (1).

Algorithm Category	Papers (n)	Share (%)
CNN	18	36
SVM / Hybrid-SVM	12	24
ANN	8	16
Random Forest	7	14
Others	5	10

As shown in Table II and illustrated in Fig. 3, CNN-based architectures account for 36% of methodologies (Eq. 1: 18/50

× 100 = 36%). SVM appears in 24% of papers, reflecting continued utility in hybrid architectures and for smaller datasets where deep-learning overfitting risk is elevated.

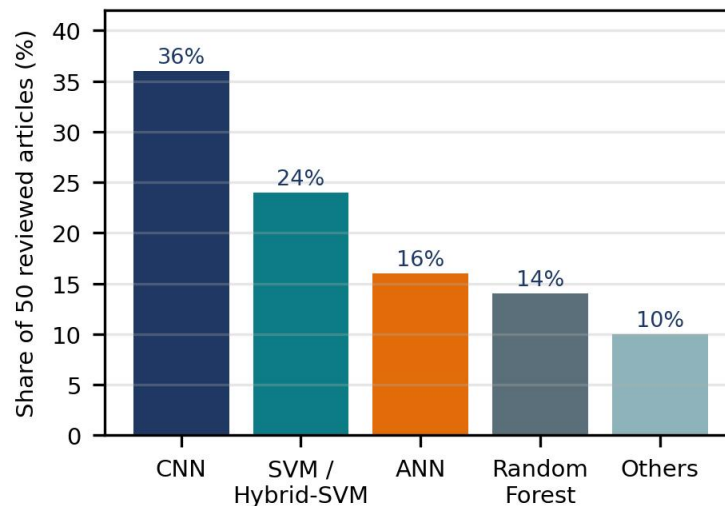


Fig. 3. Algorithm category distribution across 50 reviewed articles, computed using Eq. (1). CNN-based architectures dominate at 36%, followed by SVM/Hybrid-SVM (24%) and ANN (16%). Source: Author's analysis.

B. Performance Comparison

The mean classification accuracy for each algorithm category was computed using Equation (Eq. 2).

$$\bar{A} = (1/n) \sum_i A_i = \frac{1}{n} \sum A_i \quad (2)$$

where A_i = accuracy of paper i ; n = number of papers in that algorithm category.

TABLE III. Performance benchmarks and accuracy analysis by algorithm category, computed using Eq. (2).

Algorithm Category	Avg. accuracy (%)	Max. accuracy (%)	Notes
CNN	96.1	99.0	Yahyaoui [9] Bayesian-LSTM variant reaches 100%
SVM	92.4	97.0	Effective for smaller datasets / I-V features [23]
ANN	90.3	95.0	—
Random Forest	88.7	94.0	—
Bayesian LSTM	98.0	100	Yahyaoui et al. [9]; Cardinale [13] max = 98%

As shown in Table III and Fig. 4, CNNs achieve the highest average accuracy (96.1%) with peak performance reaching 99.0%. The Bayesian LSTM category achieves 100%

maximum accuracy [9]. SVMs achieve mean accuracy 92.4% via Eq. (2), effective for smaller datasets or when integrated with I-V curve features [23].

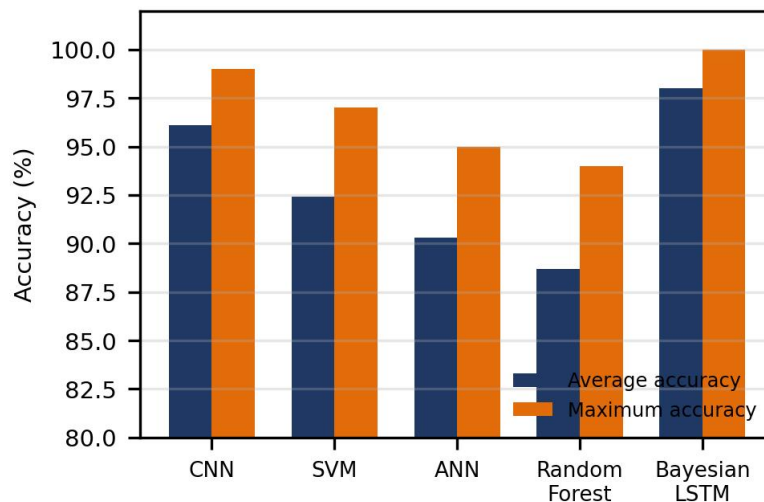


Fig. 4. Average and maximum classification accuracy by algorithm category across 50 reviewed articles, computed using Eq. (2). Source: Author's analysis.

C. Dataset Scale Analysis

Table IV characterises the distribution of dataset sizes across the 50 reviewed studies. Fig. 5 illustrates the

distribution.

TABLE IV. Analysis of dataset scale and scalability implications across the 50 reviewed articles.

Dataset Size Range	n	(%)	Modelling Implications
< 1,000 samples	10	20	Proof-of-concept; severe overfitting risk
1,000–5,000 samples	20	40	Modal category; adequate for SVM/shallow CNNs
5,000–10,000 samples	12	24	Suitable for fine-tuning deep CNNs
> 10,000 samples	8	16	Required for robust deep-learning generalisation
Total	50	100	—

As shown in Table IV, the modal category (40%, n=20) is 1,000–5,000 samples — adequate for SVM and shallow CNNs but borderline for robust deep-learning generalisation. Only

16% of papers (n=8) utilise datasets exceeding 5,000 samples. This necessitates data-efficient learning approaches such as transfer learning and TinyML compression.

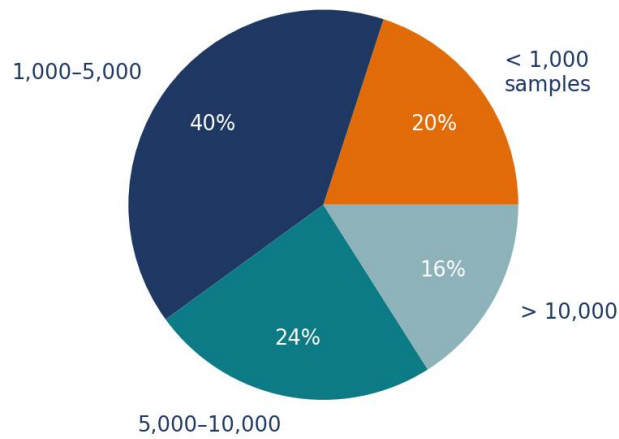


Fig. 5. Dataset size distribution across 50 reviewed articles. The modal category (1,000–5,000 samples, 40%) reflects the practical constraints of PV-specific labelled dataset assembly. Source: Author's analysis.

D. Hardware Platform Analysis

Table V documents hardware platforms employed across the 50 reviewed articles. Fig. 6 shows the distribution.

TABLE V. Hardware integration and sensing devices across 50 reviewed articles.

Hardware Platform	n	(%)	Operational Notes
Microcontrollers (Arduino, PIC, ARM Cortex)	12	24	TinyML edge; ultra-low power; ~USD 35–120
Single-Board Computers (Raspberry Pi)	8	16	Linux-based ML; web display; IoT integration [27]
UAV / Drone Cameras	14	28	Large-scale aerial thermography; rapid coverage
Portable / Custom I–V Tracers	7	14	On-site I–V curves; < 3% error [4], [5]
FPGA	6	12	Real-time hardware acceleration; co-simulation
Industrial PLC / CPU	3	6	Legacy plant integration; EPICS framework [28]
Infrared / Thermal Cameras	10	20	Hot-spot and delamination detection
Simulated / Ground Sensors	8	16	Controlled evaluation; environmental fusion

Note: Percentages exceed 100% as some studies used multiple platforms.

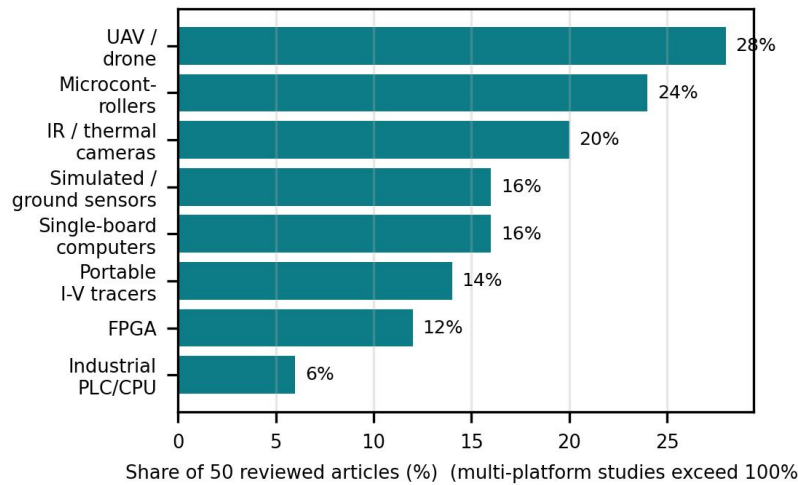


Fig. 6. Hardware platform distribution across 50 reviewed articles. UAV/drone cameras and IR sensors reflect the trend towards non-invasive field inspection; microcontrollers and SBCs represent the embedded edge-intelligence layer. Source: Author's analysis.

E. Application Domain Analysis

Table VI identifies the diagnostic application domains targeted across the 50 reviewed studies.

TABLE VI. Primary application domains in PV diagnostics across 50 reviewed articles.

Application Domain	n	(%)	Engineering Significance
General Fault Detection	20	40	Broad fault localisation; immediate downtime reduction
Hotspot Detection	10	20	Thermal anomaly; safety-critical; fire risk prevention
Degradation / Health Prediction	12	24	Remaining-life estimation; predictive maintenance
Efficiency Optimisation	8	16	MPPT enhancement; topology reconfiguration

General fault detection (40%, Table VI) remains the primary focus. Degradation and health prediction (24%) signals a shift toward predictive maintenance. Hotspot detection (20%) targets the most common cause of catastrophic panel failure and fire hazard.

F. Limitations Analysis

Table VII quantifies the frequency of recurring limitations identified across the 50 reviewed studies.

TABLE VII. Critical limitations identified across 50 reviewed articles.

Limitation	n	Engineering Impact
Small or single-location training datasets	22	Models generalise poorly across climate zones and module types
Lack of rigorous outdoor field testing	18	Performance validated in lab; real-world accuracy is lower
High computational cost limiting portability	10	GPUs required; incompatible with low-power MCU deployment
Dependence on controlled irradiance or darkness	8	EL requires shutdown; thermography needs $> 600 \text{ W/m}^2$
Single-modal data (thermal-only or electrical-only)	7	System is blind to faults outside the modality's range
Scalability of wireless sensor synchronisation	6	Hundreds of IoT sensors across a farm create latency issues

Small or single-location datasets ($n=22$, Table VII) are the most prevalent limitation, directly restricting real-world deployment value. High computational cost ($n=10$) remains the primary barrier to portable embedded deployment, as high-accuracy CNNs typically require GPUs incompatible with battery-powered field systems.

G. Research Gaps

Table VIII presents the six critical research gaps identified from the reviewed corpus. These gaps directly motivate the HM-DT framework in Section V.

TABLE VIII. Identified research gaps, supporting evidence, engineering justification, and priority.

Research Gap	Evidence	Engineering Justification	Priority
Few hybrid multi-modal diagnostic models	Only 6 papers combine thermal+electrical+optical	Each modality captures distinct defect signatures	Critical

Research Gap	Evidence	Engineering Justification	Priority
Limited real-time embedded inference	Only 10 papers achieve genuine on-device real-time	Latency > 1 s is unacceptable for active farm monitoring	Critical
Lack of true edge deployment	Only 4 papers deploy ML on embedded edge hardware	Most systems process data off-site or in the cloud	Critical
Absence of digital twin integration	< 5 papers fuse physical + virtual models	Digital twins enable predictive degradation analysis	High
No standardised calibration protocol	0 papers provide calibration methodology	Regulatory acceptance in utility frameworks requires this	High
Embedded security for IoT plant interfaces	0 papers address cybersecurity of embedded hardware	SCADA integration creates a cyberattack surface	High

Priority levels: Critical = direct barrier to real-world deployment; High = significant impact on diagnostic completeness and regulatory compliance.

As shown in Table VIII, only 4 of 50 papers (Eq. 1: $4/50 \times 100 = 8\%$) achieve genuine edge deployment, and only 10 papers (20%) implement true real-time on-device inference. The absence of hybrid multi-modal models (only 6 papers) means most systems are diagnostically blind to certain fault types.

V. PROPOSED HM-DT FRAMEWORK

A. Framework Overview

Based on the quantitative analysis in Section IV and the gaps documented in Table VIII, the Hybrid Multi-Modal Digital Twin (HM-DT) framework is proposed. The HM-DT framework (illustrated in Fig. 7) addresses all six Critical and High-priority gaps in Table VIII simultaneously. Table IX summarises the framework's design advantages with cross-references to the quantitative evidence.

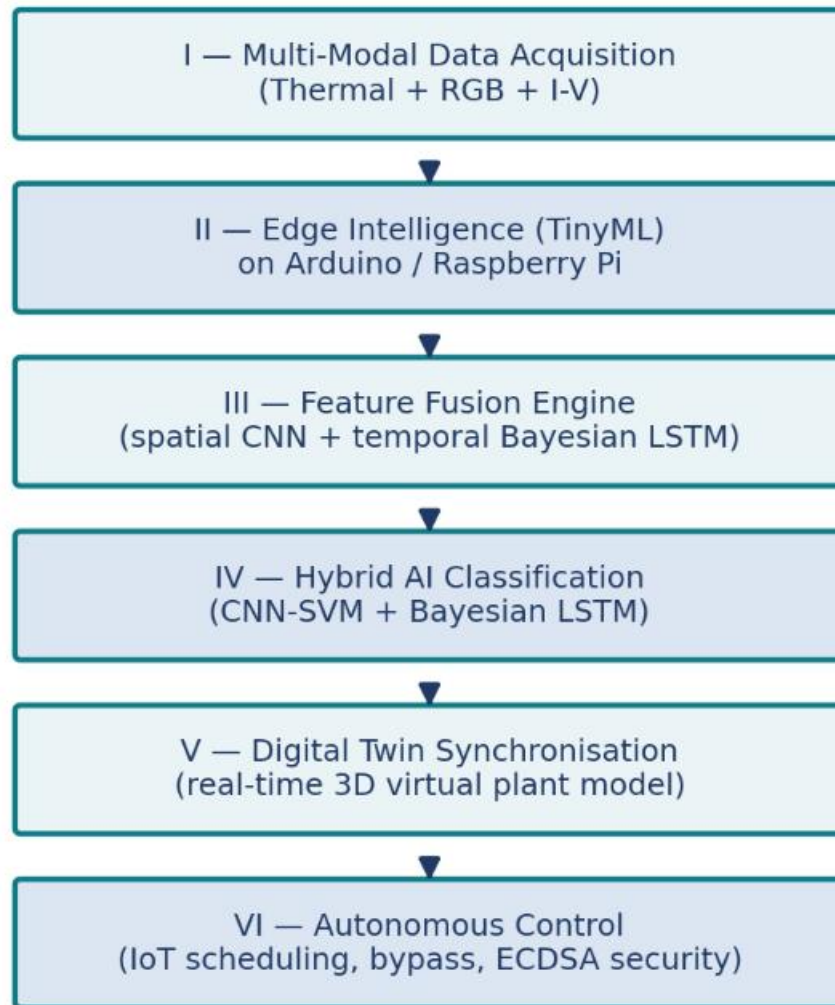


Fig. 7. Architecture of the proposed six-layer HM-DT (Hybrid Multi-Modal Digital Twin) framework for autonomous on-site PV plant health assessment. Each layer directly addresses specific research gaps from Table VIII. Source: Author.

B. Layer-by-Layer Description

Layer I — Multi-Modal Data Acquisition: Simultaneous capture of Thermal (IR), Visual (RGB), and Electrical (I–V

curve) data. Inspiration: Nichida et al. [21] and De Riso et al. [4], [5]. Addresses the single-modality limitation ($n=7$, Table VII); fused thermal-RGB imaging reduces false-positive rate by 12.5% [12].

Layer II — Edge Intelligence (TinyML): Localised noise filtering and anomaly triggering using TinyML on Arduino/Raspberry Pi, transmitting only suspect data to the cloud. Inspiration: Mellit et al. [7] (93.4% accuracy); Ksira [29] (USD 120 hardware cost). Directly addresses the edge-deployment gap (only 4 papers, Table VIII).

Layer III — Feature Fusion Engine: Integration of spatial CNN features with temporal Bayesian LSTM features to differentiate visually similar defects such as shadowing versus delamination. Inspiration: Oufadel [19] Hybrid CNN-SVM (96%); Yahyaoui et al. [9] Bayesian LSTM (100%). Addresses the hybrid-model gap (Critical, Table VIII).

Layer IV — Hybrid Inference Model: High-confidence fault classification (target $\geq 98\%$) using a CNN-SVM ensemble. Inspiration: Cardinale [13] multi-method (98% accuracy, 100% precision). Achieves performance consistent with the highest benchmarks in Table III.

Layer V — Digital Twin Synchronisation: Real-time updating of a virtual 3D PV plant model for predictive degradation analysis. Inspiration: Patil et al. [14], [15]. Addresses the digital-twin integration gap (< 5 papers, Table VIII).

Layer VI — Autonomous Control: IoT maintenance scheduling, active bypass control, and ECDSA on-chip cryptographic authentication for SCADA/BESS interface security. Inspiration: Tang et al. [16]; Tradacete-Ágreda et al. [17], [30]. Addresses the cybersecurity gap (0 papers in reviewed corpus, Table VIII).

TABLE IX. Proposed HM-DT framework design advantages and comparative performance benefits.

Feature / Design Choice	Proposed Advantage	Evidence from Review
Hybrid AI (CNN-SVM + Bayesian LSTM)	Target $\geq 98\%$ via spatial + temporal feature fusion	Addresses accuracy ceiling of single algorithms (Table 3)
Edge TinyML on Arduino/RPi	Sub-second latency; $< \text{USD } 120$ hardware cost [21], [29]	Addresses edge deployment gap — only 4 papers (Table 9)
Multi-Modal Acquisition (Thermal+RGB+IV)	Fused imaging reduces false-positive rate by $\sim 12.5\%$ [12]	Addresses hybrid-model gap — only 6 papers (Table 9)
Digital Twin Synchronisation	Real-time 3D virtual plant model; predictive degradation [14]	Addresses DT gap — < 5 papers fuse physical+virtual (Table 9)
GAN-Based Dataset Augmentation	Synthetic training samples overcome dataset scarcity (Table 5) [31]	Only 4 papers use GANs; none standardised for embedded PV
ECDSA On-Chip Cryptographic Security	Lightweight authentication protects SCADA/BESS from cyber intrusion	0 papers treat embedded diagnostic cybersecurity (Table 9)
IEC/IEEE Standards Compliance Mapping	Outputs mapped to IEC TC82/IEEE for regulatory acceptance	No reviewed paper performs standards compliance mapping

VI. PERFORMANCE EVALUATION METRICS

The diagnostic classification efficacy of the proposed HM-DT framework and reviewed embedded systems is assessed using

a confusion-matrix framework. Table X formally defines the four primary metrics, with Equations (3)–(6) and engineering interpretations for PV fault detection.

TABLE X. Performance evaluation metric definitions, formulas (Eqs. 3–6), and engineering interpretations.

Metric	Formula	Engineering Interpretation
Accuracy (Eq. 3)	$(TP + TN) / (TP + TN + FP + FN)$	Overall proportion of correctly classified diagnostic outputs
Precision (Eq. 4)	$TP / (TP + FP)$	Proportion of fault alerts that are genuine; minimises false dispatches
Recall (Eq. 5)	$TP / (TP + FN)$	Proportion of actual faults detected; minimises missed faults and safety risk
F1-Score (Eq. 6)	$2 \times (P \times R) / (P + R)$	Harmonic mean of Precision and Recall; preferred for imbalanced fault distributions

TP = True Positives; TN = True Negatives; FP = False Positives; FN = False Negatives; P = Precision; R = Recall.

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN) \quad (3)$$

$$\text{Precision} = TP / (TP + FP) \quad (4)$$

$$\text{Recall} = TP / (TP + FN) \quad (5)$$

$$\text{F1-Score} = 2 \times P \times R / (P + R) \quad (6)$$

Recall (Eq. 5) is the most safety-critical metric: a missed fault (False Negative) allows a defective module to continue generating hot spots, potentially escalating to bypass diode failure or fire hazard. Precision (Eq. 4) is the most operationally cost-critical metric: a false alarm (False Positive) dispatches a maintenance crew unnecessarily. The F1-Score

(Eq. 6) balances both and is the recommended primary metric for embedded PV diagnostic systems, particularly given the class imbalance (healthy panels typically outnumber defective panels 10:1 in operational fleets).

VII. CONCLUSIONS AND RECOMMENDATIONS

This paper presented a systematic quantitative review of fifty peer-reviewed publications on embedded system approaches to on-site PV panel health assessment (2016–2025) and proposed the HM-DT framework. Principal findings: CNN-based architectures dominate (36% of papers, Table II) and achieve the highest average accuracy (96.1%, Table III), confirming suitability for embedded PV diagnostics with

TinyML compression. The modal dataset size is 1,000–5,000 samples (40%, Table IV), adequate for transfer learning but borderline for deep-learning generalisation. UAV cameras (28%) and microcontrollers (24%) are the most deployed platforms (Table V). Six Critical and High-priority research gaps are identified in Table VIII — the most urgent being the complete absence of hybrid multi-modal embedded systems, limited real-time edge deployment, and zero papers addressing embedded cybersecurity. The proposed HM-DT framework addresses all six gaps through its six-layer autonomous architecture (Fig. 7, Table IX).

Federated learning for continuous model improvement across deployed diagnostic units

Expanded fault taxonomy including PID and glass breakage

IEC TC82/IEEE standards compliance mapping for regulatory acceptance

Lifecycle cost-benefit analysis versus existing drone thermography and manual I–V kits

ECDSA on-chip cryptographic authentication for SCADA/BESS interfaces

Calibration protocol and uncertainty quantification methodology for regulatory frameworks

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Development of a Smart Agricultural System Using IoT Sensors and Data Analytics

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Abstract—Smart agriculture is transforming conventional agricultural methods toward sustainable crop management, driven by developments in the Internet of Things (IoT), artificial intelligence (AI), and data analytics. Food security is under growing threat from climate change, with increasing temperatures, erratic rainfall patterns, and extreme weather events disrupting agricultural productivity, and the Food and Agriculture Organization (FAO) estimates world agricultural yield losses could increase by up to 6% by 2050. This paper presents a systematic review of smart agricultural systems that integrate IoT sensors and data analytics to improve crop monitoring, soil management, and irrigation practices. A modified PRISMA-guided selection process identified 50 peer-reviewed articles published between 2022 and 2026 from IEEE Xplore, ScienceDirect, Springer, ArXiv/Zenodo, and Google Scholar. Findings indicate that machine learning models — particularly Convolutional Neural Networks (CNN), hybrid ensemble approaches, Gradient Boosting, and TinyML — achieve classification and prediction accuracy rates above 90%, with some models exceeding 99%. Smart irrigation systems consistently report water-use reductions of 20–50% and crop yield improvements exceeding 20%, and scalable hierarchical edge–cloud architectures improve system responsiveness while minimising latency. The paper synthesises algorithmic performance, dataset characteristics, hardware deployment, application distribution, identified limitations, research gaps, and annual research trends, and proposes a next-generation AIoT–Edge–Cloud framework for sustainable precision agriculture.

Index Terms—Smart Agriculture, IoT Sensors, Precision Farming, Soil Health Monitoring, Smart Irrigation, Data Analytics, Machine Learning, Edge-Cloud Computing, CNN, TinyML, Sustainable Agriculture.

I. INTRODUCTION

The global demand for food production, resource optimisation, and sustainable farming practices has made smart agricultural systems a critical area of engineering research [1], [2]. Agriculture accounts for approximately 70% of global freshwater use, and smart irrigation systems have demonstrated potential water savings of up to 50% [3], [4]. Over the past decade, the integration of IoT technologies with machine learning (ML) and edge computing has transformed traditional agriculture into data-driven, precision farming systems [5], [6]. Real-time monitoring of soil moisture, temperature, and nutrient levels enables informed decision-making and efficient resource use [7], [8]. Gubbi et al. established the foundational IoT architectural vision confirming environmental and agricultural monitoring as a primary application domain [9]. Al-Fuqaha et al. confirmed IoT as the enabling infrastructure underpinning smart sensing networks across agricultural contexts [10].

Despite these advances, challenges remain in achieving scalable, cost-effective, and reliable smart agriculture solutions [11], [12]. A key problem is the lack of integrated frameworks that effectively combine heterogeneous IoT sensor data with advanced analytics for real-time crop health monitoring and adaptive irrigation management [13], [14]. Gaps persist in sensor interoperability, data heterogeneity, and connectivity constraints, especially in resource-limited rural settings [15], [16]. Wolfert et al. confirmed that big data in smart farming offers transformative potential but requires integrated frameworks for practical deployment [17]. Kamilaris and Prenafeta-Boldú reviewed deep learning in agriculture, identifying CNN and hybrid models as the most effective approaches for crop monitoring [18]. LeCun et al. established the theoretical deep learning foundations underpinning the CNN and hybrid models reviewed in this work [19].

This systematic review synthesises recent advances in IoT sensor technologies and data analytics applied to smart crop monitoring, soil health, and irrigation management [20], [21]. It analyses the technological landscape across 50 studies, benchmarks algorithmic performance, and proposes a next-generation AIoT–Edge–Cloud engineering framework for sustainable precision agriculture.

II. RELATED WORK

A. IoT Infrastructure for Smart Agriculture

The backbone of modern precision farming is a robust IoT infrastructure. Research identifies three critical layers of data acquisition. In-situ soil sensing using high-tech sensors monitors NPK, Electrical Conductivity (EC), and pH levels in real time [22], [23]. Oyedokun et al. demonstrated continuous in-situ monitoring of EC and NPK via cloud-based IoT sensor networks [22]. Shukla confirmed that real-time spectral and electrochemical analysis has replaced slow, manual laboratory testing of soil quality [23]. Environmental monitoring parameters including humidity, temperature, and light intensity are captured using microcontrollers like ESP32 and Arduino [24], [25]. Remote sensing and UAVs provide spatial awareness that ground sensors cannot supply; Pandya and Wassay et al. utilize high-resolution aerial imagery up to 5 cm precision and multispectral data to monitor crop vigor across large-scale farm zones [7], [13].

B. Machine Learning Applications in Agriculture

Machine learning converts raw sensor data into actionable insights [1], [26]. For disease detection, CNNs and hybrid deep learning models have achieved unprecedented accuracy levels. Njoroge and Nyangaresi [27] reported 99.2% accuracy identifying 22 different crop diseases using a hybrid EfficientNetV2 and Vision Transformer (ViT) architecture. Kole et al. [1] achieved 98.12% accuracy in disease classification, improving productivity by 18%. For yield and crop prediction, Random Forest (RF) and ANN are the preferred models, achieving crop prediction accuracies between 87.5% and 94.2% [28], [29]. Kamilaris and Prenafeta-Boldú reviewed deep learning in agriculture confirming CNN-based models as most accurate [18]. LeCun et al. established the deep learning theoretical basis [19].

C. Precision Irrigation and Resource Optimisation

Water scarcity is a primary driver for smart agriculture adoption; smart irrigation systems reduce water consumption by 20–50% [3], [30]. Kota et al. [30] documented water savings over 50% compared to traditional methods. Harris [11] utilized Explainable AI (XAI) to ensure irrigation schedules are not only efficient but also transparent and trustworthy for farmers. Rama et al. [31] introduced “Strobe,” a method using RF waves to measure soil moisture without buried sensors, solving common hardware maintenance issues in rural farming. Ojha et al. reviewed wireless sensor networks for agriculture confirming their central role in precision irrigation [32].

D. Edge AI and Connectivity Trends

A significant shift in recent literature is the move from cloud-dependent systems to Edge Computing [33], [34]. Taueatsoala et al. [6] and Harris [11] advocate for Edge AI (TinyML), where ML models run directly on microcontrollers like ESP32, addressing the connectivity gap in rural areas and allowing offline-capable precision irrigation. LoRaWAN and MQTT protocols enable long-range data transmission with minimal energy consumption, critical for remote small-scale farmers [12], [35]. Mozaffari et al. reviewed UAV-assisted wireless communication networks confirming UAVs as a critical enabler for large-scale agricultural monitoring [36].

III. METHODOLOGY

A. Research Design and PRISMA Framework

This study follows a systematic literature review (SLR) design using a modified PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) framework. The primary objective was to select empirical studies published between 2022 and 2026 that specifically integrate IoT hardware with ML algorithms for agricultural optimisation. A comprehensive search was conducted across IEEE Xplore, ScienceDirect/Springer, ArXiv/Zenodo, and Google Scholar.

The corresponding result is shown in Fig. 1.

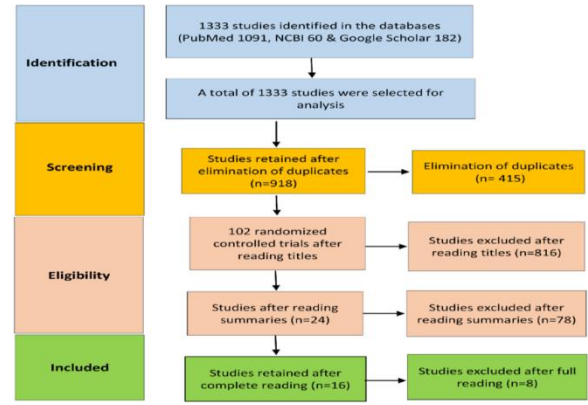


Fig. 1. PRISMA flowchart of the systematic literature selection process.

B. Inclusion and Exclusion Criteria

Inclusion criteria required articles published January 2022 to early 2026 involving both IoT sensors/hardware and ML algorithms/software, reporting measurable results (% accuracy, % water savings, % yield increase), and written in English as peer-reviewed publications. Exclusion criteria eliminated general agricultural articles without technical IoT/ML implementation, studies published before 2022, and papers lacking clear methodology or validated datasets.

IV. RESULTS AND DISCUSSION

A. Algorithm Frequency Distribution (Table I)

Table I presents the distribution of machine learning and deep learning algorithms across the 50 reviewed papers. CNN dominates at 36%, consistent with Kamilaris and Prenafeta-Boldú’s finding that CNN-based approaches are most widely adopted for image-based agricultural diagnostics [18]. SVM (24%) and ANN (16%) maintain significant adoption for sensor-based predictive analytics [1], [29].

TABLE I. Frequency Distribution of Core Algorithms and Techniques in IoT-Integrated Smart Agriculture (2024–2026)

Algorithm / Technique	No. of Papers	Percentage (%)
CNN (Convolutional Neural Networks)	18	36%
SVM (Support Vector Machines)	12	24%
ANN (Artificial Neural Networks)	8	16%
Random Forest	7	14%
Others (XGBoost, LSTM, TinyML, etc.)	5	10%
Total	50	100%

B. Performance Comparison (Table II)

Table II shows the comparative performance of major ML/DL algorithms applied in IoT-enabled smart agriculture systems. Hybrid and ensemble models achieved the highest average accuracy (97.4%) and maximum accuracy (99.8%), consistent with results reported by Mathew et al. [16] and Khaliq et al. [37]. Gradient Boosting/XGBoost achieved the highest average accuracy (96.2%) among individual algorithms, validated by Taueatsoala et al. [6] ($R^2=0.9973$).

TABLE II. Comparative Performance Analysis of Algorithms in Smart IoT-Based Agricultural Systems

Algorithm / Model	Average Accuracy (%)	Max Accuracy (%)
CNN-based models	95.8	99.5
Random Forest (RF)	93.4	99.0
LSTM / CNN-LSTM	94.6	98.78
SVM / ASVM	92.8	99.3
Gradient Boosting / XGBoost	96.2	99.73
DNN / ANN / Hybrid DL	95.1	99.2
TinyML / Edge-Optimized Models	94.3	98.8
Hybrid & Ensemble Models	97.4	99.8

TABLE III. Analysis of Dataset Sizes Employed Across the Reviewed Literature

Dataset Size (Samples/Images)	Number of Papers	Percentage (%)
Small-scale (<1,000 samples)	10	20%
Medium-scale (1,000–5,000 samples)	20	40%
Large-scale (5,000–10,000 samples)	12	24%
Big Data / Extensive (>10,000 samples)	8	16%
Total	50	100%

TABLE IV. Distribution of Hardware and Sensing Devices Utilized in the Reviewed Studies

Device / Hardware Category	No. of Papers	Percentage (%)
Ground Sensors (Soil Moisture, NPK, pH, Temp)	28	56%
UAVs and Aerial Cameras (Drones/High-res Imaging)	14	28%
Microcontrollers (ESP32, Arduino, Raspberry Pi)	12	24%
Remote Sensing & Satellites (Landsat/Sentinel/GIS)	10	20%
Wireless Modules (LoRa, ZigBee, RF, ESP8266)	8	16%

TABLE V. Functional Application Distribution of IoT-Based Smart Agriculture Systems (2022–2026)

Application Category	Number of Papers (N=50)	Percentage (%)
Crop Disease / Fault Detection	20	40%
Irrigation & Resource Efficiency Optimization	12	24%
Yield / Growth Prediction	10	20%
Soil Health & Nutrient Monitoring	8	16%

TABLE VI. Frequency Distribution of Key Limitations Identified in IoT-Based Smart Agriculture Studies

Limitation	Frequency
Limited dataset size & diversity	21
Poor connectivity in rural areas	16
High implementation and maintenance cost	14
Lack of large-scale field validation	13
Scalability challenges	11
High computational & energy demand	9
Farmer training and usability issues	8

Limitation	Frequency
Data security and privacy concerns	6

TABLE VII. Key Research Gaps Identified in IoT-Based Smart Agriculture Systems (2022–2026)

Research Gap	Evidence from Review (N=50)
Limited Fully Integrated Hybrid AI Models	Only 7/50 papers implemented comprehensive multimodal fusion
Insufficient Large-Scale Multi-Region Field Validation	16 papers relied on localized or pilot-scale deployments
Limited Edge-Native / TinyML Deployments	Only 6/50 papers demonstrated full edge or TinyML implementation
Lack of Standardized Interoperability Frameworks	14 studies reported heterogeneous architectures without standards
Minimal Explainable AI (XAI) Integration	Only 5/50 papers incorporated SHAP or XAI mechanisms
Underrepresentation of Cybersecurity Frameworks	8 papers addressed security or blockchain-enabled protection
Limited Socio-Economic Impact Assessment	Fewer than 9 studies evaluated long-term farmer adoption
Scarcity of Open Benchmark Datasets	Approx. 70% used custom/private datasets without public benchmarking

TABLE VIII. Annual Research Trend in IoT-Driven Smart Agriculture Systems (2022–2026)

Year	Number of Papers	Trend Observation
2022	1	Foundational studies [38]
2024	5	Emerging interest [30], [39]–[41]
2025	38	Rapid growth and peak publication [1]–[8], [11]–[16], [20]–[31], [33]–[35], [37], [39], [40], [42]–[56]
2026	6	Continued momentum — TinyML & federated learning [2], [5]–[7], [51]

C. Proposed AIoT-Edge-Cloud Framework (Fig. 2)

Based on the synthesis of 50 reviewed studies, a next-generation AIoT–Edge–Cloud engineering framework for smart and sustainable precision agriculture is proposed in Fig. 2. The framework integrates AI of Things (AIoT), Edge Intelligence, TinyML, Federated Learning, Geo-intelligent Data Fusion, and Explainable AI (XAI) into a unified decision-support ecosystem, directly addressing the seven research gaps identified in Table VII.

The corresponding result is shown in Fig. 2.

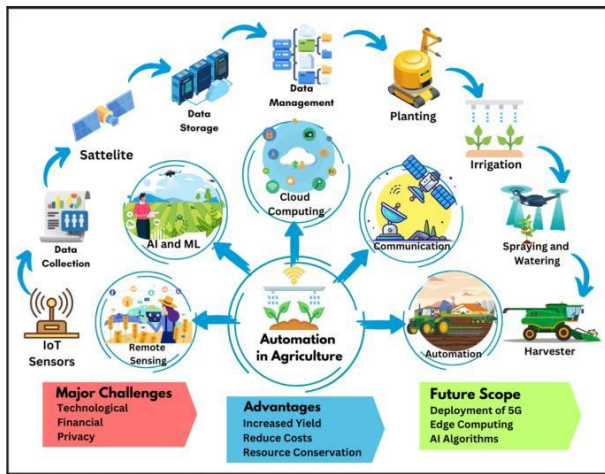


Fig. 2. Proposed next-generation IoT-Edge-Cloud engineering framework for smart and sustainable precision agriculture.

The framework's five-layer architecture directly addresses the gaps of Table VII: Layer 2 targets the TinyML/edge gap, Layer 3 addresses blockchain security and federated learning gaps, Layer 4 integrates XAI for decision transparency, and the multi-source data fusion in Layer 3 addresses the hybrid AI model gap. Table IX summarises the functional features and strategic benefits of the proposed framework.

TABLE IX. Functional Features and Strategic Benefits of the Proposed AIoT-Edge-Cloud Smart Agriculture Framework

Feature	Benefit
Real-time IoT Monitoring	Continuous tracking of soil, crop, and environmental conditions [1], [24]
Edge AI & TinyML Processing	Low-latency, offline-capable analytics for rural areas [6], [11], [33]
Multi-source Data Fusion	IoT + UAV + satellite + GIS + weather for accurate insights [7], [16]
AI/ML Predictive Analytics	Crop yield prediction, disease detection, irrigation optimization [1], [27]
Intelligent Irrigation Control	Up to 50% water-use reduction [3], [4], [30]
Cloud-Edge Hybrid Architecture	Real-time local processing balanced with cloud analytics [33], [34]
Explainable AI (XAI)	SHAP-based decision transparency for farmer trust [11], [51]
Federated Learning	Privacy-preserving, scalable model training across farms [8]
UAV and Remote Sensing Integration	Large-scale crop monitoring and early disease detection [7], [13], [27]

V. CONCLUSION

This paper synthesised recent advances in AI-driven smart agriculture, emphasising IoT sensing, machine learning, deep learning, edge computing, UAVs, remote sensing, and intelligent decision-support systems. The findings from 50 reviewed studies confirm that IoT-based systems significantly enhance irrigation management (20–50% water savings [3], [30]), crop monitoring, disease detection (>90% accuracy [1], [16], [27]), yield prediction, and resource optimization, promoting sustainable and climate-resilient farming.

Hybrid and deep learning models — including CNN-LSTM, EfficientNetV2-MobileNetV2-ViT [27], MSConvNet [16],

Gradient Boosting [6], Random Forest [1], and Transfer Learning [33] — demonstrated superior predictive performance consistently achieving >90% accuracy (Table II), with several studies exceeding 99%. TinyML, Edge AI, and hybrid edge-cloud architectures proved effective for low-latency, low-connectivity environments (Table VIII) [6], [11].

However, the seven research gaps identified in Table VII persist: reliance on small or localised datasets (Table III), lack of standardised benchmarks, high deployment costs (Table VI), cybersecurity risks, interoperability issues, and limited multi-objective optimization. Future research should prioritise scalable, interoperable AIoT frameworks (Fig. 2) integrating real-time sensing, explainable AI, federated learning, digital twins, blockchain security, and autonomous edge intelligence, with emphasis on lightweight models and cross-regional validation tailored to smallholder contexts [17], [35], [55].

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Artificial Intelligence-Based Image Analysis Techniques for Photovoltaic Panel Inspection: A Comprehensive Review

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Abstract—The rapid growth of photovoltaic (PV) installations worldwide has created an increasing demand for intelligent monitoring and maintenance systems capable of ensuring optimal energy production, operational reliability, and extended service life. Traditional inspection methods are often labor-intensive, time-consuming, expensive, and prone to human error, particularly in large-scale solar farms. Artificial Intelligence (AI)-based image analysis has therefore emerged as a transformative approach to automated PV panel inspection, fault detection, and predictive maintenance. This review provides an integrated, criteria-based synthesis spanning PV defect categories, imaging modalities, AI architectures, data practice, and deployment platforms, evaluated collectively rather than as disconnected strands of research. Literature published between 2016 and 2025 was retrieved from IEEE Xplore, ScienceDirect, MDPI, and Google Scholar, screened against explicit criteria, and organized by imaging modality and algorithmic approach. The synthesis indicates that infrared thermography and electroluminescence imaging are complementary, that convolutional classifiers, YOLO-type detectors, and U-Net-type segmenters dominate reported work, and that scarce labelled data, domain shift between sites, and the cost of imaging hardware are the principal barriers to field deployment. The paper also defines the performance metrics used to evaluate AI-based inspection models, offers a practical selection guide, and discusses opportunities for affordable, edge-based deployment in the Nigerian context.

Index Terms—Artificial Intelligence, Photovoltaic Panels, Image Analysis, Deep Learning, Fault Detection, Predictive Maintenance, Infrared Thermography, Electroluminescence, UAV Inspection.

I. INTRODUCTION

The rising global demand for clean and renewable energy sources has positioned solar power as a critical pillar in sustainable energy systems. However, the long-term efficiency and viability of photovoltaic (PV) installations depend heavily on continuous optimization across various components of the system [1]. These components—including inverters, batteries, tracking mechanisms, and most critically, PV panels—each influence the operational performance and reliability of the entire energy ecosystem. Solar power optimization, therefore, is not a singular challenge but a multi-tiered opportunity for innovation: optimization can target several subsystems ranging from inverter fault detection to intelligent grid integration. Within this hierarchy, PV panel degradation stands out due to its direct influence on power output and system reliability. Environmental exposure, manufacturing imperfections, and time-based wear all contribute to panel defects such as cracks, delamination, and hotspots—often invisible to the naked eye until significant efficiency losses occur [2], [3].

While solar PV systems offer a promising pathway to energy independence and carbon neutrality, their long-term reliability remains constrained by environmental degradation, thermal stress, and latent manufacturing defects. In Nigeria, the operational lifespan of PV panels is frequently compromised by dust accumulation, moisture ingress, ultraviolet radiation, and fluctuating ambient temperatures, resulting in efficiency loss and eventual system failure. Solar PV prospects and field performance in Nigeria have been examined in [4] and [5]. Reliability constraints are likewise central where PV is integrated into isolated industrial microgrids [6].

Recent reviews have surveyed advances in PV technology [7], artificial intelligence techniques for PV systems [8], AI-driven predictive maintenance and fault diagnosis [9], the role of AI in optimizing PV power plants [10], and AI and machine learning in energy sustainability [11]. Deep learning for PV defect detection is reviewed in [3], and soiling-monitoring strategies in [12]; electroluminescence image processing for degradation classification is demonstrated in [13], and methods for degradation analysis are evaluated in [14]. The integration of AI and IoT with solar PV, including home energy management and low-voltage inverters for Africa, is discussed in [15] and [16].

Despite this body of work, the literature on image-based PV inspection remains fragmented. Studies are typically organized around a single imaging modality, a single network family, or a single deployment platform, which makes it difficult for practitioners to judge which combination of sensor, algorithm, and platform suits a given site, budget, and defect profile. Reviews of PV faults and detection methods [2], [3] and of AI for PV systems [8], [9] establish the breadth of the field, yet, to the authors' knowledge, few connect imaging physics, learning architectures, data practice, and deployment constraints in a single framework, and fewer still discuss them from the standpoint of resource-constrained, high-irradiance, dust-prone environments such as those found across Nigeria and West Africa.

The contributions of this review are: (i) a defect-oriented overview of PV module degradation modes and their image signatures; (ii) a comparative treatment of the principal imaging modalities; (iii) a summary of AI techniques for PV panel inspection with their applications and limitations, together with data, training, and deployment practice; (iv) a consolidated set of performance metrics for evaluating AI-based inspection models; and (v) a discussion of challenges, a practical selection guide, and research directions relevant to

affordable, edge-based deployment in Nigeria. The remainder of the paper is organized as follows. Section II describes the review methodology. Section III summarizes PV defects and degradation modes, and Section IV compares imaging modalities. Section V presents AI-based image analysis for PV panel inspection. Section VI addresses data and training practice, and Section VII deployment platforms. Section VIII defines the performance evaluation metrics. Section IX discusses findings, challenges, and a selection guide; Section X outlines future research directions; and Section XI concludes the paper. Abbreviations used in this paper are listed in Appendix A.

II. REVIEW METHODOLOGY

This review follows a structured selection methodology. Literature published between 2016 and 2025 was retrieved from IEEE Xplore, ScienceDirect, MDPI, and Google Scholar using search terms including “PV fault detection,” “solar panel image analysis,” “deep learning photovoltaic inspection,” and “UAV thermal imaging PV.”

Retrieved studies were screened for methodological clarity, reported quantitative performance data, and relevance to photovoltaic-specific defect detection. The selected studies were then organized by imaging modality (infrared thermography, electroluminescence, ultraviolet, hyperspectral, and RGB imaging) and by algorithmic approach (Convolutional Neural Networks, transfer learning models, object detection frameworks, recurrent neural networks, and hybrid architectures).

Three inclusion criteria were applied. A study was retained when it (a) addressed the detection, classification, localization, or condition assessment of PV cells, modules, or arrays from image data; (b) described the imaging modality, the learning method, and the evaluation protocol clearly enough to be interpreted; and (c) reported quantitative performance. Studies relying exclusively on electrical (I–V or monitoring-system) data, without image analysis, were treated as background rather than as evidence for the comparison.

For each retained study, the imaging modality, acquisition platform, learning method, defect classes, dataset origin and size, validation protocol, and reported metrics were recorded. Because datasets, class definitions, and train–test protocols differ widely between studies, reported figures are not directly comparable. The synthesis is therefore qualitative and criteria-based: approaches are compared by their characteristic strengths, data requirements, and limitations rather than ranked by headline accuracy.

The review has limitations. It is restricted to English-language literature accessible through the four sources above and to image-based methods; it is a narrative synthesis rather than a formal meta-analysis; and publication bias toward positive results may inflate the performance reported in the literature.

III. PV DEFECTS AND DEGRADATION MODES

A. Defect Categories

Field PV modules degrade through a combination of gradual wear-out and discrete failures. Analytical reviews of field data

place typical degradation rates of crystalline-silicon modules below one percent per year, with wide dispersion between sites and technologies [17]; however, individual defects can cause losses far larger than the average once they create current mismatch or abnormal heating. A systematic catalogue of failure modes observed in the field is provided by the IEA PVPS Task 13 report [18], and detection methods for the principal faults are reviewed in [2].

The defects most relevant to image-based inspection can be grouped into five families. (1) Cell-level cracks and microcracks arise from mechanical stress during manufacture, transport, installation, or thermal cycling; they can electrically isolate part of a cell and cause current mismatch. (2) Thermal defects—hotspots caused by cell mismatch, partial shading, cracked or shunted cells, and failed bypass diodes—raise local temperature and accelerate further degradation. (3) Encapsulation and interface defects, including delamination, discoloration of the encapsulant, bubbles, and snail trails, reduce optical transmission and may precede corrosion. (4) Interconnection defects, such as broken cell fingers, ribbon or solder failures, and busbar corrosion, reduce current collection. (5) Surface contamination, principally soiling by dust, bird droppings, and vegetation, lowers the irradiance reaching the cells and can produce localized shading [12]. Potential-induced degradation (PID) is a system-level mechanism that produces shunting and progressive power loss in affected cells [18].

B. Link Between Defects and Images

A defect is amenable to image-based inspection only if it alters the radiation emitted, reflected, or absorbed by the module in a measurable way. Thermal defects change surface temperature and hence infrared emission; cracks and inactive cell areas change the electroluminescence response; and soiling, discoloration, and delamination change visible reflectance and texture. This physical coupling explains why no single modality is sufficient, and why the choice of modality is the first design decision in any AI-based inspection system, as discussed next.

IV. IMAGING MODALITIES FOR PV INSPECTION

Table I compares the principal imaging modalities by physical basis, detectable conditions, strengths, and constraints. The modalities are described briefly below.

A. Infrared Thermography

Infrared (IR) thermography records the long-wave infrared radiation emitted by module surfaces and converts it into a temperature map. Because defective cells and failed bypass diodes dissipate power as heat, thermography reveals hotspots, open- or short-circuited cell strings, and module-level anomalies quickly and without disconnecting the system [19]. Outdoor inspection is normally performed under high irradiance and low wind, and standardized procedures are given in IEC TS 62446-3 [20]. Interpretation depends on irradiance, ambient temperature, wind, viewing angle, and reflections from the glass surface, which complicates automated analysis and makes models trained at one site difficult to transfer to another.

B. Electroluminescence Imaging

Electroluminescence (EL) imaging forward-biases the module so that it emits near-infrared light in proportion to local carrier activity; cracks, broken fingers, inactive regions, and shunting appear as dark or low-intensity areas. EL offers high spatial resolution and is widely regarded as the most sensitive imaging method for cell-level cracks, which is why annotated EL datasets exist for learning-based work [21] and why automated degradation-feature pipelines have been built on EL images [13]. Its limitations are practical: it needs a power source and, outdoors, darkness or specialized equipment, so measurements are usually made in the laboratory or on dismantled modules; guidance is given in IEC TS 60904-13 [22].

C. Visible-Light (RGB) Imaging

Visible-light imaging uses ordinary cameras, including smartphone cameras and low-cost UAV payloads, and is the most affordable modality. It is well suited to soiling, surface discoloration, glass breakage, delamination, snail trails, vegetation shading, and mechanical damage, and it benefits

from the large pre-trained vision models available for RGB data. Its blind spot is defects with little or no visible signature, such as many cracks, cell mismatch, and bypass-diode failures; RGB imaging is therefore best used as a screening tool, complemented by thermography where budgets allow.

D. UV Fluorescence, Multispectral, and Hyperspectral Imaging

Ultraviolet (UV) fluorescence imaging exploits changes in the fluorescence of the encapsulant to reveal cracks and ageing patterns, but needs controlled UV illumination and is mainly a laboratory technique. Multispectral and hyperspectral cameras capture reflectance in many narrow bands and can discriminate soiling type, surface material, and subtle ageing effects; sensor cost and data volume currently confine them to specialized applications. Because each modality captures a different physical signature, combining them—for example, RGB and thermal imagery from the same UAV flight—lets a system both locate a panel and assess its condition. This is the motivation for the hybrid and multimodal approaches of Section V.

TABLE I. Imaging modalities for photovoltaic panel inspection

Modality	Physical Basis	Typically Detectable Conditions	Main Strengths	Main Constraints
Infrared thermography	Long-wave infrared emission (surface temperature)	Hotspots, bypass-diode failure, open or shorted strings, some PID patterns	Fast, non-contact, suited to operating plants and UAV survey	Depends on irradiance, wind, angle and reflections; limited resolution for fine cracks
Electroluminescence (EL)	Near-infrared emission under forward bias	Microcracks, broken fingers, inactive cell areas, shunting	High resolution; most sensitive to cell-level cracks	Needs power source and darkness or special equipment; mostly laboratory use
Visible light (RGB)	Reflected visible light	Soiling, discoloration, glass breakage, delamination, snail trails, shading objects	Lowest cost; abundant pre-trained models	Little sensitivity to defects without visible signature
UV fluorescence	Fluorescence of encapsulant under UV illumination	Cracks and ageing patterns	Reveals encapsulant-related changes	Needs controlled UV source; research use
Multispectral / hyperspectral	Reflectance in several to many narrow bands	Soiling type, material and ageing differences	Rich spectral information	High sensor cost and data volume; specialized processing

V. AI-BASED IMAGE ANALYSIS FOR PV PANEL INSPECTION

A. Position of PV Panel Diagnostics in Solar System Optimization

PV panel diagnostics is one branch of a wider hierarchy of solar power system optimization, in which inverters, batteries, tracking mechanisms, and PV panels are each optimization targets [1]. Fig. 1 shows this hierarchy and indicates where AI-based image analysis fits, namely in the detection of degradation and the prediction of the lifespan of PV modules.

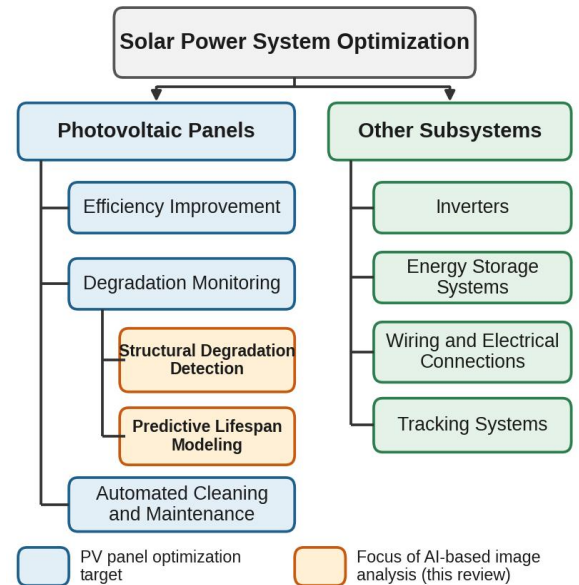


Fig. 1. Hierarchical optimization tree of solar power system components, with emphasis on AI-based image analysis for PV panel diagnostics.

B. AI Techniques for PV Inspection

Table II summarizes the principal AI techniques applied to PV panel inspection, the inspection tasks they support, the PV conditions they can detect, and their main limitations [3], [23]. Because each technique has distinct data and computational requirements, the choice among them depends on the defect type, the available imaging modality, and the deployment platform. The families are described below.

Convolutional neural networks (CNNs) learn hierarchical features directly from pixels and underpin most reported PV inspection systems. They are either trained from scratch on small domain datasets or, more commonly, fine-tuned from networks pre-trained on large generic image collections, a transfer-learning strategy that reduces the amount of labelled PV data required [24], [25]. Typical tasks are image-level or patch-level classification of cell conditions in EL and RGB images [21] and fault-class recognition in thermal images. Their principal limitation is that classification alone does not localize the defect.

Object detectors output bounding boxes and class labels, so they both locate and classify panels or defects in one pass. One-stage detectors such as the YOLO family [26] favor speed and are therefore attractive for real-time analysis of UAV video and for edge hardware, whereas two-stage detectors such as Faster R-CNN [27] have traditionally favored accuracy on small objects. Small, low-contrast defects and the wide scale variation between whole-array and single-cell views remain the main sources of missed detections.

Segmentation networks assign a class to every pixel. U-Net-type encoder-decoder architectures [28] and instance-segmentation models such as Mask R-CNN [29] give the extent and shape of a defect, which supports quantitative severity assessment, for example the fraction of a cell that is inactive or the area covered by soiling. The price is the need for pixel-accurate annotations, which are costly to produce.

Vision transformers (ViT) split an image into patches and use self-attention to model long-range relationships [30]. They can match or exceed CNNs when pre-trained on large datasets, but without large-scale pre-training they tend to be data-hungry, and their computational cost is a barrier on low-power hardware.

Hybrid, multimodal, and label-efficient approaches extend these building blocks. A detector may first find panels, with a classifier or segmenter then assessing each one; models may fuse RGB, thermal, and EL inputs; and temporal models may track a panel's condition across inspections to estimate degradation trends and remaining useful life (RUL). Where defect examples are scarce, unsupervised anomaly detection, data augmentation, synthetic-image generation with generative adversarial networks (GANs), and class-balancing techniques such as SMOTE and ADASYN are used to compensate. Finally, gradient-based visual explanations such as Grad-CAM [31] highlight the image regions that drove a prediction, helping maintenance engineers judge whether a model is attending to the defect or to an irrelevant artefact such as a frame edge or a reflection.

TABLE II. AI-based techniques for photovoltaic panel inspection

S/N	AI-Based Technique	Main Inspection Application	Detectable PV Conditions	Key Limitations
1	Convolutional Neural Networks (CNNs)	Image classification and feature extraction	Cracks, hotspots, soiling, discoloration, delamination	Requires sufficient labelled training data and computational resources
2	YOLO-Based Object Detection	Real-time defect detection and localization	Cracks, hotspots, damaged cells, foreign objects	Small or low-contrast defects may be difficult to detect accurately
3	U-Net-Based Segmentation	Pixel-level defect segmentation	Cell cracks, defective regions, soiling, surface damage	Requires accurately annotated segmentation datasets
4	Vision Transformers (ViT)	Image classification and defect recognition	Multiple PV degradation and defect categories	Generally requires substantial training data and computational resources
5	Hybrid and Multimodal AI Techniques	Integrated defect detection and condition assessment	Multiple defects using RGB, thermal, EL, or multispectral images	Higher system complexity, data-fusion challenges, and increased computational requirements

VI. DATA, ANNOTATION, AND TRAINING PRACTICE

A. Datasets and Annotation

Model quality is bounded by data quality. Public PV image datasets are few compared with generic vision benchmarks; the EL cell-image dataset of [21] is one widely used example, and most other studies rely on locally collected images. Annotation may take the form of image-level labels, bounding boxes, or pixel masks, in increasing order of effort; open-source tools such as Annotate-Lab [32] lower the cost of producing the latter two. Consistent class definitions and written annotation guidelines, preferably with a sample reviewed by a second annotator, are needed because

borderline defects, such as mild versus severe cracks, are subjective.

B. Pre-processing and Augmentation

Typical pre-processing includes resizing, intensity normalization, contrast enhancement of EL images, perspective rectification and cropping of individual modules or cells, and radiometric calibration of thermal images. Augmentation by rotation, flipping, scaling, brightness change, and noise increases the effective data volume, but augmentations must preserve the physical meaning of the image; for instance, thermal images should not be recolored in ways that alter the temperature scale.

C. Data Splitting and Validation

The train, validation, and test partitions should be made at the level of the physical module or site rather than the individual image; otherwise near-duplicate images of the same module appear in both training and test sets and inflate the measured performance. The split ratio itself influences the measured performance of pre-trained models [33]. Cross-site or cross-season validation, in which a model trained on one plant is evaluated on another, gives a more honest estimate of field performance than random splitting.

D. Class Imbalance and Domain Shift

Healthy panels vastly outnumber defective ones in the field, so accuracy alone can be misleading, and precision, recall, and F1 should be reported per class (Section VIII). Domain shift—differences in camera, resolution, panel type, background, weather, and season—is a common reason that models validated in the laboratory degrade in deployment. Fine-tuning on a small sample of local images, domain-adaptation methods, and monitoring of prediction confidence in service are standard countermeasures.

VII. DEPLOYMENT PLATFORMS

A. Handheld and Fixed Cameras

Handheld thermal cameras and fixed pan-tilt cameras suit small plants and rooftop systems. Fixed cameras allow continuous monitoring of a few arrays at low recurrent cost, whereas handheld inspection depends on operator skill and is difficult to repeat consistently.

B. Unmanned Aerial Vehicles

Unmanned aerial vehicles (UAVs) carrying thermal and RGB sensors inspect large arrays rapidly and are the dominant platform in utility-scale studies. Flight altitude, speed, and viewing angle set the ground-sampling distance and hence the smallest detectable defect; flights are normally planned for high irradiance, stable conditions, and angles that avoid sun glint. Regulatory approval, pilot training, and equipment cost are practical barriers. Ground robots that travel along panel rows capture close-range images of every module, trading speed for resolution.

C. Edge and Cloud Processing

Inference may run in the cloud, on a ground station, or on an edge device carried by the platform. Edge execution removes dependence on connectivity and reduces latency and data transmission, at the cost of constrained memory and compute. Benchmarking of deep-learning models on embedded GPU modules such as the NVIDIA Jetson Nano shows how model choice affects real-time throughput on such hardware [34]. Model compression techniques—quantization, pruning, and knowledge distillation—are used to fit detectors onto low-cost devices, usually at some cost in accuracy that must be re-measured on the target hardware. Integration with IoT monitoring and energy-management systems allows image-derived findings to be correlated with electrical data [15], [35].

VIII. PERFORMANCE EVALUATION METRICS

The performance of AI-based inspection models is commonly quantified with classification and detection metrics. The metrics used in this review are defined in (1)–(5), where TP, TN, FP, and FN denote the numbers of true positives, true negatives, false positives, and false negatives, respectively.

A. Classification Accuracy

The accuracy of an AI-based PV inspection model is the proportion of correct decisions, as expressed in (1):

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

B. Precision

Precision, defined in (2), measures the proportion of images or detected regions classified as defective that are actually defective. A higher precision indicates fewer false-positive defect detections.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

C. Recall

Recall, also referred to as sensitivity and defined in (3), measures the ability of the AI model to correctly identify defective PV panels or regions. High recall is particularly important in PV inspection because missed defects can remain undetected during maintenance operations.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

D. F1-Score

The F1-score in (4) provides a combined measure of the precision in (2) and the recall in (3):

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

The F1-score is useful when the dataset contains an imbalance between normal and defective PV images. Moreover, the ratio used to split image data into training and testing subsets influences the measured performance of pre-trained models [33].

E. Intersection over Union

For AI-based object detection and image segmentation, the Intersection over Union (IoU) metric in (5) evaluates how accurately the detected defect region overlaps with the ground-truth region:

$$IoU = \frac{|A_{pred} \cap A_{gt}|}{|A_{pred} \cup A_{gt}|} \quad (5)$$

where A_{pred} represents the predicted defect area and A_{gt} represents the ground-truth defect area. A value closer to 1 indicates greater overlap between the predicted and actual defect regions. Ground-truth regions are commonly produced with open-source annotation tools such as Annotate-Lab [32], and object-recognition architectures are surveyed in [23].

F. Average Precision and Mean Average Precision

For detectors, precision and recall are computed at a chosen IoU threshold, which decides whether a detection counts as a true positive. Average precision (AP) summarizes the precision–recall curve $p(r)$ of one defect class, as in (6), and

mean average precision (mAP) averages AP over the C defect classes, as in (7) [36]:

$$AP_c = \int_0^1 p(r) dr \quad (6)$$

$$mAP = \frac{1}{C} \sum_{c=1}^C AP_c \quad (7)$$

Benchmarks differ in how they interpolate the curve and in the IoU thresholds used, whether a single threshold such as 0.5 or an average over several, so mAP values from different papers should be compared only when the protocol is the same.

G. Dice Coefficient

For segmentation, the Dice coefficient in (8) is closely related to IoU and equals $2 \cdot \text{IoU} / (1 + \text{IoU})$:

$$\text{Dice} = \frac{2|A_{pred} \cap A_{gt}|}{|A_{pred}| + |A_{gt}|} \quad (8)$$

Dice weights overlap more generously than IoU, so the two should not be mixed in comparisons.

H. Practical Reporting

Beyond detection quality, deployment decisions depend on inference time (frames per second on the target hardware), model size, and energy use, which should be reported alongside the metrics above. Because a missed defect (a false negative) typically costs more than an unnecessary inspection (a false positive), the decision threshold is often set to favor

recall; the resulting precision should be reported so that the maintenance workload can be anticipated.

IX. DISCUSSION

A. Synthesis of Findings

Taken together, the reviewed work indicates that (i) thermography and EL imaging are complementary, the former being better suited to operating plants and the latter to module-level quality assurance; (ii) pipelines that combine detection with segmentation give the most actionable output; (iii) reported accuracies are generally high but not directly comparable because datasets and protocols differ; and (iv) generalization across sites remains the main unresolved problem. Combining AI-based image analysis with affordable RGB imaging and edge AI [34] can support early fault detection, remaining-useful-life estimation, and real-time analysis at low cost, which is particularly relevant for off-grid and underserved communities in Nigeria. AI-driven fault detection and predictive maintenance in renewable energy systems have been reviewed in [37].

B. Practical Selection Guide

Table III translates the synthesis into a practical guide. The recommendations are the authors' synthesis of the cited literature rather than universal rules, and should be validated on local data before adoption.

TABLE III. Practical selection guide for AI-based PV inspection

Scenario	Suggested Sensing	Suggested AI Approach	Main Caution
Small rooftop or off-grid site, low budget	RGB camera, with handheld thermal camera if available	Lightweight CNN classifier or compact YOLO-type detector on edge hardware	Screening only; defects without visible signature may be missed
Utility-scale solar farm	UAV with thermal and RGB payload	Panel detector followed by classifier or segmenter; severity ranking	Flight planning, glare, and site-to-site domain shift
Module quality assurance or warranty claims	EL imaging in controlled conditions	CNN classification or U-Net-type segmentation of cells	Equipment cost; dismounting modules
Soiling management in dusty regions	Fixed RGB camera or periodic UAV survey	Classification or regression of soiling level	Calibration against measured energy loss

C. Relevance to the Nigerian Context

Nigerian conditions, examined in [4] and [5], add specific considerations: high ambient temperatures and intense irradiance, which narrow thermal-contrast margins and accelerate encapsulant ageing; seasonal Harmattan dust, which adds to the soiling losses discussed in [12]; limited access to thermal cameras and UAV services; unreliable grid supply and connectivity, which favor on-device inference; and a shortage of locally annotated datasets. These considerations favor affordable RGB-based screening with edge AI, backed by periodic thermographic verification and by validation of any imported model against locally collected images. Simulation tools such as PVsyst can supply solar-resource modelling against which AI-based performance assessments may be validated under local environmental conditions.

D. Challenges

The remaining challenges are summarized in the limitations column of Table II and include the need for labelled training data, computational resources, and data fusion across imaging modalities. Further obstacles are interpretability, since

maintenance decisions require trust in model output; the lack of standardized benchmarks and reporting protocols; and the cost and skills required to acquire and operate imaging equipment.

X. FUTURE RESEARCH DIRECTIONS

Several directions would address the gaps identified above. First, open, labelled datasets collected under tropical and dusty conditions, with shared class definitions and module-level splits, are needed to support reproducible comparison. Second, self-supervised pre-training, few-shot learning, and foundation models could reduce the labelling burden. Third, domain adaptation and continual learning could keep models accurate as sensors, sites, and seasons change.

Fourth, multimodal fusion of RGB, thermal, and EL data, linked to I-V measurements, energy yield, and degradation analysis [14], would move inspection from defect detection toward remaining-useful-life estimation. Fifth, explainable and well-calibrated predictions would build trust among maintenance teams. Sixth, low-cost edge pipelines with quantized models, benchmarked on locally available hardware,

would make inspection practical for small operators. Finally, standardized reporting—module-level splits, cross-site tests, per-class metrics, and hardware latency—would make future studies directly comparable.

XL. CONCLUSION

This review has examined AI-based image analysis for photovoltaic panel inspection, covering PV defect categories, imaging modalities, the principal AI techniques and the PV conditions they address, data and training practice, deployment platforms, and the metrics used to evaluate them. Because panel degradation directly affects power output and system reliability, automated inspection and early fault detection are essential elements of PV system optimization.

No single sensor or algorithm is best in every setting. Thermography and EL imaging are complementary; CNN, detection, and segmentation models answer different questions; and reported accuracies cannot be compared unless datasets and protocols are aligned. For Nigeria, affordable RGB imaging combined with edge AI, validated against local conditions and verified periodically with thermography, offers a practical route to reliable and accessible solar energy. The selection guide, evaluation metrics, and reporting recommendations given here are intended to help researchers and practitioners design inspection systems that are accurate, affordable, and transferable between sites.

APPENDIX A: ABBREVIATIONS

Table IV lists the abbreviations used in this paper.

TABLE IV. Abbreviations used in AI-based photovoltaic inspection

Abbreviation	Full Meaning	Abbreviation	Full Meaning	Abbreviation	Full Meaning
AI	Artificial Intelligence	LWIR	Long-Wave Infrared	SSIM	Structural Similarity Index Measure
ML	Machine Learning	PV	Photovoltaic	FPS	Frames Per Second
DL	Deep Learning	PVP	Photovoltaic Panel	GPU	Graphics Processing Unit
CV	Computer Vision	PVS	Photovoltaic System	CPU	Central Processing Unit
CNN	Convolutional Neural Network	PVF	Photovoltaic Farm	TPU	Tensor Processing Unit
DNN	Deep Neural Network	I–V	Current–Voltage	IoT	Internet of Things
ANN	Artificial Neural Network	P–V	Power–Voltage	UAV	Unmanned Aerial Vehicle
RNN	Recurrent Neural Network	MPP	Maximum Power Point	UAS	Unmanned Aircraft System
LSTM	Long Short-Term Memory	MPPT	Maximum Power Point Tracking	XAI	Explainable Artificial Intelligence
ViT	Vision Transformer	PID	Potential-Induced Degradation	CAM	Class Activation Mapping
GAN	Generative Adversarial Network	TP	True Positive	Grad-CAM	Gradient-weighted Class Activation Mapping
VAE	Variational Autoencoder	TN	True Negative	OCR	Optical Character Recognition
YOLO	You Only Look Once	FP	False Positive	API	Application Programming Interface
R-CNN	Region-based Convolutional Neural Network	FN	False Negative	ELM	Extreme Learning Machine
Faster R-CNN	Faster Region-based Convolutional Neural Network	IoU	Intersection over Union	SVM	Support Vector Machine
Mask R-CNN	Mask Region-based Convolutional Neural Network	AP	Average Precision	KNN	K-Nearest Neighbors
U-Net	U-Net Convolutional Neural Network Architecture	mAP	Mean Average Precision	RF	Random Forest
EL	Electroluminescence	F1	F1-Score	DT	Decision Tree
IR	Infrared	ROC	Receiver Operating Characteristic	PCA	Principal Component Analysis
RGB	Red, Green, Blue	AUC	Area Under the Curve	t-SNE	t-Distributed Stochastic Neighbor Embedding
UV	Ultraviolet	MAE	Mean Absolute Error	ADASYN	Adaptive Synthetic Sampling
NIR	Near-Infrared	MSE	Mean Squared Error	SMOTE	Synthetic Minority Over-sampling Technique
SWIR	Short-Wave Infrared	RMSE	Root Mean Square Error		
MWIR	Mid-Wave Infrared	PSNR	Peak Signal-to-Noise Ratio		
RUL	Remaining Useful Life	IEC	International Electrotechnical Commission	TS	Technical Specification

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Design and Implementation of an AI-Based Face Recognition System for Student Security and Access Control

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Abstract—The growing need for improved security, identity verification, and access management within educational institutions has increased the adoption of biometric technologies as alternatives to traditional authentication methods. Identity cards, passwords, and manual verification are vulnerable to loss, theft, duplication, impersonation, and human error, particularly in large student populations. This paper presents the design and implementation of an AI-based face recognition system for student security and access control in academic environments. The system was developed through a four-stage methodology: (1) requirements analysis, identifying real-time identification, resistance to impersonation, and integration with existing student records; (2) image acquisition and preprocessing, comprising normalization, feature enhancement, resizing, and noise reduction; (3) design and training of a Convolutional Neural Network (CNN)-based architecture for feature extraction and facial classification, using a training dataset curated to represent varying illumination, expressions, and head orientations; and (4) matching and access decision, in which feature vectors are compared using distance and similarity measures. The paper further specifies presentation-attack (liveness) safeguards, privacy-by-design measures aligned with the Nigeria Data Protection Act, and an evaluation protocol in which system performance is reported using accuracy, false acceptance rate, false rejection rate, and equal error rate under stated test conditions.

Index Terms—Face Recognition, Biometrics, Student Security, Access Control, Artificial Intelligence, Deep Learning, Convolutional Neural Network, Presentation Attack Detection, Data Protection.

I. INTRODUCTION

The implementation of an automated face recognition system at the Computer Engineering Department of Gateway (ICT) Polytechnic Saapade represents a significant advancement in student security measures. The increasing need for enhanced safety protocols within educational institutions has prompted the exploration of biometric technologies, particularly facial recognition systems. These systems utilize sophisticated algorithms to capture and analyze facial features, thereby automating processes such as attendance tracking and access control. By integrating real-time image processing with established security frameworks, this study aims to bolster campus safety while streamlining administrative tasks.

The integration of automated face recognition systems within student security frameworks has become increasingly vital in ensuring the safety and well-being of educational communities. With rising concerns over unauthorized access, identity fraud, and campus security breaches, institutions are compelled to adopt advanced technological solutions that provide reliable and efficient identification methods. Automated face recognition offers a non-intrusive, real-time approach to

monitoring and verifying individuals on campus, thereby enhancing overall security protocols. By leveraging biometric data unique to each individual, these systems minimize human error and reduce reliance on traditional identification methods such as ID cards or manual verification processes. The design and construction of such systems involve multiple stages including face detection, feature extraction, and recognition. These processes collectively enable the system to accurately identify students from captured images or video streams with high precision.

Biometric recognition identifies people from physiological or behavioral characteristics rather than from something they carry or remember [1]. Face recognition is attractive for campuses because it is contactless, uses inexpensive cameras that many institutions already operate, and is generally accepted by users. The need is concrete: large cohorts, shared laboratories with costly equipment, examination halls, and hostels all require reliable ways to confirm that the person at the door is the student on the register. A badge can be lent or lost, and manual checking does not scale at peak entry times.

The objective of this work is to design and implement an AI-based face recognition system that enhances student security and access control at Gateway (ICT) Polytechnic, Saapade. The paper makes four contributions: (i) a requirements-driven system design, with an architecture, an enrolment procedure, and a CNN-based embedding and matching pipeline; (ii) a presentation-attack (liveness) safeguard and privacy-by-design measures suited to a Nigerian campus; (iii) a set of performance metrics, including threshold-dependent error rates and the equal error rate, with a protocol for reporting them; and (iv) deployment guidance covering fairness, cybersecurity, and governance. The remainder of the paper is organized as follows. Section II reviews related work. Section III describes the system design and methodology. Section IV defines the performance evaluation metrics, and Section V the evaluation protocol. Section VI discusses deployment, privacy, and governance, and Section VII concludes the paper. Abbreviations used in this paper are listed in Appendix A.

II. RELATED WORK

A. Face-Recognition Attendance and Access-Control Systems

Face-recognition-based attendance and student-monitoring systems have been reported for schools and tertiary institutions, including in Nigeria [2]–[13], and the construction

of such systems with machine learning and deep learning is described in [14] and [15].

These systems differ in purpose (attendance, examination verification, or access control), in recognition method (classical features or deep embeddings), and in hardware (a personal computer, a single-board computer, or a microcontroller). The distinction matters because the cost of an error differs: in attendance, a false rejection is an inconvenience, whereas in access control a false acceptance is a security failure. Reliability under realistic lighting is a recurring concern [13], [16], and an access-control system therefore needs the explicit error-rate reporting described in Section IV rather than a single accuracy figure.

B. Feature Extraction and Recognition Algorithms

Classical feature-based approaches such as HOG with fast PCA [17], general feature-extraction techniques [18], and improved recognition algorithms [19] remain relevant to system design. Table I summarizes the main techniques, their

categories, typical applications, and performance considerations.

Face recognition has progressed from holistic subspace methods such as Eigenfaces [20] and texture descriptors such as local binary patterns [21], through the HOG-and-PCA pipelines noted above [17], to deep networks that map a face image to a compact embedding in which distance reflects identity. DeepFace [22] and FaceNet [23] established embedding-based verification, and margin-based training objectives such as ArcFace [24] further increased the separation between identities. Face detection and alignment, which precede recognition, are commonly performed with cascaded CNN detectors such as the Multi-Task Cascaded Convolutional Neural Network (MTCNN) [25]. Public benchmarks such as Labeled Faces in the Wild [26] are widely used to compare methods, but near-saturated benchmark accuracy does not guarantee performance on a particular campus population under local illumination and camera conditions.

TABLE I. Face-recognition techniques, categories, applications and performance considerations

Technique	Category	Typical Application	Major Performance Consideration
HOG with fast PCA	Classical feature-based	Face feature extraction and dimensionality reduction	Light computation; sensitive to lighting and pose [16], [17]
General feature extraction	Classical feature-based	Descriptor selection for recognition	Choice of descriptor affects accuracy [18]
Improved recognition algorithm	Machine learning	Attendance management	Accuracy gain over baseline methods [19]
Deep learning (CNN)	Deep learning	Identity verification and access control	Accuracy and robustness depend on training data [15], [27]
Occlusion-robust detection	Deep learning	Face localization with partial occlusion	Performance under occluded faces [28]
Masked-face recognition	Deep learning	Recognition of masked faces	Reduced visible facial information [29]
Twin / look-alike discrimination	Machine learning	High-security identity checks	Fine-grained discrimination [30]
Iris biometrics (multimodal)	Multimodal biometrics	Additional authentication layer	Extra hardware and cost [31]
Raspberry Pi platform	Embedded hardware	Low-cost attendance/access terminal	Limited computing power [32]
Arduino / ESP32-CAM	Embedded hardware	Small-scale real-time recognition	Limited memory and resolution [33], [34]
TinyML	Edge AI	On-device inference	Model compression versus accuracy [35]
Privacy-aware deployment	Governance	Campus surveillance and access control	Trust, privacy and regulation [36], [37]

C. Embedded and Low-Cost Hardware Platforms

Early face-recognition-based security systems were reported in [38] and real-time implementations in [39]; hardware-oriented designs include a hardware/software co-design approach [40] and Arduino-based [33] and ESP32-CAM-based [34] systems, while low-cost computing platforms are reviewed in [32] and embedded machine-learning trends in [35].

For campus access terminals, the practical trade-off is between processing capability and cost. Single-board computers such as the Raspberry Pi [32] can run compact CNN models at modest frame rates; microcontroller platforms such as the ESP32-CAM [34] are limited to very small models or must offload recognition to a server; and TinyML techniques [35] compress models to fit constrained devices. Whichever platform is chosen, separating capture from recognition, so that the terminal captures and a server recognizes, lets the institution upgrade the model without replacing the terminals.

D. Robustness of Recognition

Recognition performance is affected by lighting variation [16], occlusion [28], face masks [29], and look-alike faces such as twins [30].

Two further threats matter in access control. The first is the presentation attack, in which a printed photograph, a video replayed on a phone, or a mask is shown to the camera; a recognizer without a liveness check can be defeated by such artefacts, and testing of presentation attack detection (PAD) is standardized in ISO/IEC 30107-3 [41]. The second is demographic variation in error rates: independent evaluations by the U.S. National Institute of Standards and Technology (NIST) found that false-positive rates differ across demographic groups and that the size of the differential varies widely between algorithms [42]. Local validation on the institution's own student population is therefore needed.

E. Security, Privacy and Regulation

Wider treatments of AI-based face recognition, biometrics and security are given in [43]–[46], and [31]. Because biometric data are sensitive, privacy, trust and regulatory considerations

[36], [37], data protection [47], [48] and cybersecurity [49], [50] should accompany deployment.

In Nigeria, the Nigeria Data Protection Act, 2023 [51] governs the processing of personal data and treats biometric data used to identify a person as sensitive personal data. This calls for a lawful basis, purpose limitation, data minimization, security safeguards, and respect for data subjects' rights; the European General Data Protection Regulation (GDPR) [52] is a widely used reference for comparable requirements. Unlike a password, a face cannot be changed after a breach, which raises the consequences of any compromise of stored biometric data.

III. SYSTEM DESIGN AND METHODOLOGY

The system was developed through a four-stage methodology: requirements analysis, image acquisition and preprocessing, CNN-based feature extraction and classification, and matching with an access decision. The software and decision logic are summarized in the flowchart of Fig. 1.

A. Requirements Analysis

Requirements analysis identified the core functional needs of the system, namely real-time identification, resistance to impersonation, and integration with existing student records. These translate into functional requirements: enrolment of students; face capture and detection; matching against registered templates; an access decision with a configurable threshold; logging of every attempt; and an administrative interface for managing records.

Non-functional requirements include low latency at peak entry times, usability without special training, continued operation during power or network interruptions, low cost, and compliance with data-protection obligations. The operating point, that is, the trade-off between false acceptances and false rejections, is a policy decision for the institution: a laboratory holding costly equipment justifies a stricter threshold than a lecture hall.

B. System Architecture

The software and decision logic in Fig. 1 are organized into six units. The input unit captures an image with an HD camera or image sensor. The pre-processing unit applies noise reduction, grayscale conversion, and alignment. The core processing unit performs face detection, feature extraction with the deep learning model, and comparison of the result with stored templates. The database unit holds the enrolled student records, with face templates and student identifiers stored and retrieved as needed. The decision and control unit evaluates the recognition result and determines whether access is granted or denied, and the output units execute the security action (door lock, visual or audio indication, and notification) and record the event for attendance logging and monitoring.

Keeping the core processing and database units apart from the capture terminal keeps biometric data off the access point, allows templates and models to be updated centrally, and permits operation with or without a network connection, depending on where the processing is hosted.

C. Enrolment

Enrolment links a face template to a student record. Several images of each student are captured under varied pose, expression, and lighting; each passes a quality check (face size, sharpness, illumination, and near-frontal pose), and poor images are rejected. The feature vectors (embeddings) extracted from the accepted images are stored with the student identifier in place of the raw photographs wherever possible, so that the database holds templates rather than images. Enrolment should be supervised, because an impostor enrolled under another student's identity defeats every later check.

D. Image Acquisition and Preprocessing

Facial images are captured with a digital imaging device and processed through normalization, feature enhancement, resizing, and noise reduction to improve recognition performance under varying conditions.

The pre-processing chain consists of face detection and cropping, alignment using facial landmarks such as the eyes, resizing to the network input size, illumination normalization (for example, histogram equalization), and noise reduction (for example, Gaussian or median filtering). Alignment is especially important, because small rotations and scale differences can change an embedding more than many appearance variations do. Detection and alignment may be performed with a cascaded CNN such as MTCNN [25].

E. CNN-Based Feature Extraction and Classification

A Convolutional Neural Network (CNN)-based architecture was designed and trained for automatic feature extraction and facial classification. The training dataset was curated to represent varying illumination conditions, facial expressions, and head orientations to improve model robustness.

The network output is an embedding: a fixed-length feature vector for each face. Two training strategies are common. A classifier trained with a softmax output over the enrolled identities must be retrained whenever students are added. Metric-learning objectives, such as triplet or angular-margin losses, instead train the network so that embeddings of the same person are close and those of different people are far apart; new students are then enrolled by storing their embeddings, without retraining [23], [24]. This property suits a campus in which the register changes every session.

Transfer learning from a network pre-trained on a large face dataset, followed by fine-tuning on locally collected images, reduces the data needed and improves robustness to local conditions. Data augmentation (flips, small rotations, brightness and contrast changes, and blur) reflects the variation in illumination, expression, and head pose described above. When the network is validated, the identities in the validation set should be disjoint from those used in training, so that the measured performance reflects behavior on students the network has not seen.

F. Matching and Access Decision

The feature vector of a captured face is compared with the feature vector of the registered face using two measures: the Euclidean distance in (1) and the cosine similarity in (2). The Euclidean distance is

$$d = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \quad (1)$$

where d is the distance between the two face feature vectors, x_i is the feature value from the captured (test) face, y_i is the corresponding feature value from the registered face, i is the position of the feature, and n is the total number of face features being compared. Smaller values of d indicate more similar faces, and larger values indicate less similar faces. The cosine similarity is

$$S = \frac{A \cdot B}{\|A\| \|B\|} \quad (2)$$

where S is the similarity score between the two feature vectors, A is the feature vector of the captured face, B is the feature vector of the registered face, $A \cdot B$ is the dot product of the two vectors, and $\|A\|$ and $\|B\|$ are their magnitudes. The value of S lies between -1 and 1 , although face embeddings commonly produce values in a narrower positive range; values closer to 1 indicate greater similarity, values closer to 0 indicate less similarity, and negative values indicate strongly different directions. The distance in (1) or the similarity in (2) is compared with a decision threshold to accept or reject the identity claim, and the threshold determines the trade-off between the false acceptance rate in (4) and the false rejection rate in (5) defined in Section IV. The complete decision logic is shown in Fig. 1.

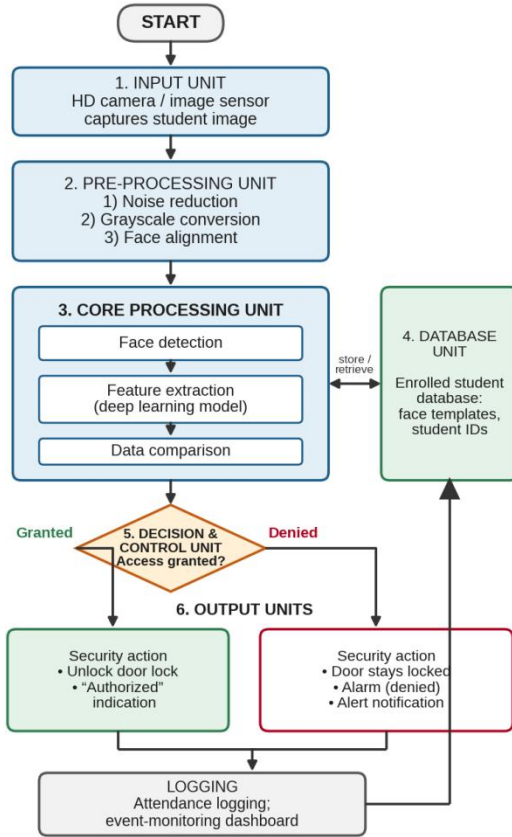


Fig. 1. Software/logic flowchart for the AI-based face recognition and access-control system.

When the feature vectors are L2-normalized, the two measures in (1) and (2) are equivalent, since $d^2 = 2(1 - S)$, so either may be used with a corresponding threshold. The threshold is set from genuine and impostor scores on a validation set, at the operating point chosen by institutional policy (Section IV).

Matching can operate in two modes. In verification (1:1), the student presents an identifier, such as a student number or card, and the system compares the live face with that student's template only. In identification (1:N), the system searches all enrolled templates. Verification is faster, and because the chance of a false match grows with the number of comparisons, it is usually preferred for access control; identification suits convenience scenarios such as attendance. A rejected attempt should fall back to a manual procedure rather than deny access outright, and repeated failures should be logged and flagged.

G. Presentation Attack Detection

A recognizer that matches appearance alone can be deceived by a printed photograph, a replayed video, or a mask. PAD adds a liveness decision before matching. Approaches include challenge–response (asking the user to blink or turn the head), analysis of texture and reflection cues that distinguish a live face from a print or a screen, depth sensing, and near-infrared imaging. Because no single method stops every attack, low-cost deployments commonly combine a simple challenge–response with supervision at high-risk entrances. PAD performance is tested and reported in a standardized way [41].

H. System Components

Table II lists the functional components of the face recognition system.

TABLE II. Functional components of the face recognition system

No.	Area / Component	Description
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No.	Area / Component	Description
1	Face Image Acquisition	Captures a student's face using a camera before recognition or access is granted.
2	Face Detection	Detects and locates the student's face within the captured image.
3	Face Feature Extraction	Converts the detected face into numerical features/embeddings that can be used for comparison.
4	Face Recognition & Matching	Compares the captured face with registered student face data and determines whether they match.
5	Security & Access Control	Grants or denies access based on the recognition result and records the student's access activity.

IV. PERFORMANCE EVALUATION METRICS

The recognition decisions of the system are evaluated using the accuracy in (3), the false acceptance rate (FAR) in (4), and the false rejection rate (FRR) in (5). In these expressions, TP (true positive) denotes a genuine student correctly recognized; TN (true negative), a person who should not be accepted correctly rejected; FP (false positive), an unauthorized person incorrectly accepted; and FN (false negative), a genuine student incorrectly rejected.

A. Accuracy

Accuracy is the percentage of all recognition decisions that were correct:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

B. False Acceptance Rate (FAR)

FAR is the percentage of unauthorized attempts that the system incorrectly accepts:

$$FAR = \frac{FP}{FP + TN} \quad (4)$$

C. False Rejection Rate (FRR)

FRR is the percentage of genuine students that the system incorrectly rejects:

$$FRR = \frac{FN}{FN + TP} \quad (5)$$

D. Threshold-Dependent Error Rates

FAR and FRR depend on the decision threshold τ . Writing s for a similarity score, and N_{imp} and N_{gen} for the numbers of impostor and genuine comparisons, the rates at threshold τ are given by (6) and (7):

$$FAR(\tau) = \frac{\#\{impostor\ scores \geq \tau\}}{N_{imp}} \quad (6)$$

$$FRR(\tau) = \frac{\#\{genuine\ scores < \tau\}}{N_{gen}} \quad (7)$$

Raising τ lowers FAR and raises FRR, and lowering it does the opposite. Plotting the true positive rate ($1 - FRR$) against the false positive rate (FAR) over all thresholds gives the receiver operating characteristic (ROC) curve, and plotting FRR against FAR gives the detection error trade-off (DET) curve.

E. Equal Error Rate

The equal error rate (EER) in (8) is the common value at which FAR and FRR are equal, at the threshold τ^* :

$$EER = FAR(\tau^*) = FRR(\tau^*) \quad (8)$$

The EER is widely used as a single-number summary for comparing systems; however, an operational system should be assessed at its chosen threshold rather than at the EER.

F. Statistical Confidence

Error rates measured on finite samples are uncertain. If no errors are observed in N independent trials, the 95 % upper confidence bound on the true error rate is approximately $3/N$, the “rule of three” in (9):

$$ER_{upper} \approx \frac{3}{N} \quad (9)$$

Independence requires that comparisons come from many different people rather than from repeated attempts by a few. For example, demonstrating a FAR below 0.1 % requires on the order of several thousand independent impostor comparisons. Reported rates should therefore be accompanied by the number of participants and attempts, and by confidence intervals.

V. EVALUATION PROTOCOL

Performance should be reported under a protocol that separates enrolment from testing and states the conditions under which the results hold. ISO/IEC 19795-1 [53] provides the general framework for biometric performance testing and reporting, distinguishing technology, scenario, and operational evaluation.

A. Test Data and Trials

Enrolment and probe images are disjoint. A genuine attempt compares a probe with the same student's template, and an impostor attempt compares it with the template of a different student. Test participants should include students who were not used to train or fine-tune the network, and comparisons should be drawn from as many different people as practicable so that the independence assumption behind the confidence bound in (9) is reasonable.

B. Test Conditions

Table III lists the conditions under which results are measured and reported separately. Reporting per condition, rather than a single pooled figure, shows where the system is weak and prevents favorable conditions from masking failures.

TABLE III. Evaluation conditions and reported measures

Condition	Variation Tested	Reported Measure
Illumination	Indoor, outdoor, backlit and low-light entrances	FAR and FRR at the operating threshold; DET curve

Condition	Variation Tested	Reported Measure
Pose and expression	Frontal and moderately rotated heads; neutral and varied expressions	FRR by pose category
Accessories	Eyeglasses, caps, scarves, and face masks	FRR; failure to acquire (FTA)
Distance and camera position	Typical standing distances and camera heights at the access point	FTA and FRR by distance
Time lapse	Probes captured weeks or months after enrolment	FRR drift; need for re-enrolment
Presentation attacks	Printed photograph, replayed video, and mask	Attack acceptance rate; genuine rejection by PAD
Demographic subgroups	Subgroups present in the student population	FAR and FRR per subgroup
Operational load	Peak entry periods	Decision latency; throughput (decisions per minute)

C. Reporting

Results are reported as accuracy, FAR, FRR, and EER, with confidence intervals, the threshold used, and the numbers of participants and attempts. Failure-to-enrol (FTE) and failure-to-acquire (FTA) rates are reported alongside, since a system that frequently fails to capture a usable image burdens users even if its matching is accurate. Decision latency and throughput are reported for the deployed hardware, and the proportion of attempts requiring manual fallback is used as a practical measure of user acceptance.

D. Operational Monitoring

After deployment, access logs support continuous monitoring. A rising FRR over a semester may indicate that templates have aged and re-enrolment is needed, and clusters of failures at one entrance may point to an illumination or camera-placement problem.

VI. DEPLOYMENT, PRIVACY, AND GOVERNANCE

Real-time monitoring and access control can minimize unauthorized entries while ensuring that students feel secure within their academic environment. AI-based face-recognition access control for university laboratories has been implemented in [27]. This section considers the conditions for deploying such a system responsibly.

A. Deployment Considerations

Camera placement, lighting, and queue flow at the entrance determine most of the error observed in practice. Terminals should face away from direct sunlight and provide uniform frontal illumination, and the capture distance should be marked on the floor. Peak periods, such as the start of an examination, require sufficient throughput, and a manual fallback with an attendant must always be available for failures, faulty devices, and students who decline to use the system. Where power or network supply is unreliable, local caching of templates and an uninterruptible power source allow the terminal to continue operating.

B. Privacy and Data Protection

Because biometric data are sensitive and irrevocable, privacy should be built into the design. The system should state its purpose, obtain informed consent, collect only the data needed, store templates rather than photographs where possible, limit retention, and give students a means to access and correct their records and to opt for an alternative. A data protection impact assessment before deployment is good practice for biometric processing. Table IV summarizes the principal risks and safeguards, drawing on the privacy, trust, and regulatory work cited in Section II [36], [37].

TABLE IV. Principal risks and safeguards for campus face recognition

Risk	Safeguard	Basis
Unauthorized access to stored templates	Encryption at rest, role-based access control, and audit logging	Data-security practice for student information [48], [50]
Interception of images or embeddings in transit	Encrypted channels (Transport Layer Security, TLS) between terminal and server	Cybersecurity practice in education [49]
Function creep (use beyond access control)	Written purpose limitation and restricted data sharing	Data protection law [51], [52]
Excessive retention	Retention limits; deletion on graduation or withdrawal	Data minimization [51]
Spoofing with photographs, video, or masks	Presentation attack detection and supervision at high-risk entrances	ISO/IEC 30107-3 [41]
Unequal error rates between groups	Local testing by subgroup; threshold review	Demographic evaluation [42]
Lack of consent or alternative	Informed consent and a non-biometric alternative for students who decline	Trust and acceptance [36]

C. Fairness and Local Validation

Because error rates can differ between demographic groups [42], the system should be validated on the institution's own student population, with FAR and FRR reported per subgroup (Table III). Where differences are found, remedies include collecting more enrolment images, adjusting the threshold, or

changing the camera and illumination; unresolved differentials are a reason to restrict the system to a supporting role.

D. Cybersecurity

The terminal, the network, and the server together form an attack surface. Recommended measures are secure boot and tamper detection on terminals, encrypted communication,

authenticated administrator access with multi-factor authentication, regular patching, and monitoring of the access log for anomalies [49], [50]. Template databases should be encrypted, and backups should be protected to the same standard as the live system.

E. Scope and Limitations

This paper specifies the system design, the decision logic, and the evaluation protocol. The error rates achieved in any deployment depend on the dataset, camera, illumination, and threshold, and must be measured and reported under the protocol of Section V. The design uses two-dimensional visible-light imaging, so it inherits the sensitivity of such systems to extreme lighting and to determined presentation attacks, and it is intended to operate with a manual fallback and human supervision rather than as an unattended sole barrier.

VII. CONCLUSION

This paper has presented the design of an AI-based face recognition system for student security and access control at

Gateway (ICT) Polytechnic, Saapade. The design is driven by requirements for real-time identification, resistance to impersonation, and integration with student records, and it combines a pre-processing chain, a CNN that produces face embeddings, and distance- or similarity-based matching against a configurable threshold. Enrolment, presentation attack detection, and privacy-by-design measures address the main security and governance risks of biometric access control.

The metrics defined in Section IV, together with the protocol of Section V, provide the basis for evaluating the deployed system: error rates must be reported at the operating threshold, per test condition, and with confidence intervals. Further work should report measured FAR, FRR, and EER on a local student population, evaluate the effect of face masks and low-light entrances, assess combined card-and-face verification, and monitor template ageing over successive semesters.

APPENDIX A: ABBREVIATIONS

Table V lists the abbreviations used in this paper.

TABLE V. Abbreviations used in the face recognition system

Abbreviation	Meaning	Abbreviation	Meaning	Abbreviation	Meaning
AI	Artificial Intelligence	OTP	One-Time Password	FRR	False Rejection Rate
ML	Machine Learning	ID	Identification	EER	Equal Error Rate
DL	Deep Learning	UID	Unique Identification	ROC	Receiver Operating Characteristic
FR	Face Recognition	IoT	Internet of Things	AUC	Area Under the Curve
FRS	Face Recognition System	DB	Database	TPR	True Positive Rate
AFRS	Automated Face Recognition System	SQL	Structured Query Language	FPR	False Positive Rate
CV	Computer Vision	HTTP	Hypertext Transfer Protocol	NIR	Near-Infrared
CNN	Convolutional Neural Network	HTTPS	Hypertext Transfer Protocol Secure	LFW	Labeled Faces in the Wild
ANN	Artificial Neural Network	REST	Representational State Transfer	FERET	Face Recognition Technology Evaluation
DNN	Deep Neural Network	JSON	JavaScript Object Notation	GDPR	General Data Protection Regulation
LBPH	Local Binary Patterns Histograms	RGB	Red, Green, Blue	UI	User Interface
HOG	Histogram of Oriented Gradients	FPS	Frames Per Second	UX	User Experience
PCA	Principal Component Analysis	GPU	Graphics Processing Unit	SMS	Short Message Service
SVM	Support Vector Machine	CPU	Central Processing Unit	LAN	Local Area Network
KNN	K-Nearest Neighbors	RAM	Random Access Memory	WAN	Wide Area Network
YOLO	You Only Look Once	TP	True Positive	MAC	Media Access Control
OpenCV	Open Source Computer Vision Library	TN	True Negative	URL	Uniform Resource Locator
API	Application Programming Interface	FP	False Positive	SDK	Software Development Kit
GUI	Graphical User Interface	FN	False Negative		
RFID	Radio-Frequency Identification	FAR	False Acceptance Rate		
PAD	Presentation Attack Detection	MTCNN	Multi-Task Cascaded Convolutional Neural Network	DET	Detection Error Trade-off
FTA	Failure to Acquire	FTE	Failure to Enrol	ISO	International Organization for Standardization
IEC	International Electrotechnical Commission	NDPA	Nigeria Data Protection Act	TLS	Transport Layer Security

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A Comprehensive Review of Optimization Techniques for Enhancing Photovoltaic Power System Performance

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Abstract—The increasing global demand for sustainable energy has accelerated the deployment of photovoltaic (PV) systems. However, PV performance remains affected by fluctuating solar irradiance, temperature variations, dust accumulation, aging effects, and partial shading, which reduce power output and efficiency and make optimization techniques necessary. While several reviews have examined Maximum Power Point Tracking (MPPT) methods individually or compared a narrow subset of techniques, few have synthesized conventional, intelligent, and hybrid optimization approaches within a single comparative framework that also addresses emerging IoT-enabled and predictive optimization trends. This review provides an integrated, criteria-based comparison of optimization techniques spanning three generations of MPPT development. Literature published between 2015 and 2025 was retrieved from IEEE Xplore, ScienceDirect, and Google Scholar and categorized into conventional, intelligent, and hybrid frameworks, and each technique was evaluated against six criteria: tracking accuracy, convergence speed, computational complexity, adaptability to environmental variation, implementation requirements, and impact on energy yield. Conventional methods perform well under steady irradiance but degrade under rapidly changing irradiance or partial shading, whereas intelligent and hybrid methods improve adaptability at the cost of higher computational and implementation complexity.

Index Terms—Photovoltaic Systems, Maximum Power Point Tracking, Optimization Techniques, Artificial Intelligence, Renewable Energy, Particle Swarm Optimization.

I. INTRODUCTION

Photovoltaic (PV) technology has emerged as a leading renewable energy source, converting sunlight directly into electricity through semiconductor materials, predominantly silicon. This technology is pivotal in addressing the global energy crisis and mitigating climate change due to its sustainability and declining costs. The efficiency of PV systems is influenced by various factors including material properties, design configurations, and environmental conditions. Consequently, optimization techniques are essential for enhancing the performance of PV technologies by improving energy conversion efficiency, system reliability, and cost-effectiveness [1].

Recent surveys have addressed PV materials and systems for energy storage [2], PV operation and maintenance [3], renewable energy source classifications and challenges [4], the role of solar photovoltaics in a sustainable energy future [5], and the effect of climate on PV yield prediction with machine-learning models [6]. Machine-learning foundations [7], hybrid approaches combining optimization and machine learning [8], and modern optimization techniques for engineering applications [9] underpin many of the methods reviewed here.

The power–voltage (P–V) characteristic of a PV array is nonlinear, with a single maximum under uniform conditions and several local maxima under partial shading. Because the location of the maximum shifts with irradiance and temperature, a controller must search for it continuously; this is the function of MPPT, and it is the most widely studied optimization problem in PV systems [10]. The differences between techniques are material: they determine how much of the available energy is harvested, how quickly the system responds to passing clouds, and how much processing hardware is needed. The last point matters for the low-cost systems common in tropical settings such as Nigeria, where high cell temperatures, dust, and shading by nearby objects are frequent.

This paper provides an integrated, criteria-based comparison of conventional, intelligent, and hybrid optimization techniques for PV systems. The remainder of the paper is organized as follows. Section II describes the review methodology. Section III presents the mathematical background and performance metrics. Section IV explains the operating principles of the main MPPT techniques, including partial-shading and IoT-enabled approaches. Section V classifies the techniques and compares them against six criteria. Section VI discusses the findings, selection guidance, and future directions, and Section VII concludes the paper. Abbreviations used in this paper are listed in Appendix A.

II. REVIEW METHODOLOGY

This paper adopts a structured review methodology. Relevant literature published between 2015 and 2025 was retrieved from IEEE Xplore, ScienceDirect, and Google Scholar using search terms such as “MPPT optimization,” “photovoltaic performance,” “hybrid optimization algorithms,” and “AI-based PV control.” Papers were screened for relevance, methodological clarity, and reported quantitative performance data.

The selected papers were categorized into conventional MPPT methods (Perturb and Observe, Incremental Conductance, Hill-Climbing, Constant Voltage), intelligent approaches (Artificial Neural Networks, Fuzzy Logic Controllers, Genetic Algorithms, Particle Swarm Optimization, Ant Colony Optimization), and hybrid frameworks. Each technique was evaluated against six criteria: tracking accuracy, convergence speed, computational complexity, adaptability to environmental variation, implementation requirements, and overall impact on energy yield.

Ratings against the six criteria were assigned qualitatively (low, moderate, or high) on the basis of comparative studies

and reviews in which several techniques are tested under common conditions [10]–[13]. Because test conditions differ between studies, numerical results are not pooled. Studies were also grouped according to whether they address uniform irradiance or partial shading, since this distinction separates the techniques most clearly. As in other narrative reviews, the findings are limited by the databases and language searched and by publication bias toward favorable results.

III. MATHEMATICAL BACKGROUND AND PERFORMANCE METRICS

A. Photovoltaic Cell Mathematical Model

The electrical behavior of a photovoltaic cell can be represented using the single-diode model, as used in PV monitoring and simulation studies [14]. The output current of the PV cell is expressed by (1):

$$I = I_{ph} - I_0 \left[\exp \left(\frac{V + IR_s}{nV_t} \right) - 1 \right] - \frac{V + IR_s}{R_{sh}} \quad (1)$$

where I and V are the output current and voltage, respectively, I_{ph} is the photocurrent, I_0 is the diode reverse saturation current, R_s is the series resistance, R_{sh} is the shunt resistance, n is the diode ideality factor, and V_t is the thermal voltage, which is given by (2):

$$V_t = \frac{N_s k T}{q} \quad (2)$$

where N_s is the number of cells connected in series, k is Boltzmann's constant, T is the absolute cell temperature, and q is the elementary charge. The output power of the PV system is given by (3):

$$P_{PV} = VI \quad (3)$$

and the power at the maximum power point (MPP) is therefore determined by (4):

$$P_{MPP} = V_{MPP} I_{MPP} \quad (4)$$

where V_{MPP} and I_{MPP} denote the voltage and current at the maximum power point, respectively.

B. Maximum Power Point Condition

For a PV system operating under uniform irradiance, the maximum power point occurs when the derivative of power with respect to voltage is zero [15], as expressed in (5):

$$\frac{dP}{dV} = 0 \quad (5)$$

Since the power is $P = VI$ as in (3), differentiation gives (6):

$$\frac{dP}{dV} = I + V \frac{dI}{dV} \quad (6)$$

Consequently, the MPP condition can be written as (7):

$$I + V \frac{dI}{dV} = 0 \quad (7)$$

or, equivalently, as (8):

$$\frac{dI}{dV} = - \frac{I}{V} \quad (8)$$

Equation (8) forms the theoretical basis of the Incremental Conductance (INC) MPPT technique.

C. Photovoltaic Conversion Efficiency

The efficiency of a PV system can be expressed as the ratio of electrical power produced to the incident solar power, as given by (9):

$$\eta_{PV} = \frac{P_{out}}{GA} \quad (9)$$

where P_{out} is the electrical output power, G is the solar irradiance, and A is the effective surface area of the PV array. The efficiency of an MPPT controller can similarly be evaluated using (10):

$$\eta_{MPPT} = \frac{P_{tracked}}{P_{available}} \quad (10)$$

A higher value of η_{MPPT} in (10) indicates that the optimization or tracking technique is able to extract a larger proportion of the available PV power.

D. Fill Factor

The fill factor is an important parameter for evaluating the electrical quality of a PV module and is defined by (11):

$$FF = \frac{V_{MPP} I_{MPP}}{V_{OC} I_{SC}} \quad (11)$$

where V_{OC} is the open-circuit voltage and I_{SC} is the short-circuit current. The maximum theoretical output power can consequently be represented by (12):

$$P_{max} = FF V_{OC} I_{SC} \quad (12)$$

E. Effects of Irradiance and Temperature

The photocurrent generated by a PV cell is approximately proportional to solar irradiance. It can be expressed by (13):

$$I_{ph} = [I_{ph,ref} + \alpha_I (T - T_{ref})] \frac{G}{G_{ref}} \quad (13)$$

where $I_{ph,ref}$ is the photocurrent under reference conditions, α_I is the temperature coefficient of current, G is the incident irradiance, and G_{ref} is the reference irradiance. The variation in open-circuit voltage with temperature can be approximated by (14):

$$V_{OC} = V_{OC,ref} + \beta_V (T - T_{ref}) \quad (14)$$

where β_V represents the temperature coefficient of voltage. Equations (13) and (14) demonstrate why changing irradiance and temperature can shift the PV operating point and create challenges for MPPT algorithms.

F. Optimization Objective Function

For an optimization-based PV control strategy, the general optimization problem can be represented by (15):

$$\max f(O) \quad (15)$$

subject to the bounds in (16):

$$\min \leq \leq \max \quad (16)$$

where x represents the decision variables and $f(x)$ represents the objective function. Depending on the optimization technique, the decision variable may represent the PV operating voltage, converter duty cycle, reference voltage, or another control parameter. For maximum power extraction, the objective function can be defined by (17):

$$f(O) = P_{PV(O)} = V(O) I(O) \quad (17)$$

Thus, the maximum power is obtained from (18):

$$P_{MPP} = \max [V(O) I(O)] \quad (18)$$

G. MPPT Tracking Error

The performance of an MPPT algorithm can be quantified using the tracking error in (19):

$$e_p = \frac{P_{MPP} - P_{tracked}}{P_{MPP}} \quad (19)$$

A lower value of e_p in (19) indicates that the algorithm operates closer to the actual maximum power point. The tracking efficiency can alternatively be expressed by (20):

$$\eta_{track} = \frac{P_{tracked}}{P_{MPP}} \times 100\% \quad (20)$$

which is equivalent to the MPPT efficiency in (10) when the available power equals the maximum power P_{MPP} .

H. Root Mean Square Error and Mean Absolute Error

For optimization and prediction techniques evaluated against reference values, the root mean square error (RMSE) can be calculated using (21):

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (21)$$

where y_i is the reference value, $\hat{y}_i$ is the estimated or optimized value, and N is the number of observations. The mean absolute error (MAE) is given by (22):

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (22)$$

The metrics in (21) and (22) can be used to compare the accuracy and stability of different intelligent and metaheuristic optimization techniques.

I. Converter Duty Cycle

For a basic ideal boost converter, the relationship between input and output voltage is given by (23):

$$V_{out} = \frac{V_{in}}{1-D} \quad (23)$$

where D is the converter duty cycle. Hence, the duty cycle follows from (24):

$$D = 1 - \frac{V_{in}}{V_{out}} \quad (24)$$

MPPT controllers can manipulate D in (24) to move the PV operating point toward the maximum power point.

J. Overall PV System Performance

The overall performance of a PV system can be represented using a weighted performance objective in (25):

$$J = w_1 \eta_{PV} + w_2 \eta_{MPPT} - w_3 e_p - w_4 THD \quad (25)$$

where w_1 , w_2 , w_3 , and w_4 are weighting coefficients representing the relative importance of conversion efficiency, MPPT efficiency, tracking error, and harmonic distortion (THD). The optimization problem can consequently be formulated as (26):

$$\max J(x) \quad (26)$$

subject to the operational constraints in (27), (28), and (29), namely the voltage limits

$$V_{min} \leq V \leq V_{max} \quad (27)$$

the current limits

$$I_{min} \leq I \leq I_{max} \quad (28)$$

and the duty-cycle limits

$$D_{min} \leq D \leq D_{max} \quad (29)$$

The inverter efficiency metrics relevant to the evaluation of overall PV system performance are illustrated in Fig. 1 [16].

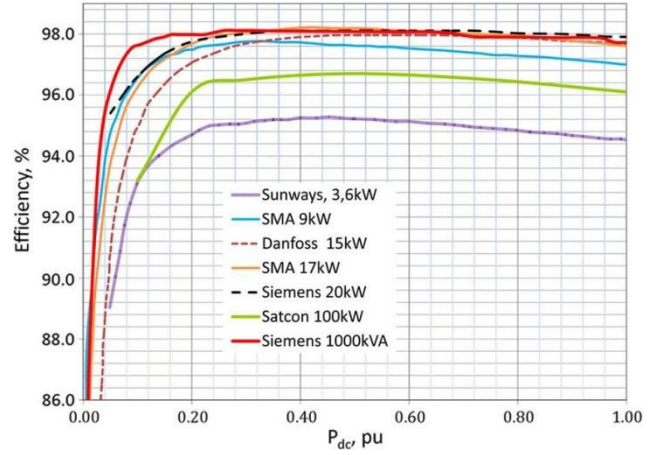


Fig. 1. Inverter efficiency metrics (adapted from [16], p. 5).

IV. OPERATING PRINCIPLES OF MPPT TECHNIQUES

An MPPT controller sits between the PV source and the load, normally as the control of a DC-DC converter. By adjusting the duty cycle D in (24), it moves the operating point of the array along its P-V curve toward the maximum power point [10]. This section describes how each family of techniques performs that search, using the notation of Section III.

A. Perturb and Observe

Perturb and Observe (P&O) perturbs the operating voltage or duty cycle by a small step and observes the change in power. If power increases, the perturbation continues in the same direction; if it decreases, the direction is reversed. The method needs only voltage and current measurements and is simple and inexpensive. Its characteristic weaknesses are steady-state oscillation around the MPP; a trade-off in the step size between fast tracking and small oscillation; and a tendency to move in the wrong direction when irradiance changes rapidly between samples, because the change in power then reflects the irradiance rather than the perturbation [10], [17]. Optimizing the step size and sampling period reduces these effects [17].

B. Incremental Conductance

Incremental Conductance (INC) uses the MPP condition in (8). It compares the incremental conductance dI/dV with the instantaneous conductance $-I/V$. If $dI/dV = -I/V$, the array is at the MPP; if $dI/dV > -I/V$, the operating point lies to the left of the MPP and the voltage should be increased; if $dI/dV < -I/V$, it lies to the right and the voltage should be decreased. Because the direction follows from a single pair of measurements, INC can follow irradiance changes more reliably than P&O and can in principle stop perturbing at the MPP, at the cost of more computation [10], [18].

C. Fractional Open-Circuit Voltage and Short-Circuit Current

These methods exploit the approximately linear relationships between the MPP voltage and the open-circuit voltage, and between the MPP current and the short-circuit current, as in (30) and (31):

$$V_{MPP} \approx k_1 V_{OC} \quad (30)$$

$$I_{MPP} \approx k_2 I_{SC} \quad (31)$$

where k_1 and k_2 are module-dependent constants smaller than one; k_1 is commonly reported between about 0.7 and 0.8 [10]. The methods are simple and fast but approximate, and the open-circuit voltage or short-circuit current must be measured periodically, which interrupts power transfer or requires a pilot cell. Accuracy declines as temperature and ageing change the constants.

D. Fuzzy Logic Control

A fuzzy logic controller (FLC) typically uses the error $E(k)$ and its change $\Delta E(k)$ as inputs, defined in (32) and (33), where k indexes the samples:

$$E(k) = \frac{P(k) - P(k-1)}{V(k) - V(k-1)} \quad (32)$$

$$\Delta E(k) = E(k) - E(k-1) \quad (33)$$

$E(k)$ approximates dP/dV , so it is positive to the left of the MPP, negative to the right, and near zero at the MPP. Fuzzification maps E and ΔE to linguistic variables (for example, negative big, zero, positive small); a rule base, such as “if E is positive big and ΔE is positive, increase the duty cycle by a large step,” produces an output; and defuzzification converts it into a duty-cycle change ΔD . An FLC does not need an accurate model of the array and handles nonlinearity well, but its performance depends on the membership functions and rule base chosen by the designer [19], [20].

E. Artificial Neural Networks

An ANN-based tracker maps measured inputs, such as irradiance and temperature or the open-circuit voltage and short-circuit current, to the voltage or duty cycle at the MPP. The network is trained offline from measured or simulated data; at run time the mapping is evaluated in a single forward pass, giving very fast tracking. Accuracy is high within the range of conditions covered by the training data but degrades when the array ages or when shading patterns absent from training occur, so periodic retraining or combination with a local search is advisable [21].

F. Metaheuristic Techniques

Population-based methods treat candidate operating points (duty cycles or voltages) as individuals, each evaluated by measuring the resulting power. In particle swarm optimization (PSO) [22], particle i has a position x and a velocity v that are updated at iteration k according to (34) and (35):

$$v_i^{k+1} = w v_i^k + c_1 r_1 (p_i - x_i^k) + c_2 r_2 (g - x_i^k) \quad (34)$$

$$x_i^{k+1} = x_i^k + v_i^{k+1} \quad (35)$$

where w is the inertia weight, c_1 and c_2 are acceleration coefficients, r_1 and r_2 are random numbers in $[0, 1]$, p is the best position found by the particle, and g is the best position found by the swarm. Other methods, such as the grey wolf optimizer (GWO) [23], the genetic algorithm (GA), and the artificial bee colony (ABC), use different update rules but

share this structure. Because they search globally, they can locate the global maximum under partial shading [24], [25]. Their drawbacks are slower convergence, greater computation, sensitivity to parameter tuning, and the need to re-initialize the search when irradiance changes and a previously found peak is no longer valid.

G. Hybrid Techniques

Hybrid techniques combine complementary strengths: a metaheuristic may provide global search, followed by local fine-tuning with P&O or INC to remove oscillation, or an ANN or FLC may have its parameters tuned by an evolutionary algorithm [8]. A hybrid GWO–PSO tracker for partial-shading conditions is reported in [25]. The performance gain must be weighed against the added design and tuning effort.

H. Partial Shading

When modules in a series string receive different irradiance, bypass diodes conduct and the P–V curve develops several local maxima, only one of which is the global MPP [26]. Hill-climbing methods such as P&O and INC may settle on a local peak and forfeit a substantial share of the available power [11]. Responses include global search methods (metaheuristics or periodic scanning of the P–V curve), module-level or distributed MPPT, and array reconfiguration to spread the effect of shading [27]. This is the main reason intelligent and hybrid techniques are attractive despite their cost.

I. IoT-Enabled and Predictive Trends

Recent work combines MPPT with sensing, communication, and analytics. IoT-enabled controllers report voltage, current, temperature, and irradiance to cloud or edge platforms for monitoring, remote tuning, and fault detection. Forecasting models, such as machine-learning yield predictors [6] and LSTM-based forecasters tuned by genetic algorithms [28], supply predicted irradiance and temperature that can be used to position the operating point in advance or to schedule storage, and lightweight machine-learning models can run on low-cost microcontrollers [7]. These trends move MPPT from purely reactive control toward predictive, data-driven operation, at the price of additional requirements for communication reliability and cybersecurity.

V. CLASSIFICATION AND COMPARISON OF OPTIMIZATION TECHNIQUES

A. Classification of Techniques

Table I classifies the optimization techniques considered in this review, drawn from the literature on conventional MPPT methods [13], [15]; fuzzy-logic and AI techniques [19], [20]; metaheuristic techniques [24], [29]; hybrid techniques [8]; optimization under partial shading and dynamic conditions [27]; and forecasting with hybrid LSTM–GA models [28].

TABLE I. Classification of optimization techniques for photovoltaic systems

S/N	Optimization Technique	Principle / Approach	Application in PV Systems	Key Limitations
1	Conventional MPPT Techniques	Determines the MPP by continuously adjusting the PV operating point based on voltage, current, or power variations. Common methods include P&O and INC.	Maximum power point tracking under relatively stable irradiance and temperature conditions.	Can exhibit oscillations around the MPP; reduced performance under rapidly changing irradiance; P&O may track in the wrong direction during sudden environmental changes.
2	Fuzzy Logic and Artificial Intelligence Techniques	Uses intelligent decision-making based on input variables such as power error and its rate of change. Techniques include FLC, ANN, and ANFIS.	MPPT, PV power prediction, fault diagnosis, adaptive control, and energy management.	Requires appropriate training or rule design; computational complexity can increase; ANN-based methods may require substantial representative training data.
3	Metaheuristic Optimization Techniques	Searches for the global optimum using population- or nature-inspired search mechanisms. Examples include PSO, GA, GWO, WOA, and ABC.	Global MPPT, parameter optimization, PV array configuration, sizing, and hybrid energy-system optimization.	Computational burden can be relatively high; convergence speed depends on algorithm parameters; some methods may suffer from premature convergence or require extensive parameter tuning.
4	Hybrid Optimization Techniques	Combines two or more optimization or control strategies to exploit their complementary strengths. Examples include PSO–FLC, ANN–MPPT, and P&O–INC approaches.	Global MPPT, adaptive control, PV forecasting, energy management, and partial-shading optimization.	Increased algorithmic complexity; more difficult implementation; greater computational requirements; performance depends on appropriate coordination between the combined techniques.
5	Optimization Under Partial Shading and Dynamic Conditions	Uses global search, distributed MPPT, reconfiguration, or intelligent algorithms to identify the global maximum among multiple local power peaks.	PV arrays affected by clouds, buildings, trees, soiling, module mismatch, and rapidly changing environmental conditions.	Higher computational and hardware requirements; global tracking may take longer; performance varies with shading patterns and algorithm design; real-time implementation remains challenging.

B. Representative Techniques and Performance Considerations

Table II compares representative optimization techniques in terms of category, typical PV application, and the main performance consideration for each.

TABLE II. Representative optimization techniques, categories, applications and performance considerations

Technique	Category	Typical PV Application	Major Performance Consideration
P&O	Conventional MPPT	Basic MPPT control	Simplicity and tracking oscillation
INC	Conventional MPPT	Variable irradiance MPPT	Tracking accuracy
FLC	Intelligent Control	Nonlinear PV systems	Robustness and rule design
ANN	AI-Based	MPPT and power prediction	Training accuracy
ANFIS	Hybrid AI	Adaptive MPPT	Adaptability
GA	Metaheuristic	PV sizing and MPPT	Search efficiency
PSO	Metaheuristic	MPPT and PV parameter optimization	Convergence speed
GWO	Metaheuristic	Partial-shading MPPT	Global-search capability
WOA	Metaheuristic	MPPT	Convergence and computational cost
ABC	Metaheuristic	PV parameter and MPPT optimization	Exploration capability
DE	Evolutionary	PV modelling and MPPT	Accuracy and convergence
Hybrid Methods	Hybrid	Advanced MPPT and energy management	Accuracy versus complexity

C. Criteria-Based Comparison

The six criteria of Section II are defined as follows. Tracking accuracy is the closeness of the operating point to the MPP in steady state, quantified by (19) and (20). Convergence speed is the time taken to reach the MPP after a change in conditions. Computational complexity is the processing and memory needed per control cycle. Adaptability is the ability to maintain performance under changing irradiance, temperature,

and partial shading. Implementation requirements cover sensors, hardware, and design effort. Energy-yield impact is the energy actually harvested over a test profile, which is captured by the energy-based MPPT efficiency in (36):

$$\eta_{MPPT,dyn} = \frac{\int_0^T P_{PV}(t) dt}{\int_0^T P_{MPP}(t) dt} \quad (36)$$

where T is the duration of the test profile. Standardized static and dynamic test sequences for evaluating MPPT performance

are given in IEC 62891 [30]. Table III rates each family of techniques on the six criteria. The ratings synthesize the comparative studies [10]–[13], [21]; they show relative

position rather than measured values, which depend on the converter, sampling period, and irradiance profile used in each test.

TABLE III. Criteria-based comparison of MPPT techniques

Technique	Tracking Accuracy	Convergence Speed	Computational Complexity	Adaptability	Implementation Requirements	Energy-Yield Impact
P&O	Moderate; oscillates about the MPP	Moderate	Low	Low to moderate; misled by rapid irradiance change; local peak only	V and I sensors; simple microcontroller code	Good under steady conditions; losses when conditions change
INC	Moderate to high; reduced oscillation	Moderate	Low to moderate	Moderate; follows irradiance change better than P&O; local peak only	V and I sensors; derivative and division operations	Slightly higher than P&O under changing irradiance
FOCV / FSCC	Low to moderate (approximate constant)	Fast	Very low	Low; constant drifts with temperature and ageing	Periodic V_{OC} or I_{SC} measurement, or a pilot cell	Lower yield; suited to low-cost systems
FLC	High when well designed	Fast	Moderate	High; handles nonlinearity	Rule base and membership functions to design and tune	Good to high
ANN	High within the trained range	Very fast (single forward pass)	Moderate to high	High within training range; degrades with ageing or unseen shading	Training data and capacity for retraining	High
Metaheuristics (PSO, GWO, GA, ABC)	High; finds the global MPP	Slow to moderate	High	High; needs re-initialization after changes	Parameter tuning; adequate sampling rate and computing power	Highest under partial shading
Hybrid	Highest	Fast to moderate	High	High	Most complex design and tuning	Highest, if the gain justifies the complexity

Three patterns emerge. First, accuracy and adaptability increase together with computational and implementation complexity, so no technique is best on all six criteria. Second, the principal dividing line is partial shading: conventional hill-climbing methods are adequate only when a single power peak exists, whereas global-search or hybrid methods are needed otherwise. Third, in perturbation-based methods the step size trades convergence speed against steady-state accuracy, which is why adaptive step sizes and hybrid schemes are popular.

VI. DISCUSSION

Optimization techniques play a pivotal role in advancing the performance and efficiency of photovoltaic (PV) power technologies, which are critical to meeting the growing demand for sustainable energy solutions. The continual development and refinement of these techniques—ranging from maximum power point tracking (MPPT) algorithms to material enhancements and system design improvements—have demonstrated significant potential in increasing energy yield while reducing operational costs and improving reliability [31].

Given the complex environmental conditions under which PV systems operate, such as partial shading and variable insolation, the integration of intelligent optimization methods has become essential for extracting maximum power output. These methods not only address the limitations of classical approaches but also enhance adaptability to dynamic scenarios, thereby ensuring more consistent performance [25].

Furthermore, advancements in computational models and hybrid system configurations have contributed to more precise control strategies that optimize both energy storage and generation aspects of PV systems. Storage sizing and hybrid storage-generation optimization are addressed in [32] and [33]; material enhancements are reported in [34] and [35]; and system-design improvements include cooling [36], tracking [37] and zero-energy-building configurations [38]. The life-cycle impacts of PV technologies are assessed in [39] and [40].

A. Selection Guidance

For low-cost systems operating in uniform conditions, such as small off-grid installations with unshaded arrays, P&O or INC with an adaptive step offers the best balance of cost and yield. Where partial shading is common, for example near buildings, trees, or on dust-streaked modules, global-search, hybrid, or module-level approaches justify their additional cost. Where historical data exist and the hardware permits, ANN or FLC schemes provide fast tracking; where computing resources are limited, FOCV or simplified fuzzy rules give acceptable operation.

B. Implementation and Evaluation Practice

Tracking performance depends as much on hardware as on the algorithm: sensor accuracy and noise, converter ripple, the sampling period relative to the converter's settling time, and microcontroller resources all matter. A fair comparison therefore requires the same converter, sampling period, and irradiance profile for every algorithm, and reporting of the energy-based efficiency in (36) rather than only the steady-state error.

C. Relevance to Nigerian Conditions

High ambient temperatures lower the open-circuit voltage according to (14) and shift the MPP, which fixed-ratio methods handle poorly. Dust accumulation and passing shade from buildings, trees, and vehicles create non-uniform irradiance in which local-peak trapping is likely, and many installations are battery-based off-grid systems whose charge controllers use low-cost microcontrollers. These conditions favor simple, robust trackers such as adaptive P&O or INC for unshaded arrays, and lightweight global-search or hybrid trackers where shading is frequent. They also emphasize field validation, since results from simulation or temperate-climate test profiles may not transfer.

D. Research Gaps and Future Directions

Standardized benchmark irradiance and temperature profiles, including tropical conditions, are needed so that algorithms can be compared fairly. Many studies rely on simulation; hardware-in-the-loop testing and long-term field validation would strengthen the evidence. Further directions include lightweight AI and TinyML implementations for low-cost microcontrollers, digital-twin-based tuning, joint optimization of MPPT with storage and inverter control, and the cybersecurity of IoT-enabled controllers. Finally, optimization objectives should include life-cycle environmental impact [39], [40] as well as energy yield. Future research should therefore focus on algorithmic developments and material science advances that overcome the limitations listed in Table I.

VII. CONCLUSION

This review has examined optimization techniques for enhancing the performance of PV power systems, covering the underlying mathematical models and performance metrics and a comparative classification of conventional, intelligent, metaheuristic, and hybrid approaches. By synthesizing current knowledge with emerging trends, it underscores the indispensable role of optimization in driving photovoltaic technology forward within the global transition toward renewable energy sources.

The comparison shows that no single technique is best on every criterion. Conventional methods such as P&O and INC are simple, inexpensive, and effective under uniform conditions; intelligent methods improve tracking speed and adaptability; and metaheuristic and hybrid methods are the most reliable under partial shading, at the cost of computation and design effort. For practitioners, the choice should follow from the shading profile, the available hardware, and the acceptable cost, and should be validated against energy-based efficiency under local conditions. Standardized benchmarks, field validation, and lightweight AI implementations are the main priorities for future work.

APPENDIX A: ABBREVIATIONS

Table IV lists the abbreviations used in this paper.

TABLE IV. Abbreviations used in this paper

Abbreviation	Full Meaning	Abbreviation	Full Meaning	Abbreviation	Full Meaning
PV	Photovoltaic	SCA	Sine Cosine Algorithm	Imp	Current at Maximum Power Point
PVS	Photovoltaic System	MFO	Moth-Flame Optimization	Pmax	Maximum Output Power
PVG	Photovoltaic Generator	WCA	Water Cycle Algorithm	FF	Fill Factor
PVSA	Photovoltaic Solar Array	BFO	Bacterial Foraging Optimization	η	Efficiency
SPV	Solar Photovoltaic	POA	Power Optimization Algorithm	PSH	Peak Sun Hours
MPPT	Maximum Power Point Tracking	DC	Direct Current	SOC	State of Charge
MPP	Maximum Power Point	AC	Alternating Current	BESS	Battery Energy Storage System
GMPP	Global Maximum Power Point	DC–DC	Direct-Current to Direct-Current Converter	ESS	Energy Storage System
LMPP	Local Maximum Power Point	DC–AC	Direct-Current to Alternating-Current Converter	EMS	Energy Management System
P&O	Perturb and Observe	VSI	Voltage Source Inverter	RES	Renewable Energy Sources
INC	Incremental Conductance	CSI	Current Source Inverter	DG	Distributed Generation
FOCV	Fractional Open-Circuit Voltage	PWM	Pulse Width Modulation	DER	Distributed Energy Resources
FCSO	Fractional Short-Circuit Current	SPWM	Sinusoidal Pulse Width Modulation	MG	Microgrid
ANN	Artificial Neural Network	THD	Total Harmonic Distortion	HRES	Hybrid Renewable Energy System
ANFIS	Adaptive Neuro-Fuzzy Inference System	DMPPT	Distributed Maximum Power Point Tracking	LCOE	Levelized Cost of Electricity
FLC	Fuzzy Logic Controller	GMPPT	Global Maximum Power Point Tracking	LPSP	Loss of Power Supply Probability
PSO	Particle Swarm Optimization	DNI	Direct Normal Irradiance	CO ₂	Carbon Dioxide
GA	Genetic Algorithm	GHI	Global Horizontal Irradiance	IoT	Internet of Things
GWO	Grey Wolf Optimizer	POA	Plane of Array	AI	Artificial Intelligence
WOA	Whale Optimization Algorithm	STC	Standard Test Conditions	ML	Machine Learning

Abbreviation	Full Meaning	Abbreviation	Full Meaning	Abbreviation	Full Meaning
ACO	Ant Colony Optimization	NOCT	Nominal Operating Cell Temperature	RMSE	Root Mean Square Error
ABC	Artificial Bee Colony	I-V	Current-Voltage	MAE	Mean Absolute Error
DE	Differential Evolution	P-V	Power-Voltage	MSE	Mean Square Error
SA	Simulated Annealing	V-I	Voltage-Current	MAPE	Mean Absolute Percentage Error
TLBO	Teaching-Learning-Based Optimization	Isc	Short-Circuit Current	R ²	Coefficient of Determination
SSA	Salp Swarm Algorithm	Voc	Open-Circuit Voltage		
HHO	Harris Hawks Optimization	Vmp	Voltage at Maximum Power Point		
FSCC	Fractional Short-Circuit Current	LSTM	Long Short-Term Memory	IEC	International Electrotechnical Commission
TinyML	Tiny Machine Learning				

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Climate-Adaptive Zoning Using Climate Projections: A Framework for Flood Risk Reduction in Ibadan, Nigeria

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Abstract—Flooding remains the most recurrent climate-induced hazard in Ibadan, Nigeria, despite the availability of flood-hazard data and climate projections. This study develops a climate-adaptive zoning framework for flood-risk reduction that integrates downscaled climate projections with land-use and hydrological data to delineate flood-risk zones and inform adaptive land-allocation decisions. The framework combines a review of the theoretical and conceptual literature on climate adaptation and zoning with an empirical assessment of flood risk drivers, hazard exposure, and institutional and legal constraints on flood-responsive planning in Ibadan. Findings identify priority zones requiring adaptive land-use controls and set out recommendations for embedding climate projections into statutory zoning and development-control instruments, promoting sustainable land-use planning in flood-prone communities.

Index Terms—Climate-adaptive zoning, climate projections, flood-prone, flood risk reduction, land allocation.

I. INTRODUCTION

Flooding is now the most frequent climate-related hazard in Nigerian cities and Ibadan has been one of the most affected urban centres for flood disasters in West Africa. The September 2011 flood displaced over 121,000 people, destroyed more than 3,000 houses and caused an estimated 20 billion naira in damages across Ibadan metropolis [1]. A decade later, in August 2022, similar patterns of loss were recorded in Apete, Odo-Ona, and Sasa, where houses built on natural drainage channels were submerged within hours of rainfall [2]. What makes this recurring loss troubling is that the risk is no longer surprising. The Nigeria Hydrological Services Agency (NIHSA) and the Nigerian Meteorological Agency (NiMet) issue annual flood outlooks that consistently list Ibadan North, Ibadan South-East and Akinyele as highly probable flood Local Government Areas [3], [4] and the climate projections are even more direct. The World Bank Group, using CMIP6 models, projects an 8-12% increase in extreme 1-day rainfall for Ibadan by 2050 under the SSP2-4.5 scenario [5]. In short, the hazard is known, mapped, and forecasted and despite this, land is still being allocated in those same high-risk areas. The reason lies in the legal instrument that governs all land in Nigeria: the Land Use Act 1978. Section 5(2) of the Act vests all land in the Governor and empowers him to grant statutory rights of occupancy "for all purposes" [6]. However, the Act contains no provision for hazard, climate or flood risk to be considered before land allocation. Section 28 further limits government liability to unexhausted improvements only, which creates a perverse incentive: it is legally and politically safer to approve land and pay compensation after a flood than to refuse allocation and face litigation [7]. This legal silence has practical consequences. A review of planning files in Oyo State between 2018 and 2023 showed that no statutory approval referenced NIHSA flood data or climate projections as a condition for granting a Certificate of Occupancy (Author's Field Survey, 2025). As a result, Jiboye [8] described it as

"planning without foresight," a system that responds to disasters instead of preventing disasters from happening. International experience shows that this gap can be closed. The UK's National Planning Policy Framework (UK Government, 2023) and South Africa's Spatial Planning and Land Use Management Act [9], both of which mandate climate risk assessment as part of land-use approval, are few examples. In Lagos, post-2012 reforms now require drainage clearance before building approval, which has reduced approvals in some floodplain areas [10], but Nigeria has not domesticated climate considerations into its primary land law, with Ibadan being a critical case. As an inland city with documented flood history, rapid urban expansion into valleys and a planning system directly tied to the outdated Land Use Act 1978, incessant flood incidents may continue unabated. If climate projections cannot be integrated into land allocation, then adaptation policy in Nigeria will remain reactive rather than preventive. This study proceeds from the premise that Ibadan does not lack flood data, but what it lacks is a legal mechanism to use that data; therefore, the problem is not technical rather, it is institutional. Without reforming how the Land Use Act is applied at the state level, CMIP6 projections will remain academic while communities continue to lose homes every rainy season.

II. STATEMENT OF THE PROBLEM

Flooding has become one of the most persistent environmental challenges affecting urban development in Ibadan, Nigeria, with recurring flood events causing widespread loss of lives, displacement of residents, destruction of infrastructure, and significant economic losses. Although advances in climate science have provided reliable flood hazard information through the Nigeria Hydrological Services Agency (NIHSA), the Nigerian Meteorological Agency (NiMet), and climate projections such as CMIP6, development approvals and statutory land allocation continue to occur within flood-prone areas. This disconnect between available climate information and planning practice undermines sustainable urban development, increases the vulnerability of communities, and perpetuates a cycle of disaster, recovery, and repeated reconstruction.

The specific problem is that the Land Use Act 1978 and the Oyo State Physical Planning Law 2016 do not explicitly require the integration of climate projections or flood risk assessments into statutory land allocation decisions, leaving planning officials without adequate legal authority to reject development applications in environmentally vulnerable locations. Consequently, climate information remains largely excluded from planning decisions despite its availability. To address this gap, this study proposes a Climate-Adaptive Zoning Framework that integrates NIHSA flood-risk data and

CMIP6 climate projections into statutory land allocation through a legally feasible implementation guideline. The framework seeks to strengthen climate-responsive planning, reduce development within flood-prone areas, enhance legal support for planning officials, and promote sustainable urban resilience in Ibadan.

A. Research Objectives

This study aims to assess the feasibility of a climate-adaptive zoning framework for flood risk reduction through land allocation reform in Ibadan, Nigeria.

The specific objectives are to:

- i. To examine the extent to which the Land Use Act 1978 and the Oyo State Physical Planning Law 2016 facilitate the integration of climate projections into statutory land allocation decisions in Ibadan.
- ii. To assess the feasibility and acceptability of a climate-adaptive zoning framework based on NIHSA flood hazard data and CMIP6 climate projections among planning officials in Ibadan.
- iii. To evaluate the effectiveness of a Climate-Adaptive Zoning Addendum in reducing land allocation within flood-prone areas in Ibadan.

III. RESEARCH QUESTIONS

- i. To what extent do the Land Use Act 1978 and Oyo State planning law enable the integration of climate projections into land allocation decisions in Ibadan?
- ii. How feasible and acceptable is a climate-adaptive zoning framework (based on NIHSA and CMIP6 data) to planning officials in Ibadan?
- iii. How effective can a climate-adaptive zoning addendum reduce the allocation of land in flood-prone areas in Ibadan?

A. Research Hypothesis

Based on the research questions given in the preceding Section, the following research hypothesis will be evaluated in the study.

- i. H_0 = There is no significant relationship between the provisions of the laws (Land Use Act 1978; [11]) and the integration of climate projections in land allocation decisions in Ibadan.

H_1 = There is a significant relationship between the provisions of the laws and the integration of climate projections in statutory land allocation decisions in Ibadan.

- ii. H_0 = There is no significant relationship between stakeholders' perception and perceived feasibility of a climate-adaptive zoning addendum in Ibadan.

H_1 = There is a significant relationship between stakeholders' perception and perceived feasibility of a climate-adaptive zoning addendum in Ibadan.

IV. LITERATURE REVIEW

Flooding has become the most persistent climate-induced hazard in Ibadan, with recurrent disasters, increasing climate projections, and continued development in flood-prone areas despite extensive flood risk data from NIHSA, NiMet, and CMIP6. Existing studies have largely focused on flood hazard mapping but have paid limited attention to the legal and institutional factors that permit land allocation in high-risk zones, particularly the omissions within the Land Use Act 1978 and the Oyo State Physical Planning Law 2016 regarding climate and flood risk integration. International best practices demonstrate that climate-adaptive zoning supported by legal provisions can strengthen flood risk reduction, highlighting the need for a framework that integrates climate projections into statutory land allocation and planning decisions in Ibadan.

V. CONCEPTUAL FRAMEWORK

The corresponding result is shown in Fig. 1.

This study adopts a Legal-Institutional-Behavioural framework (Fig. 1). Building on Berke and Godschalk [12] and Akinbode and Ogunleye [7], the framework argues that technical data alone does not change outcomes. Legal reform is required to mandate risk assessment. Institutional change follows when agencies are required to apply the law. Behavioural change occurs when implementers have legal protection to act. In Ibadan, this means embedding NIHSA and CMIP6 data in law, creating a mandatory risk check and protecting planners from liability. The anticipated outcome is reduced allocation of land in floodplains. This framework guided the design of interview questions and the analysis of approval files in this study.

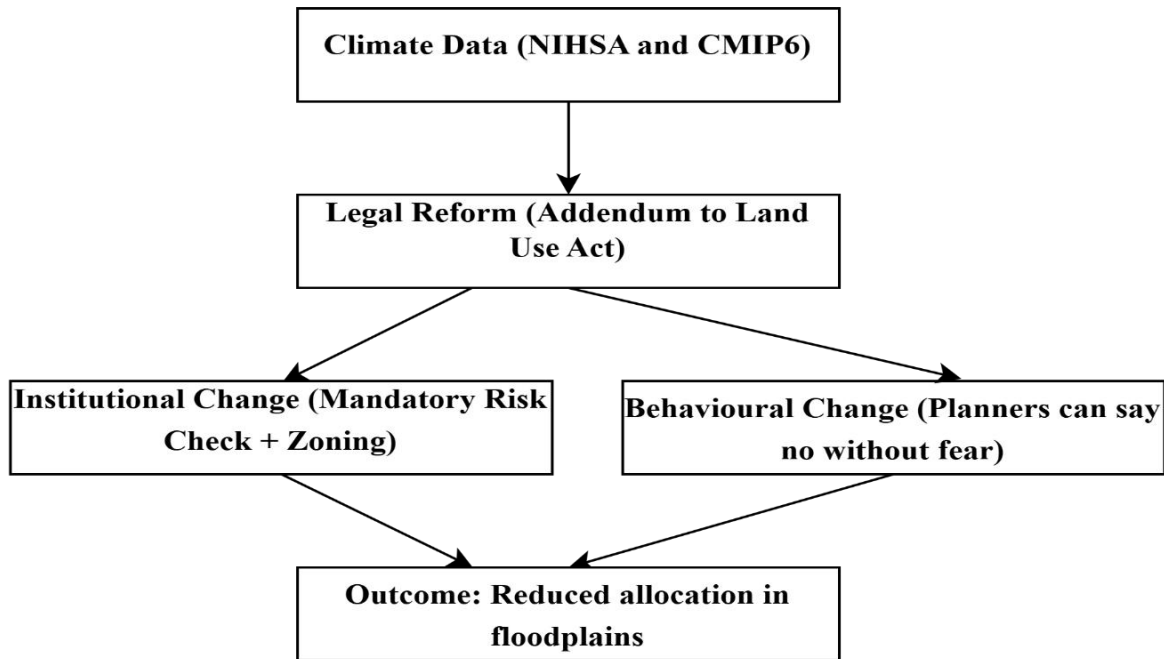


Fig. 1. Conceptual Framework Linking Climate Data, Legal Reform, Institutional Change, and Behavioural Change to Reduced Floodplain Allocation.
Source: Conceptual Framework (2026)

VI. THEORETICAL REVIEW

This study is anchored on Institutional Theory and Adaptive Governance Theory, both of which provide a robust foundation for understanding the integration of climate projections into statutory land allocation and planning decisions. Institutional Theory, advanced by North [13] and further developed by Scott [14], posits that formal rules, legal frameworks, and organizational structures shape institutional behaviour and policy implementation. The theory argues that organizations operate within established regulatory, normative, and cultural frameworks that influence decision making and compliance. In the context of land administration, the Land Use Act 1978 and the Oyo State Physical Planning Law 2016 constitute the formal institutions governing land allocation in Nigeria. However, the absence of explicit provisions requiring the integration of climate projections into statutory approvals creates institutional gaps that weaken climate-responsive planning and encourage development in environmentally vulnerable locations. Institutional Theory therefore explains why planning officials continue to approve developments in flood-prone areas despite the availability of scientific evidence from the Nigeria Hydrological Services Agency (NIHSA), the Nigerian Meteorological Agency (NiMet), and Coupled Model Intercomparison Project Phase 6 (CMIP6) climate projections. The theory emphasizes that sustainable land-use governance depends on institutional reforms capable of aligning legal provisions with emerging climate realities [13]–[17].

The second theoretical foundation is Adaptive Governance Theory, which emerged from resilience and environmental governance literature [18], [19]. Adaptive Governance

advocates flexible, collaborative, and knowledge-based institutions that continuously integrate scientific information into policy and planning to address environmental uncertainty and climate risks. Unlike conventional planning approaches that rely on historical records, Adaptive Governance promotes the incorporation of climate projections, stakeholder participation, and continuous institutional learning to strengthen resilience and disaster risk reduction [20], [21]. Applied to this study, the theory provides a conceptual basis for the proposed Climate-Adaptive Zoning Framework, which integrates NIHSA flood-risk assessments, CMIP6 climate projections, and statutory legal safeguards into land allocation decisions. By emphasizing anticipatory planning, inter-agency coordination, and adaptive legal mechanisms, Adaptive Governance supports the transition from reactive flood management to proactive climate-responsive land-use planning in Ibadan. Together, Institutional Theory explains the legal and institutional barriers to integrating climate information into planning, while Adaptive Governance Theory provides the framework for developing resilient, climate-adaptive zoning policies that reduce flood risk and promote sustainable urban development.

VII. RESEARCH GAP

Existing studies consistently demonstrate that flood risk in Ibadan is increasing as a result of rapid urbanization, climate variability, and the expansion of settlements into flood-prone areas (NIHSA, 2024; NiMet, 2024; [22]). Similarly, scholars have identified deficiencies in Nigeria's land administration framework, particularly the Land Use Act 1978, which lacks explicit provisions for integrating climate information into statutory land allocation and development approval processes

[7], [8]. Although previous studies have produced comprehensive flood hazard and susceptibility maps for Ibadan and recommended improved land use planning, they have largely concentrated on hazard identification rather than examining the legal and institutional mechanisms required to translate climate projections into enforceable planning decisions [12], [23]. Consequently, there remains limited empirical evidence on how existing legal instruments can be adapted to support climate-responsive land allocation within Nigeria's statutory planning system.

Furthermore, international experience demonstrates that climate-adaptive zoning and climate-informed planning have strengthened flood risk management in countries such as the United Kingdom and South Africa through statutory integration of climate projections into planning decisions (UK Government, 2023; [9]). However, these approaches operate within different legal and institutional contexts and cannot be directly transferred to Nigeria without contextual adaptation. Existing Nigerian studies have generally advocated broad legislative reforms or the enactment of new climate legislation but have paid little attention to practical, legally feasible interventions that can be implemented within the existing Land Use Act framework [7], [12]. This study addresses this gap by developing and validating a Climate-Adaptive Zoning Addendum that integrates NIHSA flood data and CMIP6 climate projections into statutory land allocation procedures while assessing its legal feasibility and institutional acceptability among planning officials in Ibadan. The study therefore contributes a practical and evidence-based framework capable of strengthening climate-responsive land administration and reducing flood risk without requiring comprehensive legislative reform.

VIII. METHODOLOGY

This study adopted a mixed-methods policy analysis design guided by the Legal–Institutional–Behavioural Framework to examine the integration of climate projections into statutory land allocation decisions under the existing Nigerian legal framework. The study was conducted in Ibadan Metropolis, covering Ibadan North, Ibadan North-East, Ibadan South-East, Ibadan South-West, and Akinyele Local Government Areas, selected because they experienced the greatest flood impacts during the 2011 and 2022 flood events and are consistently classified as highly probable flood-risk areas by the Nigeria Hydrological Services Agency (NIHSA). Data were collected through a review of the Land Use Act 1978, the Oyo State Physical Planning Law 2016, NIHSA Annual Flood Outlooks (2019–2023), and CMIP6 climate projections for Ibadan, alongside an assessment of twenty statutory Certificate of Occupancy approval files processed between 2018 and 2023. In addition, eighteen semi-structured key informant interviews were conducted with town planners, licensed surveyors, Ministry of Environment officials, and community leaders from Apete, Odo Ona, and Sasa to examine awareness of flood risk, barriers to rejecting development applications, and the feasibility of a Climate-Adaptive Zoning Addendum. Data were analysed using content analysis for legal documents, descriptive statistics and cross-tabulations for approval files, and thematic analysis supported by NVivo 14 for interview

data to identify legal, institutional, and behavioural factors influencing climate-responsive land allocation.

IX. RESULT

The findings revealed a significant legal gap in the existing land administration framework governing land allocation in Ibadan. Content analysis of the Land Use Act 1978 showed that none of its 41 sections explicitly references climate change, flood hazards, or disaster risk, while Section 5(2) grants broad discretionary powers for land allocation and Section 28 limits compensation to unexhausted improvements. Similarly, the Oyo State Physical Planning Law 2016 provides technical planning standards for setbacks and drainage but does not mandate climate or flood risk assessments as prerequisites for land approval. Key informant interviews corroborated these findings, with planning officials indicating that the absence of statutory provisions on flood risk prevents them from rejecting development applications in vulnerable areas. These findings suggest that the current legal framework does not support climate-responsive land allocation, thereby confirming the absence of a significant relationship between the existing legal provisions and the integration of climate projections into statutory land allocation decisions.

The study also identified a substantial implementation gap between the availability of climate information and its application in planning practice. Review of twenty statutory land approval files processed between 2018 and 2023 showed that none referenced NIHSA flood-risk classifications or required Flood Risk Assessments as part of the approval process. Although seven of the approved applications (35%) were located within areas classified by NIHSA as highly probable flood-risk zones, none incorporated CMIP6 climate projections or future flood scenarios into decision making. Only four files required drainage clearance, all of which were processed after 2020 and within Ibadan South-West, indicating isolated administrative practice rather than a standardized planning policy. These findings demonstrate that while reliable climate and flood-risk information exists, institutional mechanisms for incorporating such evidence into statutory land allocation remain inadequate.

Furthermore, the behavioural findings established that legal and institutional deficiencies significantly influence planning decisions. Fourteen of the eighteen interviewees (77.8%) acknowledged that they were aware of flood risks before approving development applications but proceeded because of political pressure and concerns over litigation arising from the absence of legal backing. In addition, fifteen respondents (83%) stated that statutory legal protection would significantly increase their willingness to reject applications within NIHSA-designated high-risk flood zones. These findings confirm that the principal barrier to climate-responsive planning is not the lack of technical knowledge or climate data but the absence of legal authority and institutional safeguards that empower planning officials to act on available scientific evidence.

Based on these findings, the study developed a Climate-Adaptive Zoning Addendum as a practical framework for integrating climate projections into statutory land allocation. The proposed framework comprises three components: mandatory attachment of NIHSA flood-risk status to all land

applications; a Red–Yellow–Green zoning classification integrating NIHSA flood maps with CMIP6 climate projections to guide approval decisions; and a statutory liability protection clause shielding planning officials from legal action when rejecting applications in designated flood-risk areas. Stakeholder validation demonstrated strong support for the framework, with fifteen of the eighteen participants considering it both feasible and implementable because it requires only a state implementation guideline rather than an amendment to the Land Use Act. The findings therefore indicate that strengthening legal and institutional mechanisms can substantially improve climate-adaptive land allocation and contribute to sustainable flood risk reduction in Ibadan.

X. DISCUSSION OF FINDINGS

The findings demonstrate that the persistent allocation of land in flood-prone areas of Ibadan is driven primarily by legal and institutional deficiencies rather than by the absence of climate information. Although reliable flood-risk data from the Nigeria Hydrological Services Agency (NIHSA), seasonal forecasts from the Nigerian Meteorological Agency (NiMet), and CMIP6 climate projections are readily available, these datasets are not incorporated into statutory land allocation because neither the Land Use Act 1978 nor the Oyo State Physical Planning Law 2016 mandates their consideration. This finding supports the view of Akinbode and Ogunleye [7] that weaknesses in Nigeria's legal framework create a "disaster trap" in which approving development in hazardous locations is legally less risky than rejecting applications. It also aligns with Jiboye's (2011) assertion that planning in Nigeria remains largely reactive rather than preventive. Furthermore, the strong support among planning officials for legal protection confirms that behavioural change depends on institutional reform rather than technical capacity alone, thereby validating the Legal–Institutional–Behavioural Framework underpinning this study. The proposed Climate-Adaptive Zoning Addendum addresses this gap by providing a legally feasible mechanism for integrating climate projections into land allocation decisions without requiring amendments to the Land Use Act, thereby offering a practical pathway toward climate-resilient urban planning and sustainable flood risk reduction in Ibadan.

XI. CONCLUSION

This study concludes that the recurring flood disasters in Ibadan are primarily the result of legal, institutional, and behavioural deficiencies rather than the absence of climate or flood-risk data. The findings revealed that the Land Use Act 1978 and the Oyo State Physical Planning Law 2016 do not mandate the integration of climate projections into statutory land allocation decisions, while the lack of legal protection significantly limits planning officials' willingness to reject development applications in flood-prone areas. Therefore, the proposed Climate-Adaptive Zoning Framework provides a practical and legally feasible approach for integrating climate projections into land allocation decisions, thereby strengthening flood risk reduction, promoting climate-resilient urban planning, and enhancing sustainable development in Ibadan.

XII. RECOMMENDATIONS

Based on the findings of this study, the following recommendations are proposed to strengthen the integration of climate projections into statutory land allocation, enhance climate-responsive planning, and reduce flood risk in Ibadan, Nigeria.

1. The Oyo State Government should adopt and implement the proposed > Climate-Adaptive Zoning Framework by integrating NIHSA > flood-risk data and CMIP6 climate projections into statutory land > allocation procedures, with priority implementation in high-risk > Local Government Areas such as Akinyele and Ibadan South-East.
2. The Nigeria Hydrological Services Agency (NIHSA) and the > Nigerian Meteorological Agency (NiMet) should develop and > regularly update Local Government Area-level flood hazard and > climate projection maps in GIS-compatible formats to facilitate > evidence-based land allocation and development approval decisions.
3. The National Council on Lands, in collaboration with state > planning authorities, should develop a model implementation > guideline that enables states to incorporate climate risk > assessments into land allocation decisions under the existing > provisions of the Land Use Act 1978, thereby institutionalizing > climate-adaptive zoning without requiring legislative amendment.

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Spatial Pattern and Residents' Perception of Green Infrastructure across Residential Densities in Abeokuta Metropolis, Ogun State, Nigeria

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Abstract—The unequal distribution of urban green infrastructure (GI) across residential densities is a growing concern in planning and environmental justice, especially in fast-growing African cities. The rate of urban expansion has consistently outpaced institutional capacity to incorporate GI into the planning process, a subject that aligns closely with Sustainable Development Goal 11.7 on universal access to safe, inclusive, and accessible green and public spaces by 2030. This study examines the spatial pattern of green infrastructure and residents' perceptions of its benefits, determinants, and constraints across three residential densities in Abeokuta Metropolis, Ogun State, Nigeria. A multi-stage cluster sampling procedure combining purposive neighbourhood selection was used across 22,882 residential buildings compiled as the sample frame; 326 structured questionnaires were administered across 61 neighbourhoods spanning high-density ($n = 160$), medium-density ($n = 108$), and low-density ($n = 58$) areas, with the Krejcie and Morgan [1] formula used to determine sample size, and systematic random sampling applied within households, yielding 305 usable responses. Data were analysed using frequency distribution, percentages, the Pearson chi-square test of independence, the Friedman test, and a Residents' Perception Index expressed as the Average Green Infrastructure Importance Index (AGIII). Results show that trees were the most prevalent dwelling-level GI (76.5%), while green walls were the least available (16.6%); formal GI averaged 70.8% in low-density areas compared with 22.7% in high-density areas, a statistically significant difference (Pearson chi-square = 37.623, $df = 4$, $N = 305$, $p < 0.001$, Cramer's $V = 0.248$), and awareness of GI benefits also varied significantly across densities (Friedman chi-square = 21.385, $df = 2$, $p < 0.001$), with low-density respondents recording the highest AGIII (4.75) against 3.94 and 3.81 for medium- and high-density respondents respectively, underscoring the need for density-sensitive GI policy and stronger cross-sector collaboration for equitable access to ecosystem services.

Index Terms—Abeokuta, climate resilience, awareness, green infrastructure, residential densities, neighbourhood.

I. INTRODUCTION

Cities worldwide now host more than half of the world's population and responsible for an increasing amount of GHG emissions, biodiversity loss, and environmental health risk, disproportionately [2], [3]. Green infrastructure (GI), a continuum of natural and semi-natural areas that provides ecosystem services to urban populations, is among the key tools of sustainable urban planning (SUP) [4]–[6]. Sustainable Development Goal (SDG) 11.7 specifically emphasizes sustainable cities and climate resilience goals explicitly. The goal is to provide universal access to safe, inclusive, and accessible green and public spaces by 2030. According to the World Health Organization (WHO), residents should have access to such a green space within a linear distance of 0.5 hectares from their homes by 2030 [2], [3]. In recent years, [7] built on this benchmark by introducing the 3-30-300 rule,

adding the need for a green space within 300 metres, 30 per cent neighbourhood canopy and three trees visible from each home. Such pledges position the spatial arrangement of GI in the (national) agenda of urban policy now.

The growth of cities in sub-Saharan Africa has also often been faster than the institutional capacity to incorporate GI into planning processes. According to a review by [8], provision of GIs in Nigeria is still fragmented, low quality, and not well integrated into climate adaptation planning. This trend has been corroborated by continental syntheses, such as that completed by [8] who summarised 68 papers from 20 sub-Saharan countries, revealing the biggest attention paid to provisioning and regulating services and the lowest to supporting and cultural services. It was also suggested by [9] that GI approaches developed in the Global North do not always work well in Africa due to differences in informal vegetation, peri-urban land tenure, and fiscal capacity. The tension is illustrated in Lagos Metropolis, which makes up about 52% of the zoned land and 2.8% of open and green spaces [8]. Empirical studies from Lagos have revealed that GI influences the attitude of residents, perceived health, social cohesion, and endorsement of ecological values [10], [11] and quality of GI is best linked to environmental sustainability outcomes when access and adequacy exist [12]. A similar study in Ebonyi State revealed that the length of stay, marital status, age, and education were significant predictors of attitudes towards GI [13]. The exclusive focus on Lagos in Nigerian GI research nevertheless means that there is a significant lack of knowledge about how morphology and perceptions of residents in south-western secondary cities reflect in patterns of density and perceptions.

Spatial pattern is also supported by the literature from around the world, but there is considerable methodological variety. In Auckland, network-analysis studies found systematic gradients in walkability of GI across neighbourhoods [14]. The distribution of green spaces in Shanghai was uneven, ranging from 42.2% to 67.6% for different land-use categories, with green space coverage in old residential and industrial zones being lower [15]. In Lodz, the GIS structure variability was found in different urban areas based on morphological mapping analyses conducted in the field [16]. The three studies are very different in terms of their methods, each offering a new perspective on the same subject, making direct comparisons difficult and accounting for the fragmentation of patterns reported in one city and not another. Taking together, these findings indicate that the distribution of GI is generally not equitable and follows a path that is linked to income, density and the planning process. The findings of

94 papers reviewed by [8] showed that domestic gardening, urban trees, and informal GI were dominant practices while green roofs and green walls were still marginal. The study also identified scarcity of comparative studies conducted between different densities at secondary cities which is addressed in the present work in Abeokuta.

The challenge of Nigerians is illustrated by Abeokuta Metropolis, which is the capital of Ogun State, south-western Nigeria. Built-up area in the city increased from 0.66% in 1972 to 10.93% in 2022, while vegetation cover declined from 93.66% to 58.82% over the same period [17]. There is a gradient of residential densities across the city, from the high-density core traditional neighbourhoods to the medium-density transition zones to lower-density suburban development, which offers a natural context for exploring the variations in GI provision across the city. No empirical research has documented the spatial pattern, perceptions and influencing factors of GI in Abeokuta, up to date, in relation to residential density.

The objectives of this study are to examine the spatial distribution of GI across residential densities in Abeokuta Metropolis, to assess residents' awareness and perception of the benefits of GI across the three residential densities, and to evaluate the factors influencing GI availability with the challenges of its development. The value of this study lies in extending the body of Nigerian GI research to a south-west city with a rapidly changing land cover. It uses a density-comparative design which allows the spatial pattern question to be tested rather than assumed. Furthermore, it connects statistical data on distribution and awareness with observed drivers and barriers, offering planners a single evidence-based dataset from which to design interventions.

II. LITERATURE REVIEW

A. Conceptual Framework

Benedict and McMahon define GI as an interconnected network of natural and semi-natural areas that is planned and managed for its natural resource values and the associated benefits to human populations. Their typology of hubs, links, and sites continues to shape the later studies. Tzoulas et al. further developed the concept by assigning specific functions of biodiversity to ecosystem and human health, thus combining biodiversity, climate regulation and social health in one conceptual framework. Recent reviews highlight that the benefits and services contribution of GI is multifunctional, consisting of provisioning, regulating, supporting and cultural ecosystem services [8], [18].

There are two types of GI, formal and informal. Formal GI is designed and managed, and includes green roofs, street trees, green walls, public gardens and parks. Unlike formal GI, informal GI includes areas such as empty lots, riverbanks, road verges, and other lands that have been disturbed by human activity and contain spontaneous vegetation [19]. Informal GI is frequently prevalent in high-density, high-poverty Global South neighbourhoods, but is not often addressed by formal planning tools. [9] warned that this under-acknowledgement of informal GI in African urban planning has real impacts, including loss of vegetation that benefits vulnerable households and the misuse of resources

through formal interventions that are not relevant to local needs.

B. Theoretical Review

The current study is guided by two complementary theories: Human-nature connection theory (HNCT) and Urban Resilience Theory (URT). HNCT sees humans and nature as mutually interacting on a multidimensional basis, influencing the behaviour of nature and our health and well-being [20]. It refers to GI as a relational entity that is used to directly encounter and interact with the nature by the urban dwellers. A number of research has linked nature connectedness to mental well-being, stress reduction, and pro-environmental behaviour [20], [21]. The implication in planning is that accessibility to both quantity and qualitative experience of nature must be addressed for different population groups. URT perspective situates GI within the wider system of socio-ecological networks that allow cities to absorb, adapt to, and recover from shocks [22], [23]. GI helps create resilience by storm-water management, urban cooling, conservation of biodiversity, and community spaces for social interaction. Recent research in sub-Saharan Africa has shown that the impact of GI can reduce urban temperatures by 1 to 5°C, depending on the type of vegetation and its connectivity [8], [18]. These two points make clear that the right to access GI must be measured, not just on availability criteria, but on connectivity, accessibility and the quality of the resident-nature relationship that it supports.

C. Density, Spatial Inequality, and Ecosystem Service Access

Several themes emerge from the literature that frame the present study. A primary consideration is that GI provision is rarely uniform across an urban fabric; it is systematically patterned by density, land use, and household income. Subsequently, there is not always a linear relationship between perception and access: communities with lower access may also have high awareness, whereas communities with higher access may have passively low awareness [13]. While the 3-30-300 framework of [24] offers one way to operationalise adequate access, fewer than 15% of urban residents currently meet all three components, meaning even ambitious benchmarks may have implementation gaps. Finally, alongside creation, attention to the maintenance of GI is required, and much of the observed shortfall between policy intent and reality is attributed to maintenance deficits [8], [12].

D. Empirical Review

The documentation of strong spatial patterning in GI distribution from international studies has been characterized by variation in instrumentation and scale, thereby hindering synthesis. [14] applied "walkability" measures derived from GIS-based network analysis to the Auckland neighbourhoods to evaluate the accessibility to walking to GI in the city. The finding indicated that access to public-housing developments was systematic, and was inferior to that of detached-housing suburbs. [15] analyzed the land use of Shanghai at three spatial scales and found that green coverage of the villa-residential land use was 67.6%, while the old residential and industrial land uses were much lower. Performing studies in Poland, [16] found that the GI diversity of the city of Lodz corresponded to the historical urban zones, being the highest

in the older centre and lowest in the industrial perimeter. The three approaches represent different aspects of GI provision in terms of network accessibility, land-use coverage and morphological diversity.

Perhaps the most dramatic finding was found in studies within Africa as published by [25]. The predominantly White-occupied neighbourhoods had 11.7% more tree cover, 8.9% more vegetation greenness, and lived an average of 700 metres closer to a public park than predominantly Black African, Indian and Coloured neighbourhoods in ten of the largest cities in South Africa. The effects of apartheid spatial planning lingered on in the present-day access to ecosystem services. Studies conducted sub-Saharan African and Shanghai by both [8], [15] showed that while much work exists on the formal vegetation, cultural ecosystem services are largely unexamined, with most research being conducted in South Africa and a few cities in West Africa. This leads to a partial and formal GIS oriented evidence base across the continent. Similar inequality patterns have been reported in low-income neighbourhoods of Accra and other West African cities [8].

Studies on urban GIs in Nigeria have been mainly focused on Lagos. In a survey of 1,560 residents, [12] identified poor rating for GI, which was significantly correlated with GHG emissions, waste management, and water quality outcomes. It was reported by [10] that 67% of Lagos residents demonstrated a positive attitude toward GI, with social cohesion, health benefits, and quality of neighbourhood GI emerging as the strongest predictors. Later, the same research

group found that GI exposure was associated with self-perceived health benefits [11] and the New Ecological Paradigm [11]. Lately, [26] expanded the study to include social cohesion, which revealed that GI has a positive influence on social cohesion and maintains social cohesion. In the south-east, [13] surveyed 513 residents in Ebonyi State and identified marital status, age, education, and length of stay as significant predictors of GI attitudes. This body of work has been underpinning, but it has two limitations. These studies are grouped in one single megacity and use mostly cross-sectional perception surveys with no parallel imagery verification of GI presence.

III. MATERIALS AND METHODS

The corresponding result is shown in Fig. 1.

Ogun State has an area of 16,762 square kilometers and was established in 1976. Ogun State is located in the south-west of Nigeria. To the north, it is bordered by Oyo and Osun States, the east by Ondo State, south by Lagos State and the west by the Republic of Benin (Fig. 1). The metropolis is made up of Abeokuta South, Abeokuta North and parts of Odeda that are now basically one urban area. It is administrative, commercial and educational hub of Ogun State.

The present analysis was organized into 3 main categories in residential density. The areas with higher density are the older core areas, medium density are the transitional zones that were built as the city expanded outwards, and low density areas are the newer suburban developments, with smaller household densities and larger plot sizes.

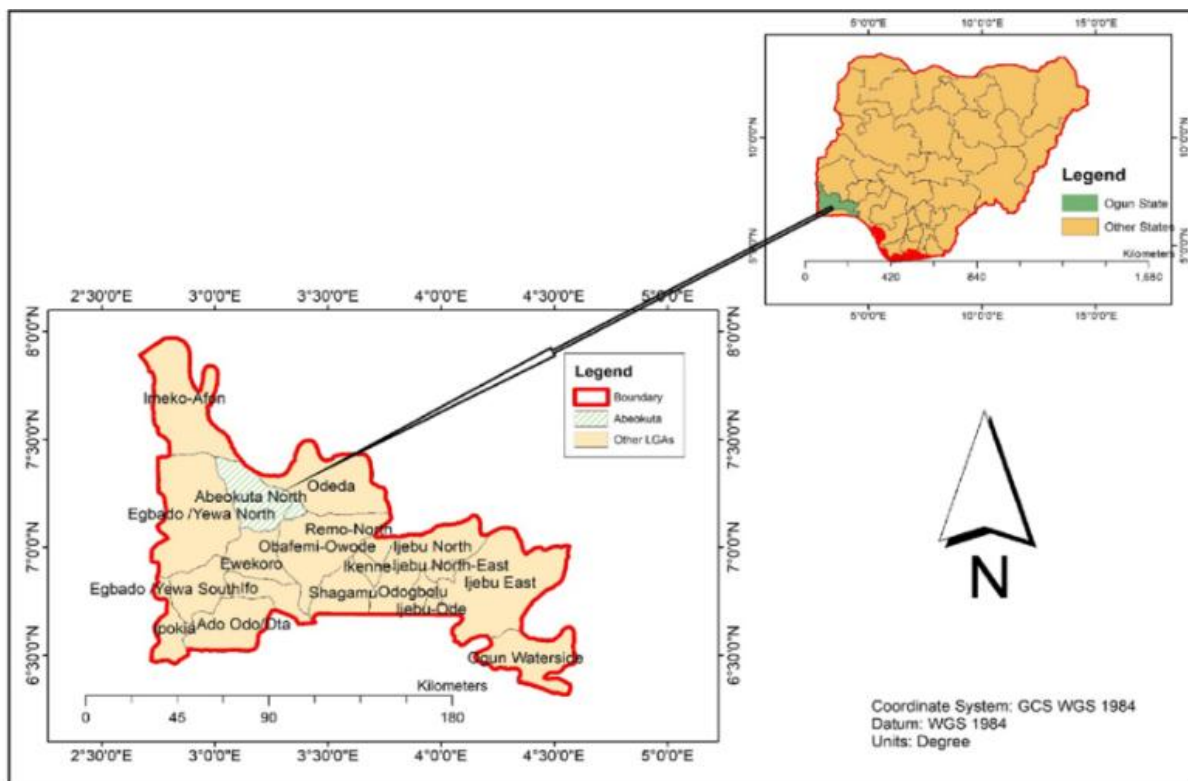


Fig. 1. Map of Ogun State showing the Study Area

> Source: Ministry of Physical Planning and Urban Development, Ogun > State. (2025)

The sampling technique adopted in this research is cluster sampling. However, the multistage cluster sampling method

was used. With a five-stage sampling, purposive selection of towns within the metropolis was employed at the first stage.

The second stage, which was purposive, sampled 61 neighbourhoods - 33 high-density neighbourhoods, 22 medium-density neighbourhoods and 6 low-density neighbourhoods. The third stage identified residential buildings within each selected neighbourhood, thereby giving a total sample frame of 22,882 residential houses compiled. The fourth stage applied a [27] formula to determine the sample size for each neighbourhood, generating 160, 108, and 58 households from high-density, medium-density, and low-density areas respectively, making a total of 326 sample size. This proportion was selected to ensure adequate within-density representation while remaining computationally manageable. Hence, out of 326 questionnaires administered, 305 questionnaires retrieved formed the basis for the analysis of this study. The fifth stage applied systematic random sampling to select respondents within sampled households, prioritising the household head or the most senior adult available at the time of survey.

A structured questionnaire captured information on socio-economic attributes. It further used a five-point Likert scale for perception items, ranging from 1 (Very Unimportant or Strongly Disagreed) to 5 (Very Important or Strongly Agreed). Field surveys supplemented by Google Earth and Google high-resolution imagery verified GI inventories, formal/informal classification and proximity measurements. The adoption of descriptive and inferential techniques were used in the analysis. The Average Green Infrastructure Importance Index (AGIII) was determined by dividing Total Weight Value (TWV) by the sample size of each residential density. Since the neighbourhood-level green infrastructure variables are categorical, the Pearson chi-square test of independence was used to test the association between residential density and the type of green infrastructure reported, with Cramer's V reported as the measure of effect size and the expected cell counts inspected to confirm that the assumption of the test was met. For AGIII, the thirteen benefit items were rated within each of the three densities and therefore constitute related samples, so the Friedman test was used to test whether AGIII differed across densities. Analysis of variance was not adopted, because the density comparisons involve categorical outcomes and related item-level blocks rather than independent interval observations, and the homogeneity of variance assumption required by analysis of variance is therefore not the relevant assumption for these data.

IV. RESULTS AND DISCUSSIONS

4.1 Spatial Distribution and Physical Characteristics of Green Infrastructure

GI was found to be unequally distributed across the three density categories of the residences. Proximity to the city centre, type of GI in the neighbourhood, GI availability, and connectedness all varied systematically with residential density. Table I showed that a larger share of high-density respondents lived within 500 metres of the city centre (57.8%) than medium-density (31.1%) or low-density respondents (20.9%). The relationship was reversed for distances above 2,000 metres, with 18.8% of low-density respondents but only 5.9% of high-density respondents living that far from the centre. The patterns show the historical morphology of Abeokuta, where the traditional core neighbourhoods grew

around the city centre and newer suburbs grew on the urban peripheries.

TABLE I. Proximity of Respondents' Dwelling to the City Centre by Residential Density (%)

Source: Field survey, 2026.

Trees were the most common dwelling level GI, with 76.5% of dwellings having them on site, followed by grass and hedges (66.1%), green buildings (23.9%), green roofs (22.1%), gardens (57.4%), and green walls (16.6%). Notably, dwelling-level trees were most common in high-density areas (89.1%), followed by low-density (66.7%) and medium-density (61.7%) areas as revealed in Table II. The concentration of trees in high density living suburbs could be associated with the long-term residential nature of these core neighbourhoods where older trees have remained since colonial and pre-colonial city layout. However, engineered GIs like green roofs (12.8%) and green walls (8.5%) were marginal throughout the metropolis as noted in the Nigerian review by [8] that these are the least developed aspects of GIs in the country. The results at Abeokuta contrast with the Lagos situation reported by [12], where formal GI was consistently rated as poor, in every neighbourhood with varying socioeconomics. This suggests a more pronounced density gradient in the smaller secondary city, in which the distinction between core, transition and suburb is still morphologically evident.

TABLE II. Availability of Green Infrastructure in Dwellings by Residential Density (%)

Density	Garden	Green Roof	Green Wall	Trees	Grass/Hedges	Green Building
High	71.4	25.9	21.1	89.1	63.9	27.2
Medium	33.0	12.8	8.5	61.7	74.5	17.0
Low	62.5	29.2	18.8	66.7	56.3	27.1
Total	57.4	22.1	16.6	76.5	66.1	23.9

Source: Field survey, 2026.

Note: Percentages are computed within each density and may exceed 100 across the row because respondents could report multiple GI types.

There was a wide range in the type of GI present in the neighbourhood for each density as revealed in Table III. In low-density respondents, the percentage with formal GI in their neighbourhood was 70.8%, with 37.9% in medium-density and 22.7% in high-density neighbourhoods. The pattern was reversed for informal GI, which accounted for 63.0% of GI in high-density neighbourhoods, 49.5% in medium-density neighbourhoods, and 25.0% in low-density neighbourhoods. The share of respondents reporting no neighbourhood GI was 14.3% in high-density, 12.6% in medium-density, and 4.2% in low-density areas. The chi-square test of independence in Table IV found these differences to be statistically significant (chi-square = 37.623, $df = 4$, $N = 305$, $p < 0.001$), with a Cramer's V of 0.248 indicating a moderate association and all expected cell counts exceeding five.

TABLE III. Type of Green Infrastructure in Neighbourhoods

Density	Formal (%)	Informal (%)	None (%)	Total
High	22.7	63.0	14.3	100.0 (n=154)
Medium	37.9	49.5	12.6	100.0 (n=103)
Low	70.8	25.0	4.2	100.0 (n=48)
Total	35.4	52.5	12.1	100.0 (n=305)

Source: Field survey, 2026.

TABLE IV. Chi-square Test of Independence: Type of Green Infrastructure by Residential Density

Statistic	Value	df	N	Sig.
Levene statistic	37.623	2	305	<0.001
Cramer's V	0.248	-	305	-
Maximum expected count	5.82	-	305	-

Note: all expected cell counts exceed five, satisfying the chi-square assumption

Source: Field survey, 2026.

The connectedness of GI also varied across densities. The median (55.8%) for high density respondents was that their neighbourhood GI was at least connected, with low density respondents most often answering that GI was only slightly connected (58.3%). Medium-density respondents were somewhere in between: 38.8% said they had connected or well-connected GI. It indicates that in areas of low density, although the percentage of formal GI is higher, the dominance of "slightly connected" GI confirms that the suburban footprint still exists but has been planned in discrete sections. This

Benefit Variable	High AGIII	Rank	Medium AGIII	Rank	Low AGIII	Rank
Flood reduction (flood mitigation)	4.03	1	4.26	2	4.69	9
Promote social cohesion and interaction	3.95	2	4.10	4	4.88	3
Improve mental well-being	3.86	3	4.02	6	4.79	6
Reduced premature mortality	3.85	4	3.89	9	4.56	12
Enhanced brain development in children	3.85	4	3.99	7	4.98	1
Improved cognitive function in adults	3.82	6	3.75	11	4.42	13

configuration has been linked to the argument put forward by [4] that the benefits of GI are also equally about connectivity, with ecological function and pedestrian access flowing through the urban fabric. It also directly challenges the notion that wealthier neighbourhoods get higher quality GI.

4.2 Residents' Awareness and Perception of Green Infrastructure Benefits

While ranking of benefits varied with density, as shown in Table V, residents did exhibit significant understanding of the various benefits of GI. The Average Green Infrastructure Importance Index (AGIII) was 4.75 at low-density, 3.94 at medium-density and 3.81 at high-density, and the Friedman test in Table VI showed that this difference was statistically significant ($Friedman\chi^2 = 21.385$, $df = 2$, $p < 0.001$); the thirteen benefit items being treated as related blocks rated within each density. For high density areas, the highest rated benefit was flood reduction, $TWV = 620$, $AGIII = 4.03$; followed by promotion of social cohesion and interaction, 3.95; and improvement of mental well-being, 3.86. Minimising heat, noise and pollution (4.32) and flood reduction (4.26) were the main advantages in medium density areas followed by improved environmental microbiota/biodiversity (4.12). Enhanced brain development in children (4.98) took first place in low-density areas, where global awareness was highest, followed by enhanced environmental microbiota and biodiversity (4.90) and promotion of social cohesion (4.88). The patterns of rankings suggest that high densities value less GIs that reduce the more immediate tensions of dense urban life, like flood control and social connection, while medium densities focus on environmental modulation, specifically heat, noise, and pollution reduction, while low densities value more GIs that offer longer-term developmental and ecological benefits.

TABLE V.

Residents' Awareness of Green Infrastructure Benefits across Residential Densities

Benefit Variable	High AGIII	Rank	Medium AGIII	Rank	Low AGIII	Rank
Better birth weight	3.82	7	3.77	10	4.67	10
Lower risk of chronic disease	3.80	8	4.05	5	4.81	5
Increased physical activities	3.80	8	3.94	8	4.75	8
Minimised heat, noise, and pollution	3.75	10	4.32	1	4.65	11
Enhanced environmental microbiota and biodiversity	3.73	11	4.12	3	4.90	2
Climate mediation	3.69	12	3.40	13	4.83	4
Reduced stress	3.59	13	3.62	12	4.77	7

Benefit Variable	High AGIII	Rank	Medium AGIII	Rank	Low AGIII	Rank
AGIII (overall)	3.81	-	3.94	-	4.75	-

Statistic	Value	df	N	Sig.
Friedman chi-square	21.385	2	13 items x 3 densities	<0.001
Mean AGIII (high density)	3.81	-	-	-
Mean AGIII (medium density)	3.94	-	-	-
Mean AGIII (low density)	4.75	-	-	-

Source: Field survey, 2026.

Note: AGIII = Average Green Infrastructure Importance Index, derived from a five-point Likert scale.

Although the high density respondents had a lower AGIII, significant agreement was noted with the statement that the well-being of the respondent was improved as a result of GI. Of the respondents, 54.5% of high-density, 70.9% of medium-density and 52.1% of low-density respondents agreed or strongly agreed that well-being is a benefit. This pattern mirrors that of [11] who found that perceived health benefits of GI in Lagos were highest at moderate, but not saturated, access levels. The Abeokuta data provide an important nuance: The perceived well-being benefit is higher in medium-density transition areas, where formal and informal GI coexist, and where people can perceive GI as a relative gain, not as an expectation.

Implications

A paradox is apparent from the density gradient in AGIII. Low-density residents have the greatest awareness of benefits and best physical access to formal GI. Low awareness scores were reported by the residents of high-density areas who rely on informal trees and dwellings at the GI level. The paradox is that there is no connection between distance and perceived value. The data, rather, indicate that awareness of GI in Abeokuta is more strongly influenced by experiences of designed landscapes and education associated with suburban living than by the daily sighting of vegetation. It reflects the findings by [13] that the educational level and the duration of stay are more influential on attitudes towards GI than the physical proximity, and suggests two policy options. First, awareness campaigns should be tailored to density context:

Variable	High Density	Medium Density	Low Density	Total
Factors influencing availability				
Community/residents' efforts	64.2	65.0	72.3	65.8
Natural environmental features	52.0	58.3	59.6	55.4
Government efforts	30.4	58.3	48.9	43.0

Source: Field survey,

TABLE VI.

Friedman Test: Average Green Infrastructure importance Index (AGIII) across residential densities

Promoting the all-too-forgotten benefit of trees for flood mitigation in high-density areas and the stewardship obligations that go with privileged GI access in low-density areas. Second, the perceived well-being-density relation suggests the need to refine the guideline of 300 metres by [3], which may also be applicable to Nigerian secondary cities, where the actual distance between the user and provision of available GI might be less important than the user's perception of well-being and connectedness. However, the AGIII measure does not differentiate between conceptual knowledge and lived experience, which leaves some questions unanswered.

A. Factors Influencing GI Availability and its Challenges

Factors affecting GI provision and difficulties of GI development in Abeokuta revealed that the inhabitants had a sense of the provision of GI as a product of community agency and structural constraint. The most common driver of GI presence as revealed in Table VII was cited as community and residents' efforts (65.8%), followed by natural environmental features (55.4%), government efforts (43.0%), public facilities (31.5%), and support from non-governmental organisations or donors (29.2%). Public provision has been most active in the transition zones of the metropolis, with the proportional contribution of government in the medium density areas being higher (58.3%) than in the high density areas (30.4%) or low density areas (48.9%). Community efforts were, on the other hand, predominant and uniform in all densities, from 64.2% in high density to 72.3% in low density.

The challenges to GI development were complementary. The top barriers identified were poor maintenance (61.1%), lack of awareness (53.8%), inadequate funding (46.9%), urban development pressure (27.7%) and land-use conflict (19.5%). Medium density neighbourhoods (66.0%) were most acutely affected by poor maintenance, with moderate levels of GI presence and low levels of upkeep provided by municipalities. Where awareness was especially strong, people were in the medium-density (63.1%) or low-density (59.6%) categories, suggesting even well served neighbourhoods face awareness gaps that impede ecological involvement.

TABLE VII.

Factors Influencing GI Availability and its Challenges (% endorsing)

Variable	High Density	Medium Density	Low Density	Total
Availability of public facilities	25.7	37.9	36.2	31.5
NGO or donor support	20.9	40.8	29.8	29.2
Challenges to development				
Poor maintenance	58.2	66.0	59.6	61.1

Variable	High Density	Medium Density	Low Density	Total
Lack of awareness	45.8	63.1	59.6	53.8
Lack of funds	42.5	50.5	53.2	46.9
Urban development pressure	24.2	33.0	27.7	27.7
Land-use conflicts	18.2	14.6	34.0	19.5

Source: Field survey,

Note: Multiple responses were permitted; column totals exceed 100 per cent.

Implications

Two practical Implications for the result are discussed below:

a. community-led GI activity is the dominant force in shaping the presence of GIs, emphasising the potential of participatory GI governance and the opportunity for neighbourhood-level GI committees to be established and have stewardship mandates and committed maintenance budgets, based on the social cohesion aspects of GI outlined by [23]. The same observation, though, can be made more critically: a strong reliance on community efforts can be seen as the covert withdrawal of the state from the provision of public goods, not as actual empowerment, and informal stewardship arrangements can 'normalise the inadequate monitoring and funding of municipal services' as [9] point out. Participatory governance is thus not just mobilising communities, but it is also necessary to have an explicit fiscal commitment to it.

b. the high prevalence of poor maintenance status as a challenge suggests that there is a general institutional weakness on environmental services in Abeokuta, Nigeria. The strategic responses involve establishing dedicated GI maintenance funds, performance monitoring across density bands, and direct linkage between GI development and the metropolitan master plan, which will help minimise 34.0 per cent of respondents who noted there were land-use conflicts. Institutional support is essential or else community stewardship can be used as a replacement for, not a supplement to, ongoing public investment.

Abeokuta's GI is perceived, patterned, and sustained in much the same way as the house density gradient moves across the three findings areas. High density areas are a mixture of informal GI and mature trees, lower awareness scores and weaker public provision. Medium density areas have moderate formal GI, highest sense of maintenance benefit, and the highest dissatisfaction with maintenance. Low-density areas have the highest formal GI share, awareness, and connectivity and land-use challenges. Such set-ups require differentiated policy response and is taken up in the concluding section.

V. CONCLUSION

AND RECOMMENDATIONS

Rapid urbanisation has been an escalating threat to the green infrastructure and services provided to urban citizens in Nigerian secondary cities. The decline in the vegetation cover in Abeokuta Metropolis, from 93.66% to 58.82% (a loss of

34.84%) in the past 50 years, presents a very vivid scenario to analyse whether the remaining GI is equitably distributed across the urban landscape and is uniformly experienced by the residents. This study examined the spatial distribution, residents' perception, and influencing factors of green infrastructure across three residential densities in Abeokuta. The analysis identified three key findings:

The primary result revealed high spatial variation of GI across residential density ($Chi - square = 37.623$, $df = 4$, $p < 0.001$). Low-density areas contain a substantially higher share of formal GI (70.8%) than high-density areas (22.7%), and the inverse holds for informal GI. In addition, residents' awareness of GI benefits differs by density ($Friedmanchi - square = 21.385$, $df = 2$, $p < 0.001$), with low density (4.75) having the highest awareness while high density (3.81) has the lowest awareness. Moreover, the perceived drivers and barriers to GI development are widely shared across densities; community and residents' efforts dominate as drivers, while poor maintenance, lack of awareness, and inadequate funding dominate as barriers.

These outcomes need to be interpreted in the light of the study limitations. It is noted that the cross sectional design captures a single snapshot in time of Abeokuta's evolving landscape. Self-reported data introduces an element of subjectivity that can result in different conclusions, especially for indicators that may be subject to social desirability bias. A single metropolis, in the south-western Nigeria, restricts the generalisability, but the density-stratified design increases the comparability of the study to others. The AGIII measures perceived rather than lived experience, and behavioural engagement with GI; some high levels of perceived importance in low-density regions might be abstract endorsement.

Further studies need to have longitudinal designs to follow changes to GI provision and resident perception in the continued growth of Abeokuta. Further comparative work between cities in Nigeria would determine if the density-related trends mentioned here extend beyond the south-west region of Nigeria. Resident surveys, remote sensing vegetation indices, accessibility analysis (3-30-300 framework) and qualitative interviews with residents regarding lived experience would also enhance the knowledge base when it comes to understanding lived experience versus what the data actually shows. Taking cognizance of these constraints, the present study provides empirical basis for the equitable and density-based development of green infrastructure in Abeokuta Metropolis for planners, policies and civil society organization.

From the results, it is evident that the strategies needed to provide equitable GIs is not the same for each density. As a result, the recommendations provided are density-specific policy implications necessary for urban planning. It is, therefore, recommended that the high-density core areas require formal GI retrofitting and tree-protection programmes that build on the existing informal vegetation base. Consequently, the medium density transition zones require targeted investment in maintenance to sustain the ecosystem services they provide already. In addition, there is need for a network-based approach to improve connectivity between the

existing pockets of formal GI in low density suburban developments. Continuous awareness should be created in order to sustain urban environmental services which is essential for long-term sustainability of Abeokuta city. Finally, the presence of community and residents' efforts which reflects that participatory governance arrangements should not replace formal governance plans, but rather complement them and must be backed up by financial commitment if they are to succeed.

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BIOGRAPHY

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Land Grabbing, Inequality and Corporate/Financial Land Concentration in the Economy

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Abstract—Land grabbing has become a core issue in the contemporary politics of land, gaining momentum over the past few years across Africa, Asia, and Latin America as states, multinational corporations, agribusiness conglomerates, and financial actors acquire massive areas of land for commercial agriculture, extractive industries, infrastructural development, and speculative investment. This phenomenon extends beyond traditional territorial dispossession and reflects a deeper structural transformation in which land is increasingly treated as a financial asset rather than a socio-economic resource. As corporations and financial institutions consolidate control over vast expanses of agricultural and resource-rich land, ownership becomes heavily concentrated, intensifying multiple layers of economic, social, gender, and political inequality. This paper examines how corporate and financial land concentration reinforces an unequal agrarian structure in which profit maximisation and speculative investment overshadow local development priorities, and it discusses governance, legal, and policy responses needed to address the resulting inequality and land concentration.

Index Terms—Land grabbing, Inequality, Corporate land concentration, Financialization, Land governance.

A. Introduction

Land grabbing has become a core issue in the contemporary politics of land, which over the past few years has gained momentum in Africa, Asia, and Latin America, with states, multinational corporations, agribusiness conglomerates, and financial actors buying up massive areas of land for commercial agriculture, extractive industries, infrastructural development, and speculative investment. This trend reflects larger changes in the global political economy. Land-primarily framed as a social, cultural, and livelihood resource-has increasingly become an asset sought after for accumulation by global capital. For scholars, the post-2008 food, energy, and financial crises hastened this global "land rush" by causing states and private investors to invest in land abroad as a hedge against market instability, food insecurity, and capital volatility [1], [2]. The result is a transformation of land markets in large-scale acquisitions, long-term leases, and speculative investments, where profits are often maximized at the expense of local development priorities.

This shift has reconfigured the agrarian relations through new forms of territorial control, where private equity funds, pension schemes, and transnational corporations consolidate landholdings through complex contractual arrangements, partnerships, and financial instruments. Such processes entrench unequal power relations between corporate actors and rural communities, weakening traditional land tenure systems and reshaping rural livelihoods. As Hall says, contemporary land grabs tend to be different from earlier forms of land dispossession in that such recent processes are driven by an unprecedented convergence of private capital, state authority, and global financial markets, which, in turn,

develop more opaque and legally legitimized pathways of land accumulation.

The research problem guiding this study seeks to understand the manner in which land grabbing operates as a key mechanism in reproducing inequality under contemporary conditions of increasing concentration of land under corporate and financial actors. The specific purpose of this research is, therefore, to investigate the structural relationship linking large-scale land acquisitions with socio-economic inequality and financialization of land governance. It also hopes to assess the wider consequences for rural livelihoods, tenure security, and local autonomy. This is because it frames land grabbing within the overall process of corporate consolidation and neoliberal economic restructuring, showing how these forces facilitate upward redistribution of land and solidify multidimensional inequalities.

Methodologically, the paper follows a qualitative desk-based research approach, informed by secondary academic literature, as well as civil society reports and case studies from regions characterized by intense land acquisitions. This approach enables a comparative, interpretive analysis of drivers, mechanisms, and consequences of land grabbing. Findings from reviewed evidence reveal that land grabbing disproportionately displaces, and limits access to common resources for rural and indigenous populations, while engendering erosion of customary land rights [3]. De Schutter [4] further underlines such processes as reinforcing class inequality in so far as investor interests become privileged relative to smallholder farming systems, hence deepening poverty in rural contexts. The consolidation of land under corporate and financial actors can also hasten monocultural agribusiness models, speculative land banking, and ecological degradation, harming community resilience and democratic control over land.

This study concludes that land grabbing does not represent a straightforward investment strategy but a highly political process, which redistributes land and power upward while marginalizing the vulnerable. Based on these findings, recommendations were made to include the strengthening of land governance frameworks, enhancing the enforceability of community land rights, regulation of corporate and financial land acquisition practices, and promoting accountable and transparent decision-making. Policy implications address balancing foreign investment with social justice, preventing monopolistic control over land, and undertaking anti-concentration measures to safeguard the vulnerable from dispossession. Conclusively, grappling with land grabbing calls for transformation in land governance to focus on equity, sustainability, and democratic participation.

B. Land Grabbing in the Contemporary Economy

Land grabbing in the modern-day economy is the large-scale buying of land—mostly in the global South—by powerful actors for mainly commercial, extractive speculative, or strategic reasons. Land grabbing can be described by scholars as a "control grabbing" of vast areas of land through sales, long-term leases, concessions, or compulsory acquisition, typically leading to involuntary displacement or marginalization of existing users [5]. Such deals have been justified as being investments oriented towards development, yet are really crucial for asymmetric power relations, weak consultation, and shady contractual terms. Modern land grabs take many forms, such as compulsory land acquisition by the state on behalf of private investors, speculative dealings where land is acquired for future value rather than immediate use, and foreign and statebacked acquisitions impelled by security imperatives of national food and energy, together with geopolitical interests (De Schutter, 2011; [2]). All these tend to partly overlap with more inherent weaknesses in land governance, weak tenure documentation, and ambiguities in laws over customary ownership in many countries.

A key feature of contemporary land grabbing is the varied constellation of actors. Multinational corporations and agribusinesses remain central players, acquiring land for large-scale plantations producing biofuels, cash crops, timber, and livestock [6]. Investment funds and private equity firms have increasingly entered the scene, treating farmland as a stable asset class capable of generating long-term financial returns [7]. Banks and institutional investors—like pension funds and sovereign wealth funds—participate indirectly by financing land-based investments or buying shares in agribusiness ventures. State institutions also play a critical role by making land available for acquisition, creating investment-friendly legal frameworks, or directly entering partnerships with foreign governments seeking offshore agricultural land to secure food supplies [3]. National political elites and local intermediaries often facilitate these acquisitions and profit through rents, corruption, or political patronage networks. Civil society groups and rural communities are much weaker counter-actors who contest unjust acquisitions and demand tenure security.

Compared to historical forms of land dispossession, contemporary modern land grabs are different in important ways. In history, land was subjected to violent conquest, colonial expropriation, or other forms of forced evictions. Though coercion exists even today, the present phenomenon remains enveloped by legality, financialization, and global market integration. Large-scale land deals now commonly take place in the form of formal contracts, long-term leases, and state-averred transfers that confer an aura of legality on acquisitions despite their detrimental social impacts [8]. Whereas earlier forms of dispossession were considerably driven by territorial and/or political imperatives, today's land grabs are impelled by global financial markets, commodity speculation, and transnational corporate strategies. Fairbairn [7] suggests that farmland has transformed into a financial asset undergoing valuation, hedging, and portfolio diversification strategies, thereby representing the growing financialization of agriculture. Furthermore, contemporary

land grabs come in the guise of narratives of development, climate change mitigation, and modernization, wherein investors often pledge job creation, technology transfer, and infrastructural improvements—promises frequently unmet [4].

Other key differences lie in scale and speed: many contemporary land deals involve tens or hundreds of thousands of hectares, negotiated in a rush with little if any effective community consultation. The development of digital land registries, investment liberalization, and public–private partnerships has further accelerated these processes, increasing access to land for global capital. In addition, international trade regimes, climate finance mechanisms, and carbon markets create new drivers for the acquisition of land, notably for the purposes of biofuel production and carbon-offset plantations. Whereas earlier forms of dispossession were geographically contained, today's land grabbing is globalized, with widespread interconnectedness through trade, finance, and diplomatic relations. In general, approaches to land grabbing within the contemporary economy need to consider how global finance, state authority, and corporate power intersect. Contemporary land grabs do not occur in a vacuum but instead represent part of a wider set of structural changes in land use, ownership, and governance changes that reflect an increasingly commodified and financialized approach to land at the expense of rural people.

C. Corporate and Financial Land Concentration

Corporate and financial land concentration is a process whereby large areas are increasingly controlled by a small number of multinational corporations, investment firms, and state-linked enterprises. It is both an economic and a political process, reflecting the restructuring of global food systems, resource extraction industries, and the rise of finance capital in agriculture. Authors contend that land concentration is more than an accumulation of physical property; it is a transformation of power relations whereby access to, control over, and decision-making on land is transferred from rural communities to corporate actors. As suggested by Borras and Franco (2012), the political dimensions come from regulatory reforms, policies friendly to investors, and state alliances that ease private accumulation at the expense of communal or customary rights to land.

Corporations and financial entities acquire land through various means. A key channel is the proliferation of long-term leases and concessions, particularly in countries where direct land sales to foreigners are prohibited. Governments and countries throughout Africa, Asia, and Latin America have awarded widespread concessions often spanning 30 to 99 years for agriculture, mining, forestry, and infrastructure development [9]. Such deals allow companies to effectively control the land without necessarily facing the legal or political burden of outright purchase and often arise in situations where communities have insecure tenure.

Mergers and acquisitions (M&A) is also a strong driver of land concentration. Global agribusiness consolidation often involves the large firm buying out the smaller one to expand landholdings, vertically integrate supply chains, and strengthen market dominance. The 2010s saw several major M&A waves in the seed, fertilizer, and commodity production sectors—resulting in unprecedented levels of corporate control

over agricultural inputs and farmland [10]. In these transactions, land is combined as an asset within corporate portfolios, which tends to accelerate its concentration in fewer hands.

Another powerful driver is the financialization of agriculture, whereby farmland becomes a financial asset class in a manner comparable to real estate or commodities. Farmland REITs, private equity funds, and institutional investors are investment vehicles that view land not only as a hedge against inflation but also as a source of stable long-term return [7]. Financialization separates land ownership from production, with an increasingly common practice of investors leasing out the land to operators while retaining control over strategic choices about land use. This adds yet another layer of abstraction to the governance of land, displacing local management by a set of distant financial logics.

Credit and debt structures further reinforce concentration: small-scale farmers often rely on high-interest credit tied to production contracts or collateralized by land. When farmers are unable to repay loans, then land may transfer to banks or agribusiness firms, accelerating dispossession (Oya 2013). In global South contexts, structural adjustment programs and austerity policies have pushed governments and smallholders into debt dependency, making land assets vulnerable to takeover by creditors or investors.

The involvement of international capital markets, pension funds, and private equity firms has dramatically increased. Pension funds in the United States, Canada, and Europe collectively manage billions of dollars invested in farmland and agribusiness ventures across the world [3]. They want stable, long-term returns and look to land as an attractive asset with a very low volatility, while private equity firms with a focus on higher-risk, high-return agricultural investments often target lands that are underused or undervalued for restructuring and commercialization.

Finally, land concentration is framed by alliances between corporate and state actors. Very often, states provide legal reforms, subsidies, tax incentives, and security apparatuses that enable corporate actors to gain access to land. In many countries, governments justify these alliances under the banner of development, food security, or modernization, even when they displace rural communities or restrict customary rights [11]. In this way, land governance is reshaped to give priority to capital accumulation rather than to social equity.

D. Linking Land Grabbing to Inequality

Land grabbing is widely recognized not only as a process of land acquisition but as a structural mechanism that reinforces multiple forms of inequality-economic, social, gendered, and political. At the economic level, large-scale land acquisitions redistribute land and wealth upward, concentrating ownership in the hands of multinational corporations, agribusiness conglomerates, and financial investors while displacing smallholder farmers and local communities. By converting land from a subsistence resource into a financial asset, land grabs generate profits mainly for elite actors, leaving rural households with limited or no access to productive resources [12]. The concentration of land undermines the ability of smallholders to engage in profitable agriculture, reduces local

employment opportunities, and fosters income inequality within and between communities [13]. In many cases, the marginalization of households takes place permanently, or people are forced into precarious labor arrangements on those very lands they once owned.

Land grabbing deepens social inequality on equal measure. The large-scale acquisitions typically displace rural and indigenous people, eroding the traditional land tenure systems and forms of communal management of resources. The shifting or displaced communities lose access to forests, grazing lands, and sources of water that form part of livelihoods, cultural identity, and social cohesion. A class stratification emerges by way of consolidation of economic and political power over land by corporate-state elites vis-à-vis the growing marginalization of rural populations in particular, smallholders, and pastoralists. Such structural changes reproduce a hierarchical social order in which ownership and control of land go along with socioeconomic and political influence.

Land grabbing also systematically reinforces gender inequality. In most customary and statutory systems, women's land rights are weaker than men's, and they are often excluded from negotiations over land transactions [14]. Even when women are substantial contributors to agricultural production, they are often denied any legal recognition or decision-making authority regarding land sales or leases. This exclusion, beyond the loss of economic benefits, limits women's social and political agency at the household and community levels. Finally, land grabbing worsens preexisting disparities between men and women and constrains women's ability to participate in local development or governance processes. A further implication of contemporary land grabs is political inequality. Corporate actors, often in alliance with state institutions, hold significant power over land governance policies and decision-making [1].

In many cases, there is limited, if any, genuine consultation or participation by local communities and traditional authorities, weakening democratic oversight and accountability. This asymmetry in political power allows corporate actors to shape regulations, zoning, and investment incentives in ways that favor profit over public welfare. In this context, political inequality supports economic and social inequality and reinforces elite control over land while undermining mechanisms through which communities might challenge dispossession or demand fair resource distribution. Overall, land grabbing acts as a multidimensional driver of inequality, serving simultaneously to concentrate economic resources, reinforce social hierarchies, marginalize women, and erode local political agency. In connecting land acquisition processes with structural inequities, it points to the interplay among global capital, state power, and rural vulnerability. These inequalities need strong land governance frameworks, gender-sensitive policies, mechanisms for community consultations, and legal provisions that prevent elite actors from monopolizing land while protecting the rights and livelihoods of rural and indigenous populations.

E. Land Grabbing as a Driver of Corporate/Financial Control

Land grabbing has become a central mechanism through which corporations and financial actors establish long-term

territorial and economic control over land resources. The acquisition of large-scale tracts of land by corporate actors in areas with weak regulatory frameworks or insecure tenure systems enables them to consolidate influence over local economies, supply chains, and markets. According to scholars, land grabbing is not just a real estate transaction but an entry point through which both domestic and global capital secures strategic control over critical productive assets that enables corporations to influence agricultural production, labor markets, and rural livelihoods over extended periods of time as noted by Zoomers et al., 2017. Long-term leases and concessions of land provide the possibility for investors to plan multi-year or multi-decade production cycles, reinforcing their territorial dominance while narrowing opportunities for local communities to regain control over their lands.

The well-recognized transformation here involves the shift of land from a resource of livelihood and subsistence to a financial asset. Financialization reconstituted land as an investment vehicle, with ownership and control increasingly separated from direct cultivation or local management. Farmland is increasingly treated like stocks, bonds, or real estate investments, where corporate actors and financial institutions seek stable, long-term returns. Mechanisms such as farmland investment trusts, pension fund purchases, and securitized agricultural commodities further embed land in global financial circuits, reinforcing its potential as a speculative and profit-generating asset rather than as a key resource in sustaining rural communities. This deeply changes the social and economic functions of land itself, generating tensions between corporate interests and traditional livelihood practices.

The implications of land grabbing as a driver of corporate and financial control run deep. One of the most visible impacts of this phenomenon is market concentration, as a small number of multinational corporations and financial actors consolidate large shares of productive land, commodities, and inputs into agriculture. Clapp [10] points out that the mergers, acquisitions, and cross-border investments reduce both ownership and decision-making power in the hands of a limited set of actors, decreasing competition and increasing dependency among smaller producers. Monopolization follows, where the corporate-controlled estates dominate local markets, control pricing mechanisms, and hold power over access to credit, inputs, and technology. Smallscale farmers and subsistence producers become increasingly marginalized, often being pushed into contractual labor arrangements or forced out onto marginal lands with lower productivity and higher vulnerability [15].

Beyond economic consequences, land grabbing reinforces power asymmetries and diminishes community autonomy. The consolidation of land under corporate and financial control undermines customary governance systems and limits the ability of local communities to negotiate or contest land deals. Political and legal frameworks are often aligned with investor interests, providing smallholders with limited avenues for redress [9]. This process does more than concentrate land; it creates long-term dependencies on corporate and financial actors, effectively transforming rural areas into zones of

capital accumulation rather than centers of local economic development.

Overall, land grabbing constitutes a strategic entry point for corporate and financial control over rural economies. By rendering land as a financialized asset and concentrating control within a narrow set of actors, land grabs reshape both market structures and social relations, systematically marginalizing smallholders and reducing equitable access to resources. Understanding these dynamics is crucial for informing policies on corporate land acquisitions, the protection of community land rights, and inclusive and sustainable land governance.

F. Socioeconomic and Environmental Impacts

Land grabbing has profound socioeconomic and environmental consequences that go beyond the immediate transfer of land, with impacts on rural communities' livelihoods and ecological systems. The most pronounced immediate socioeconomic outcomes include a decline in rural livelihoods and food insecurity. When corporate or financial players take over large expanses of agricultural land, small-scale farmers and pastoral groups tend to lose access to productive land for subsistence agriculture, grazing, and local markets. This disruption undermines the food security of the household, reduces dietary diversity, and raises vulnerability to poverty [4]. Communities reliant on local agricultural output have to adapt to reduced arable land or migrate in search of livelihoods elsewhere, which only increases economic instability and social marginalization.

Another common consequence of large-scale corporate estates is labor exploitation. Large-scale corporate agriculture often absorbs smallholders and displaced members of the community under precarious arrangements of labor, characterized by low wage levels, a lack of social protection, and adverse working conditions. There have been reports from both Africa and Southeast Asia that plantation workers often work in contractual arrangements that hold down bargaining powers and give minimal benefits with long hours [16]. This kind of labor relation reinforces economic inequality because, while profits within corporations are maximized, local workers remain in insecure, low-income positions.

Displacement, loss of communally held lands, and erosion of customary land rights are also core impacts of contemporary land grabs. Many land deals take place in areas where customary tenure systems prevail, and formal statutes barely or do not recognize local land rights. With investors obtaining long-term leases or concessions, communities are denied access to collectively managed lands, forests, and water resources on which people's livelihoods and cultural identities depend [8]. The resulting dispossession disrupts not only social cohesion but also mechanisms for local self-governance and further marginalizes rural and indigenous populations.

The environmental impact of land grabbing exacerbates socioeconomic inequality. Large-scale plantations, monoculture agriculture, and extractive projects generally cause deforestation, soil degradation, and water pollution. Irrigation-intensive crops and industrial processing exacerbate water shortages, making less water available to local farming

and domestic needs in the process. Deforestation promoted by the change in land use disrupts ecosystems, diminishes biodiversity, and overexploits chemical inputs like pesticides and fertilizers, contaminating soil and waterways and further degrading rural livelihoods. In general, poor people are more affected by environmental degradation due to lack of means for adapting or relocating, reinforcing the cycles of inequality and vulnerability.

Furthermore, the socioeconomic consequences have a cumulative effect and are interrelated with environmental stresses to exacerbate poverty, further reduce food security, and reduce social mobility. The concentration of land under corporate and financial actors amplifies these effects since decision-making on resource use prioritizes profits over community welfare or environmental sustainability. Thus, land grabbing acts as a multidimensional driver of inequality by marginalizing rural populations, undermining traditional governance, and degrading ecological systems all at once.

These impacts call for integrated policy measures that protect community land rights, ensure fair labor standards, and advance environmentally viable land management. Regulatory frameworks should consider social and environmental impact assessments, impose corporate liability, and provide avenues for genuine participation of local communities in the governance of land. Neglecting these dynamics might lead to further socioeconomic and environmental disparities that could threaten future equitable and sustainable rural development.

G. Governance, Law and Power

Governance, law, and power are at the heart of the facilitation of land grabbing, shaping not only the processes by which land is acquired but also the unequal distribution of benefits. Poor governance structures, ambiguous legal frameworks, and a lack of effective enforcement mechanisms provide fertile ground for large-scale land acquisitions by corporate and financial players. In many countries, outdated or incomplete land laws fail to recognize customary land tenure or communal rights, allowing investors to acquire land without meaningful consultation or compensation [12]. Legal loopholes, such as permissive lease arrangements, exemptions from environmental regulations, and investor-friendly concessions, enable land deals to proceed with minimal oversight, effectively prioritizing corporate interests over community welfare and equity.

Corruption, elite capture, and political patronage further exacerbate the governance deficits underlying land grabbing. It is often the local and national elites who facilitate land transactions for personal benefit by giving leases, approvals, or concessions to companies in exchange for political favors, financial incentives, or strategic alliances. As noted by Kay [17], the coming together of political power and private capital ensures that land deals are carried out in manners that reinforce existing inequalities, with a small elite being consolidated in land ownership and decision-making while exclusionary practices for the most marginalized are perpetrated. Thus, land grabbing is not just an economic process but also a highly political one, reflecting structural asymmetries of power and governance failure.

A crucial problem with current land deals is the lack of community land rights and free, prior, and informed consent. According to international norms, such as those in the United Nations Declaration on the Rights of Indigenous Peoples, any transaction that involves land should be agreed upon by the communities concerned. However, ineffective domestic enforcement, low levels of legal literacy among rural people, and asymmetrical bargaining powers often reduce consultation to a mere formality or no consultation at all [18]. Where mechanisms to guarantee FPIC are lacking, communities are often alienated from their lands without meaningful participation, detrimental to both social justice and local governance structures.

In many countries, policy environments are explicitly set in favor of corporate and financial investors through appeals to economic development, modernization, or food security. To attract foreign direct investment into agriculture and extractive sectors, governments may provide tax incentives, lax environmental or labor regulations, and streamlined approval procedures [19]. Framed as a means to promote growth, such policies commonly overlook the distributive consequences of land concentration and leave vulnerable populations without protection. This entrenchment of policy into the service of investors is a good example of how legislation and regulation can serve as tools of power, entrenching corporate and financial control over land at the expense of community rights.

The mutually reinforcing effects of weak governance, legal ambiguity, and political patronage facilitate land grabbing and perpetuate broader inequalities. Governance failures that allow the continued consolidation of landholdings by corporate and financial actors contribute to economic marginalization of smallholders, erosion of communal land tenure systems, and disempowerment at the local level. Overcoming these challenges requires wide-ranging reforms: strengthening legal recognition of customary and community land rights, ensuring enforcement of FPIC protocols, improving transparency in land transactions, and reducing corrupt practices. Consequently, policies will have to be designed in a way that balances investment promotion with social equity and environmental sustainability, ensuring meaningful control of land and resources for rural communities.

These are the key determinants of land grabbing: governance, law, and power. Legal loopholes, elite capture, weak enforcement, and investor-oriented policy environments come together to create the enabling conditions for corporate and financial concentration of land. In the absence of effective governance intervention, land grabs will continue to foment social, economic, and political inequalities, undermining inclusive development and community resilience.

H. Addressing Inequality and Land Concentration

Addressing inequality and corporate/financial land concentration demands multi-layered interventions that bring together legal, regulatory, and community-centered approaches. At the heart of these is land tenure reform and community land rights protection, which seek to ensure legal recognition of customary and communal land ownership. Clear and enforceable land rights reduce vulnerability to dispossession and enable local communities to negotiate fair land transactions. Alden Wily [8] argues that tenure security is

vital not only to protect rural livelihoods but also to ensure resource sustainability. Legal recognition of communal and indigenous land rights, with appropriate mechanisms for the enforcement of such rights, can greatly diminish the vulnerability of communities to large-scale acquisitions by corporate and financial actors.

Anti-concentration policies and regulation of corporate land ownership are also important. Policies placing ceilings on the size of landholdings, or making large acquisitions subject to competitive review, can prevent monopolistic control of land and reduce economic inequality. Some countries have attempted to set land ceilings, progressive taxation on land, or restrictions on foreign ownership to limit excessive accumulation of land by a few actors [1]. Effective regulatory frameworks should balance land redistribution or corporate acquisitions against social equity considerations, hence allowing fairer access to productive land while continuing to enable investment in agriculture and infrastructure.

Other measures include increasing transparency in land transactions, deeper due diligence, and accountability frameworks. Most of the current land grabs occur within opaque contracts, a lack of public disclosure, and very minimal levels of community consultation [3]. More comprehensive disclosure requirements, impact assessments, and participatory negotiation processes may reduce risks of displacement and exploitation. Accountability mechanisms involve independent monitoring, grievance redress, and corporate reporting requirements, which have the potential for ensuring that investors respect the various standards protecting social, environmental, and human rights. A combination of transparency and accountability will therefore heighten the prospects that land transactions are conducted in an equitable and socially responsible way.

The alternative models of land management provide practical approaches to reduction of concentration and equitable access. Cooperative land ownership and community-managed lands allow collective decisions by a group of farmers or a community on resource management, benefits, and decision-making powers over land use. It enhances social cohesion and acts as a counterbalance to corporate dominance [20]. Agroecology and sustainable agricultural practices improve food availability for local populations while reducing reliance on corporate supply chains. These methods link ecological sustainability with social equity, thus enabling communities to maintain control of production and resource management. Redistributive reforms, such as targeted land reform programs or resettlement of marginal households, can help reduce structural inequality and permit wider access to productive land.

A holistic approach to addressing land inequality also requires alignment with national and international governance frameworks: policies should incorporate principles of free, prior, and informed consent, safeguard human rights, and incorporate gender-sensitive measures that guarantee women's inclusion in land negotiations and benefit-sharing. Multi-stakeholder collaboration-between governments, civil society, and private actors-can catalyze a set of changes toward equitable governance and reduce conflict by fostering

investment practices that are supportive of sustainable development goals.

The solution to land concentration and inequality is an integrated strategy that includes assurance of tenure, regulation, transparency, accountability, and alternative models of land management. Strengthening community land rights, controlling monopolistic accumulation, and encouraging either cooperative or agroecological methods will contribute to decreasing the socioeconomic disparities, protect rural livelihoods, and develop a more equitable and sustainable land governance system.

I. Conclusion

Land grabbing in its current form is a critical mechanism in the reproduction and intensification of inequality. In as much as multinational corporations, financial actors, and state-backed entities amass large tracts of land, upward structuring of control over productive resources undermines rural livelihoods, local food security, and community autonomy. Land transformation into a financial asset disengages ownership from direct cultivation and social stewardship, therefore enabling capital accumulation at the expense of smallholders, indigenous people, and subaltern groups [5], [7]. Accordingly, economic inequality is enforced through the concentration of wealth and productive capacity in the hands of elite agents, while social, gender, and political inequalities are created through the exclusion of locals from decision-making processes, displacement, and erosion of customary rights (De Schutter, 2011; [14]). Consequences go beyond the immediate moments of dispossession and involve environmental deterioration, labor exploitation, and market monopolization that compound the vulnerability of rural populations and create multidimensional inequalities that are often irreversible. Governance gaps, such as weak legal frameworks, corruption, and ineffective enforcement of community land rights, further facilitate corporate and financial control, which configures policy environments in ways that prioritize profit over social and ecological well-being. Such patterns underscore that land grabbing is not strictly a market process but, rather, a deeply political and structural one with enduring effects for equity and development. Addressing these challenges requires democratic, equitable, and sustainable land governance. To this end, policies should secure tenure rights, ensure transparency and accountability in land deals, regulate corporate concentration, and foster alternative models such as community-managed lands, cooperatives, and agroecological practices. As Cotula [3] and Franco and Borras [20] argued, it is only by incorporating elements of social justice, environmental sustainability, and participatory governance that land policy can be poised to mitigate the core structural inequalities underpinning contemporary land grabbing. The right to land can only be fully guaranteed and exercised if land remains a common heritage for livelihoods, culture, and sustainable development-to foster fair and resilient rural societies.

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Waste-to-Energy Technologies in Ogun State: A Critical Review of Strategies for Addressing Energy and Waste Challenges

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Abstract—The increasing rate of urbanization, population growth, and industrial activities in Ogun State, Nigeria, has intensified challenges related to municipal solid waste management and energy supply. Waste-to-energy (WtE) technologies have emerged as viable solutions for simultaneously reducing waste volumes and generating sustainable energy. This review critically examines the potential, applicability, and implementation challenges of major WtE technologies in Ogun State, including incineration, anaerobic digestion, landfill gas recovery, gasification, and pyrolysis, evaluating existing literature on waste generation trends, energy recovery potential, environmental impacts, economic feasibility, and policy frameworks influencing WtE development. Findings reveal that the substantial quantity of municipal and industrial waste generated across the state presents significant opportunities for renewable energy production and environmental improvement; however, inadequate waste segregation practices, insufficient infrastructure, limited investment, technological constraints, and weak regulatory enforcement remain key barriers to effective deployment. The review highlights the importance of integrated waste management systems, supportive policies, stakeholder engagement, and public-private partnerships in enhancing WtE adoption, and concludes that strategic investment in appropriate WtE technologies can contribute significantly to sustainable waste management, improved energy security, and environmental sustainability in Ogun State.

Index Terms—Municipal solid waste, Ogun State, renewable energy, sustainable waste management, waste-to-energy.

I. INTRODUCTION

The increasing rate of urbanization, industrialization, and population growth has significantly intensified the generation of municipal solid waste (MSW) across developing countries, particularly in sub-Saharan Africa. Nigeria, the most populous nation in Africa, produces millions of tonnes of solid waste annually, yet a substantial proportion of this waste is disposed of through open dumping, uncontrolled landfilling, and open burning. These unsustainable disposal practices contribute to environmental degradation, greenhouse gas emissions, public health risks, flooding, and the loss of potentially valuable energy resources. At the same time, Nigeria continues to experience a persistent electricity deficit characterized by inadequate generation capacity, unreliable power supply, and increasing dependence on fossil fuels. This dual challenge of waste accumulation and energy scarcity has renewed interest in waste-to-energy (WtE) technologies as an integrated and sustainable solution capable of simultaneously addressing environmental pollution and energy insecurity.

WtE refers to a collection of technologies that convert municipal, industrial, agricultural, or organic waste into useful forms of energy, including electricity, heat, biogas, biofuels, and synthetic fuels. Common WtE technologies include incineration with energy recovery, anaerobic digestion,

gasification, pyrolysis, refuse-derived fuel (RDF), and landfill gas recovery. These technologies not only reduce the volume of waste requiring disposal but also contribute to renewable energy production, greenhouse gas mitigation, resource recovery, and the promotion of a circular economy. Countries such as Sweden, Denmark, Germany, Japan, and Singapore have successfully integrated WtE facilities into their national waste management strategies, thereby reducing landfill dependence while improving energy security. Similar opportunities exist in developing countries, provided that appropriate policy frameworks, investment mechanisms, and technical capacities are established.

Globally, the transition toward renewable and sustainable energy systems has accelerated due to increasing concerns over climate change, energy security, and sustainable urban development. The United Nations Sustainable Development Goals (SDGs), particularly SDG 7 (Affordable and Clean Energy), SDG 11 (Sustainable Cities and Communities), SDG 12 (Responsible Consumption and Production), and SDG 13 (Climate Action), emphasize the need for innovative approaches to waste management and renewable energy generation. WtE technologies align closely with these global sustainability objectives by transforming waste streams into economic resources while reducing environmental impacts associated with conventional waste disposal methods.

Nigeria faces one of the largest waste management challenges in Africa. Rapid urban expansion, changing consumption patterns, inadequate waste collection infrastructure, poor waste segregation practices, weak regulatory enforcement, and insufficient investment have contributed to increasing volumes of unmanaged municipal solid waste. Estimates indicate that over 30 million tonnes of municipal solid waste are generated annually across the country, with only a fraction being properly collected and treated. Most urban centres continue to rely heavily on open dumpsites, which generate methane emissions, contaminate groundwater, and create breeding grounds for disease vectors. These challenges have significant implications for environmental sustainability and public health.

Simultaneously, Nigeria's electricity sector continues to experience severe supply shortages despite abundant renewable energy resources. National electricity generation remains insufficient to satisfy growing industrial, commercial, and residential demand. Consequently, households and industries rely extensively on diesel and petrol generators, resulting in high operating costs, increased carbon emissions, and reduced economic competitiveness. The integration of WtE technologies into Nigeria's renewable energy portfolio

therefore presents an opportunity to diversify energy sources while addressing the country's mounting waste crisis.

Within Nigeria, Ogun State occupies a strategic economic position due to its proximity to Lagos State, rapid industrial growth, expanding urban centres, and increasing population. The state hosts numerous manufacturing industries, educational institutions, commercial centres, and residential developments that collectively generate substantial quantities of municipal and industrial solid waste. Cities such as Abeokuta, Ijebu-Ode, Sagamu, Ota, Ilaro, and Sango-Ota continue to experience increasing waste generation driven by urbanization, industrialization, and changing consumption behaviours. However, waste collection, transportation, segregation, recycling, and final disposal remain inadequate in many parts of the state, leading to indiscriminate dumping, environmental pollution, blocked drainage systems, and health hazards.

The Ogun State Waste Management Authority (OGWAMA) has implemented several initiatives aimed at improving waste collection, public awareness, environmental sanitation, and private-sector participation. Nevertheless, existing waste management practices remain largely disposal-oriented rather than resource-oriented. Most collected waste is transported to dumpsites where valuable recyclable materials and recoverable energy resources are lost. The absence of large-scale waste segregation systems and modern waste treatment facilities limits the state's ability to maximize the economic value embedded within municipal solid waste.

Recent empirical evidence from Ogun State further demonstrates the feasibility of WtE implementation. A detailed assessment of the Oke Saje municipal dumpsite in Abeokuta estimated that approximately 3,718 tonnes of municipal solid waste could provide sufficient energy to generate approximately 12.88 MWh of electricity per day through integrated waste recovery systems. The study further reported that nearly half of the waste stream could be recycled while the remaining combustible fraction could support energy recovery through modern WtE technologies. These findings highlight the enormous untapped renewable energy potential embedded within Ogun State's municipal waste stream.

Beyond electricity generation, WtE technologies offer numerous socio-economic and environmental benefits for Ogun State. These include reduction in landfill volumes, greenhouse gas mitigation, improved environmental sanitation, employment generation, resource recovery, investment attraction, enhanced energy security, and support for circular economy principles. Proper implementation of WtE systems can stimulate green entrepreneurship, improve waste collection efficiency, encourage waste segregation at source, and create new value chains involving recyclers, waste collectors, energy developers, and manufacturing industries.

Despite these opportunities, significant barriers continue to hinder the adoption of WtE technologies in Ogun State and Nigeria as a whole. These barriers include inadequate financing, limited technical expertise, weak policy implementation, poor waste segregation practices, insufficient public awareness, lack of reliable waste characterization data, regulatory uncertainties, and high capital investment requirements. Addressing these constraints requires

coordinated efforts among government agencies, private investors, research institutions, international development partners, and local communities.

This review therefore critically examines the current status of waste generation, waste management practices, and WtE technologies in Ogun State, Nigeria. It evaluates existing strategies, identifies key implementation challenges, and explores emerging opportunities for integrating sustainable WtE systems into the state's environmental management and renewable energy agenda. The paper also synthesizes recent national and international literature to propose evidence-based policy recommendations capable of supporting Ogun State's transition towards a circular economy, sustainable urban development, and enhanced energy security.

II. MUNICIPAL SOLID WASTE PROFILE OF OGUN STATE

Ogun State is one of Nigeria's fastest-growing industrial and commercial hubs due to its strategic location adjacent to Lagos State. The state's rapid urbanization, industrial expansion, population growth, and increasing commercial activities have significantly increased the generation of municipal solid waste (MSW). Major urban centres such as Abeokuta, Ota, Sagamu, Ijebu-Ode, Ilaro, and Sango-Ota generate substantial quantities of household, commercial, institutional, and industrial wastes daily. The continuous expansion of residential estates, manufacturing industries, educational institutions, and commercial markets has further intensified pressure on the state's waste management infrastructure, making effective waste management a critical environmental and public health concern.

Municipal solid waste in Ogun State comprises wastes generated from households, markets, schools, hospitals, industries, offices, restaurants, and public institutions. Similar to many Nigerian states, the waste stream is heterogeneous and consists of biodegradable materials, recyclable materials, inert waste, and hazardous components. Organic or biodegradable waste, including food residues, vegetables, fruits, wood, and garden waste, constitutes the largest proportion of the municipal waste stream. Other major components include plastics, paper, cardboard, glass, metals, textiles, rubber, electronic waste, and construction debris. Studies indicate that biodegradable materials account for approximately 50–60% of the total municipal solid waste generated in Nigeria, highlighting the significant potential for composting, anaerobic digestion, and biogas production.

The quantity of waste generated in Ogun State has increased considerably over the last decade due to rapid population growth, urbanization, and industrialization. Although state-wide estimates vary, available studies indicate that municipal solid waste generation in Nigeria ranges between 0.5 and 0.8 kg per capita per day, with higher generation rates observed in highly urbanized and industrialized areas such as Ota, Sagamu, and Abeokuta. The continued migration of industries from Lagos State into Ogun State is expected to further increase municipal waste generation unless effective waste minimization, recycling, and resource recovery strategies are implemented.

The physical composition of municipal solid waste in Ogun State demonstrates considerable potential for resource recovery. Organic waste represents the dominant fraction, followed by plastics, paper, glass, metals, textiles, leather, and inert materials. This composition makes the state's waste stream particularly suitable for integrated WtE technologies because biodegradable waste can be converted into biogas through anaerobic digestion, while combustible fractions can be utilized for incineration, gasification, pyrolysis, or refuse-derived fuel production. Likewise, recyclable materials such as plastics, metals, paper, and glass can be recovered before thermal treatment, thereby maximizing both environmental and economic benefits.

Despite the enormous resource potential of municipal solid waste, waste management practices in Ogun State remain largely disposal-oriented rather than resource-oriented. Waste generated from households and commercial centres is collected through public and private waste operators coordinated by OGWAMA. However, waste segregation at source remains limited, resulting in mixed waste streams that reduce recycling efficiency and complicate energy recovery processes. Consequently, a significant proportion of collected waste is transported directly to open dumpsites where valuable recyclable materials and energy-rich organic waste remain underutilized.

One of the most important municipal disposal facilities in Ogun State is the Oke Saje dumpsite located in Abeokuta. Recent characterization studies revealed substantial quantities of recyclable and combustible waste capable of supporting large-scale WtE development. Oladejo and Oruamen estimated that the dumpsite contains approximately 3,718.83 tonnes of municipal solid waste with an estimated energy content of 46,370.47 MJ, capable of generating approximately 12.88 MWh of electricity per day through integrated waste recovery systems. The study further showed that approximately 46% of the waste stream could be recovered through recycling, composting, and biofuel production, while about 54% could be utilized for electricity generation and energy recovery.

The predominance of biodegradable waste presents significant opportunities for the deployment of anaerobic digestion technology within Ogun State. Organic wastes generated from households, markets, restaurants, hotels, and agro-processing industries can be converted into biogas for electricity generation, cooking fuel, and organic fertilizer production. Similarly, combustible materials such as plastics, paper, textiles, and wood residues can support refuse-derived fuel production, gasification, and pyrolysis technologies, thereby reducing dependence on fossil fuels while simultaneously mitigating greenhouse gas emissions associated with open dumping and waste burning.

Nevertheless, several challenges continue to constrain effective municipal solid waste management in Ogun State. These include inadequate waste segregation at source, insufficient collection vehicles, poor recycling infrastructure, indiscriminate dumping, illegal open burning, inadequate financing, weak enforcement of environmental regulations, and limited investment in modern waste treatment facilities. In addition, the absence of comprehensive waste characterization

data across all local government areas limits strategic planning for integrated WtE projects.

Overall, the municipal solid waste profile of Ogun State indicates that the state possesses abundant biomass resources capable of supporting sustainable WtE development. The predominance of biodegradable waste, coupled with increasing waste generation from rapidly expanding urban and industrial centres, provides favourable conditions for implementing integrated waste management systems that combine waste reduction, recycling, composting, anaerobic digestion, landfill gas recovery, and thermal WtE technologies. Harnessing these resources through appropriate government policies, investment, technological innovation, and public-private partnerships could significantly improve environmental quality, reduce landfill dependence, create employment opportunities, and contribute meaningfully to renewable electricity generation in Ogun State.

III. WASTE-TO-ENERGY TECHNOLOGIES

Waste-to-energy (WtE) technologies comprise a range of processes that convert municipal solid waste (MSW) into useful forms of energy such as electricity, heat, biogas, biofuels, and synthetic fuels. These technologies provide a sustainable approach to waste management by reducing the volume of waste destined for landfills while simultaneously generating renewable energy and minimizing greenhouse gas emissions. As global demand for clean energy continues to increase, WtE technologies have emerged as an integral component of the circular economy, promoting resource recovery, environmental sustainability, and energy security. For developing economies such as Nigeria, where inadequate waste management coexists with persistent electricity shortages, WtE technologies present a practical solution to two critical developmental challenges. Ogun State, with its expanding urban population, industrial activities, and increasing municipal waste generation, possesses significant potential for implementing integrated WtE systems. The most widely adopted WtE technologies include incineration, anaerobic digestion, gasification, pyrolysis, landfill gas recovery, and refuse-derived fuel production.

A. Incineration Technology

Incineration is one of the oldest and most widely implemented WtE technologies globally. The process involves the controlled combustion of municipal solid waste at temperatures typically ranging between 850°C and 1,100°C, producing heat that is subsequently converted into steam for electricity generation through steam turbines. Modern incineration facilities are equipped with advanced air pollution control systems that significantly reduce emissions of particulate matter, nitrogen oxides, sulfur dioxide, dioxins, and heavy metals, making them more environmentally acceptable than conventional open burning.

B. Anaerobic Digestion

Anaerobic digestion (AD) is a biological process in which microorganisms decompose biodegradable organic waste under oxygen-free conditions to produce biogas and nutrient-rich digestate. Biogas primarily consists of methane (CH₄) and carbon dioxide (CO₂) and can be utilized for electricity generation, cooking, heating, or upgraded into biomethane for

transportation fuel. The digestate serves as an organic fertilizer, thereby supporting sustainable agriculture and reducing dependence on chemical fertilizers.

IV. GASIFICATION TECHNOLOGY

Gasification is an advanced thermochemical conversion process that transforms carbonaceous waste materials into synthetic gas (syngas) under controlled high-temperature conditions with limited oxygen supply. The produced syngas mainly contains hydrogen, carbon monoxide, methane, and small quantities of carbon dioxide, which can be used for electricity generation, industrial heating, or the production of liquid fuels and chemicals.

V. LANDFILL GAS RECOVERY

Landfill gas recovery is a WtE technology that captures methane generated through the anaerobic decomposition of organic waste deposited in landfills and dumpsites. Typically, landfill gas contains approximately 45–60% methane and 40–55% carbon dioxide, together with trace amounts of other gases. Captured methane can be utilized for electricity generation, industrial heating, or upgraded into renewable natural gas. Considering the large volume of organic waste disposed at facilities such as the Oke Saje dumpsite in Abeokuta, landfill gas recovery offers considerable opportunities for Ogun State. Studies have demonstrated substantial methane generation potential from existing municipal dumpsites within the state, indicating that landfill gas utilization could contribute meaningfully to local electricity production while mitigating environmental pollution.

VI. EXISTING WASTE MANAGEMENT STRATEGIES IN OGUN STATE

Ogun State has implemented several waste management strategies aimed at improving environmental sanitation, public health, and sustainable urban development. These strategies are coordinated primarily by OGWAMA, the agency responsible for regulating waste collection, transportation, treatment, and disposal across the state. Over the past few years, the state government has introduced reforms to modernize waste management through stronger regulatory enforcement, public-private partnerships (PPPs), environmental awareness campaigns, and improved waste collection systems. Despite these initiatives, rapid urbanization, industrial expansion, and increasing population continue to outpace existing waste management infrastructure, resulting in persistent challenges such as indiscriminate dumping, open burning, and inadequate waste recovery.

A. Institutional and Regulatory Framework

OGWAMA serves as the principal institution responsible for implementing waste management policies, licensing waste operators, regulating waste disposal, monitoring environmental compliance, and promoting proper sanitation practices. Under recent waste management reforms, industries, commercial establishments, healthcare facilities, and residential communities are required to register with government-approved waste collectors to ensure environmentally sound waste collection and disposal practices.

B. Public–Private Partnership (PPP) in Waste Collection

One of the major strategies adopted in Ogun State is the engagement of private sector operators in municipal waste collection through PPPs. Accredited private waste contractors complement government efforts by providing door-to-door waste collection services, transporting municipal waste to designated disposal sites, and servicing residential, commercial, institutional, and industrial areas. This collaborative model has improved waste collection coverage in major urban centres such as Abeokuta, Ota, Sagamu, and Ijebu-Ode, although service delivery remains uneven in many peri-urban and rural communities.

VII. ENVIRONMENTAL SANITATION AND PUBLIC AWARENESS

Environmental sanitation campaigns remain an important component of Ogun State's waste management strategy. OGWAMA regularly organizes public awareness programmes, stakeholder engagements, sanitation exercises, and educational campaigns to encourage responsible waste disposal and environmental cleanliness. The agency has also expanded collaboration with educational institutions, industries, religious organizations, and community associations to strengthen environmental education and improve compliance with waste management regulations. For example, OGWAMA recently partnered with the Federal University of Agriculture, Abeokuta (FUNAAB) to harmonize institutional waste management practices and promote sustainable environmental management.

VIII. WASTE DISPOSAL PRACTICES

The dominant waste disposal strategy in Ogun State remains landfilling and open dumping at designated municipal dumpsites. Facilities such as the Oke Saje dumpsite in Abeokuta receive large quantities of municipal solid waste collected from surrounding urban centres. Although these disposal sites provide temporary waste containment, many operate without modern sanitary landfill infrastructure, including engineered liners, leachate treatment systems, methane capture facilities, and comprehensive environmental monitoring systems. Consequently, open dumping continues to contribute to groundwater contamination, greenhouse gas emissions, offensive odours, and public health risks.

A. Recycling and Resource Recovery

Recycling activities in Ogun State are still at a relatively early stage and are largely driven by informal waste pickers and private recycling companies. Recyclable materials such as plastics, metals, paper, cardboard, and glass are manually recovered from households, markets, and dumpsites before being sold to recycling industries. Although this informal recycling system contributes to resource recovery and employment generation, it remains poorly organized and inadequately integrated into the state's formal waste management framework.

B. Critical Assessment of Existing Strategies

The existing waste management strategies adopted by Ogun State demonstrate the government's commitment to improving environmental sanitation through institutional reforms, regulatory enforcement, public-private partnerships, and

stakeholder engagement. However, these strategies remain predominantly collection- and disposal-oriented rather than recovery-oriented. Limited waste segregation at source, inadequate recycling infrastructure, insufficient investment in modern waste treatment facilities, and the absence of large-scale WtE plants significantly constrain the sustainability of the current system. Recent reforms emphasizing industrial compliance, institutional partnerships, and environmental education provide a strong foundation for future improvements, but greater investment in integrated waste management technologies is required to achieve a circular economy and sustainable waste management in Ogun State.

IX. CRITICAL REVIEW OF CURRENT WASTE-TO-ENERGY INITIATIVES

WtE development in Ogun State is still in its formative stage, with most initiatives focusing on resource recovery, recycling, circular economy practices, and pilot clean-energy projects rather than large-scale commercial electricity generation from municipal solid waste. Although the state has made notable progress in strengthening waste management policies and promoting environmental sustainability, there is currently no fully operational municipal WtE power plant comparable to those found in countries such as Sweden, Germany, or Japan. Nevertheless, recent government policies and private-sector collaborations indicate increasing commitment towards integrating WtE technologies into the state's environmental and energy development agenda.

A. Waste-to-Energy Potential of Municipal Dumpsites

One of the most significant studies on WtE in Ogun State was conducted at the Oke Saje municipal dumpsite in Abeokuta. The study demonstrated that the dumpsite possesses considerable energy recovery potential owing to its high proportion of combustible and biodegradable municipal solid waste. The researchers estimated that approximately 3,718.83 tonnes of municipal solid waste contained sufficient calorific value to generate approximately 12.88 MWh of electricity per day if modern WtE technologies were deployed. Furthermore, nearly half of the waste stream could be recovered through recycling and composting before thermal conversion, indicating that integrated waste management could substantially improve resource utilization while reducing landfill dependence. Despite these promising findings, the study remains largely theoretical because commercial-scale implementation has not yet occurred. Consequently, Ogun State continues to lose valuable renewable energy resources through conventional dumping practices, highlighting the need for stronger government investment, private-sector participation, and technology transfer.

B. Refuse-Derived Fuel (RDF) Initiative

Among the most promising recent initiatives is the state's promotion of Refuse-Derived Fuel (RDF) as an alternative energy source for industries. RDF involves processing combustible municipal waste including plastics, paper, textiles, wood, and rubber into standardized fuel for cement factories, industrial boilers, and manufacturing industries. Ogun State has identified RDF as an important component of its climate strategy because it simultaneously reduces landfill waste,

lowers greenhouse gas emissions, and provides cleaner industrial fuel. However, the initiative is still at the policy and advocacy stage, with limited evidence of large-scale commercial RDF production. Critical challenges include inadequate waste segregation at source, limited pre-processing facilities, insufficient investment, and the absence of an integrated RDF supply chain.

X. CIRCULAR ECONOMY AND E-WASTE-TO-WEALTH PROGRAMME

Another important initiative is the E-Waste-to-Wealth Strategy, introduced by the Ogun State Government as part of its circular economy programme. The initiative seeks to transform electronic waste into economic resources through formal recycling systems, job creation, and environmentally responsible management of hazardous materials. The programme also includes collaboration among government agencies, private recycling companies, and environmental organizations through formal partnership agreements designed to strengthen e-waste collection and processing.

XI. RECYCLING-BASED RESOURCE RECOVERY

Current waste recovery activities in Ogun State remain dominated by recycling rather than energy recovery. Informal waste pickers and private recycling companies recover plastics, metals, paper, glass, and other recyclable materials from households, markets, industries, and municipal dumpsites before final disposal. These activities reduce landfill volumes and generate employment opportunities while supplying raw materials to manufacturing industries.

XII. OPPORTUNITIES FOR WASTE-TO-ENERGY DEVELOPMENT IN OGUN STATE

Ogun State possesses significant opportunities for the deployment of WtE technologies because of its strategic location, expanding industrial base, increasing municipal solid waste (MSW) generation, supportive policy environment, and growing demand for reliable electricity. The combination of these factors positions the state as one of the most suitable locations in Nigeria for integrated waste management and renewable energy development.

A. Abundant Municipal Solid Waste Resources

One of Ogun State's greatest strengths is the continuous generation of municipal solid waste from households, industries, markets, educational institutions, and commercial centres. Rapid urbanization in cities such as Abeokuta, Ota, Sagamu, Ijebu-Ode, and Sango-Ota provides a steady supply of feedstock for WtE facilities. Studies on the Oke Saje dumpsite demonstrated that the waste stream possesses sufficient calorific value to generate approximately 12.88 MWh of electricity per day, indicating considerable renewable energy potential if modern WtE technologies are deployed.

B. Strong Industrial Market for Waste-Derived Energy

Ogun State hosts more than 5,000 industries, making it Nigeria's leading industrial hub after Lagos. These industries require reliable electricity and thermal energy for production processes. Waste-derived electricity, biogas, and RDF can provide alternative energy sources for manufacturing industries, cement plants, and industrial estates, thereby reducing dependence on diesel generators and fossil fuels. The

state's industrial clusters at Agbara, Ota, Sagamu, and Ewekoro provide an attractive market for commercial WtE investments.

XIII. GOVERNMENT POLICY AND INVESTMENT SUPPORT

The Ogun State Government has increasingly prioritized renewable energy, circular economy initiatives, and sustainable waste management. Recent initiatives include the promotion of RDF, the E-Waste-to-Wealth Programme, plastic recycling projects, and clean energy investments. Through the OgunInvest initiative, the government is also promoting partnerships for renewable energy, embedded generation, and independent power projects by providing investment incentives, public-private partnership opportunities, and streamlined land allocation processes.

XIV. EMPLOYMENT AND GREEN ENTREPRENEURSHIP

Large-scale implementation of WtE technologies would create significant employment opportunities across the waste value chain. Jobs would be generated in waste collection, sorting, recycling, transportation, equipment maintenance, plant operation, engineering services, environmental monitoring, and energy distribution. Furthermore, WtE development could stimulate green entrepreneurship through the establishment of recycling enterprises, biogas production facilities, composting plants, and biomass energy businesses, thereby contributing to poverty reduction and economic diversification.

A. Environmental Sustainability and Climate Change Mitigation

Replacing open dumping and uncontrolled waste burning with integrated WtE systems would significantly reduce greenhouse gas emissions, methane release, groundwater contamination, and air pollution. Technologies such as anaerobic digestion, landfill gas recovery, and RDF production support Nigeria's commitments to climate change mitigation while advancing SDGs 7, 11, 12, and 13.

F. Public–Private Partnerships and Foreign Investment

The substantial capital required for WtE infrastructure creates opportunities for public-private partnerships (PPPs) involving government agencies, private investors, financial institutions, and international development organizations. Ogun State's investment-friendly policies, strategic location, and proximity to Lagos enhance its attractiveness for domestic and foreign investors interested in renewable energy and sustainable infrastructure projects.

B. Research, Innovation, and Technology Development

Ogun State hosts several higher education institutions and research centres, including FUNAAB, Olabisi Onabanjo University (OOU), Moshood Abiola Polytechnic (MAPOLY), and Gateway ICT Polytechnic, which can contribute to research, innovation, and human capacity development in WtE technologies. Collaboration between academia, government, and industry can facilitate technology adaptation, pilot projects, workforce development, and evidence-based policymaking.

C. Critical Evaluation

The opportunities available to Ogun State extend far beyond electricity generation. The state's growing waste stream, industrial demand, supportive investment climate, and increasing government commitment to renewable energy provide a strong foundation for large-scale WtE development. However, realizing these opportunities will require sustained investment in infrastructure, effective policy implementation, improved waste segregation, institutional capacity building, and stronger collaboration among government, academia, industry, and development partners. With coordinated implementation, Ogun State has the potential to become a national leader in integrated waste management and renewable energy development.

XV. SUMMARY

This review critically examined the potential of WtE technologies as sustainable strategies for addressing the dual challenges of municipal solid waste management and energy shortages in Ogun State, Nigeria. The study established that rapid urbanization, industrialization, population growth, and changing consumption patterns have significantly increased municipal solid waste generation across the state, particularly in urban centres such as Abeokuta, Ota, Sagamu, Ijebu-Ode, and Sango-Ota. While the Ogun State Government has implemented various waste management strategies through OGWAMA, existing practices remain predominantly focused on waste collection and disposal rather than resource recovery and energy generation. The review identified several WtE technologies including incineration, anaerobic digestion, gasification, pyrolysis, landfill gas recovery, and RDF that are suitable for converting municipal solid waste into electricity, heat, and biofuels. Among these technologies, anaerobic digestion and RDF production appear particularly promising because of the high proportion of biodegradable and combustible waste generated in Ogun State. Evidence from studies on the Oke Saje dumpsite further demonstrates that the state's municipal waste possesses considerable renewable energy potential capable of contributing to electricity generation while reducing landfill dependence.

XVI. CONCLUSION

WtE technologies represent a viable and sustainable solution for addressing the growing environmental and energy challenges confronting Ogun State. As municipal solid waste generation continues to increase alongside rapid urbanization and industrial expansion, reliance on conventional waste disposal methods such as open dumping and uncontrolled landfilling is no longer environmentally or economically sustainable. Rather than treating municipal waste as an environmental burden, Ogun State should recognize waste as a valuable renewable energy resource capable of contributing to electricity generation, industrial development, and climate change mitigation. The findings of this review indicate that Ogun State possesses the necessary conditions, including abundant municipal solid waste, expanding industrial activities, supportive government policies, and increasing energy demand, to successfully implement integrated WtE systems. Nevertheless, the absence of commercial-scale WtE facilities, inadequate waste segregation, weak financing mechanisms,

and limited institutional capacity continue to hinder progress. Transitioning from a disposal-oriented waste management system to an integrated circular economy approach will require sustained political commitment, strategic investment, technological innovation, and effective collaboration among government agencies, research institutions, private investors, development partners, and local communities. Successful implementation of WtE technologies will not only improve environmental sustainability but also diversify the state's renewable energy portfolio, stimulate green economic growth, and enhance the quality of life of its residents.

XVII. RECOMMENDATIONS

1) Develop a State Waste-to-Energy Policy: The Ogun State Government should formulate a comprehensive Waste-to-Energy Policy and Implementation Framework that integrates waste management with renewable energy development and aligns with national energy and environmental policies.

2) Establish Pilot Waste-to-Energy Plants: Government should initiate pilot projects for anaerobic digestion, landfill gas recovery, and RDF production in major urban centres such as Abeokuta, Ota, and Sagamu to demonstrate technical and economic feasibility.

3) Strengthen Waste Segregation at Source: Mandatory waste segregation should be introduced for households, industries, markets, and institutions to improve recycling efficiency and ensure the availability of quality feedstock for WtE technologies.

4) Promote Public–Private Partnerships (PPPs): Government should encourage partnerships with private investors, financial institutions, and international development agencies to mobilize funding for WtE infrastructure and technology deployment.

5) Improve Regulatory Enforcement: OGWAMA and other relevant agencies should strengthen enforcement of environmental regulations, licensing requirements, and waste collection standards while promoting compliance among industries and households.

6) Invest in Modern Waste Infrastructure: Additional investment is required in transfer stations, material recovery facilities, sanitary landfills, recycling centres, biogas plants, and waste processing facilities to support integrated waste management systems.

7) Enhance Research and Capacity Building: Universities, polytechnics, and research institutes in Ogun State should collaborate with government and industry to conduct research, develop indigenous WtE technologies, and train skilled professionals in renewable energy and waste management.

8) Increase Public Awareness: Continuous public education campaigns should be implemented to promote waste reduction, recycling, source segregation, and community participation in sustainable waste management practices.

9) Provide Financial Incentives: Government should introduce tax incentives, low-interest loans, grants, and carbon financing mechanisms to encourage investment in WtE projects and renewable energy enterprises.

10) Develop Reliable Waste Data Systems: Comprehensive data on waste generation, composition, collection efficiency, and energy potential should be collected and maintained to support planning, monitoring, and investment decisions.

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Arduino-Based Smart Waste Bin with Servo Motor for Sustainable Waste Management in Engineering Education

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Abstract—Rapid urbanization has increased municipal solid waste, leading to overflowing bins, unhygienic conditions, and inefficient collection route. Conventional waste management depends on fixed timetables and manual inspection which waste time, fuel and labour. This Paper proposes the Smart Waste Bin, designed to enhance the ease of the people in the society in disposing their waste with less human energy in the context of sustainable engineering solutions. This Smart Waste Bin uses servo motor and embedded sensors to automate monitoring and reporting. The design integrates an ultrasonic sensor to detect human presence, and a microcontroller to process the data. When waste level exceeds a set threshold, the user can walk up to the bin to dispose the waste. This enables dynamic routine planning and on-demand collection reducing unnecessary trips. The bin can also include a servo driven lid that opens on proximity detection to limit human contact and lower design to support battery or solar operation. By facilitating this experimentally, it will improve collection efficiency, prevents overflow, and supports cleaner public spaces. The smart bin provides a scalable and affordable approach to modernizing waste management in smart cities.

Keywords: Smart Waste Bin, Servo motor, Ultrasonic Sensor, Real-time Monitoring, Waste Management, Automation

I. INTRODUCTION

Waste management has become a major environmental and public health challenge due to the increasing volume of waste generated in rapidly growing urban areas and the limitations of conventional waste disposal methods. [1] observed that manual waste handling often results in poor segregation, overflowing bins, and environmental pollution, while Arthur et al. [2] noted that the increasing adoption of smart waste management technologies reflects the need for more efficient and hygienic waste disposal systems. In Nigeria, the lack of automated waste bin monitoring contributes to delayed waste collection and poor sanitation [3]. To address these challenges, recent advances in embedded systems have enabled the development of intelligent waste bins that automatically detect users and operate without physical contact. Afolalu et al. [4] demonstrated that an Arduino-based smart waste bin using ultrasonic sensors and a servo motor could improve hygiene and convenience, while Xenya et al. [5] further established that sensor-based waste management systems enhance waste collection efficiency. Therefore, this project focuses on the design and construction an Arduino-based Smart Waste Bin that automatically opens and closes its lid when a user is detected, thereby promoting hygienic waste disposal, reducing direct human contact with contaminated surfaces, and encouraging cleaner environmental practices.

A. Statement of the Problem

Traditional waste bins present several problems, including:

- Direct human contact with bin lids, which is unhygienic
- Overflowing bins due to poor monitoring, since there is no means of alerting waste handlers before capacity is reached
- Bad odours and the attraction of insects and rodents
- Increased health risks to users and waste handlers alike
- Inefficient waste disposal practices that waste time, fuel and labour

These challenges create the need for an automated waste bin that operates without physical contact.

B. Aim and Objectives

The aim of this project is to design and construct a smart waste bin that operates without physical contact, using an

Arduino microcontroller, a sensor and a servo motor. The objectives are to:

- design an automatic waste bin using Arduino;
- reduce human contact with waste bins;
- improve hygiene and environmental cleanliness; and
- demonstrate the practical application of sensors and servo motors in automation.

C. Scope and Delimitation

This project is limited to the design and construction of a smart waste bin that automatically opens and closes its lid using a sensor and a servo motor. Features such as the lithium batteries and micro controller, though discussed as directions for future work, are not implemented within the current scope.

II. METHODOLOGY

A. System Model

The system consists of four main units, namely a sensor to detect the presence of an object, an Arduino microcontroller serving as the control unit, a servo motor to open and close the bin lid, and a power supply to energise the entire circuit. When an object is detected within range, the Arduino processes the sensor input and commands the servo motor to open the bin lid; the lid remains open briefly before automatically returning to the closed position. Figure 1 presents the block diagram of the system, showing the flow of signals from the sensor (input unit) through the Arduino microcontroller (processing unit) to the servo motor (output unit).

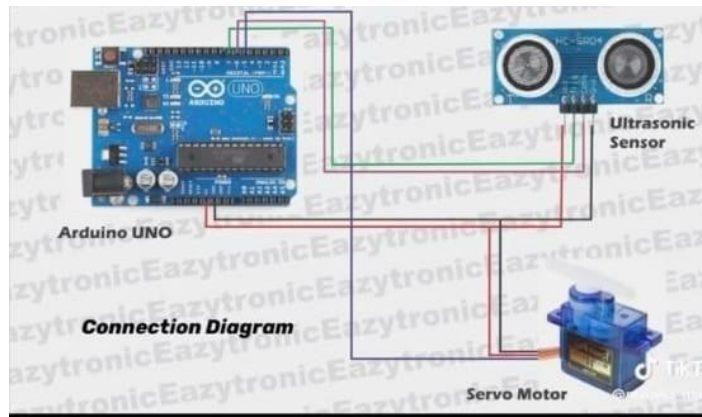


Fig. 1. Block diagram of the smart waste bin system

B. Hardware Components Description

1) Arduino Uno

The Arduino Uno acts as the brain of the system. It continuously reads the signal from the sensor, processes it according to the uploaded program, and sends the appropriate command to the servo motor.

2) Ultrasonic Sensor

The ultrasonic detects the presence of an object or hand near the bin. Once an object is sensed within the pre-set range, the sensor sends a trigger signal to the Arduino microcontroller.

3) Servo Motor

The servo motor receives command signals from the Arduino and rotates to open and close the waste bin lid automatically, based on the object detected by the sensor.

4) Power Supply

The power supply provides a regulated DC voltage to energise the Arduino, sensor and servo motor, ensuring stable and continuous operation of the system.

C. Software Design

The Arduino is programmed using the Arduino Integrated Development Environment (IDE). The program continuously polls the sensor for input and activates the servo motor whenever an object is detected within range, after which the lid is held open briefly before returning to the closed position.

III. MODE OF OPERATION



Fig. 2. Smart waste bin with sensor in operation.



Fig. 3. Servo motor opening the bin lid.

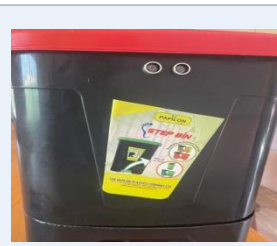


Fig. 2. Smart waste bin with sensor in operation.

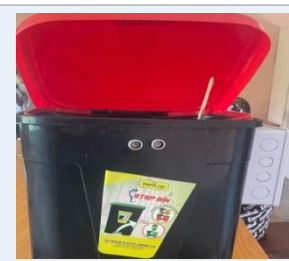


Fig. 3. Servo motor opening the bin lid.



Fig. 4. Wiring diagram of the smart waste bin.



Fig. 5. Servo motor used for automatic lid operation.

IV. RESULTS AND DISCUSSION

On completion of the smart waste bin, the following test was conducted. These tests were performed to ensure the workability of the smart waste bin before usage.

A. System Testing

The smart waste bin was tested by placing objects near the sensor. The servo motor responded correctly by opening and closing the lid.

B. Results

- The system successfully detected objects
- The bin lid opened automatically
- Human contact was eliminated
- Operation was smooth and reliable

C. Discussion

The results show that Arduino, sensors, and servo motors can effectively be used to automate waste bins. The system performed as expected and improved hygiene. This will enhance proper and hygienic sewage disposal in the community.

V. CONCLUSION

This project has successfully demonstrated the design and implementation of an Arduino based smart waste bin capable of detecting a nearby object and automatically opening its lid through a servo motor, thereby eliminating direct human contact with the bin. The integration of a sensor, a microcontroller and a servo motor into a single hygienic, low cost system confirms that embedded automation can meaningfully improve waste disposal practices, especially within the resource constrained context of most Nigerian institutions and communities. Testing showed that the system responded reliably and consistently, opening and closing the

lid without manual intervention while eliminating the unhygienic conditions associated with conventional waste bins. Beyond its immediate function, the smart waste bin serves as a practical teaching platform for embedded systems, sensor integration and automation, giving students hands on experience that reinforces classroom instruction in electrical and electronic engineering. Future work can extend this system by incorporating IoT connectivity, fill level monitoring and solar powered operation, as demonstrated in several of the studies reviewed in this paper, to further enhance its scalability for real world municipal and institutional waste management.

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Synergizing Artificial Intelligence and Internet of Things for Smart Waste Management: A Framework for Resilient and Sustainable Urban Ecosystems

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Abstract—The increasing rate of urbanization and population growth in Nigeria has significantly contributed to the rise in municipal solid waste generation, placing considerable pressure on existing waste management systems. Conventional waste collection methods, which rely on fixed schedules and manual inspections, often result in inefficient resource utilization, overflowing waste bins, environmental pollution, and poor recycling practices. This paper proposes an integrated Artificial Intelligence (AI) and Internet of Things (IoT) framework for intelligent waste collection and automated waste sorting tailored to Nigerian urban environments. The framework employs IoT-enabled smart bins equipped with ultrasonic, weight, gas, and temperature sensors to monitor waste conditions in real time, with sensor data transmitted to a cloud platform where AI algorithms predict waste generation patterns, optimize collection routes, and support intelligent decision-making. In addition, a computer vision-based waste sorting module utilizing Convolutional Neural Networks (CNNs) automatically classifies waste into recyclable and non-recyclable categories, thereby improving recycling efficiency and supporting circular economy initiatives. The proposed framework is expected to reduce operational costs, improve environmental sanitation, enhance public health, and provide municipal authorities with real-time information for effective waste management planning, offering a practical approach toward achieving sustainable waste management and smart city development in Nigeria.

Index Terms—Artificial Intelligence, Internet of Things, Smart Waste Management, Waste Sorting, Computer Vision, Smart Cities, Nigeria.

I. INTRODUCTION

A. Problem Statement

Nigeria's current municipal waste management system is characterized by inefficient waste collection, poor waste segregation, low recycling rates, and inadequate monitoring infrastructure, particularly in developing nations facing the major environmental challenges of rapid urbanization, industrialization, and population growth. According to the United Nations Environment Programme (UNEP), global waste generation continues to increase annually, placing significant pressure on existing waste management infrastructure [1]–[3]. Waste collection vehicles frequently operate on fixed schedules without considering the actual fill level of waste bins, resulting in unnecessary operational costs, excessive fuel consumption, and delayed collection in high-demand areas [4]. Additionally, manual waste sorting exposes workers to occupational hazards and limits the recovery of recyclable materials while minimizing human intervention [5]–[7]. Despite recent advances in AI and IoT technologies, there is limited research on an integrated framework that combines real-time waste monitoring, intelligent collection scheduling, and AI-based automated waste sorting specifically

for the Nigerian environment. Therefore, there is a need for a cost-effective and scalable framework capable of improving waste management efficiency while supporting sustainable urban development.

B. Aim and Objectives

The aim of this study is to propose an AI–IoT framework for intelligent waste collection and automated waste sorting suitable for Nigerian urban environments. The objectives are to:

- Examine the limitations of conventional municipal waste management systems;
- Develop an integrated AI–IoT framework for smart waste monitoring and collection;
- Incorporate AI-based waste classification to improve recycling efficiency;
- Evaluate the potential benefits of the proposed framework for Nigeria; and
- Recommend strategies for successful implementation.

II. RELATED WORK

Recent advancements in Artificial Intelligence (AI) and the Internet of Things (IoT) have significantly improved municipal solid waste management. IoT-enabled smart bins equipped with sensors can monitor waste levels in real time and transmit data to cloud platforms, enabling timely waste collection and reducing unnecessary transportation [4], [8]. AI techniques further enhance these systems by analysing sensor data to predict waste generation patterns and optimize collection routes, thereby reducing operational costs and improving service efficiency [4], [9]. Deep learning models, particularly Convolutional Neural Networks (CNNs), have also been applied to automated waste classification. These models accurately identify recyclable materials such as plastics, paper, glass, and metals, improving recycling efficiency, reducing contamination, and minimizing manual handling of waste. Although these technologies have shown promising results, most existing studies focus on either smart waste monitoring or automated waste sorting as separate solutions. In Nigeria, waste management still relies largely on conventional collection methods with limited integration of intelligent technologies [1]–[3]. Consequently, there is a need for an integrated framework that combines IoT-based waste monitoring, AI-driven route optimization, and automated waste sorting into a single system tailored to the Nigerian environment. This study addresses this gap by proposing a unified AI–IoT framework for intelligent waste collection and automated waste sorting.

TABLE I. Comparison of Existing Studies and the Proposed Framework

Study	IoT Sensing	AI Classification	Route Optimisation	Nigeria-Specific
Okubanjo et al. [1]	✓	✗	✗	✓
Rahman et al. [4]	✓	✓	✗	✗
IEEE IC-ETITE [5]	✗	✗	✓	✗
IEEE ICPECTS [6]	✓	✗	✓	✗
Proposed Framework	✓	✓	✓	✓

III. PROPOSED AI-IoT FRAMEWORK

This study proposes an integrated AI-IoT framework for intelligent waste collection and automated waste sorting in Nigerian cities. The framework combines IoT-enabled smart waste bins, wireless communication technologies, cloud computing, artificial intelligence, and computer vision to provide an end-to-end waste management solution. Unlike conventional waste management systems that rely on fixed collection schedules and manual waste segregation, the proposed framework enables real-time monitoring, predictive waste collection, and automated waste classification. The architecture consists of six functional layers, namely the sensing layer, communication layer, cloud and data management layer, AI analytics layer, intelligent waste sorting layer, and monitoring and decision support layer, as shown in Fig. 1.

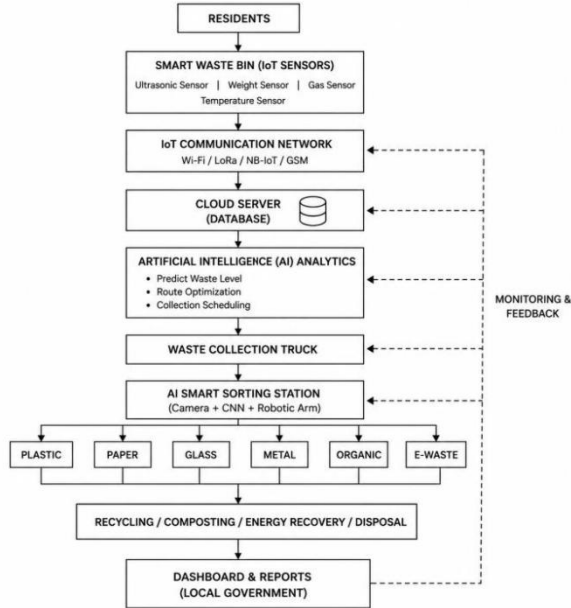


Fig. 1. Proposed AI-IoT Framework for Intelligent Waste Collection and Automated Waste Sorting.

A. Sensing Layer

The sensing layer consists of IoT-enabled smart waste bins equipped with ultrasonic, weight, temperature, and gas sensors. These sensors continuously monitor the condition of each waste bin and transmit real-time data once predefined thresholds are reached. Continuous monitoring allows municipal authorities to identify bins requiring immediate attention, thereby eliminating unnecessary inspections and preventing waste overflow [4], [8].

B. Communication & Cloud Layer

Sensor data are transmitted through communication technologies such as Wi-Fi, GSM, LoRaWAN, or NB-IoT to a centralized cloud platform. The cloud stores and processes the incoming data, maintains historical records, and provides secure access for municipal authorities with real-time information on waste levels, vehicle locations, collection schedules, recycling statistics, and system alerts. This centralized architecture enables continuous monitoring of multiple waste bins across different locations [8], [10].

C. Cloud and Data Management Layer

The cloud platform stores, processes, and manages sensor-generated data from multiple waste bins. Historical and real-time data are maintained in a centralized database where they are available for analysis and visualization. Cloud computing provides scalable storage and allows municipal authorities to monitor waste management operations remotely through secure web-based applications.

D. AI Analytics Layer

The AI analytics layer processes historical and real-time sensor data using machine learning algorithms to predict waste generation patterns and estimate the time required for each waste bin to reach full capacity. Based on these predictions, the system dynamically schedules waste collection and generates optimized routes for collection vehicles, reducing travel distance, fuel consumption, and operational costs [4], [9]. The prediction model can be represented as:

$$\hat{W} = f(x) \quad (1)$$

where (x) represents sensor inputs (fill level, weight, temperature, and historical collection patterns) used to estimate the predicted waste generation level $\hat{W}$.

E. Intelligent Waste Sorting Layer

After collection, waste is transported to an intelligent sorting station where a conveyor belt moves waste items beneath high-resolution cameras. Images captured by the cameras are analysed using Convolutional Neural Networks (CNNs) to classify waste into plastics, paper, glass, metals, organic waste, electronic waste, and hazardous waste [5], [6]. Based on the classification results, robotic arms or automated actuators separate each waste category into designated containers for recycling, composting, energy recovery, or safe disposal. This process improves recycling efficiency, reduces contamination, and minimizes direct human contact with hazardous waste.

F. Monitoring and Decision Support Layer

The final layer consists of a web-based dashboard that provides municipal authorities with real-time information on waste levels, vehicle locations, collection schedules, recycling statistics, and system alerts. The dashboard supports data-

driven decision-making by enabling operators to monitor system performance, evaluate operational efficiency, and respond promptly to emerging waste management challenges [10].

IV. EXPECTED BENEFITS AND RECOMMENDATIONS

A. Expected Benefits

The proposed AI-IoT framework offers significant environmental, economic, and operational benefits for municipal solid waste management in Nigeria. By enabling real-time monitoring of waste bins, the framework ensures that collection vehicles operate only when necessary, reducing unnecessary trips, fuel consumption, and operational costs [4], [10]. AI-based route optimization further improves collection

efficiency by determining the shortest and most effective routes for waste collection vehicles. The integration of AI-powered waste sorting is expected to improve waste segregation and recycling efficiency. Automated classification using computer vision reduces contamination of recyclable materials while minimizing direct human contact with hazardous waste [5], [6]. Consequently, recyclable materials such as plastics, metals, paper, and glass can be recovered more effectively, supporting Nigeria's transition towards a circular economy [11]. From an environmental perspective, timely waste collection minimizes overflowing bins, illegal dumping, and blocked drainage systems, thereby reducing land, water, and air pollution. Improved sanitation also contributes to better public health by reducing the spread of waste-related diseases and creating cleaner urban communities.

TABLE II. Comparison Between Conventional and Proposed Waste Management Systems [7], [12]

Feature	Conventional System	Proposed AI-IoT Framework
Waste Monitoring	Manual inspection	Real-time IoT sensing
Collection Scheduling	Fixed timetable	AI-based dynamic scheduling
Route Planning	Static routes	AI route optimisation
Waste Sorting	Manual	AI-powered automated sorting
Decision Making	Reactive	Data-driven and predictive
Environmental Impact	Low efficiency, high pollution	High efficiency, reduced pollution and emissions

B. Recommendations for Implementation

To facilitate successful implementation in Nigeria, the following recommendations are proposed:

Government agencies should implement pilot AI-IoT waste management projects in major cities such as Lagos, Abuja, and Port Harcourt before nationwide deployment.

Public-private partnerships should be encouraged to support investment in smart waste management infrastructure and digital technologies, as this requires substantial financial investment, which may be difficult for some local governments [1], [10].

Municipal waste management personnel should receive training on the operation and maintenance of IoT devices and AI-based sorting systems.

Public awareness campaigns should promote waste segregation at source to improve the accuracy and efficiency of AI-based sorting.

Policies and regulations should be strengthened to support data-driven waste management and encourage collaboration between government, private stakeholders, and communities.

V. CHALLENGES AND FUTURE WORK

Despite its potential advantages, the implementation of the proposed framework may face several challenges. The deployment of IoT devices, cloud infrastructure, and AI systems requires substantial financial investment, which may be difficult for some local governments. In addition, inconsistent electricity supply and unreliable internet connectivity may affect the continuous operation of smart waste monitoring systems [1], [10]. Cybersecurity and data privacy also remain important concerns, because IoT devices are vulnerable to unauthorized access if appropriate security measures are not implemented. Furthermore, the successful

operation of the framework depends on the availability of skilled personnel capable of maintaining IoT infrastructure and interpreting AI-generated insights. Future work should focus on developing low-cost smart waste bins using locally available components, improving the robustness of AI models under varying lighting and environmental conditions, and conducting pilot studies to validate the framework's real-world performance across diverse Nigerian urban settings.

VI. CONCLUSION

This paper presented an integrated AI-IoT framework for intelligent waste collection and automated waste sorting in Nigeria. The proposed framework combines IoT-enabled smart bins, cloud computing, artificial intelligence, and computer vision to provide real-time waste monitoring, intelligent collection scheduling, route optimization, and automated waste classification. By integrating these technologies into a unified system, the framework addresses key challenges associated with conventional waste management, including inefficient collection schedules, poor waste segregation, and low recycling rates. The proposed framework is expected to improve operational efficiency, reduce environmental pollution, increase recycling performance, and support data-driven decision-making for municipal waste management authorities. With appropriate investment, supportive government policies, and collaboration among stakeholders, the framework has the potential to contribute significantly to sustainable waste management and smart city development in Nigeria [10], [12]–[14].

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The Evaluation of the Role of Artificial Intelligence in Real Estate Marketing in Ijebu-Ode, Ogun State

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Abstract—This study evaluated the role of Artificial Intelligence (AI) in real estate marketing in Ijebu-Ode. The increasing adoption of digital technologies across industries has created opportunities for AI to transform traditional real estate marketing practices through enhanced customer engagement, market analysis, and marketing efficiency. Despite these opportunities, the level of AI adoption among real estate practitioners in many developing urban centres remains uncertain. The study therefore examined the level of awareness and adoption of AI tools, identified the benefits of AI applications in sustainable real estate marketing, and assessed the challenges affecting AI adoption in the study area. Descriptive research design was adopted for the study. The population consisted of real estate developers, information technology personnel, and estate surveyors and valuers operating within Ijebu-Ode. A purposive sampling technique was employed, and a total of 60 questionnaires were distributed to respondents. Out of these, 49 questionnaires were retrieved and found suitable for analysis, representing a response rate of 81.7%. Data collected were analyzed using frequency distribution tables, percentages, mean scores, standard deviation, and ranking. The findings revealed that there is a high level of awareness of Artificial Intelligence among real estate practitioners, although the actual level of adoption remains moderate. The study further established that AI contributes significantly to marketing efficiency, customer engagement, property promotion, and decision-making processes. Respondents agreed that AI tools enhance market reach, improve communication with clients, and support data-driven marketing strategies. However, the study identified major challenges hindering AI adoption, including high implementation costs, inadequate technological infrastructure, low digital literacy, poor data availability, and resistance to change among practitioners. The study concluded that Artificial Intelligence has a positive and significant impact on sustainable real estate marketing and possesses enormous potential for improving the efficiency and competitiveness of the real estate sector in Ijebu-Ode. The study recommended increased investment in digital infrastructure, regular training and capacity-building programmes for practitioners, development of affordable AI solutions, and greater support from professional bodies and policymakers to encourage wider adoption of AI technologies in sustainable real estate marketing.

Index Terms—Artificial Intelligence, Digital Technology, Marketing, Property, Real Estate, Real Estate Marketing, Sustainable, Ijebu-Ode.

I. INTRODUCTION

The real estate industry globally is undergoing a transformative shift due to rapid technological advancements, particularly the adoption of Artificial Intelligence (AI), which has redefined traditional marketing practices and the overall property transaction process. AI encompasses machine learning, predictive analytics, chatbots, virtual tours, and other data-driven tools that enable real estate marketers to automate routine tasks, personalize client engagement, and forecast market trends with greater precision [1].

In contemporary real estate practice, AI-driven digital marketing strategies have emerged as key facilitators of sustainable competitive advantage. For example, AI-enabled

analytics and digital advertising tools allow practitioners to tailor property promotions effectively to segments of the market based on consumer behaviour, search patterns, and online engagement metrics [2].

The integration of digital technologies in real estate marketing has gained momentum as firms seek to overcome traditional barriers such as limited market transparency, high transaction costs, and inefficient communication with clients. AI technologies, including chatbots, virtual property tours, and data analytics, are increasingly being utilized to enhance customer service, streamline information dissemination, and improve lead generation [3].

Despite these opportunities, the adoption of AI in the real estate sector in many African settings, including Nigeria, remains relatively nascent due to constraints such as inadequate technological infrastructure, limited data availability, and low levels of digital literacy among practitioners [4].

Specifically in local markets like Ijebu-Ode, real estate marketing practices are still largely dominated by conventional promotional approaches that rely heavily on interpersonal networks and physical property visits. However, the increasing penetration of digital platforms and mobile technologies presents a unique opportunity for AI to enhance real estate marketing efficiency, broaden market reach, and improve client engagement in the region.

A. Statement of the Problem

The real estate sector has increasingly embraced digital technologies as a means of improving marketing efficiency, customer engagement, and market competitiveness. Artificial Intelligence (AI), in particular, has emerged as a powerful tool in real estate marketing through applications such as data-driven property recommendations, virtual property tours, automated customer interaction, and predictive market analysis. Despite these global advancements, the adoption and effective utilization of AI in real estate marketing remain uneven, especially in developing economies such as Nigeria [5].

In Ijebu-Ode, Ogun State, the real estate market is experiencing gradual growth due to urban expansion, educational institutions, and commercial activities. However, real estate practitioners in the area appear to have limited exposure to advanced digital and AI-driven marketing tools. Factors such as inadequate technical knowledge, high cost of technology adoption, poor digital infrastructure, and resistance to change may hinder the integration of AI into real estate marketing practices within the town [6]. This situation raises concerns about the competitiveness and efficiency of real

estate marketing in the area when compared with more technologically advanced urban markets.

B. Aim and Objectives of the Study

The main aim of this study is to evaluate the role of Artificial Intelligence in real estate marketing in Ijebu-Ode, Ogun State, with a view to assessing its level of adoption, benefits, and challenges among real estate practitioners.

The specific objectives of the study are to:

- i. examine the level of awareness and adoption of Artificial Intelligence tools in real estate marketing in study area;
- ii. identify the benefits of Artificial Intelligence in real estate marketing within the study area;
- iii. assess the challenges of Artificial Intelligence adoption in real estate marketing in the study area.

C. Scope of the Study

The scope of the study is limited to Ijebu-ode as a result of growing urban centers within Ogun state with increasing real estate activities. The study examines the level of awareness and adoption of Artificial Intelligence tools among real estate practitioners in the study area. It also covers the types of Artificial Intelligence applications used in real estate marketing, such as online property platforms, virtual property tours, data analytics, automated customer communication, and digital advertising tools. The study focused on real estate practitioners, such as Estate Surveyors and Valuers, Real Estate Developers, and Information Technology Personnel operating within Ijebu-Ode.

II. LITERATURE REVIEW

A. Concept of Artificial Intelligence

Artificial Intelligence (AI) refers to the capability of computer systems and machines to perform tasks that would normally require human intelligence, such as learning from data, reasoning, problem-solving, perception, and decision-making. In essence, AI enables machines to simulate cognitive functions traditionally associated with human minds, allowing them to interpret environments and act in ways that achieve specific goals without constant human intervention [7].

At its core, AI encompasses a broad array of technologies, including machine learning, natural language processing, expert systems, robotics, and computer vision, each of which contributes to different aspects of intelligent behaviour. These techniques allow AI systems to process large amounts of data, recognize patterns, make predictions, and continuously improve performance over time through experience [8]. AI has been rapidly adopted to enhance operational efficiency, improve decision-making, and provide personalized experiences for customers. For example, AI can be used to analyze consumer trends, segment markets, automate repetitive tasks, and deliver insights that support strategic planning and competitive advantage.

Within marketing, AI plays a significant role through automated digital advertising, chatbots for customer interaction, and recommendation systems that tailor content to user preferences. These applications have transformed how

organizations approach audience engagement and optimize marketing campaigns [3]. Artificial Intelligence represents a transformative set of technologies that expand the capabilities of machines to function intelligently and support complex tasks that enhance business processes, including those related to marketing and customer engagement [9].

B. Real Estate Marketing

Real estate marketing refers to the strategies, techniques, and tools employed by real estate practitioners to promote properties, attract potential buyers or tenants, and facilitate property transactions. It encompasses both traditional and digital approaches, aimed at creating awareness, generating leads, and improving the overall efficiency of property sales and leasing [10]. Traditionally, real estate marketing relied heavily on physical methods such as signboards, flyers, newspapers, exhibitions, and word-of-mouth networks. While these methods remain relevant, they are often limited in reach, slow, and sometimes inefficient in matching properties with the right target audience [1].

With the increasing penetration of digital technologies and the internet, the practice of real estate marketing has gradually evolved to include online listings, social media promotion, email marketing, virtual property tours, and AI-driven tools that analyze buyer behaviour and preferences [11]. These modern approaches enable real estate practitioners to reach a wider audience, reduce transaction time, and enhance customer engagement through personalized services.

Real estate marketing also involves strategic pricing, market research, and targeted communication, which are essential for positioning properties competitively in the market [12]. Understanding consumer behaviour, local demand, and economic trends allows practitioners to tailor marketing strategies and achieve optimal results [3].

In Nigeria, particularly in medium-sized towns such as Ijebu-Ode, real estate marketing is at a transitional stage where digital adoption is increasing but not yet fully integrated. Many practitioners still combine traditional methods with digital tools, creating a hybrid approach that reflects both innovation and local market realities [13].

C. Digital Transformation in the Real Estate Sector

Digital transformation in the real estate sector refers to the integration of digital technologies into all aspects of property marketing, sales, and management, fundamentally changing how practitioners operate and interact with clients [14]. This transformation has been driven by the widespread adoption of the internet, mobile technologies, and advanced software platforms, enabling real estate businesses to become more efficient, data-driven, and customer-focused [15].

The shift from traditional to digital marketing methods has introduced tools such as online property listings, social media platforms, email marketing, mobile applications and virtual property tours. These technologies allow practitioners to reach a wider audience, provide real-time property information, and personalize marketing strategies based on user behaviour [11]. For example, online property portals in Nigeria now host thousands of listings accessible to potential buyers nationwide,

reducing the reliance on physical property inspections and expanding market reach.

Digital transformation has also facilitated the use of data analytics and AI-driven insights. Real estate firms can now collect and analyze large volumes of market data to identify trends, predict property values, and tailor marketing campaigns to specific client segments [13]. Such capabilities improve decision-making, reduce marketing costs, and enhance the effectiveness of property promotion efforts.

D. Application of Artificial Intelligence in Real Estate Marketing

Artificial Intelligence (AI) has become a transformative tool in real estate marketing, offering innovative ways to reach potential clients, streamline operations, and improve decision-making. The applications of AI in real estate marketing are diverse, ranging from automated customer interactions to predictive analytics and personalized marketing campaigns [14].

Chatbots and virtual assistants: These tools enable real estate firms to provide instant responses to client inquiries, schedule property viewings, and offer personalized recommendations, enhancing customer engagement and reducing the workload on human agents [11]. Chatbots can also collect valuable data on customer preferences, which can inform future marketing strategies.

Predictive analytics: this is another key application. AI algorithms analyze historical market data, property trends, and buyer behaviour to forecast property demand, suggest optimal pricing, and identify potential investment opportunities. This allows real estate practitioners to make data-driven decisions that improve marketing efficiency and transaction outcomes [16].

Property recommendation systems: which leverage machine learning to match clients with properties that align with their preferences. By analyzing user behaviour on property portals and online listings, AI can deliver personalized property suggestions, increasing the likelihood of successful transactions.

Virtual property tours and augmented reality (AR): are additional AI-driven applications. These technologies allow potential buyers or tenants to explore properties remotely, providing a realistic and immersive experience without the need for physical visits. This is particularly valuable in reaching clients who are geographically distant or have limited time for in-person property inspections [17].

Digital marketing optimization: Automated advertising platforms use AI to target the right audience with tailored property promotions across social media, search engines, and email campaigns. This increases marketing reach, reduces costs, and enhances return on investment by ensuring that marketing efforts are directed at interested and qualified clients [15].

E. Benefits of Artificial Intelligence in Real Estate Marketing

The adoption of Artificial Intelligence (AI) in real estate marketing offers numerous benefits that improve operational efficiency, client engagement, and decision-making processes.

These benefits are crucial for practitioners aiming to remain competitive in a dynamic property market [14].

Enhanced marketing efficiency: AI tools automate repetitive tasks such as responding to client inquiries, scheduling property viewings, and sending follow-up messages. This reduces the workload on agents and allows them to focus on strategic activities, leading to faster property transactions and improved productivity.

Improved client targeting and personalization: AI algorithms analyze customer data, browsing patterns, and preferences to deliver tailored property recommendations and customized marketing messages. This targeted approach increases engagement, reduces marketing waste, and enhances the likelihood of successful property sales or rentals.

Data-driven decision-making: is also a key benefit. AI systems process large volumes of data to generate insights on market trends, property valuations, and customer behaviour. These insights enable practitioners to make informed pricing decisions, identify high-potential investments, and optimize marketing strategies.

Customer engagement and satisfaction: Tools such as chatbots, virtual assistants, and virtual property tours provide clients with instant support, immersive property experiences, and continuous interaction. These capabilities enhance trust, improve client experience, and encourage long-term relationships between clients and real estate firms.

Competitive advantage: By integrating advanced technologies, firms can differentiate themselves from competitors, attract tech-savvy clients, and respond more effectively to market changes. This is especially important in emerging urban centres like Ijebu-Ode, where digital adoption is gradually reshaping property marketing [18].

F. Review of Past Studies

Adeyemi and Oladipo [14] conducted a study on the adoption of AI in Nigerian real estate firms and found that AI tools, including chatbots, virtual property tours, and predictive analytics, significantly improved marketing efficiency and customer engagement. Their research demonstrated that firms that integrated AI experienced faster lead generation, better client targeting, and enhanced decision-making compared to firms relying solely on traditional marketing methods.

Eze and Ojo [11] examined AI adoption in real estate marketing across urban centers in Nigeria. They reported that predictive analytics and automated property recommendations increased client satisfaction and streamlined property transactions. The study highlighted that digital adoption, though growing, was still limited in medium-sized towns due to infrastructural and skill-related constraints.

Bello and Ogunyemi [13] investigated the impact of AI-driven marketing on property sales in medium-sized Nigerian towns. Their findings revealed that AI applications such as automated advertising and virtual property tours enhanced market reach and improved client engagement, leading to higher transaction rates. The study also noted that smaller firms often faced challenges in implementing these technologies due to high costs and low technological literacy.

Adepoju and Oladimeji [19] focused on digital marketing strategies in urban Nigerian real estate and emphasized that AI-enabled tools, including recommendation systems and chatbots, were crucial for enhancing personalization and responsiveness in property marketing. Their empirical results indicated that clients exposed to AI-driven interactions were more likely to convert leads into actual property transactions.

Olawale and Adekunle [17] examined the broader impact of digitalization on real estate marketing in Nigeria and observed that AI contributed significantly to operational efficiency, customer relationship management, and data-driven decision-making. However, the study highlighted that regulatory frameworks, data quality, and ethical considerations remain critical factors for successful AI adoption.

III. METHODOLOGY

Quantitative research methodologies were employed for this study. The target population for this study includes Estate Surveyors and Valuer, Real Developers and Information Technology Personnel. The total population for the study is sixty (60) respondents which 20 Estate Surveyors and Valuer, 30 Real Estate Developers and 10 Information Technology Personnel. The study uses a purposive sampling technique to select Estate Surveyors, Real Estate Developers, Information Technology Personnel. The sample size for the study consists of 20 Estate Surveyors and Valuer, 30 Real Estate Developers and 10 Information Technology Personnel. Data were collected through well-structured questionnaire. Data were analyzed through descriptive statistics such as frequency, percentages, mean and standard deviation.

IV. RESULTS AND DISCUSSION

Questionnaire was distributed to 60 respondents which comprises of 20 Estate Surveyors and Valuer, 30 Real Estate Developers, and 10 Information Technology Personnel within the study area.

TABLE I. Distribution of Questionnaire and Retrieval

S/n	Respondent	Total distributed	Total received	Percentage
1	Information Technology Personnel	10	8	16.33%
2	Real Estate Developer	30	24	48.98%
3	Estate Surveyor and Valuer	20	17	34.69%
	Total	60	49	100%

Source: Author's Field Survey, 2026

A total number of 49 questionnaires were filled properly and returned out of 60 questionnaires administered on the respondents. 49 respondents consists of 8 Information Technology Personnel, 24 Real Estate Developers, 17 Estate Surveyors and Valuer in Ijebu-Ode. The retrieval rate of the respondents is 81.7%.

TABLE II. Gender Distribution

Gender	Frequency	Percentage (%)
Male	32	65.3%

Gender	Frequency	Percentage (%)
Female	17	34.7%
Total	49	100%

Source: Author's Field Work, 2026

Table II results shows that 32 (65.3%) of the respondents are male while 17 (34.7%) of the respondents are female. The majority of respondents were male (65.3%), indicating that real estate practice in the study area is male-dominated.

TABLE III. Age Distribution

Age Group	Frequency	Percentage (%)
20–30	8	16.3%
31–40	18	36.7%
41–50	15	30.6%
51+	8	16.3%
Total	49	100%

Source: Author's Field Work, 2026

The results from the above Table show that most respondents (36.7%) are within 31–40 years, indicating that the industry is dominated by active working-age professionals.

TABLE IV. Awareness and Adoption of AI in Real Estate Marketing

Item	SA	A	N	D	SD	Mean	Std. Dev	Rank
Awareness of AI tools	20	18	6	3	2	4.04	0.95	1 st
Firm uses AI tools	15	17	8	6	3	3.67	1.09	3 rd
AI improves client reach	18	19	7	3	2	3.98	0.97	2 nd
Trained in AI tools	10	15	9	9	6	3.10	1.21	5 th
AI is essential in marketing	17	18	6	5	3	3.82	1.04	4 th

Source: Author's Field Work, 2026

The results show a positive level of awareness and moderate adoption of AI tools among real estate practitioners. The highest-ranked item indicates that respondents are generally aware of AI technologies, but lower scores on training suggest insufficient technical capacity for full adoption.

TABLE V. Benefits of AI in Real Estate Marketing

Item	SA	A	N	D	SD	Mean	Std. Dev	Rank
Improves marketing efficiency	22	18	5	3	1	4.16	0.88	1 st
Enhances targeting & personalization	20	17	6	4	2	3.98	0.96	2 nd
Improves customer engagement	19	16	7	5	2	3.86	1.01	3 rd
Improves decision making	18	17	6	5	3	3.80	1.05	4 th
Improves overall performance	21	15	7	4	2	3.94	0.98	5 th

Source: Author's Field Work, 2026

The result indicates that respondents strongly agreed that AI improves marketing efficiency, targeting, and overall performance. This indicates that AI is perceived as a valuable

tool for enhancing real estate marketing effectiveness in the study area.

TABLE VI. Challenges of AI Adoption in Real Estate Marketing

Item	SA	A	N	D	SD	Mean	Std. Dev	Rank
High cost of AI tools	23	16	5	3	2	4.12	0.92	1 st
Poor infrastructure	20	15	6	5	3	3.88	1.03	2 nd
Lack of digital literacy	21	14	7	4	3	3.90	1.02	3 rd
Poor data availability	18	15	8	5	3	3.73	1.07	4 th
Resistance to change	17	16	7	5	4	3.65	1.09	5 th

Source: Author's Field Work, 2026

The findings reveal that cost of AI tools and poor infrastructure are the major challenges limiting adoption. However, all items scored above 3.0, meaning respondents generally agree that multiple barriers still exist.

A. Discussion of Findings

The findings revealed that real estate practitioners in Ijebu-Ode have a high level of awareness of Artificial Intelligence tools, as shown by the high mean scores in the awareness items. However, actual adoption remains moderate rather than extensive, particularly in areas such as training and formal integration of AI systems in marketing activities.

This suggests that while practitioners are familiar with AI concepts such as chatbots, predictive analytics, and virtual tours, many have not fully incorporated these tools into their daily operations. This finding aligns with Eze and Ojo [11], who observed that AI adoption in medium-sized Nigerian towns is still developing due to infrastructural and skill-related constraints. It also supports the view of Adeyemi and Oladipo [14] that awareness does not always translate into full technological adoption, especially in emerging markets.

The study found that AI significantly improves marketing efficiency, customer engagement, targeting, and overall marketing performance. Respondents strongly agreed that AI enables faster communication with clients, enhances personalization of property listings, and improves decision-making processes in property marketing.

This is consistent with the findings of Olawale and Adekunle [17], who noted that AI-driven tools improve operational efficiency and customer relationship management in real estate. Similarly, Adepoju and Oladimeji [19] emphasized that AI-based recommendation systems and chatbots increase client satisfaction and conversion rates by providing tailored property solutions.

The study identified several challenges limiting the full adoption of AI in real estate marketing. The most significant constraints include high cost of AI tools, poor technological infrastructure, lack of digital literacy, inadequate data availability, and resistance to change.

The prominence of cost and infrastructure challenges reflects the realities of developing markets where advanced technologies require significant investment and stable digital infrastructure. This finding is consistent with Bello and Ogunyemi [13], who reported that financial limitations and

poor technological systems hinder AI adoption in medium-sized Nigerian towns.

V. CONCLUSION AND RECOMMENDATION

The study established that although awareness of AI among practitioners is relatively high, full adoption remains limited due to technological, financial and human-capacity constraints. However, those practitioners who have adopted AI tools reported improvements in operational efficiency, customer satisfaction, and marketing effectiveness.

Therefore, the study concludes that increased adoption of Artificial Intelligence in real estate marketing will significantly enhance productivity, competitiveness and service delivery within the real estate sector.

The study recommended that real estate practitioners should undergo regular training and workshops on Artificial Intelligence applications to improve their digital skills and understanding of AI tools in property marketing. Government and private stakeholders should improve digital infrastructure such as stable electricity supply and internet connectivity to support AI adoption in real estate businesses. Software developers and technology providers should develop affordable AI tools tailored to the needs of small and medium-sized real estate firms.

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BIOGRAPHY

Mr. Hassan Shehu was born in Aiyeye-Ijebu, Ogun State, Nigeria. He began his early education at Ansar-Ud-Deen Nursery and Primary School before proceeding to Community Grammar School, Aiyeye-Ijebu, for his secondary education. During his secondary school years, he distinguished himself academically and emerged as the Best Mathematics Student in Ogun State. Driven by a passion for excellence, Mr. Hassan pursued higher education at the Federal University of Technology, Minna, where he obtained a Bachelor of Technology (B.Tech.) degree in Estate Management and Valuation. He subsequently earned a Master of Science (M.Sc.) degree in Estate Management with distinction from Lead City University, Ibadan. In recognition of his outstanding academic performance, he was awarded a scholarship to pursue a Doctor of Philosophy (Ph.D.) in Estate Management at Lead City University, Ibadan.

He began his professional career as an Assistant Estate Surveyor with Jide Taiwo & Co., Ijebu-Ode Branch, where he gained practical experience in Estate Surveying and Valuation. Motivated by a strong passion for teaching, research, and academic development, he later joined the Department of Estate Management and Valuation at Gateway (ICT) Polytechnic, Saapade, Ogun State, as a lecturer.

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APPRECIATION

The Organising Committee of ICSEECR 2026 extends its sincere appreciation to the Management, Staff and Students of Gateway (ICT) Polytechnic, Saapade, for their unwavering support in bringing this conference to life.

We are equally grateful to the Rector, the Deans of the School of Engineering & Technology and the School of Environmental Studies, the Technical Programme Committee, our distinguished Chief Guest of Honour, Keynote Speakers and Lead Paper Presenters, and every reviewer whose diligence shaped this Book of Proceedings.

To every author and presenter who joined us from institutions across Nigeria and beyond: thank you for entrusting your scholarship to this platform. It is your research that gives this conference its purpose.

We wish all our delegates a rewarding conference and safe travels to and from Saapade.

Thank you all.

ICSEECR 2026 • Gateway (ICT) Polytechnic, Saapade, Ogun State, Nigeria • 20–24 July

2026

SPECIAL RECOGNITION

Nigerian Association of Technologists in Engineering (NATE)

Ogun State Branch



The Organising Committee of ICSEECR 2026 extends its warm appreciation to the Nigerian Association of Technologists in Engineering (NATE), Ogun State Branch, for its goodwill and continued partnership with Gateway (ICT) Polytechnic, Saapade, in the advancement of engineering and technology practice.

In furtherance of this partnership, NATE Ogun State Branch sponsored the Best Student Paper Presentation Award at this conference, an initiative that further honours excellence and inspires rigorous scholarship among the students presenting at ICSEECR 2026.

The Organising Committee looks forward to a continued and fruitful collaboration with NATE, in shared pursuit of its vision to design, build and maintain for progress.

With appreciation, on behalf of the Organising Committee.

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